

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

CS T32 ELECTRONIC DEVICES AND CIRCUITS

III-SEMESTER

Prepared By

T.KAVIARASAN. M.Tech (Assistant Professor)

Department Of Electronics and Communication Engineering

CS T32 ELECTRONIC DEVICES AND CIRCUITS

UNIT I

Diode and its Applications : PN junction diode, Diode equivalent circuit, Diode as a switch – transition and diffusion capacitance – reverse recovery time, Zener diode, Applications of diode – AND/OR gates using diodes, Clippers and clampers – Voltage doubler and tripler – Voltage regulation – Series and shunt voltage regulators.

UNIT II

Biasing and Modeling for BJT and FET : Biasing and operating point, BJT Bias circuits – Fixed-bias, Emitter stabilized bias, Voltage divider bias and DC bias with voltage feedback, FET biasing – Fixed-bias, Self-bias, Voltage-divider bias, MOSFET biasing. Transistor modeling – Important parameters of BJT- h - parameter model of BJT (CE only) – Important parameters of JFET, Small signal model of JFET and MOSFET

UNIT III

RC-coupled amplifier, Operation and Frequency response, Power amplifier – Series fed and transformer coupled Class A amplifiers, Class B amplifier, Circuit and Operation, conversion efficiency, amplifier distortion, Class C and D amplifiers. Concept of feedback- Negative and positive feedback, Barkhausen Criterion – Wien bridge oscillators, Hartley, Colpitts and crystal oscillator – Frequency stability.

UNIT IV

Operational Amplifier: Introduction to op-amp, Characteristics of op-amp. – Differential and common mode operation, op-amp parameters – Applications: Inverting and non-inverting amplifier, summer, subtractor, differentiator, integrator, comparator, analog multiplier, second order low pass and high pass active filters.

UNIT V

Special Devices: Varactor diode, Tunnel diode, PIN diode, LED, LCD, Seven segment displays, Opto-isolator. UJT – Characteristics and equivalent circuit – intrinsic standoff ratio – UJT relaxation oscillator, SCR – Two transistor model, DIAC and TRIAC – Operation, Characteristics and their applications.

Text Books:

1. Robert L. Boylestad and Louis Nashelsky, “Electronic Devices and Circuit Theory” , Pearson Education, Tenth Edition, 2009.
2. Jacob Millman, C. Halkias and Satyabrata Jit, “Electronic Devices and Circuits” , Tata McGraw Hill, Third Edition, 2010.

Reference Books:

1. Jacob Millman and Arvin Grabel, “MicroElectronics” , Tata McGraw Hill, Second Edition, 2008.
2. David A. Bell, “Electronic Devices and Circuits” , Oxford University Press, Fifth Edition, 2008.

Websites:

1. www.ecee.colorado.edu, www-inst.eecs.berkeley.edu.

UNIT-1

Topic	Page. No
1.1 PN junction diode	5
1.1.1 Forward Bias Characteristics	5
1.1.2 Reverse Bias Characteristics	7
1.1.3 Diode Current Equation	7
1.1.4 Diode equivalent circuit and as a switch	8
1.1.5 Transition capacitance	9
1.1.6 Diffusion capacitance	10
1.1.7 Diode Switching Times	11
1.1.8 Applications	13
1.2 Zener diode	13
1.2.1 Characteristics of Zener Diode	14
1.2.2 Equivalent Circuit	15
1.2.3 Breakdown Mechanisms	16
1.2.3.1 Zener breakdown	16
1.2.3.2 Avalanche Breakdown	16
1.2.6 Applications	17
1.2.7 OR Gate Using Diodes	17
1.2.8 AND Gate Using Diodes	18
1.3 CLIPPERS	19
1.3.1 Positive Clipper:	19
1.3.2 Negative Clipper	20
1.3.3 Biased Clipper:	21
1.4 CLAMPERS	21
1.4.1 Biased Clampers:	25
1.5 VOLTAGE MULTIPLIERS:	25
1.5.1 Voltage doubler	25
1.5.2 Voltage Tripler	27
1.5.3 Voltage Quadrupler	28
1.5.4 Advantages	28

1.5.5 Limitations	28
1.5.6 Applications	28
1.6 VOLTAGE REGULATION	29
1.6.1 Line Regulation:	30
1.6.2 Load Regulation:	30
1.6.3 Types of Voltage Regulators	31
1.6.3.1 Series Voltage Regulators	31
1.6.3.2 Shunt Voltage Regulator	32

Department of Computer Science and Engineering

Subject Name: ELECTRONIC DEVICES AND CIRCUITS

Subject Code: CS-T32

Prepared by:

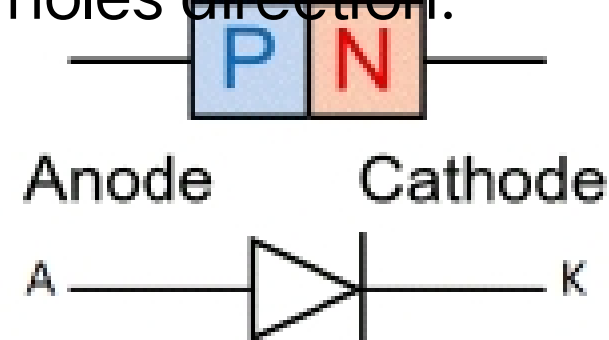
Mr. T.KAVIARASAN. M.Tech (Assistant
Professor)

UNIT-I

Diode and its Applications : PN junction diode, Diode equivalent circuit, Diode as a switch – transition and diffusion capacitance – reverse recovery time, Zener diode, Applications of diode – AND/OR gates using diodes, Clippers and clampers – Voltage doubler and tripler – Voltage regulation – Series and shunt voltage regulators.

1.1 PN JUNCTION DIODE:

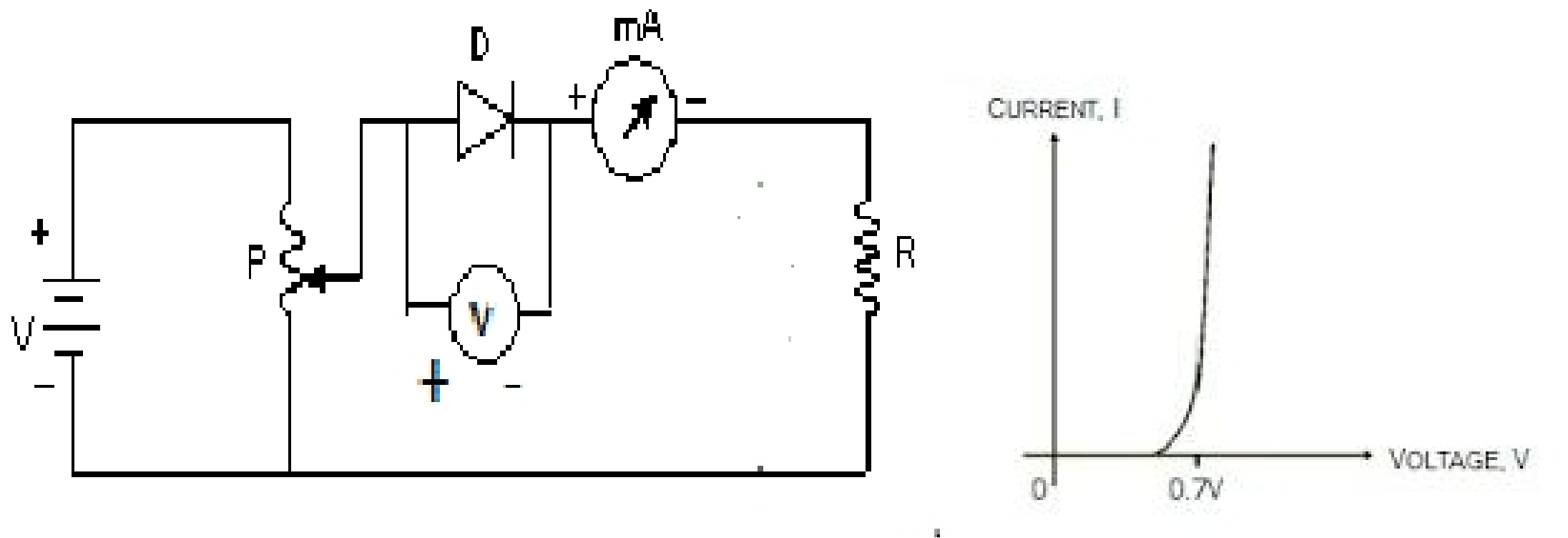
- ▶ It consists of PN junction formed either as germanium or silicon crystals.
- ▶ It has two terminals namely anode and cathode as shown in figure
- ▶ The arrow head represents the direction of current flow, when it is forward biased, current flows towards the holes direction.



1.1.1 Forward Bias Characteristic:

- ▶ Figure shows the circuit for forward characteristics of a diode
- ▶ DC voltage V_{AA} is connected to the diode through the potentiometer (P) and resistance (R)

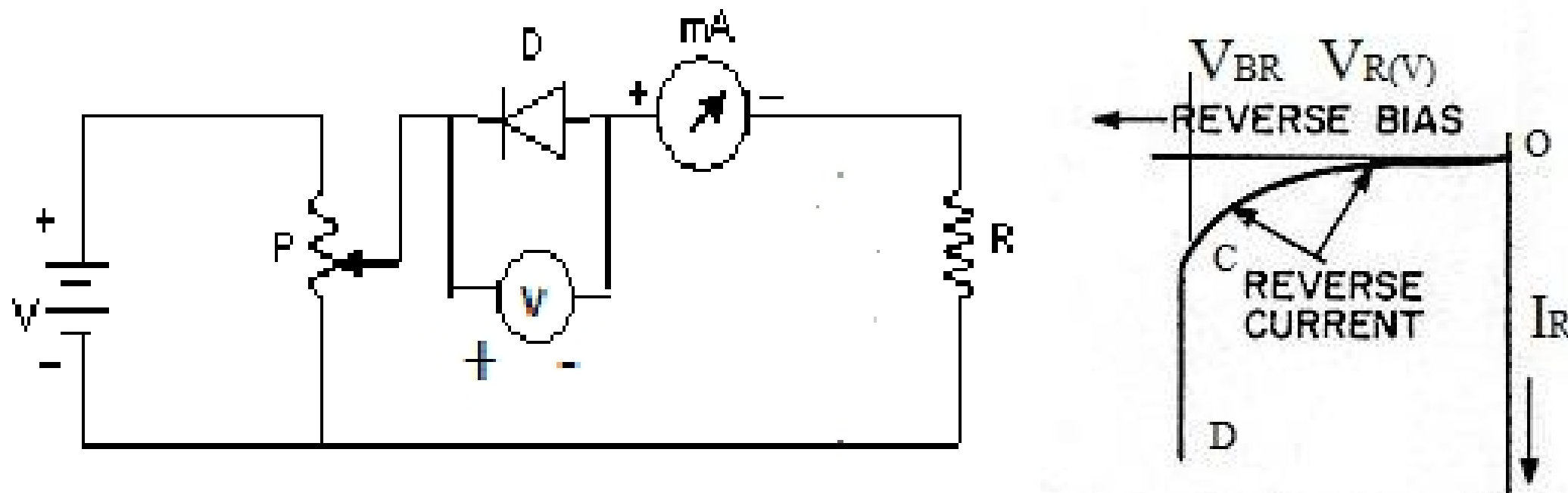
- ▶ Potentiometer is used for varying the voltage applied across the diode
- ▶ Resistance is used limit the current through the diode
- ▶ Voltmeter used to measure the voltage across the diode, whereas the milliammeter measures the current in the circuit
- ▶ The positive terminal of the voltage is connected to the anode and the negative terminal of the voltage is connected to the cathode
- ▶ Diode is in forward biased, by gradually increase the voltage is small steps of about 0.1V and read the corresponding values the diode current
- ▶ The graph is plotted between voltage across the diode with diode current(OAB) is called the forward characteristics of a silicon PN junction diode



- ▶ There is no diode current till the point A is reached, because that the external voltage is being opposed by the junction voltage 0.7 for silicon and 0.3 for germanium
- ▶ Further increasing voltage above A, the diode current increases rapidly
- ▶ Note the voltage of 1V produces the forward current of about (20-50)mA
- ▶ If the applied voltage should not be increased beyond a safe limit, otherwise the diode is likely burn out
- ▶ Diode starts conducting, the voltage is referred as knee voltage or threshold voltage or cut in voltage (V_k or V_g)
- ▶ 0.6V for silicon and 0.2V for germanium
- ▶ The knee voltage may be obtained from the graph by extending the AB backward, till it meet the horizontal axis
- ▶ It is equal to the knee voltage

1.1.2 Reverse Bias Characteristics:

- ▶ It is similar to the forward characteristics, expect two changes namely the diode terminals are reverse biased and the mille ammeter is replaced by the micro ammeter
- ▶ Circuit for reverse characteristics of a diode is shown in figure



- ▶ Negative terminals of the voltage is connected to the anode and a positive terminal is connected to the cathode of a diode
- ▶ The applied reverse voltage is gradually increased above zero and the values of diode current are recorded at each step
- ▶ The curve OCD is called reverse characteristics of the diode
- ▶ The applied reverse voltage is gradually increased above zero and the values of diode current are recorded at each step
- ▶ The curve OCD is called reverse characteristic of the diode. When the applied reverse voltage is below breakdown voltage (V_{BR}), the diode current is small and remains constant
- ▶ This current is called reverse saturation current (I_o), 1nA for silicon and 1 μ A for germanium diode
- ▶ When the reverse voltage is increased to a sufficiently large value, the diode reverse current is increases as rapidly (see CD curve)
- ▶ This voltage is referred as Breakdown voltage (V_{BR}).

1.1.3 Diode Current Equation

- ▶ The mathematical equation, it describes the forward and reverse characteristic of a semiconductor diode is called the diode current equation
- ▶ The current equation for a forward biased diode is given by

$$I = I_o (e^{V/(\eta V T)} - 1)$$

- ▶ Where I - Forward or reverse diode current

I_0 – Reverse saturation current

V – External voltage, positive for forward and negative for reverse biased

η – Constant (1 for Ge and 2 for Si for below the knee voltage) and (1 for both Ge and Si which higher levels of diode current)

V_T – Volt- equivalent of temperature $\frac{T}{11600}$

T = absolute temperature

- ▶ At room temperature(300K), $V_T = 26\text{mV}$

- ▶ On substituting in „I“ we get $I = I_0 e^{\frac{40V}{V_T}} - 1$

- ▶ As below knee voltage, I is based on η is given by

$$I = I_0 e^{\frac{40V}{V_T}} - 1 \quad [\eta = 1 \text{ for Ge}]$$

$$I = I_0 e^{\frac{20V}{V_T}} - 1 \quad [\eta = 2 \text{ for Si}]$$

- ▶ If applied voltage is greater than unity,

$$I = I_0 e^{\frac{40V}{V_T}} = I_0 e^{\frac{20V}{V_T}}$$

- ▶ The current equation for reverse biased, by changing the sign of the applied voltage(V) and the diode current is given by,

$$I = I_0 (e^{-V/(\eta V_T)} - 1)$$

If $V \gg V_T$, $e^{-V/(\eta V_T)} \ll 1$ Therefore $I = I_0$ (i.e) diode current is equal to reverse saturation current as the external voltage below the breakdown voltage.

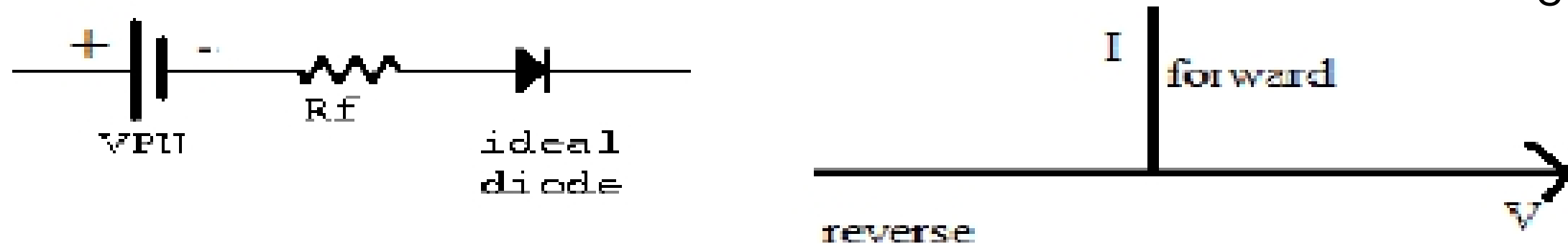
1.1.4 Diode Equivalent Circuit and As a Switch:

- ▶ The diode has a very important property that permits only unidirectional conduction (i.e) conducts only in forward direction when forward biased and conducts very poorly in reverse biased
- ▶ An ideal diode should conduct instantaneously when forward biased and should not conduct at all when reverse biased, the VI characteristics of such an ideal diode would be shown in figure. An ideal diode acts as a switch
- ▶ But in a practical diode, there is a voltage drop across a diode and it start conducting heavily after its cut in voltage(V_k) is exceeded by the forward biased voltage
- ▶ Similarly it is not a perfect conductor with zero resistance in forward biased condition and it has some finite resistance in forward biased condition

- ▶ The voltage drop across forward biased is equal to cut in voltage and the drop due to its forward resistance and due to forward current ($I_f R_f$)

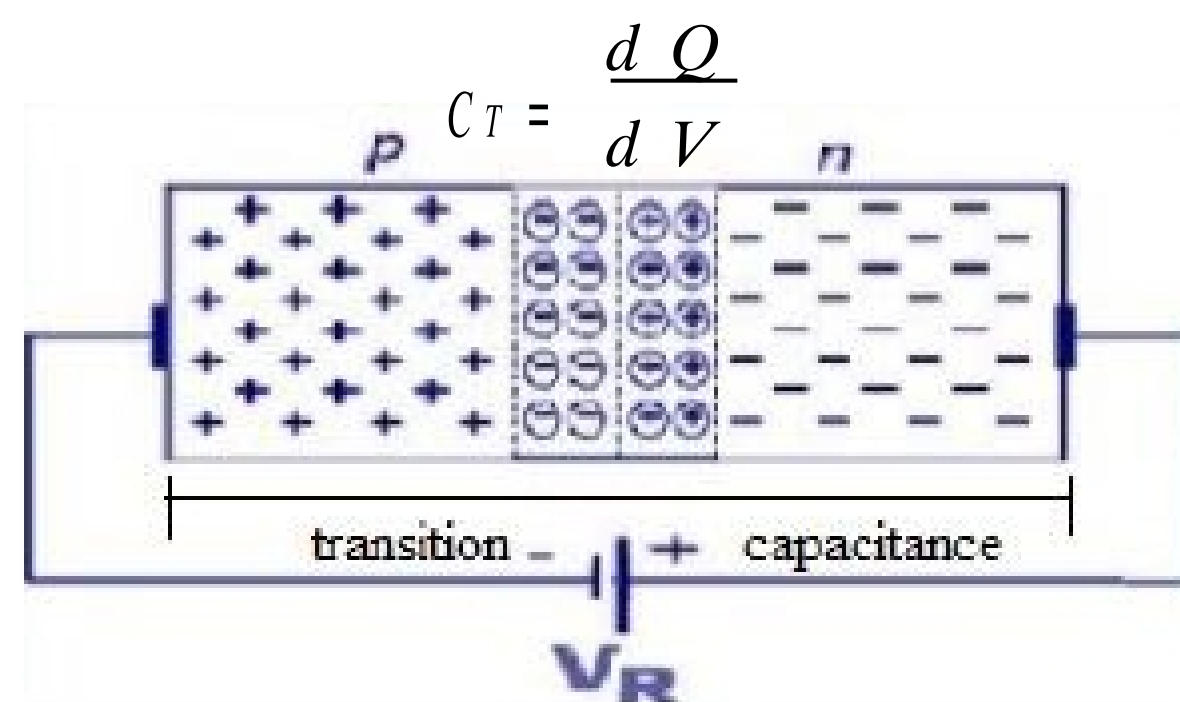
$$V = V_k + I_f R_f$$

- ▶ So a diode is replaced by a metal with a battery equal to cut in voltage of diode, the forward resistance of a diode in series with an ideal diode as shown in figure.



1.1.5 Transition Capacitance(C_T):

- ▶ Consider a reverse biased PN junction diode as shown in figure. When the diode is reverse biased, reverse current flow due to minority carrier. Majority carrier (electrons in N-region and holes in P-region) move away from the junction. This increases the width of depletion region. It increases as reverse bias voltage increases the charged particle move away from the junction there exists a change in charge with respect to applied reverse voltage. So change in dQ w.r.t change in dV is referred as capacitive effect. Such capacitance under reverse biased condition is called transition or space charge are barrier or depletion layer capacitance and denoted as C_T . It is given by



- ▶ It is not allowed but depends on magnitude of the reverse voltage. If the W width of the depletion region, then the C_T is given by,

$$C_T = \frac{\epsilon A}{W}$$

Where A - area of cross section of the junction

ϵ – Permittivity of the semiconductor

1.1.6 Diffusion Capacitance:

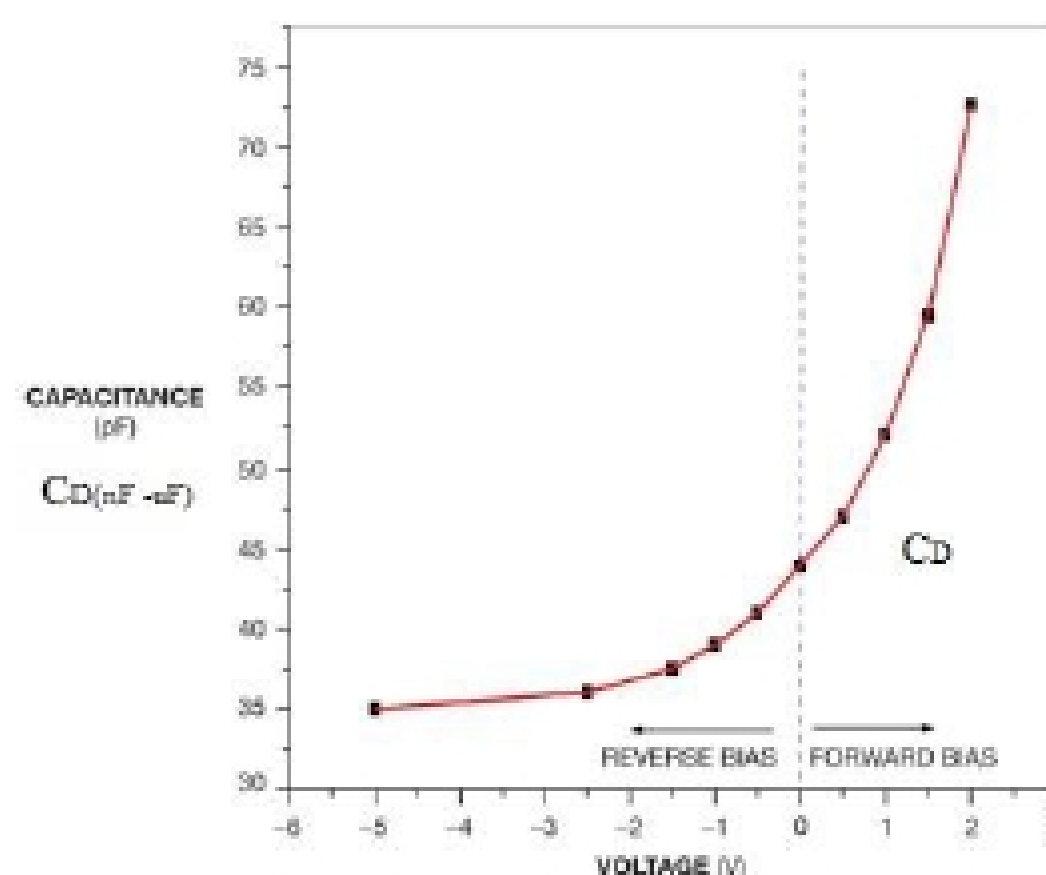
- ▶ During forward biased, diffusion or storage capacitance is exist denoted as C_D
- ▶ In forward biased the width of the depletion region, decreases holes from P-side get diffused in N-side while electrons from N-region move into P-side.
- ▶ As the applied voltage increases, concentration of injected charged particles increases, this rate of change of injected charge with applied voltage is defined as a capacitance called diffusion capacitance

$$C_D = \frac{dQ}{dV}$$

- ▶ It can be determined by the expression

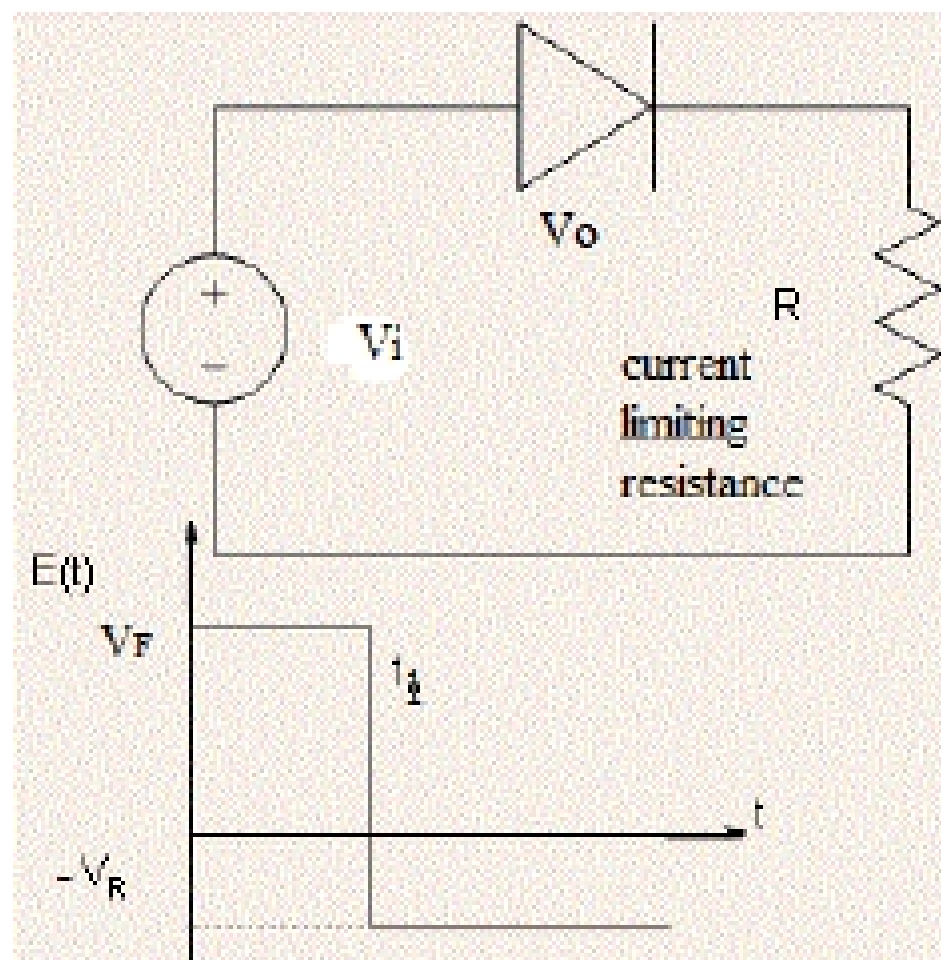
$$C_D = \frac{I}{\eta V_T}$$

- ▶ Where I mean life thrice for holes
- ▶ So C_D is proportional to the current
- ▶ For forward biased conditions, the value of C_D is of the order of nano farads to micro farads while C_T is of the order of pico farads (i.e) $C_D \gg C_T$
- ▶ C_D appears in parallel with a forward resistance which is very small, so the constant which is function of product of the forward resistance and C_D is also very small for ordinary signals
- ▶ C_D versus forward applied voltage graph is shown in figure

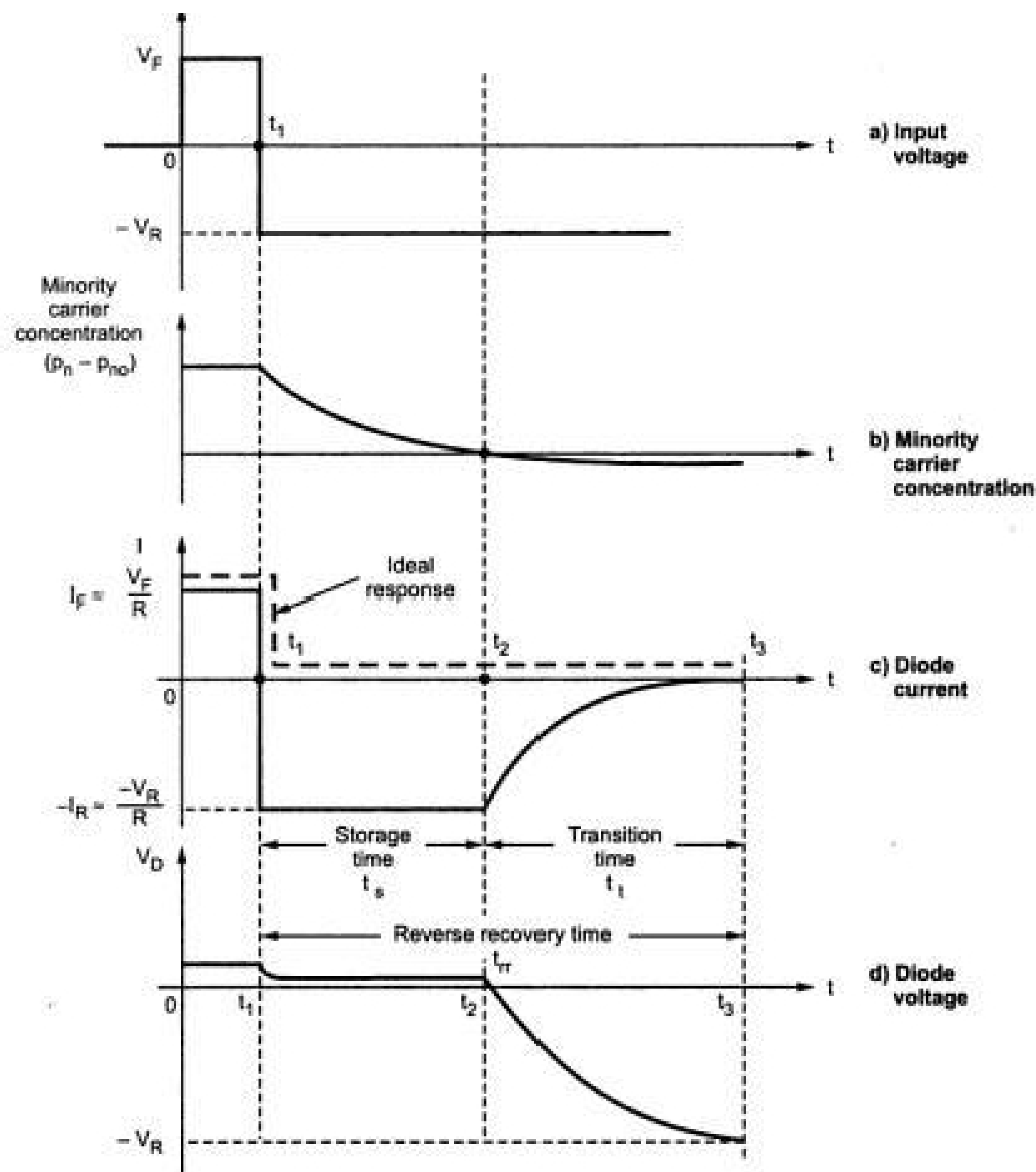


1.1.7 Diode Switching Times:

- To understand the various switching time, consider simple diode circuit and an input waveform is shown in figure



- The following reacts will take place due to the nature of the applied voltage
- Event 1: Till time t , the forward voltage applied is V_F and diode is forward biased. The value of R is large enough such that drop across forward biased diode is very small compared to drop across R . The forward current is then $I_F = V_F/R$ neglecting forward resistance of diode.
- Event 2: At time t , the applied voltage is suddenly reversed and reverse voltage of $-V_R$ is applied to the circuit. Ideally diode also must become OFF from ON state instantly. But this does not happen instantly. The number of minority carriers takes time to reduce from $p_n - p_{no}$ to zero at the junction as shown in figure. Due to this, at t , current just reverses and remains at that reversed value $-I_R$ till the minority carrier concentration reduces to zero. This current is given by $-I_R = -V_R/R$. This continuous to flow till time T_2 .
- Event 3: From t_2 onwards, the diode voltage starts to reverse and the diode current starts decreasing as shown in figure. At $t=t_3$, diode state completely gets reversed and attains stead state in reverse biased condition



Key Points

- ▶ During time t_1 to t_2 , the minority charge remains stored and decrease slowly to zero. This time is called storage time denoted as " t_s "
- ▶ The time from t_2 to t_3 , time required by the diode current to reduce to its reverse saturation value is called transition time denoted as " t_t "
- ▶ For quick switching from ON to OFF state, the reverse recovery time should be as small as practicable
- ▶ To have fast switching ON to OFF of a diode, the transition capacitance should be as small as possible
- ▶ The total time required by the diode which is the storage time and transition time, to recover completely from the change of state is called reverse recovery time of the diode and denoted as " t_{rr} ". It is an important consideration in high speed switching applications.

$$t_{rr} = t_s + t_t$$

- ▶ It depends on the RC time constant where C-transition capacitance of a diode. Therefore C_t plays an important role in the switching circuits using diodes
- ▶ Most commercially available switching diodes have „ t_{rr} “ ranging from a few nanoseconds to one microsecond. However, diodes are specially manufactured having „ t_{rr} “ as small as only a few picoseconds
- ▶ The total switching time „ t_{rr} “ puts the limit on the maximum operating frequency of the diode. Hence „ t_{rr} “ is an important used to minimize the effect of the reverse current, the time period of the operating frequency must be at least ten times „ t_{rr} “

$$T = 10 t_{rr} \quad \text{or} \quad f_{\max} = 1/T = 1/10t_{rr}$$

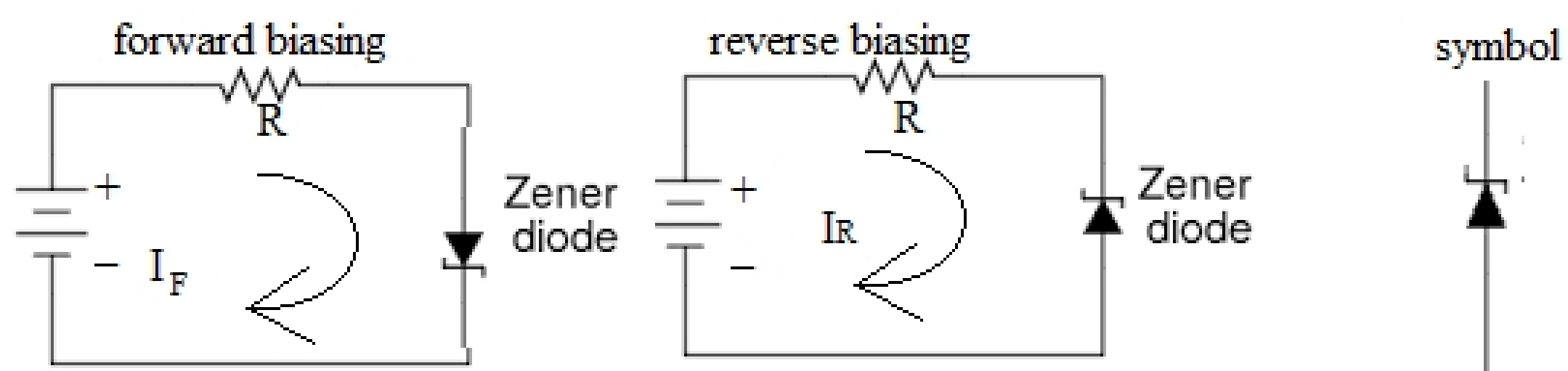
Where $f_{\max}$ – maximum operating frequency

1.1.8 Applications:

- ▶ Rectifiers or power diodes in DC power supplies
- ▶ Signal diodes in communication circuits
- ▶ Zener diodes in voltage switching circuits
- ▶ As varactor diodes in radio and TV receivers
- ▶ Switch in logic circuits used in computers

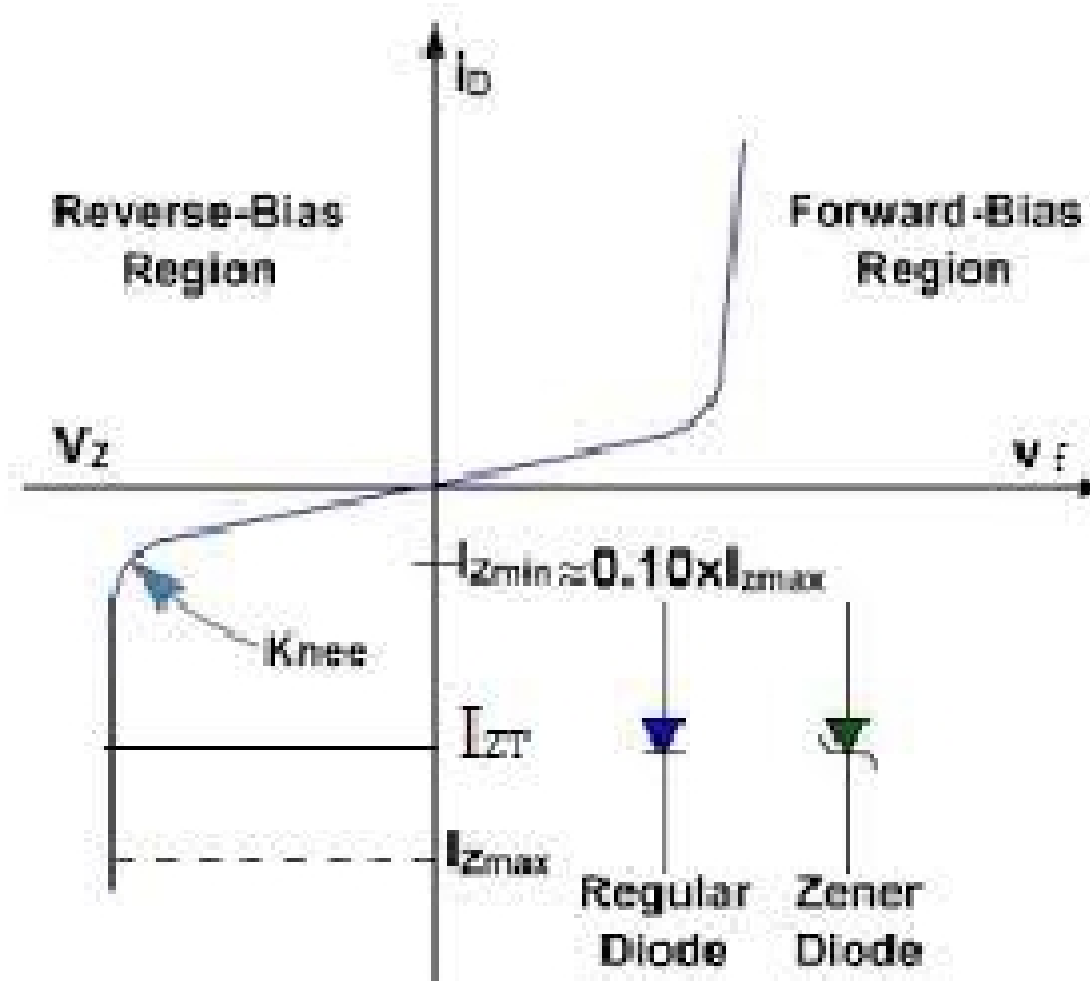
1.2 ZENER DIODE:

- ▶ It is a silicon PN junction semiconductor diode, which is generally operated in its reverse breakdown region
- ▶ Zener diodes are fabricated with precise breakdown voltages, by controlling the doping level during manufacturing
- ▶ It has a breakdown voltage range from (3-200)V
- ▶ Its symbol and forward, reverse biased modes are shown in figure.
- ▶ Practically zener diodes are used in reverse biased mode



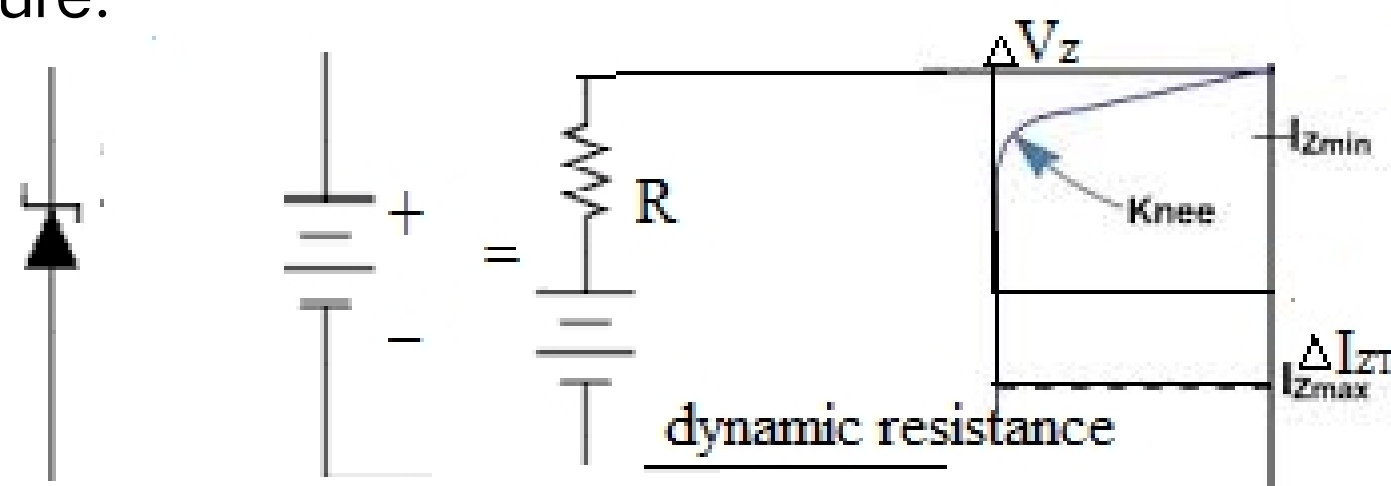
1.2.1 Characteristics of Zener Diode

- ▶ In the forward biased condition, the normal rectifier diode and the zener diode operate in similar fashion. But the zener diode is designed, to be operated in the reverse biased condition. In reverse biased condition, the diode carries reverse saturation current till the reverse voltage applied is less than the reverse breakdown voltage. When the reverse voltage exceeds reverse breakdown voltage, the current through it changes directionally but the voltage across it remains almost constant. Such a breakdown region is a normal operating region for a zener diode
- ▶ The breakdown characteristic for a zener diode is significantly important, as it is an operating region for the diode. When the reverse voltage applied to a zener diode is increased, initially the current through it is a very small, of the order of few μA or less. This is the reverse leakage current of the diode, denoted by I_o . At a certain reverse voltage, current through zener diode increases rapidly. The change from a low to large value of current is very sharp and well defined. Such a sharp change in the reverse characteristics is called knee or zener knee of the curve. At this knee, a breakdown is said to occur in the device. The reverse bias voltage at which the breakdown occurs is called zener breakdown voltage denoted as V_z .
- ▶ It is set by carefully controlling the doping level during manufacturing process. The current corresponding to a knee point is called zener knee current and it is a minimum current zener must carry to operate in reverse breakdown region. It is denoted as I_{CMIN} .
- ▶ From the bottom of the knee, the zener breakdown voltage remains about constant, through it increases slightly as the zener current I_z , increases. The current at which the nominal zener breakdown voltage is specified is called zener test current, denoted as I_{ZT} . This value and correspond zener voltage V_z are specified on datasheets. Every zener diode has a capacity to carry current. As current increases the power dissipated $P_z = V_z I_z$ increases. If the dissipation increases beyond certain value the device may get damaged.
- ▶ The maximum current a zener diode can carry safely is called zener maximum current is denoted I_{Zmax} . In practical circuit to limit the zener current between I_{zmin} and I_{zmin} a current limiting resistors used in series with zener diode. The V-I characteristics is shown in figure.



1.2.2 Equivalent Circuit:

- ▶ When the breakdown occurs then I_Z may increase from I_{Zmin} to I_{Zmax} but voltage across zener remains constant. Hence actually the internal zener impedance decreases as current increases in zener region. But this impedance is very small. Hence ideal zener diode is indicated by a battery of voltage V_Z , which remains fairly constant in the zener region. It is shown in figure.



- ▶ Practically though very small, zener has its internal resistance. In the region, this resistance is called dynamic resistance of the zener denoted as Z_Z . practically zener region is not exactly vertical. The small change in zener current ΔV_Z to ΔI_Z is called zener resistance Z_Z . This is shown in figure. Hence practical equivalent circuit is also shown in figure.
- ▶ From the graph, dynamic resistance is defined as

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{1}{\frac{\Delta I_Z}{\Delta V_Z}} = \frac{1}{[\text{slope of the reverse characteristic in zener region}]}$$

- ▶ It is almost constant over the full range of zener region and is of the order of few tens of ohms

1.2.3 Breakdown Mechanisms:

- ▶ There are two distinct mechanisms due to which breakdown may occur in the zener diode are follows

1.2.3.1 Zener breakdown:

- ▶ The voltage of 6V or less cause a field across the depletion region, at PN junction of the order of $3 \times 10^5 \text{V/cm}$. an electric field of such high magnitude exerts a large force on the valence electrons of the atom, tending to separate them from their respective nuclei. Actual breaking of the covalent bonds occurs when the electric field intensity becomes more than $3 \times 10^5 \text{V/cm}$. hence, electron- hole pairs are generated in large numbers and a sudden increase of current is observed. If a limiting resistance in the circuit does not prevent the current from increasing to high value, the diode may be destroyed because of exercise heating at the junction. In this, the value of breakdown voltage decreases as PN junction temperature increases, thus such diodes have negative temperature co-efficient.

1.2.3.2 Avalanche Breakdown:

- ▶ The applied reverse bias causes a small reverse current I_0 to flow in the device. This is due to the moment of minority charge particles, via electrons from the p- material and holes from n-materials. The polarity of reverse bias voltage is such that only the minority charge particles are able to cross the PN junction, while the majority charge particles are able to cross the PN junction, while the majority charge particles move away from the junction. As the applied reverse bias voltage becomes larger, the minority carriers increasingly accelerate. There are collision between these particles and electrons involved in the covalent bonds of the crystal structure.
- ▶ If the applied voltage is such that the travelling electrons do not have high velocity, then the collision takes some energy away from them, altering their velocity. If the applied voltage is increased, the velocity and hence kinetic energy ($\text{KE} = \frac{mv^2}{2}$) of electron increases. If such an electron dashes against an electron involved in covalent bond, then the collision gives bond valence electron enough energy to enable it to break its covalent bond Thus, one electron by collision creates an electron-hole pair. These secondary particles are also accelerated and participate in collision that generate new electron- hole

pairs. This phenomenon is known as carrier multiplication. Electron-hole pairs are generated by quickly and in such large number that there is an apparent avalanche or self sustained multiplication process. At this stage junction is said to be in breakdown and current starts increasing rapidly. To limit such current below I_{zmax} current limiting resistor is necessary.

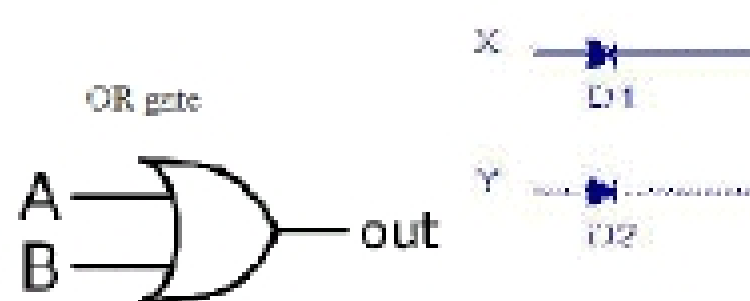
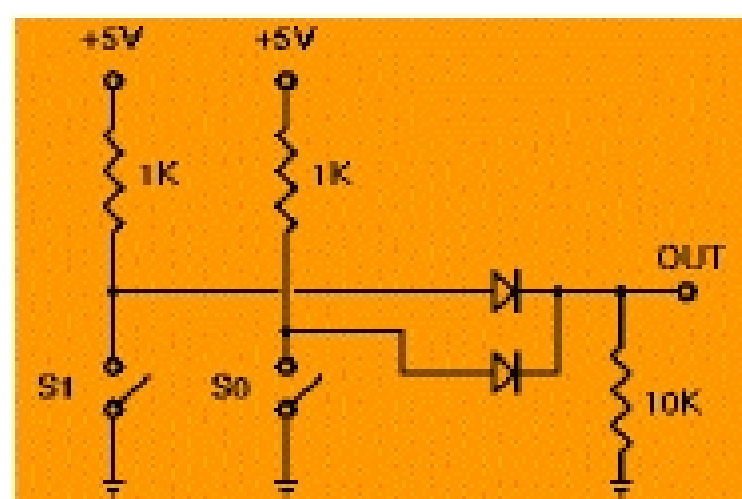
- ▶ Zener diodes having voltage above 6V exhibit avalanche breakdown and such diodes have positive temperature co-efficient. The value of breakdown voltage increases as PN junction temperature increases. Thus it has positive temperature co-efficient

1.2.6 Applications:

- ▶ Voltage regulator.
- ▶ Fixed reference voltage in transistor biasing circuits.
- ▶ Peak clippers or limiters in wave shaping circuits.
- ▶ For meter protection against damage from accidental applications.

1.2.7 OR Gate Using Diodes

- ▶ Figure shows the two-input OR gate using diode and simplified form is also shown. From truth table
- ▶ If $A=0$ and $B=0$, both diodes are non-conducting and $X=0$
- ▶ If $A=0$ and $B=1$, diode D_2 will conduct and output $x=1$
- ▶ If $A=1$ and $B=0$, diode D_1 will conduct and output $X=1$
- ▶ If $A=1$ and $B=1$ both diodes will conduct and output $X=1$



INPUTS		OUTPUT
A	B	$X=A+B$
0	0	0
0	1	1
1	0	1
1	1	1

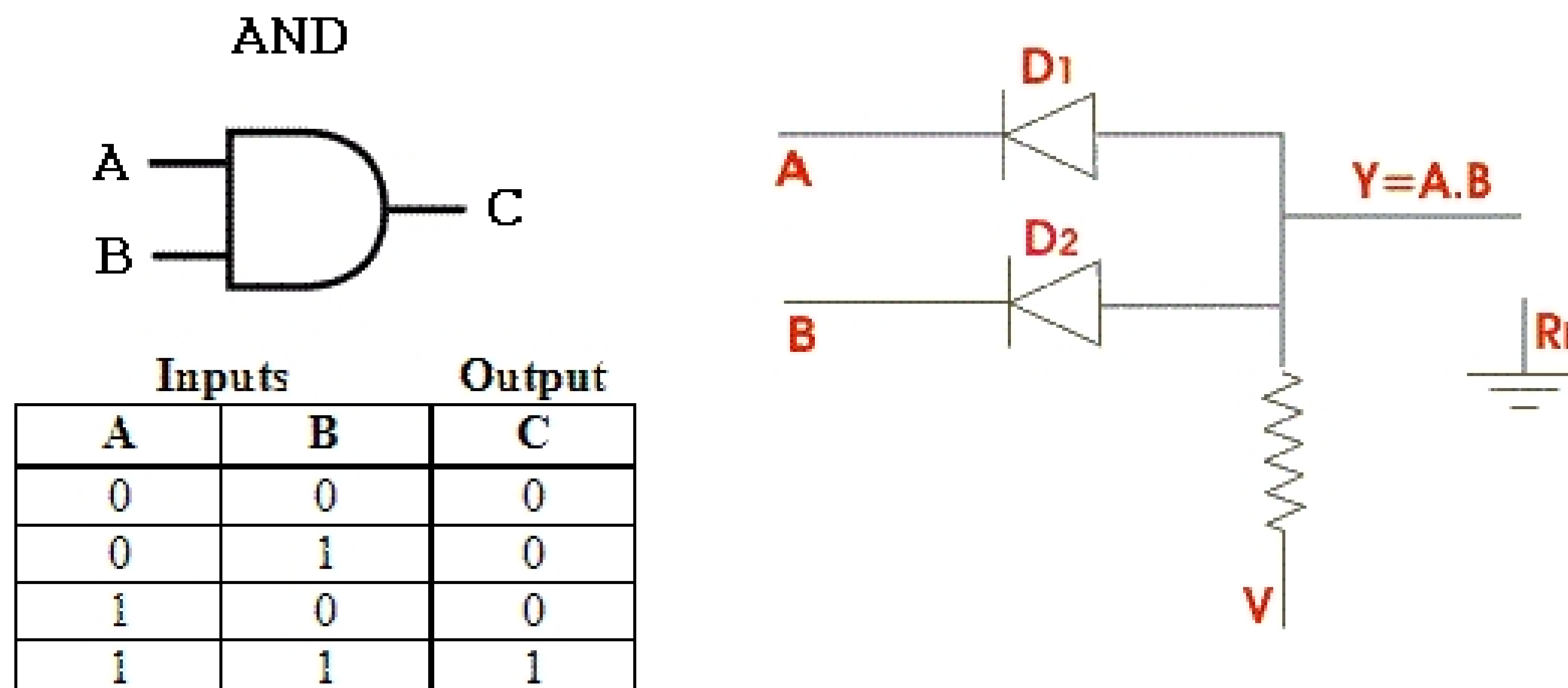
Operation:

- ▶ When both A and B are at zero levels (i.e) low the output voltage is low because the diodes are non-conducting. Therefore X is low, when A is low and B is high lower diode is forward biased so that current flows through R_L and output X is high. The same explanation follows when A is high and B is low. In the last case, when both A and B are

high, both the diodes (D1 and D2) are conducting as they are forward biased and hence the output X is high.

1.2.8 AND Gate Using Diodes:

- ▶ Figure shows two input AND gate and the two diodes are connected in parallel. The circuit operation is very simple to understand from the truth table is also shown.



Operation:

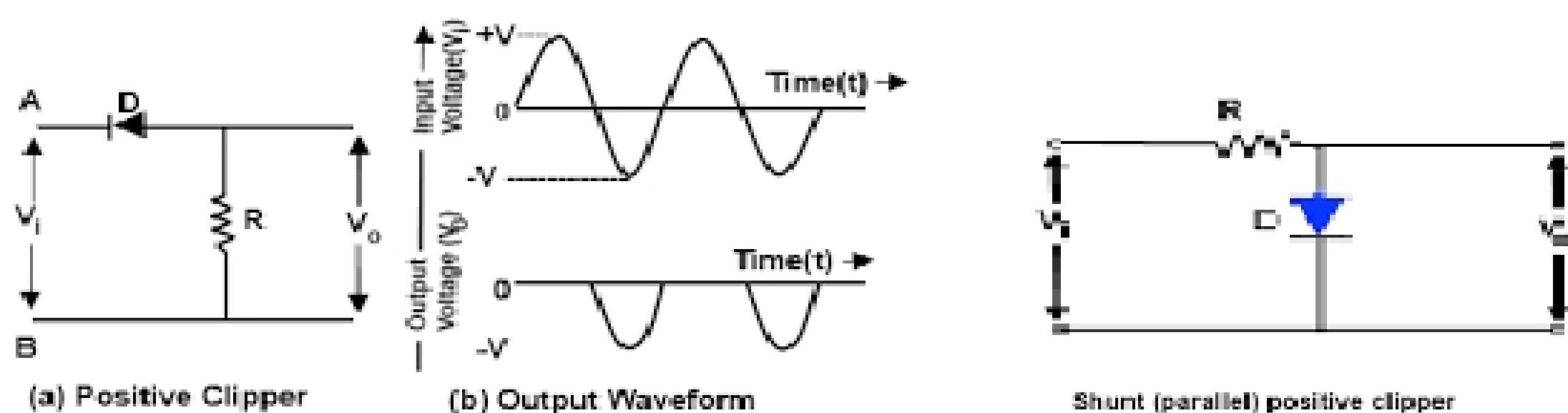
- ▶ When A and B are Zero, the cathode of each diode is grounded. Therefore the positive supply forward-biases both the diodes in parallel. Forward biased means that both the diodes are conducting, and have low resistance and output voltage between anode and ground will be nearly zero. The whole voltage V_{cc} will be dropped across R_L only. Hence output is also zero.
- ▶ When A is low and B is high (5V), the diode D1 is grounded and D2 is reverse-biased. Due to reverse biasing, the diode D2 stops conducting. The diode D1 will now conduct due to forward biasing. Therefore the output will be zero as seen in previous point.
- ▶ When A is high and B is low. In this case, the output is short circuited to ground through diode D2. Hence the output is zero
- ▶ When A and B are high, both the diodes are non-conducting since the diodes are off, there is no current through R_L . the output is equal to the supply voltage $V_{cc} = 5V$. therefore, output is high.
- ▶ Thus the output will result only when the inputs A and B are present simultaneously (ie) $A=B=1$

1.3 CLIPPERS:

- ▶ It is a non-linear wave shaping circuits use the switching properties of diodes and transistors to realize various functions
- ▶ The waveform is shaped by removing (clipping) a certain portion of the input signal voltage above or below a certain level is called as clipping circuit or supply clipper.
- ▶ It is used in digital computers, radars, radio and television receivers, etc.
- ▶ It may also be used to limit the amplitude of the input signal voltage as required in several applications also known as amplitude limiter. It has four types namely,
 1. Positive clipper
 2. Negative clipper
 3. Biased clippers
 4. Combinational clippers

1.3.1 Positive Clipper:

- ▶ It is shown in figure. It consists of a diode D and resistor R with output taken across the resistor. The diode acts as an ideal switch between the source and the load. It acts as a closed switch, when the input voltage is negative(i.e) $V_i < 0$ and as an open switch when the input voltage is positive(i.e) $V_i \geq 0$. The purpose of resistor(R) is to limit the current through diode, when it acts as a closed switch. The operation may be explained.



Operation:

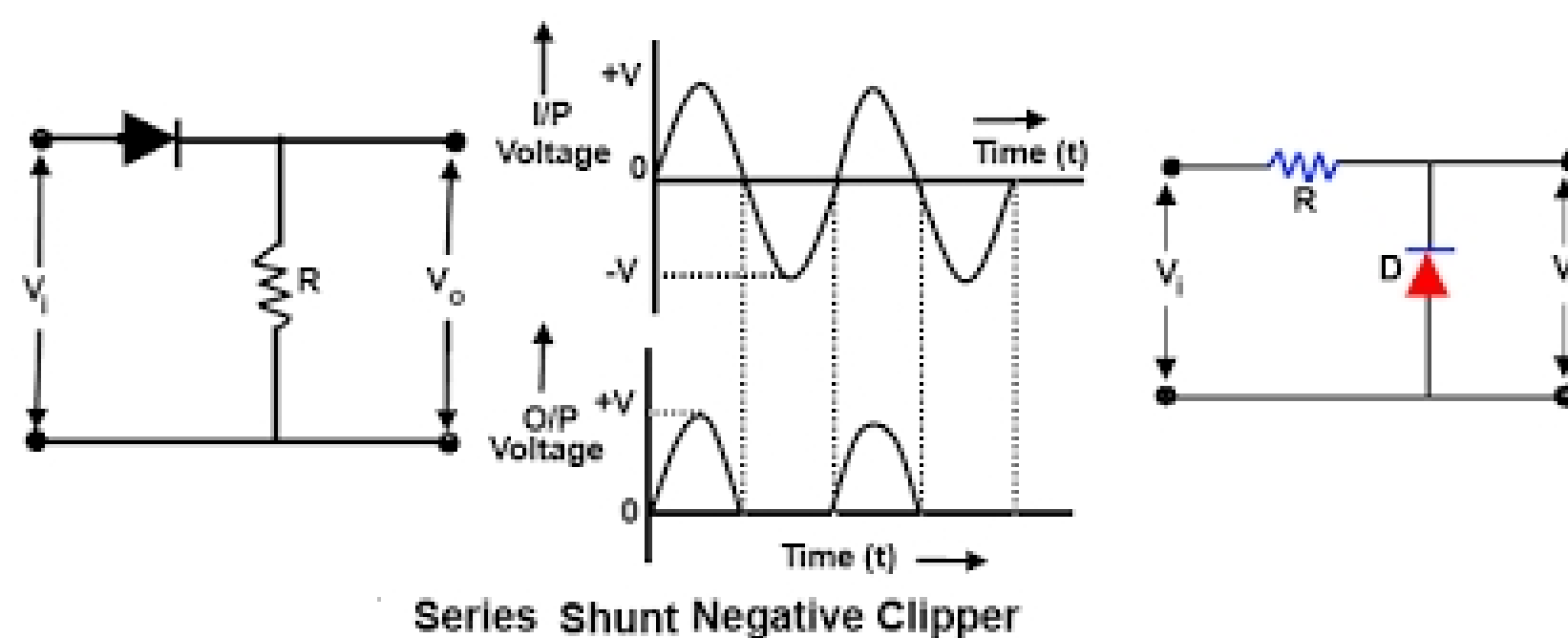
- ▶ From graph, during the positive half cycle of the input voltage, the terminal A is positive with respect to B. this reverse biases the diode across the diode and one across the resistor. As a result, there is no output voltage during the positive half cycle of the input voltage. During the negative half cycle of the input voltage, the terminal B is positive with respect to A. Therefore, it forward biases the diode and it acts as an closed switch.

Thus there is no voltage drop across diode and all the input voltage is dropped across the resistor.

- ▶ The output waveform of the positive clipper has allowed to pass the negative half cycle of the input of voltage and clipped the positive half cycle completely. So it is a positive clipper. Also diode acts a series switch between the source and load. So it is a series positive clipper.
- ▶ It is a possible to connect a diode as a shunt switch between source and load is shown in figure. Here, the diode acts as a closed switch when the input voltage is positive (i.e) $V_i \geq 0$ and as an open switch when the input voltage is negative (i.e) $V_i < 0$. It is known as shunt positive clipper and the output waveform is similar that of a series clipper.

1.3.2 Negative Clipper:

- ▶ It is shown in figure. The diode is connected in a direction opposite to that of a positive clipper. The function of resistor(R) is to limit the current when the diode acts as a closed switch for a positive input voltage (i.e) $V_i \geq 0$ and as an open switch for negative input voltage (i.e) $V_i < 0$. The operation is discussed below.



Operation:

- ▶ The output waveform is shown in figure. During the positive half cycle of the voltage, the terminal A is positive with respect to the terminal B. Therefore, the diode is forward biased and it acts as a closed switch. As a result, all the input voltage appears across the resistor as shown in figure. During the negative half cycle of the input voltage, terminal B is positive with respect to A, diode is reverse biased and it acts as an open switch. Thus there is no voltage drop across the resistor during this cycle and the output waveform is shown.

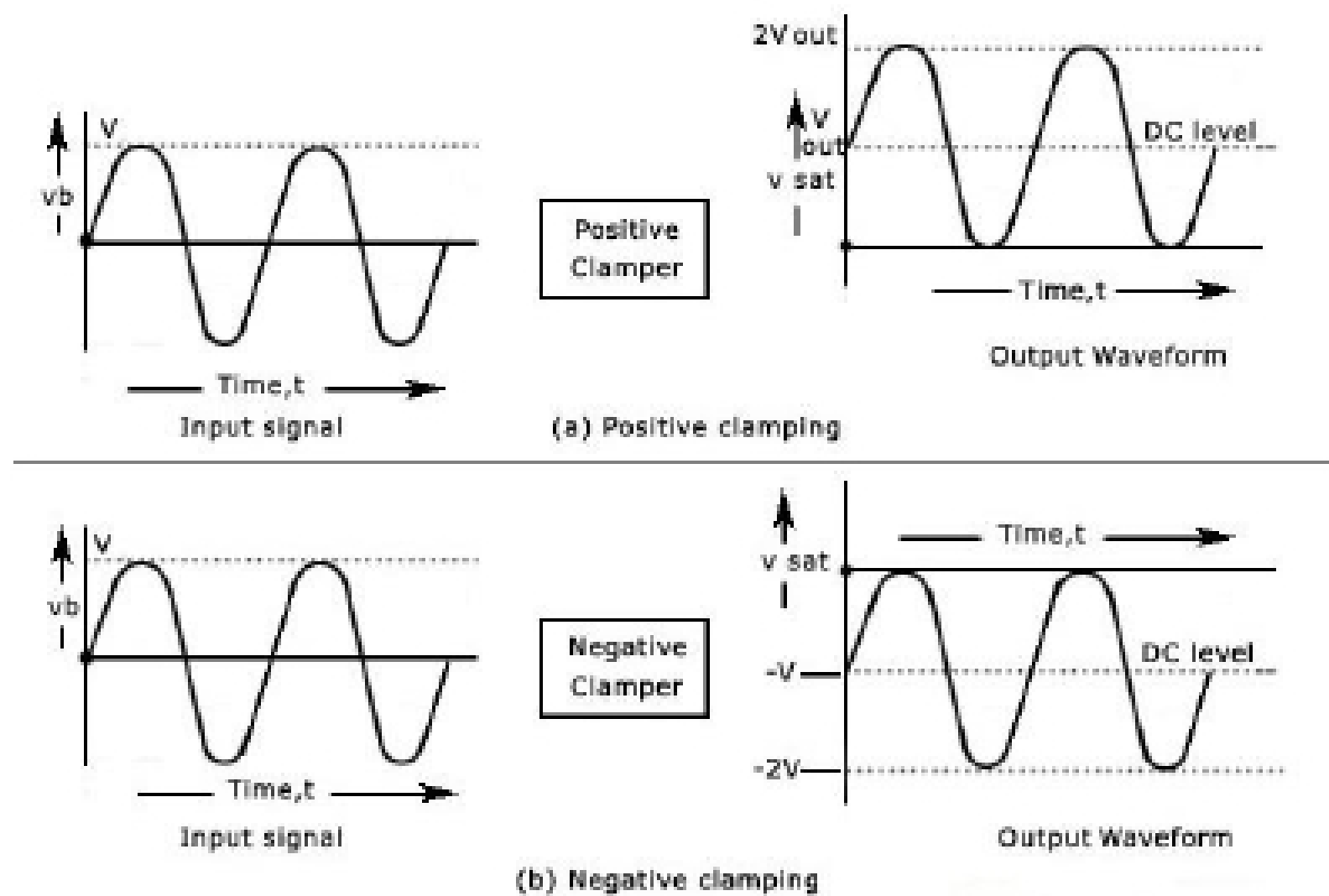
- ▶ The negative clipper has allowed to pass the negative half cycle of the input voltage are clipped and it is called series negative clipper, because diode is connected series switch between the source and the load.
- ▶ It is possible to connect a diode as shunt switch between the source and the load as shown in figure. Here the diode acts as a closed switch for a negative input voltage (i.e) $V_i < 0$ and as an open switch for a positive input voltage (i.e) $V_i \geq 0$. So it is called shunt negative clipper and output waveform is similar to that of series clipper.

1.3.3 Biased Clipper:

- ▶ A clipping circuit, which has a provision for the adjustment of clipping levels, is called a biased clipper. The name bias is designated because the adjustment of the clipping level is achieved by adding a bias voltage in series with the diode or resistor.
- ▶ The biased circuits for several clippers are shown in figure. First series and shunt positive clipper, here the clipping takes place during the positive cycle when the input voltage is greater than battery voltage (i.e) $V_i \geq V_B$. Thus the clipping level can be shifted up or down by varying the bias voltage V_B .
- ▶ If the polarity is reversed, then it is a series and shunt negative clipper, here the input signal lying above the voltage level ($-V_B$) is clipped and it also shown in figure
- ▶ The negative biased clipper is shown in figure (i.e) both series and shunt biased negative clippers. Here the clipping takes place during the negative half cycle only with the input voltage $V_i \leq -V_B$. If the bias voltage is reversed, the entire signal below the voltage level V_B has been clipped off. It is also shown and the corresponding output waveforms are also shown in figure.

1.4 CLAMPERS:

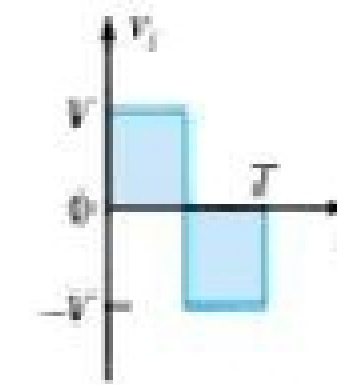
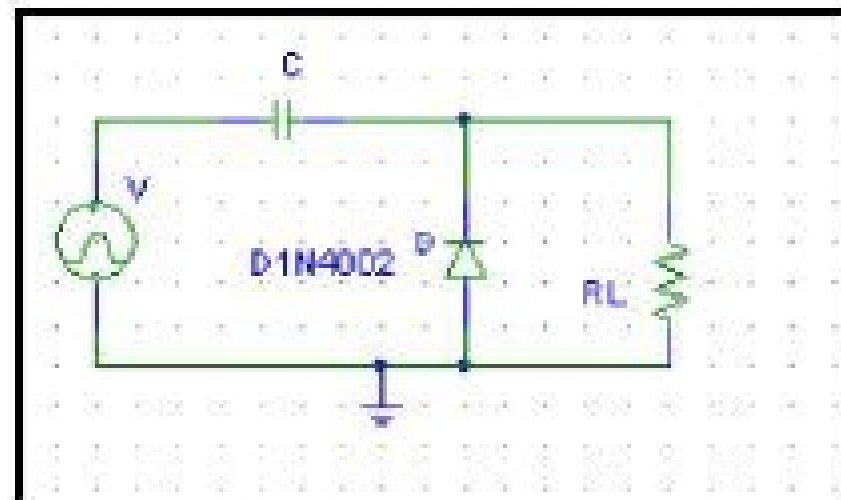
- ▶ The circuit which the waveform can be shifted in such a way so that a particular part (positive or negative peak) is maintained at a specified voltage level is called a clamping circuit or clamper. It restores dc level to an ac signal. So it is also known as DC restorer. It used in TV receives to restore the original DC reference to video signal.
- ▶ Consider an AC signal waveform is shown in figure of $\pm 5V$. This signal waveform is lifted up so as to touch the horizontal axis. It is said to have a DC level of +5V and DC level of -5V is for negatively clamped at 0V.



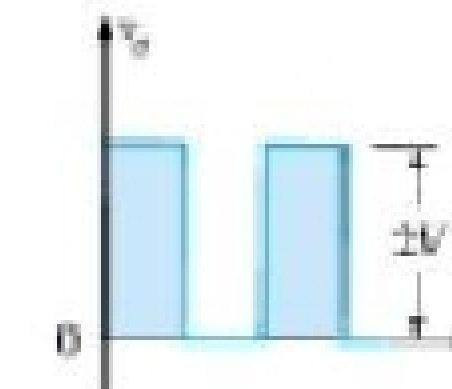
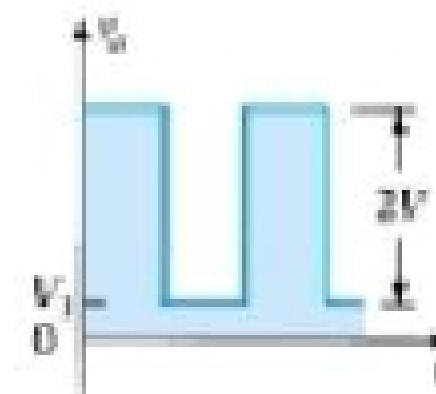
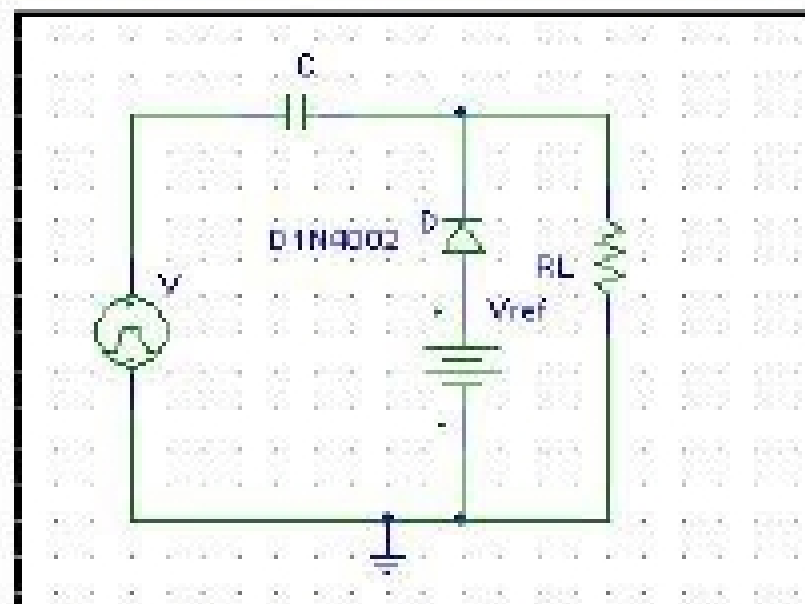
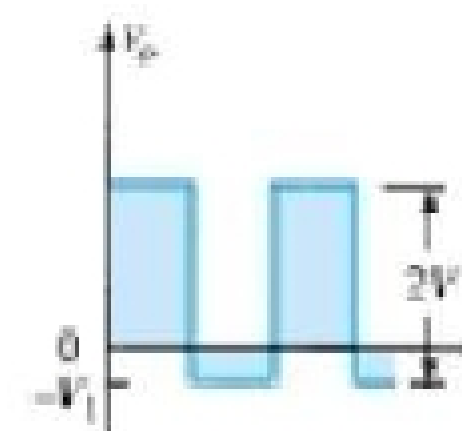
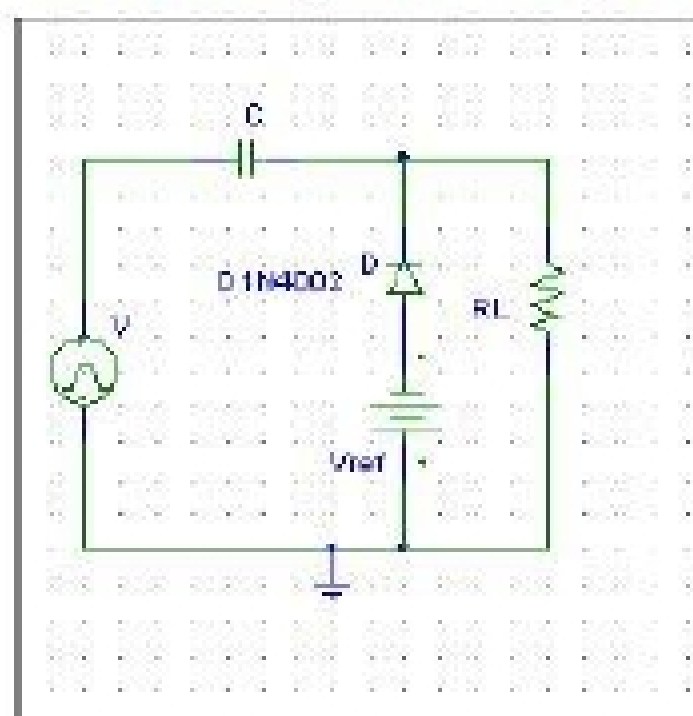
- The way of clamping the input signal, positively at 0V, is use circuit as shown in figure. It consists of a diode and a capacitor and is popularly known as diode clamper. The clamping action of the circuit may be understood from the fact that during the negative half cycle of the circuit may be understood from the fact that during the negative half cycle of the input voltage (V_i), the diode is forward biased and the current flows through the circuit. As a result, capacitor (C) is charged to a voltage equal to the negative peak value ($-V_m$). Once the capacitor is fully charged $-V_m$, it cannot discharge because the diode cannot conduct in the reverse direction. It means that this capacitor acts as a battery with an e.m.f equal to the $-V_m$. The polarity of this voltage is such that it adds to the input signal. Therefore, the output voltage is equal to sum the input signal and the capacitor voltage (V_m) i.e $V_o = V_i + V_m$. A DC signal voltage equal to V_m is added to input signal. This causes the waveform to clamp positively at 0V

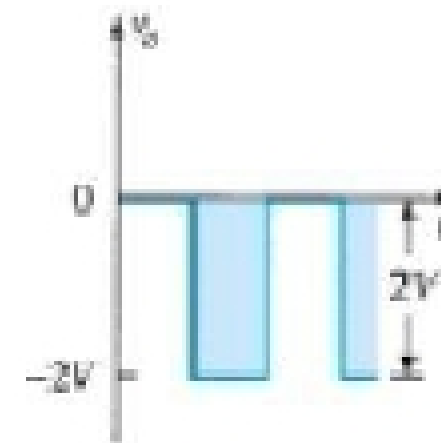
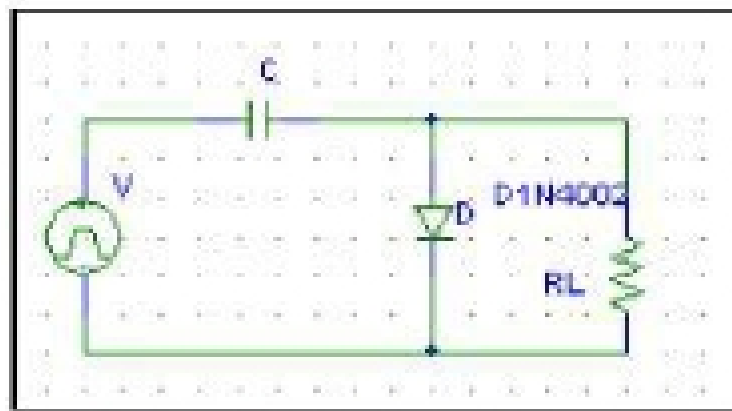
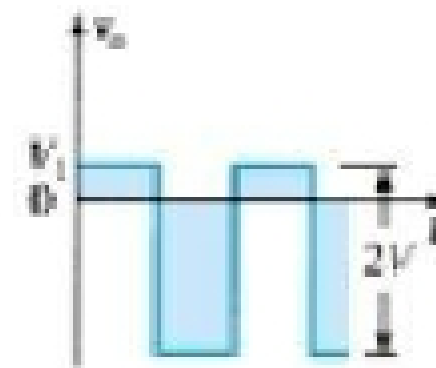
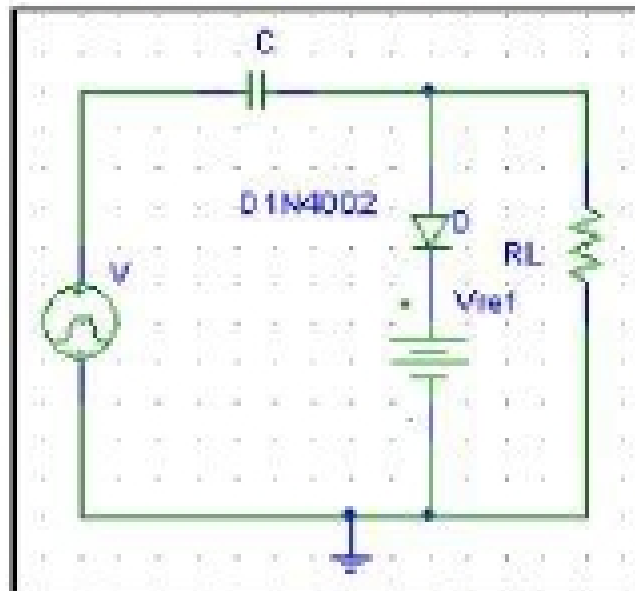
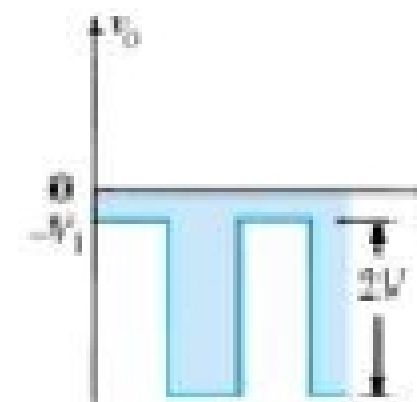
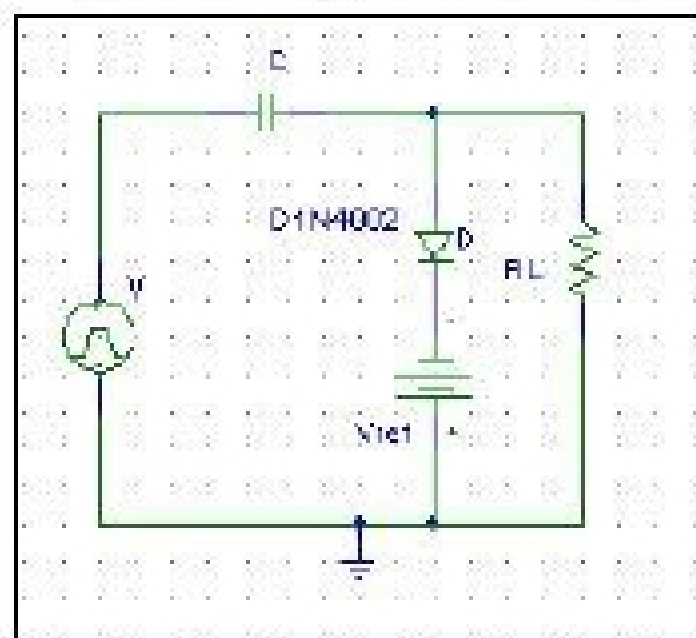
CIRCUIT DIAGRAM:**MODEL GRAPH:**

Input (for all clampers)

**Positive clampers:**

Output

**Positive clampers with positive reference voltage:****Positive clampers with negative reference voltage:**

Negative clippers:**Negative clippers with positive reference voltage:****Negative clippers with negative reference voltage:**

- Now suppose it is required to clamp the input signal waveform negatively at 0V. this can be done by recovering the diode conditions as shown in figure. Here the diode is forward biased during the positive half-cycle of the input signal. Therefore the capacitor is charged to a voltage equal to the positive peak of the input signal (V_m), the polarity of this voltage is such that it is subtracted from the input signal (V_i). Hence the output voltage $V_o = V_i - V_m$
- It is evident from the above relation that DC voltage equal to $-V_m$ is added to the input signal. This causes the waveform to clamp negatively at 0V.

1.4.1 Biased Clampers:

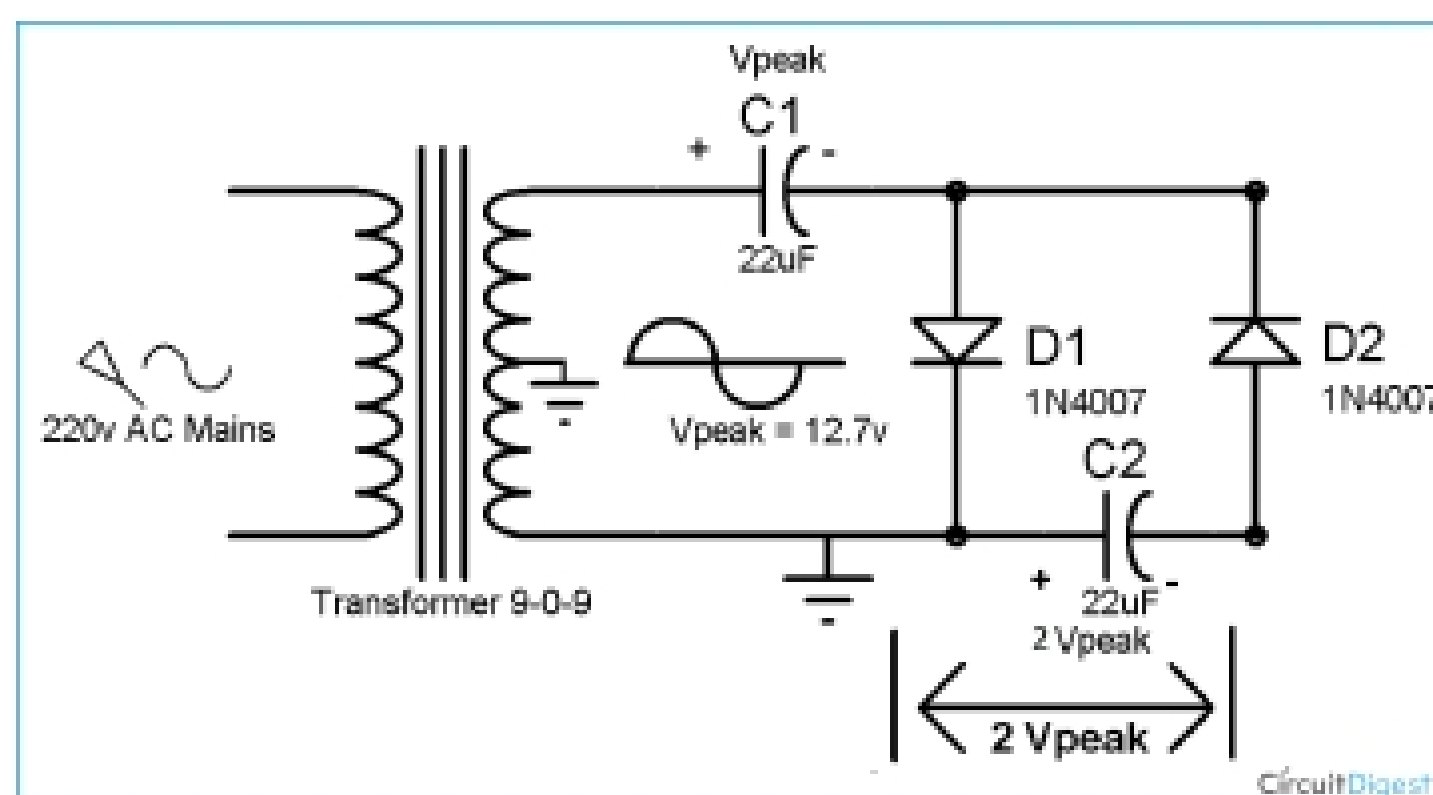
- ▶ Like a biased clipper, it is also positive to have a biased clamper. A biased clamper means that the clamping can be done at any voltage level other than zero. Here a battery of V_{is} is added in such a way that the clamping takes place positively and V_1 in corresponding output. There are four types of biased clampers are
 - ✓ Positive clampers with positive biased
 - ✓ Positive clampers with negative biased
 - ✓ Negative clampers with negative biased
 - ✓ Negative clampers with Positive biased

1.5 VOLTAGE MULTIPLIERS:

- ▶ It is the circuit which produces the output DC voltage whose value is multiple of peak AC input voltage (i.e) $2V_m$, $3V_m$, $4V_m$ and so on. These circuits are used as a power supply for high voltage or low current device like cathode ray oscilloscope.
- ▶ There are several voltage multiplier circuits are
 - ✓ Voltage Doubler
 - ✓ Voltage Tripler
 - ✓ Voltage quadrupler
- ▶ Each circuit is further classified as
 - ✓ Half wave multiplier
 - ✓ Full wave multiplier

1.5.1 Voltage doubler:

- ▶ The figure shows the half wave voltage doubler which gives DC voltage output equal to $2V_m$, where V_m is the maximum value of the input AC voltage.

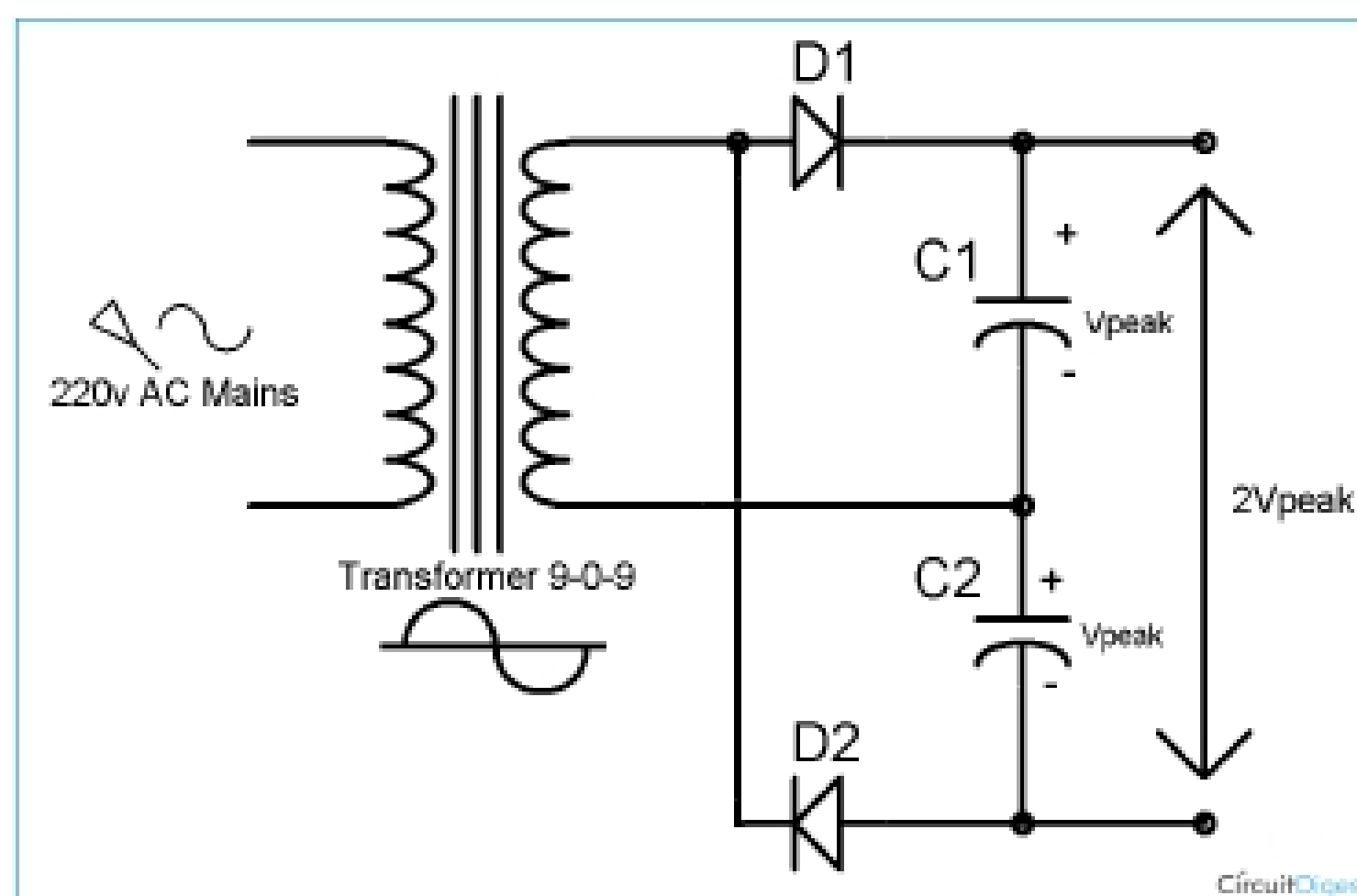


Operation:

- ▶ During positive half cycle of V_i , D1 will be forward biased and the diode D2 will be reverse biased. The capacitor C1 will get charged to V_m . The charging of capacitor C1 is very fast. As D2 is reverse biased, next part of the circuit remains disconnected from the circuit.
- ▶ During negative half cycle, the diode D1 will be reverse biased and the diode D2 will be forward biased. So capacitor C2 will get charged to $2V_m$ with the polarities as shown in figure. This $2V_m$ as the V_m on the C1 adds to the input voltage, the capacitor C1 retains its voltage V_m all the time, once charged. The peak value of input is V_m with polarities such that the point A is negative and B is positive.

Full Wave:

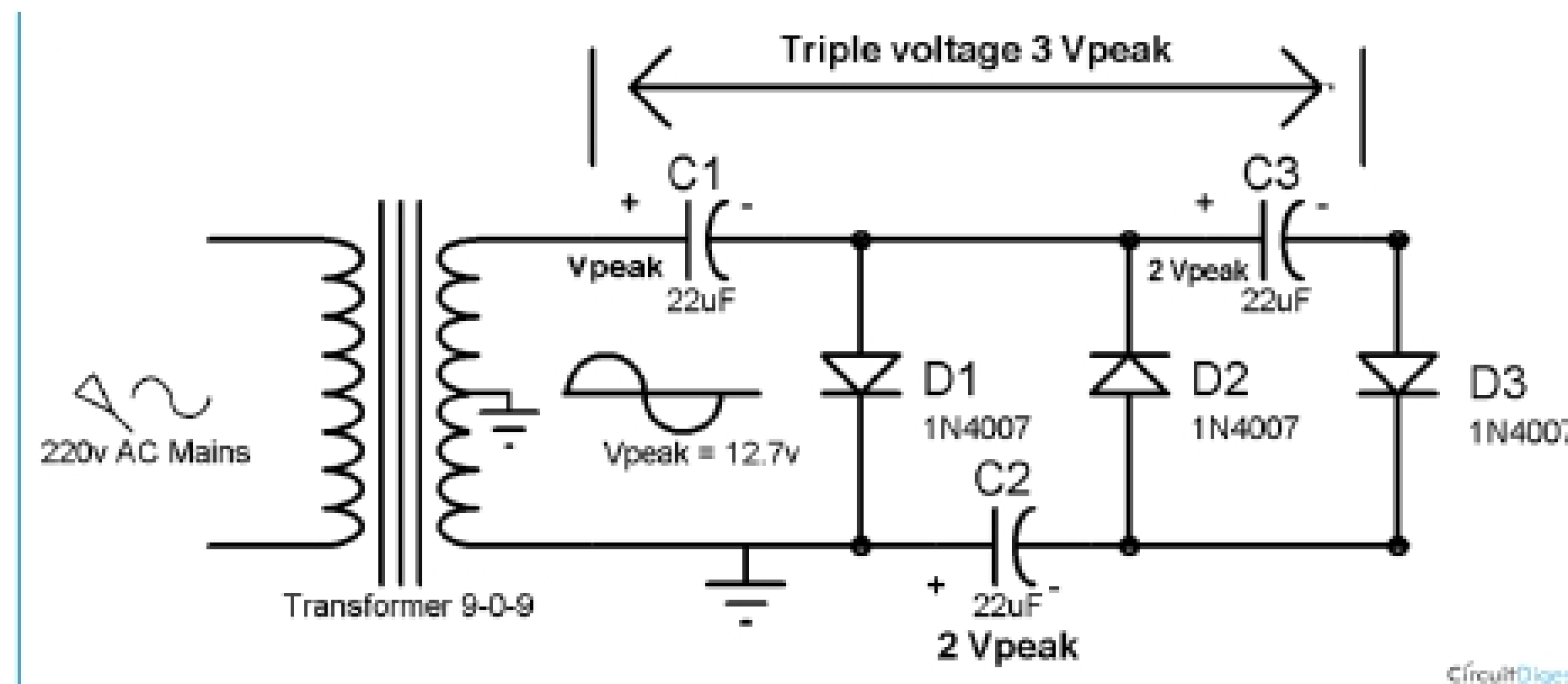
- ▶ Full Wave voltage doubler is as shown in the figure

Operation:

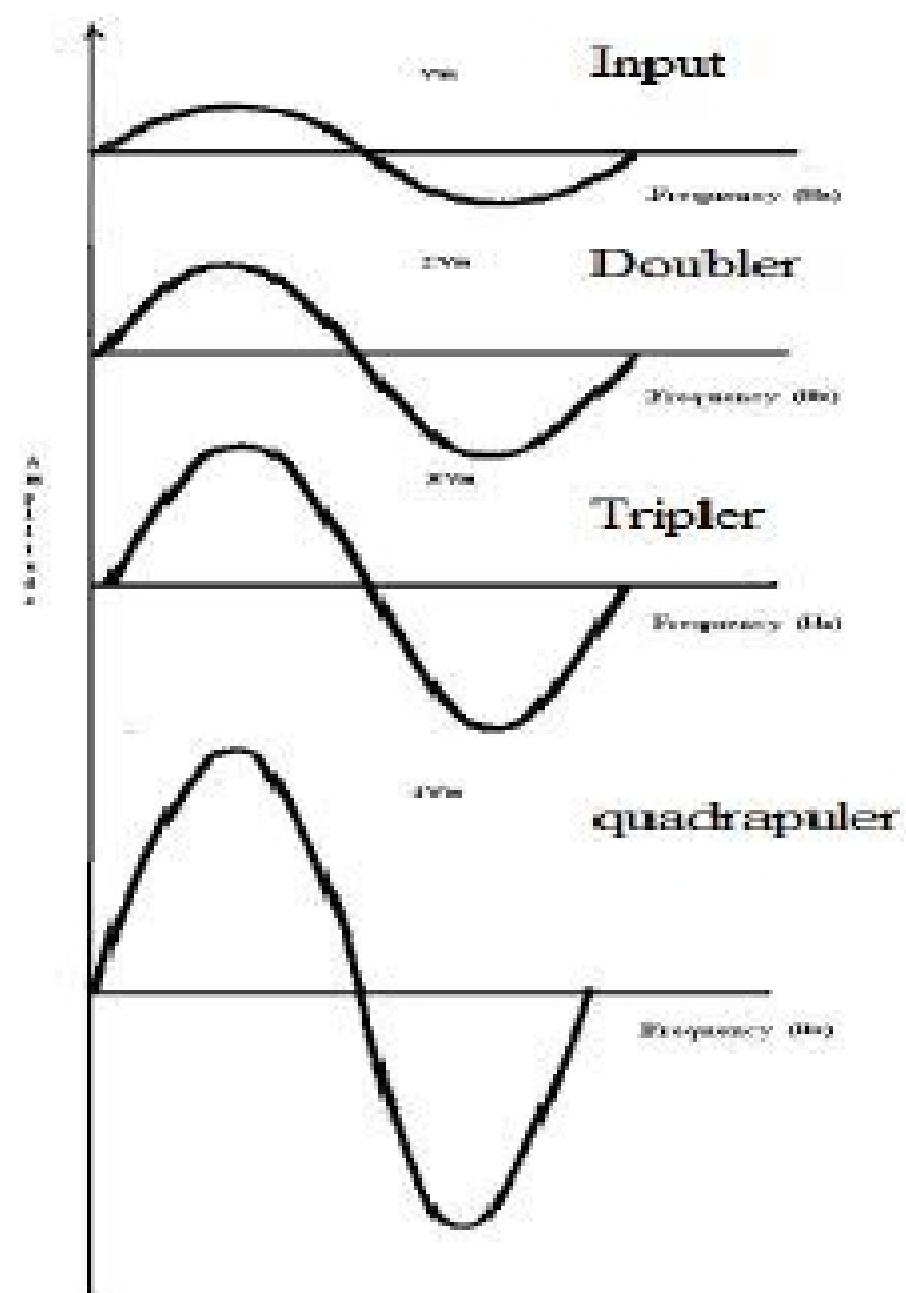
- ▶ During the positive half cycle of the secondary voltage of the transformer the diode D1 is forward biased and capacitor C1 gets charged equal to V_m , the peak of the secondary transformer voltage assuming ideal diodes.
- ▶ During negative half cycle of the secondary voltage of the transformer, the diode D2 is forward biased while diode D1 is reverse biased. The capacitor C2 gets charged equal to V_m , with polarity as shown in the figure

1.5.2 Voltage Tripler:

- It can be obtained by adding one more diode capacitor section to the half cycle wave doubler circuit is shown in the figure.

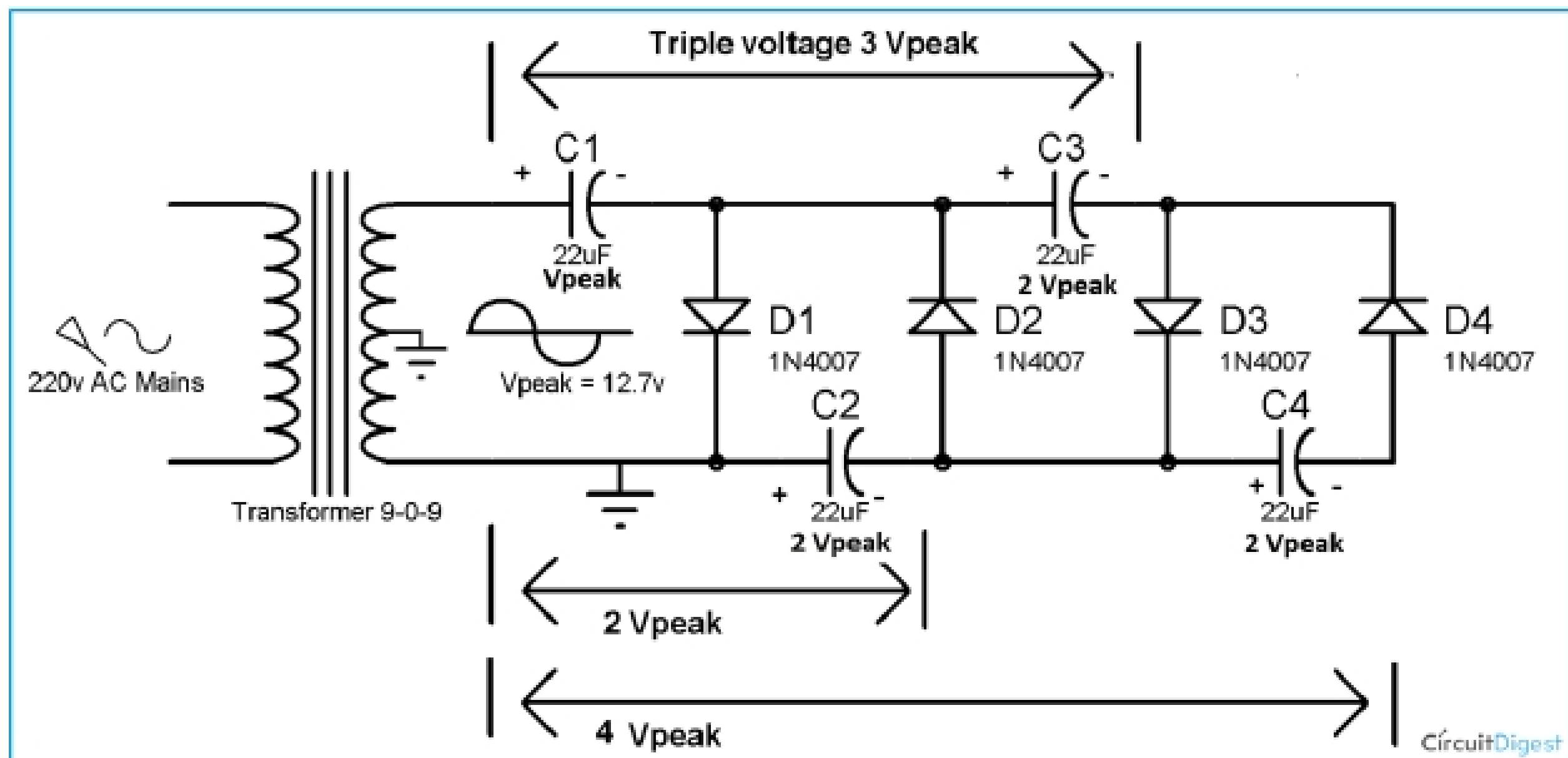


- During first positive half cycle, the diode D_1 becomes forward biased, while the diode D_2 and D_3 are reverse biased. The capacitor C_1 charges to V_m with polarities as shown. During the negative half cycle diode D_2 is forward biased. The capacitor C_2 charges to $2V_m$ with polarities as shown. During next positive half cycle, the diode D_3 is forward biased as C_2 retains its charge and voltage to $2V_m$ through forward biased D_3 with the polarities as shown in the figure.



1.5.3 Voltage Quadrupler:

- ▶ The addition of another diode capacitor section to voltage tripler gives a voltage quadrupler is shown in the figure.



- ▶ The capacitor C4 charges to $2V_m$ with polarity shown in the negative half cycle through D4 which is $4V_m$.

1.5.4 Advantages:

- ▶ Increase peak rectified voltage without increase in the rate of input transformer
- ▶ It is economical and centre tap transformer not requires.

1.5.5 Limitations:

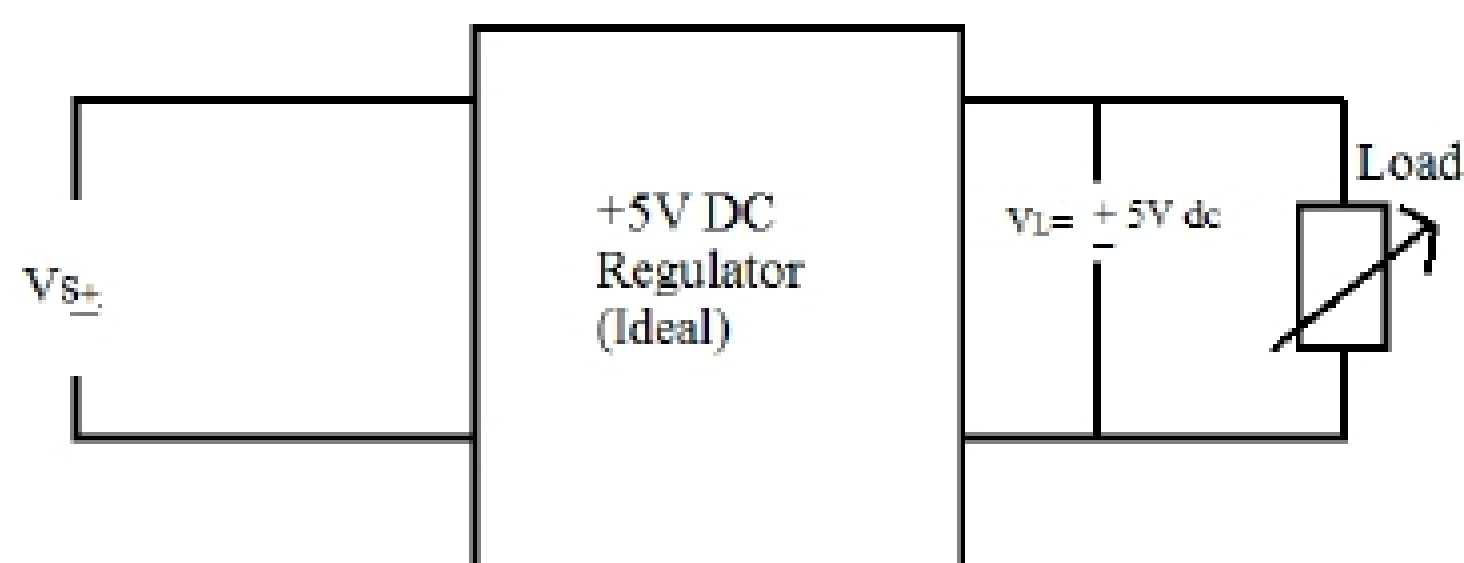
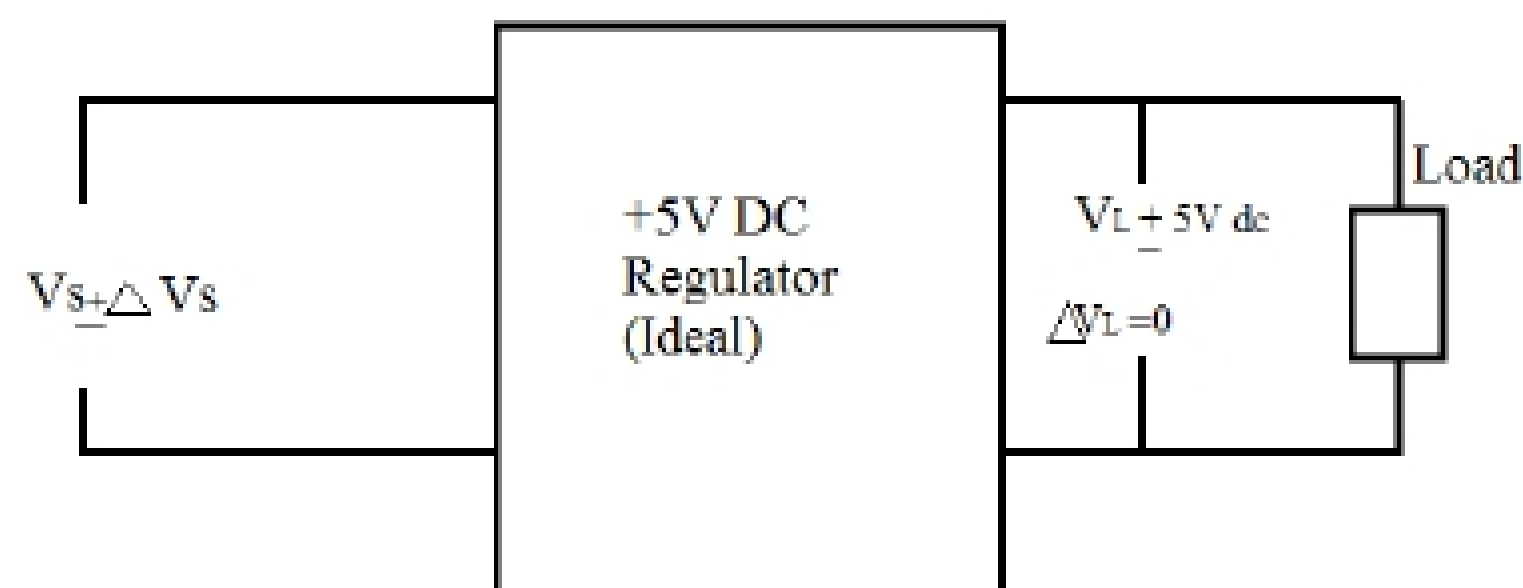
- ▶ More and more stages of diode capacitors are added to a cahin the output voltage keeps on falling substantially.

1.5.6 Applications:

- ▶ High voltage low current applications
- ▶ TV receivers
- ▶ Cathode ray tube (CRT) anode voltages.

1.6 VOLTAGE REGULATION:

- ▶ An ideal voltage regulator will maintain a constant DC output voltage regardless of changes in either its input voltage or its load current demand. For example consider a +5V DC regulator as shown in figure. A change in the regulator input voltage shown as ΔV_s . It would not appear at the regulator output. It may be carefully noted that ΔV_s could be either a change in the steady state DC value of V_s or could be some ripple voltage. In either case, the change in V_L for the ideal regulator output would be 0V.
- ▶ However during this discussion, we have assumed that the value of input voltage to the voltage regulator (V_s) does not decrease below the value required to maintain the operations of the voltage regulator. As a rule, a regulator cannot have 5V dc output if the input is less than 5V DC.



- ▶ Figure shows how an ideal voltage regulator would respond to a change in load current demand. The ideal voltage regulator maintains a constant output voltage (i.e) $V_L = 0$ despite the change in load current demand. To specify the quality of regulation for a voltage regulator

- ✓ Line regulation
- ✓ Load regulation

1.6.1 Line Regulation:

- ▶ A change in input voltage to a voltage regulator will cause a change in its output or load voltage. The line regulation rating of a voltage regulator indicates the change in output voltage that will occur per unit change in the input voltage. It is given by

$$\text{Line regulation} = \frac{\Delta V_L \text{ mV}}{\Delta V_s \text{ V}} \text{ or } \frac{\mu V}{V}$$

- ▶ Where ΔV_L – change in output voltage usually in mV or μV
- ▶ ΔV_s – change in input voltage usually in Volts

1.6.2 Load Regulation:

- ▶ In practice the voltage regulator will also experience a slight change in output voltage when there is a change in load current demand. The load regulation indicates the change in output voltage and will occur per unit change in load current. Mathematically,

$$\text{Load regulation} = \frac{V_{NL} - V_{FL} \text{ } \mu V}{\Delta I_L \text{ } \mu A}$$

Where V_{NL} – No load output voltage

V_{FL} – Full load output voltage

ΔI_L – Change in load current demand

- ▶ Ideally speaking, whether it is a line or load regulation, the lower the two values are the better is the quality of regulation.
- ▶ Some people prefer another way of expressing voltage regulation as given below. Here, the voltage regulation is expressed as the ability of the regulator to maintain a constant output voltage in spite of AC input voltage fluctuations and changes in load resistance. Mathematically,

$$\% \text{ of regulation} = \frac{V_{NL} - V_{FL}}{\Delta I_L} * 100$$

Where V_{NL} – No load output voltage

V_{FL} – Full load output voltage

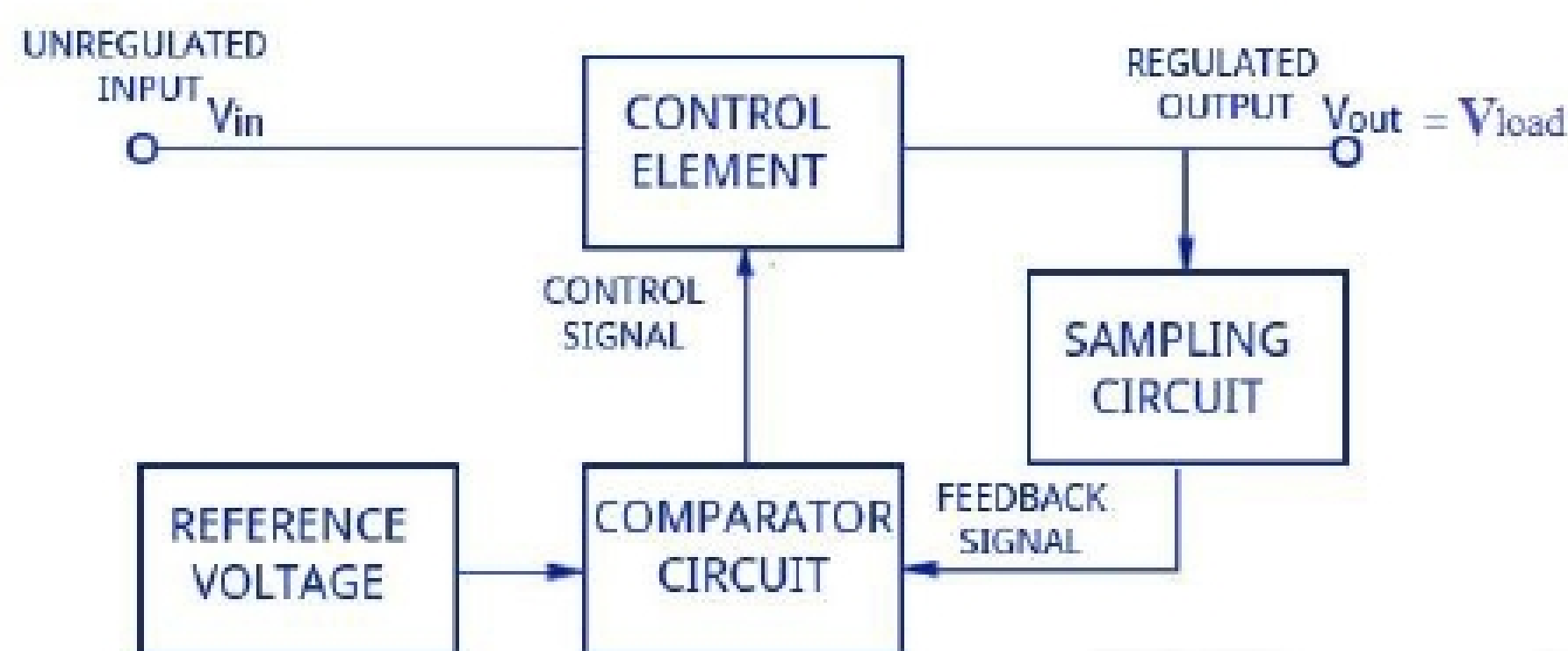
1.6.3 Types of Voltage Regulators:

- ▶ There are two main types of voltage regulators are
 - ✓ Series voltage regulator
 - ✓ Shunt voltage regulator

1.6.3.1 Series Voltage Regulators:

- ▶ If a voltage regulator circuit, the control element is connected in series with the load, the circuit is called series voltage regulator circuit. The block diagram is shown in figure.

SERIES VOLTAGE REGULATOR - BLOCK DIAGRAM

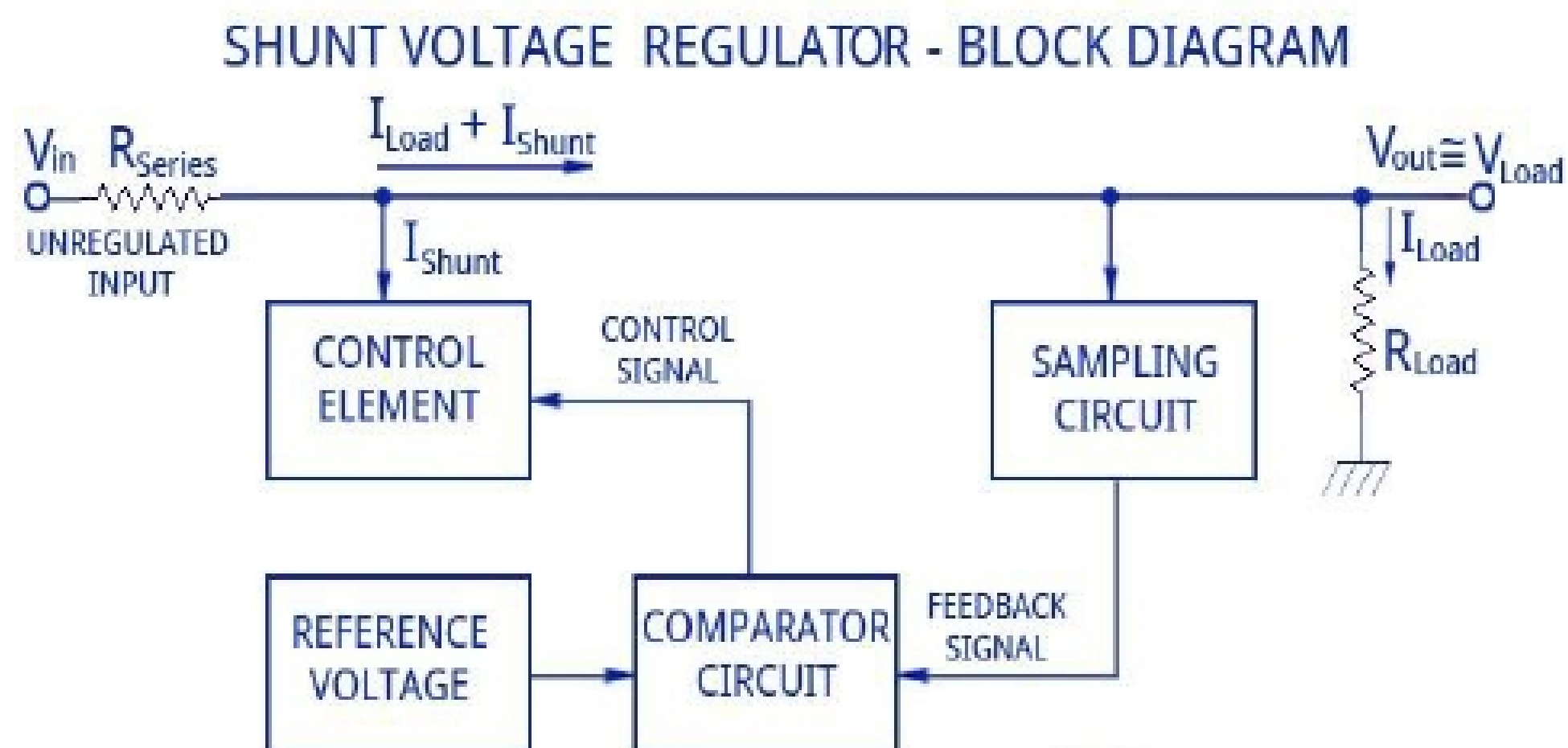


- ▶ The unregulated DC voltage is the input to the circuit. The control element, controls the amount of the input voltage, that gets to the output. The sampling circuit provides the necessary feedback signal. The comparator circuit compares the feedback with the reference voltage to generate the appropriate control signal.
- ▶ For example, if the load voltage tries to increase the comparator generates a control signal based on the feedback information. This control signal causes the control element to decrease the amount of the output voltage. Thus it is maintained constant. The control element which regulates the load voltage, based on the control signal is in series with load and hence it is called series voltage regulator.
- ▶ Here the entire current passes through the control element and hence control element is high current. Low voltage resting component. As input current and load current are same, the efficiency depends on output voltage. It provides good regulation than shunt regulators. It can be used for fixed voltage as well as variable voltage requirements. To

compensate for the drop across the control element, input voltage V_{in} must be at least 2 to 3V more than output voltage.

1.6.3.2 Shunt Voltage Regulator:

- The heart of any voltage regulator circuit is a control element. If such a control element is connected in shunt with the load, the regulator circuit is called shunt voltage regulator. The block diagram of shunt voltage regulator circuit is shown below.



- The unregulated input voltage V_{in} , tries to provide the load current, But part of the current is taken by the control element, to maintain the constant voltage across the load. If there is any change in the load voltage, the sampling circuit provides a feedback signal to a comparator circuit. It compares the feedback with the reference voltage and generates a control signal which decides the amount of current required to be shunted to keep the load voltage constant. For example, if voltage (V_L) increases then comparator circuit decides the control signal based on the feedback information, which draws increased shunt current „ I_{sh} “ value. Due to this , the load current I_L decreases and hence the load voltage decreased to its normal. Thus control element maintain the constant output voltage by shunting the current, hence the regulator is called voltage shunt regulator circuit.
- From block diagram, only part of the load current required to be divided, passed through the control element. Thus the control element is low current, high voltage rating component. The efficiency depends on the load current I_L . Hence shunt regulators are not performed for varying load conditions.

UNIT-2

Topic	Page No
2.1 Need For Biasing	34
2.2 Ope rating Point And Stability Factor	35
2.3 Fixed Bias Circuit	35
2.4 Emitter Stabilized Bias Circuit	37
2.5 Voltage Divider Bias	39
2.6 Dc Bias With Voltage Feedback	40
2.7 FET Biasing	41
2.7.1 Fixed Bias Circuit	42
2.7.2 Voltage Divider Bias	42
2.7.3 Self Bias Circuit	44
2.8 MOSFET	45
2.8.1 Common Source Circuit For EMOSFET With Source Resistor	46
2.9 Transistor Modeling	47
2.10 H – Parameter Model	50
2.10.1 Common Emitter Amplifier	51
2.11 Important Parameters Of JFET	55

Department of Computer Science and Engineering

Subject Name: ELECTRONIC DEVICES AND CIRCUITS

Prepared by:

Mr. T.KAVIARASAN. M.Tech (Assistant
Professor)

UNIT II

Biasing and Modeling for BJT and FET : Biasing and operating point, BJT Bias circuits - Fixed-bias, Emitter stabilized bias, Voltage divider bias and DC bias with voltage feedback, FET biasing - Fixed-bias, Self-bias, Voltage-divider bias, MOSFET biasing. Transistor modeling - Important parameters of BJT- h - parameter model of BJT (CE only) - Important parameters of JFET, Small signal model of JFET and MOSFET

2.1 NEED FOR BIASING:

- The transistor can be operated in three regions: Cut off, active and saturation by applying proper biasing conditions as shown in the table.

Region of operation	Emitter base junction	Collector base junction
Cut off	Reverse biased	Reverse biased
Active	Forward biased	Reverse biased
Saturation	Forward biased	Forward biased

- In order to operate transistor in the desired region we have to apply external DC voltages of Correct Polarity and magnitude to the two junctions of the transistor. This is nothing but the biasing of the transistor. Because DC voltages are used to bias the transistor, biasing is known as DC biasing of the transistor.

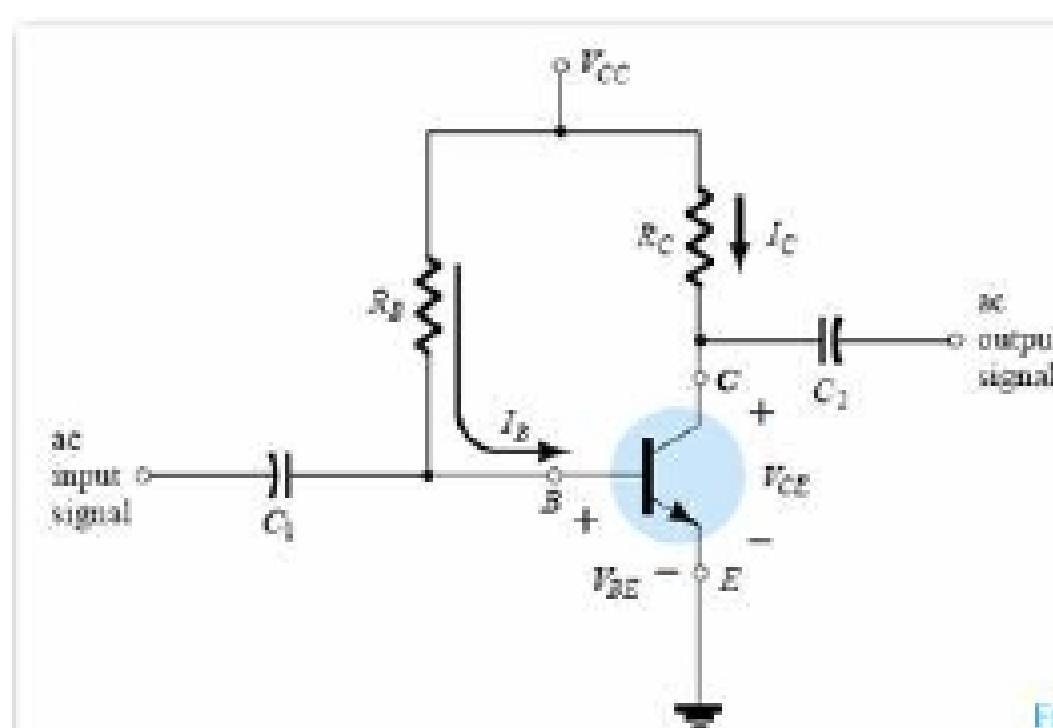
- ▶ In transistor amplifier circuit, Output signal power is always greater than input signal power. Now how this amplification of power is achieved. The DC source supplies the power to the transistor circuit to get the output signal power greater than input signal power.

2.2 OPERATING POINT AND STABILITY FACTOR:

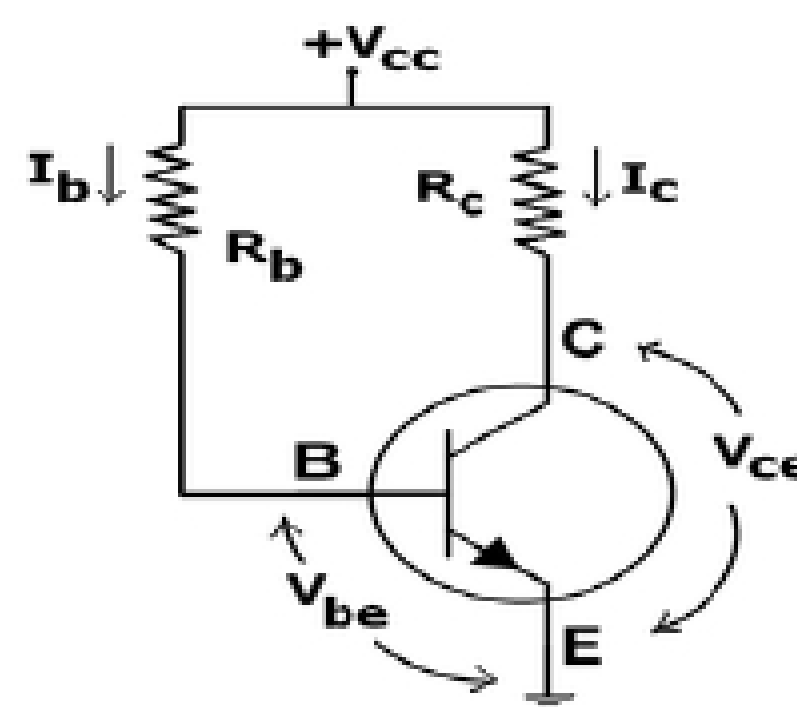
- ▶ When we bias a transistor we establish a certain voltage and current condition for the transistor. These conditions are known as operating condition or dc operating or quiescent point
- ▶ It must be stable for proper operation of the transistor. However it shifts with changes in transistor parameters such as β , I_{CO} and V_{BE} . As transistor parameters are temperature dependent, the operating point also varied with changes in temperature.
- ▶ The stability factor indicates the degree of change in operating point due to variation in β , I_{CO} and V_{BE} .

2.3 FIXED BIAS CIRCUIT:

- ▶ The figure shows the fixed bias circuit. It is the simplest DC bias configuration. For the DC analysis it can replace capacitor with an open circuit because the reactance of a capacitor for dc is
- ▶ $X_C = 1/2\pi f C = 1/2\pi (0)C = \text{Infinity } (\infty)$. The dc equivalent of fixed bias circuit is also shown in figure.



Fixed Bias circuit



DC Equivalent

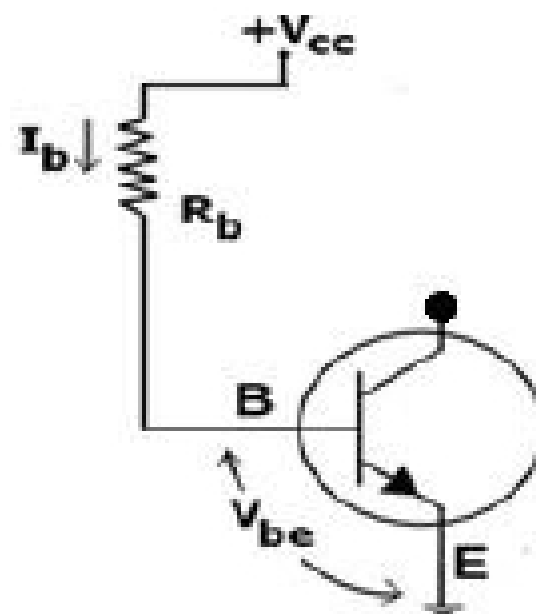
Base Circuit:

- ▶ Let us consider the base circuit as shown in figure. Applying Kirchhoff's voltage law to the base circuit we get,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

Solving for the current I_B

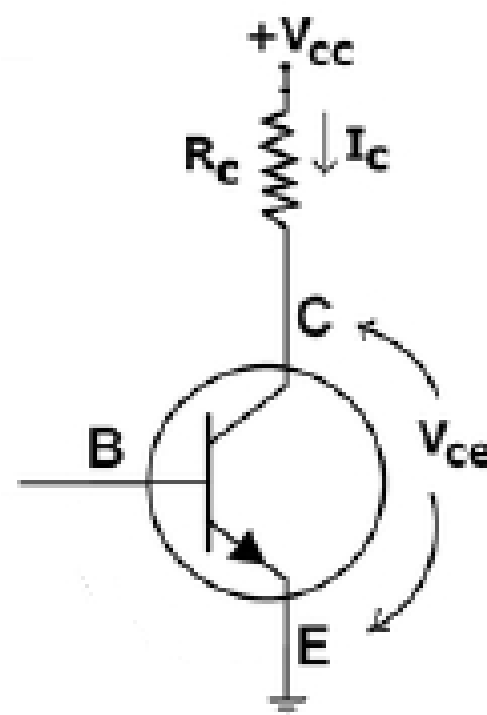
$$I_B = (V_{CC} - V_{BE}) / R_B$$



Base Circuit

Collector circuit:

- We now consider the collector circuit as shown in figure.



- Applying Kirchhoff's voltage law to the collector circuit we get,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C \text{ (or) } I_C = (V_{CC} - V_{CE}) / R_C$$

- The Magnitude of the collector is given by

$$I_C = \beta I_B$$

- It is important to note that since the base circuit is controlled by the value of R_B and I_C is related to I_B by a constant β , the magnitude of I_C will not affect the level of I_B or I_C as long as we remain in the active region of the device. However, the change in R_C will change the value of V_{CE} .

$$V_{CE} = V_C - V_E$$

- Where V_C – collector voltage

V_E – Emitter voltage

Similarly,

$$V_{BE} = V_B - V_E$$

Where V_B – Base voltage

In this circuit, $V_E = 0$

Therefore $V_{BE} = V_B$

And $V_{CE} = V_C$

Stability factor:

- Stabilization techniques give different biasing circuits which are discussed in the following sections. In order to compare the stability provided by these circuits, one term is raised called stability factor.
- Which indicates degree of change in operating points due to variation in temperature? Since there are three variables which are temperature dependent, the stability factors are,

$$\begin{aligned}
 S &= \left. \frac{\partial I_C}{\partial I_{CO}} \right|_{V_{BE}, \beta \text{ constant}} & \text{or} & \quad S = \left. \frac{\Delta I_C}{\Delta I_{CO}} \right|_{V_{BE}, \beta \text{ constant}} \\
 S' &= \left. \frac{\partial I_C}{\partial V_{BE}} \right|_{I_{CO}, \beta \text{ constant}} & \text{or} & \quad S' = \left. \frac{\Delta I_C}{\Delta V_{BE}} \right|_{I_{CO}, \beta \text{ constant}} \\
 S'' &= \left. \frac{\partial I_C}{\partial \beta} \right|_{I_{CO}, V_{BE} \text{ constant}} & \text{or} & \quad S'' = \left. \frac{\Delta I_C}{\Delta \beta} \right|_{I_{CO}, V_{BE} \text{ constant}}
 \end{aligned}$$

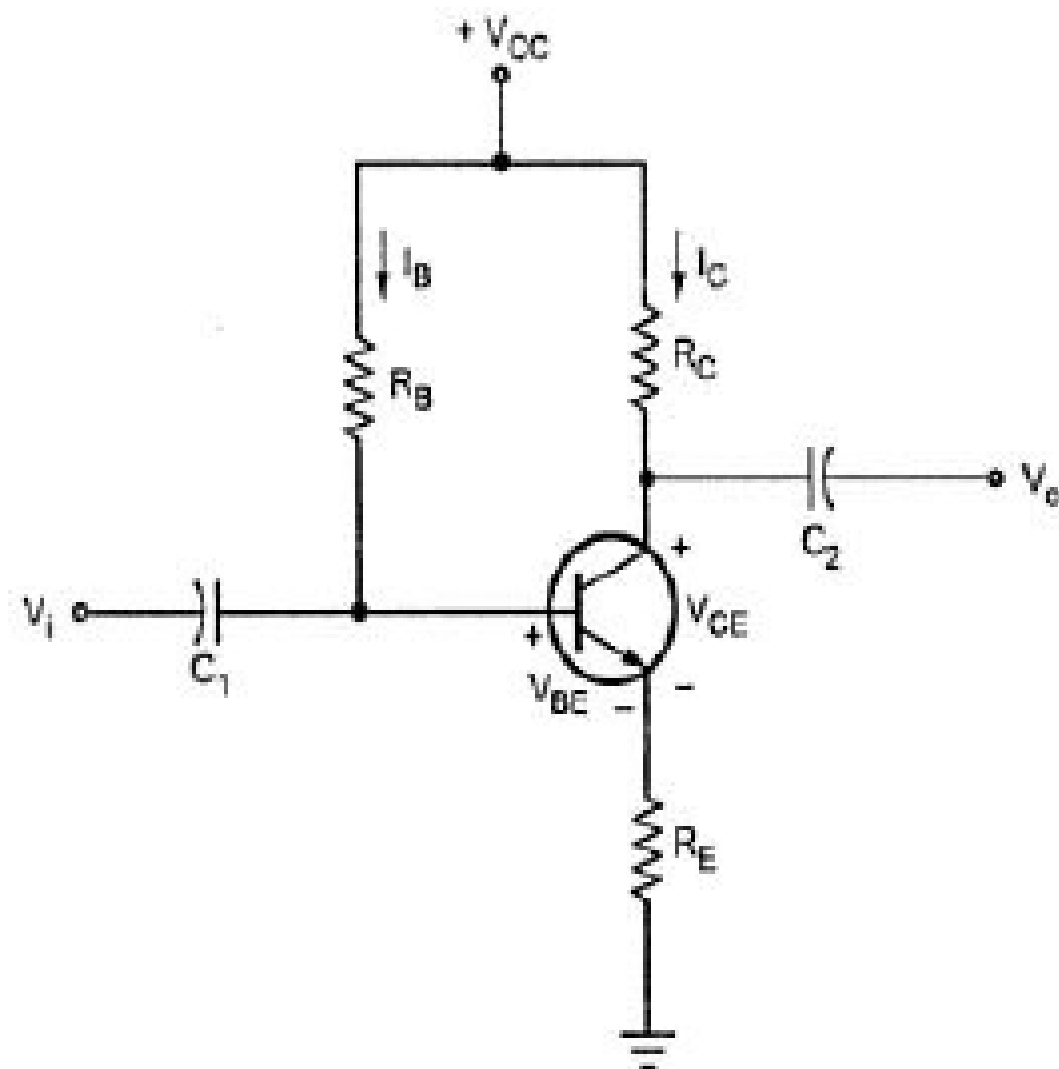
$$S = \frac{(1 + \beta)}{1 - \beta (\partial I_B / \partial I_C)}$$

Key points:

- Ideally stability factor should be perfectly zero to keep operating point stable.
- Practically stability factor should have the value as minimum as possible.
- Thermal stability of a circuit is assessed by deriving a stability factor „ S “.

2.4 EMITTER STABILISED BIAS CIRCUIT:

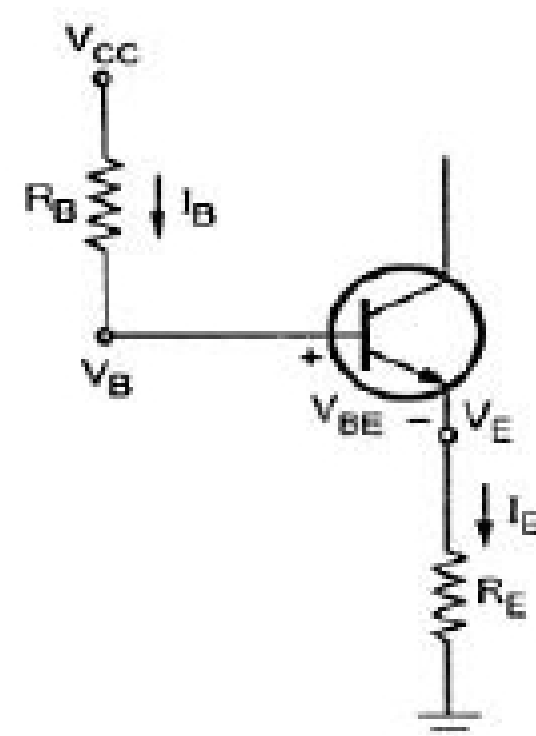
- To improve the stability of the biasing circuit over the fixed bias circuit, the emitter resistance is connected in the stabilized bias circuit is shown in the figure.



Emitter bias circuit

Base Circuit:

- Let us consider the base circuit shown in the figure

**Base circuit**

Applying KVL, we get

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

But, $I_E = (1 + \beta) I_B$

Then, $V_{CC} - V_{BE} = I_B R_B + (1 + \beta) I_B R_E$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E}$$

If $\beta \gg 1$,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_E}$$

- Note that only difference between the equation for I_B and that obtained for fixed bias configuration is the term βR_E .

$$V_B = V_{BE} + V_E \text{ (or) } V_{CC} - I_B R_B$$

Therefore, $V_E = I_E R_E$

$$V_B = V_{BE} + I_E R_E.$$

Collector circuit:

- Consider the collector circuit is shown in figure.

Applying KVL we get,

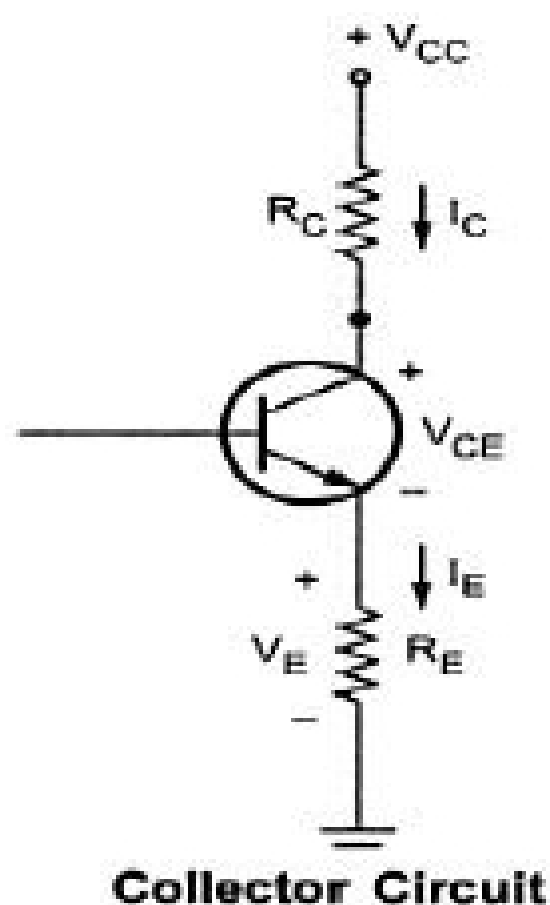
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CC} + I_C R_C - I_E R_E = V_{CE}$$

$$V_E = I_E R_E$$

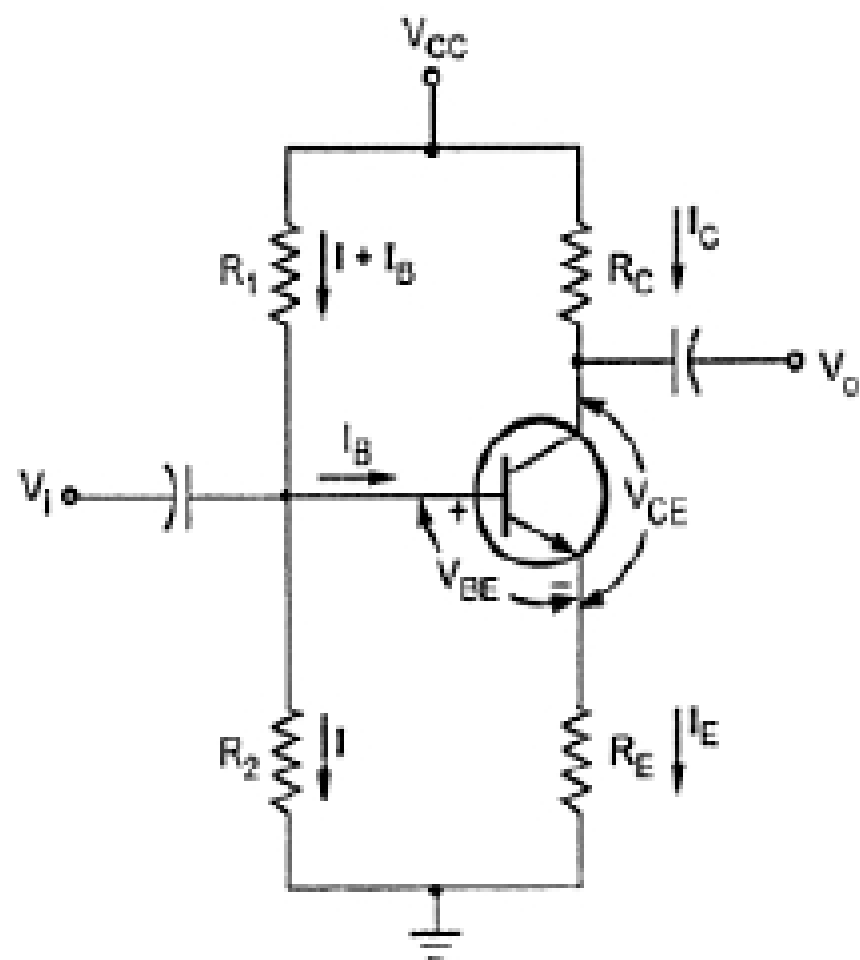
$$V_{CE} = V_{CC} - I_C R_C - V_E$$

$$V_C = V_{CC} - I_C R_C$$

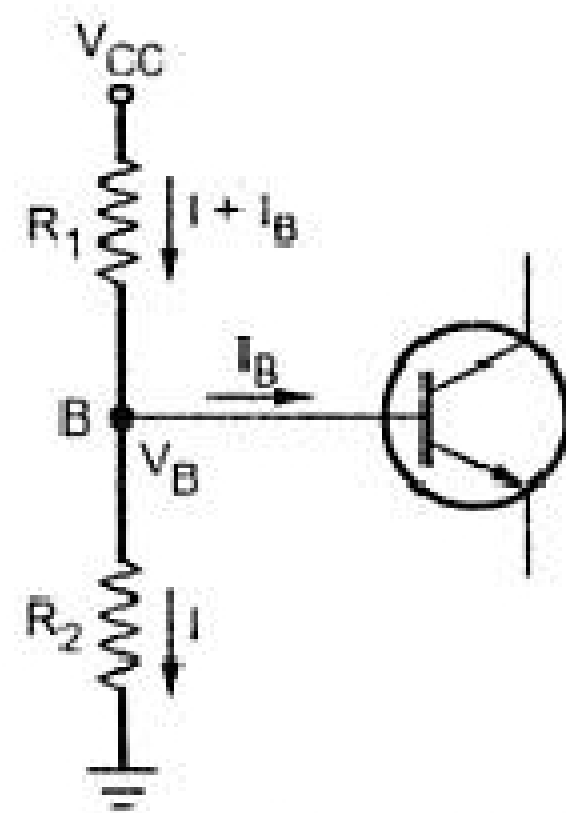


2.5 VOLTAGE DIVIDER BIAS:

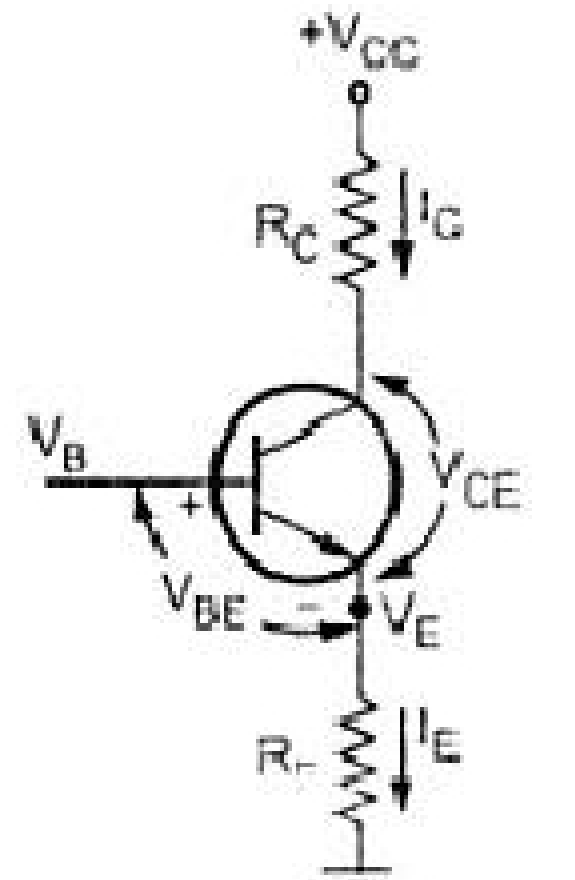
- Figure shows the voltage divider bias circuit. Here the biasing is provided by three resistors: R_1 , R_2 , and R_E . The resistor R_1 and R_2 acts as a potential divider giving a fixed voltage to point B which is base. If the collector current also increases due to the change in the temperature or change in β , the emitter current I_E also increases, reducing the voltage difference between base and emitter (V_{BE}) and the voltage drop across R_E increases. Due to reduction in V_{BE} , the base current I_B and hence the collector current I_C also reduces. Therefore, we can say that negative feedback exists in the emitter bias circuit. This reduction in collector circuit compensates for the original change in I_C .



Voltage divider bias circuit



Base circuit



Collector Circuit

Base circuit:

- From figure, voltage across R2 is the base voltage V_B .
- Applying the voltage divider rule to find V_B , we get

$$V_B = (R_2 (I) / R_1 (I+I_B) + (R_2(I)) * V_{CC}$$

$$V_B = (R_2 / R_1 + R_2) * V_{CC}$$

Collector Circuit:

- From figure, voltage across R_E (V_E) can be obtained as

$$V_E = I_E R_E = V_B - V_{BE}$$

Therefore, $I_E = V_B - V_{BE} / R_E$

- Applying KVL to the collector circuit, we get

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

Therefore, $V_{CE} = V_{CC} - I_C R_C - I_E R_E$

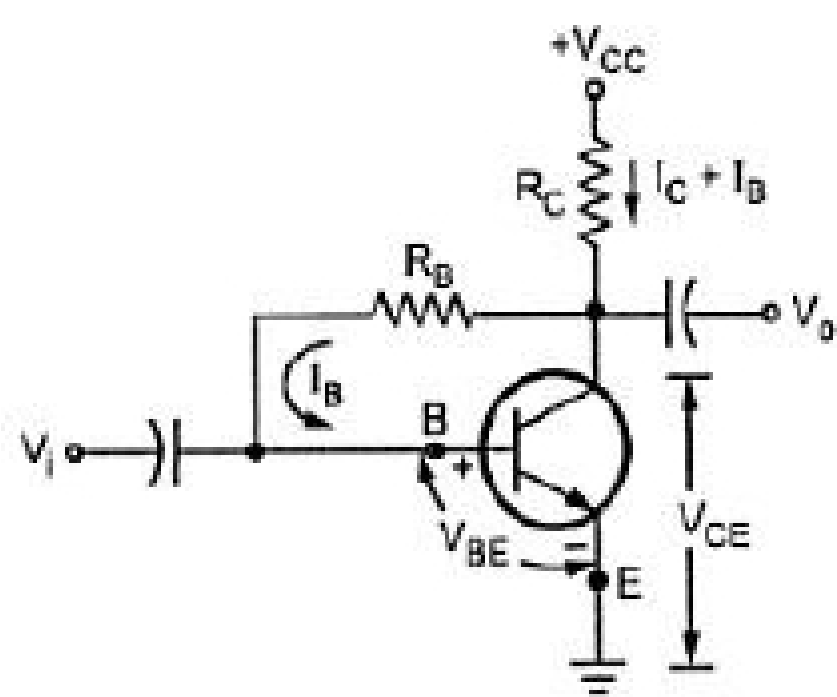
2.6 DC BIAS WITH VOLTAGE FEEDBACK:

- An improved level of stability can also be obtained by introducing a feedback path from collector to base as shown in figure. Although the Q-Point is not totally independent of beta, the sensitivity to changes in beta or temperature variations is normally less than encountered for the fixed bias or emitter biased configuration. The analysis will be performed by fixed analyzing base circuit with the result applied to the collector circuit.

Base Circuit:

- The base circuit for the voltage feedback configuration is shown in figure.

Apply KVL for the circuit.



$$V_{CC} - (I_B + I_C) R_C - I_B R_B - V_{BE} = 0$$

$$V_{CC} = (R_B + R_C) I_B + I_C R_C + V_{BE}$$

$$= (R_B + R_C) I_B + \beta I_B R_C + V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_C}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_C}$$

Dc bias with voltage feedback

Collector circuit:

Applying KVL for the loop in clockwise direction.

$$I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Since $I_C \approx I_E$ and $I_E \approx I_C$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Therefore, $I_{CO} = V_{CC} / R_C + R_E$

2.7 FET BIASING:

- ▶ Like BJT, the parameters of FET are also temperature dependent. In FET as temperature increases drain resistance also increases, reducing the drain current. Thus unlike BJT, thermal runaway does not occur with FET. However, the wide difference in maximum and minimum transfer characteristics make it necessary to keep drain current I_D stable at its quiescent value. The general relationship that can be applied to the DC analysis of all FET amplifier are:

$$I_G = 0$$

$$I_D = I_S$$

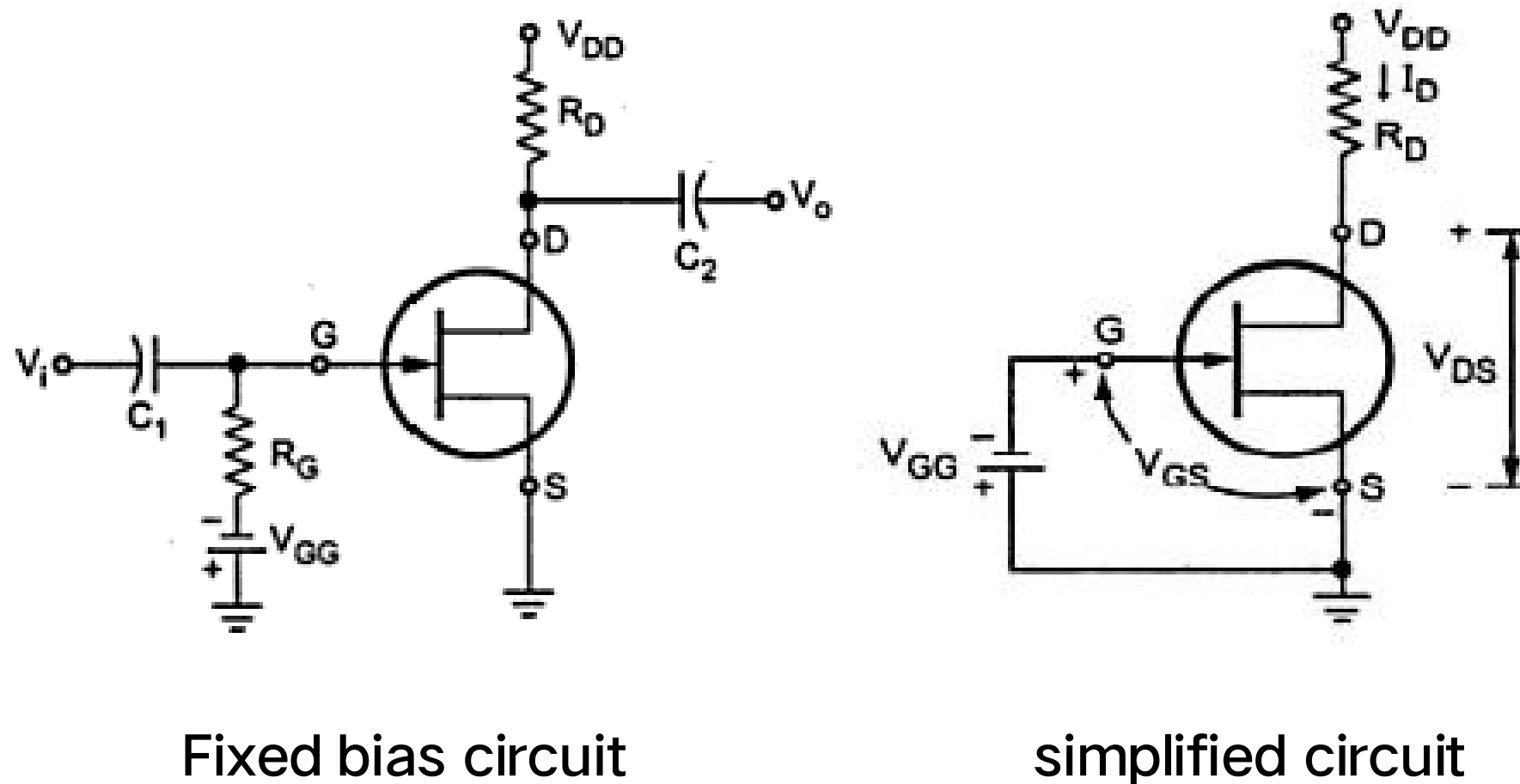
$$I_D = I_{DSS} (1 - (V_{GS} / V_P))^2$$

Biasing circuit:

- ▶ Fixed biasing
- ▶ Self biasing
- ▶ Voltage divider biasing

2.7.1 FIXED BIAS CIRCUIT:

- Figure shows the fixed bias circuit for the n-channel JFET. This is the simplest biasing arrangement. To make gate-source junction reverse biased, a separate supply V_{GG} is connected such that gate is more negative than the source. For the DC analysis coupling capacitors are open circuit. The current through R_G is I_G which is zero. This permits R_G to be replaced by short circuit equivalent, simplifying the fixed bias circuit as shown in the figure.



Fixed bias circuit

simplified circuit

Step 1: calculate V_{GS}

We know that for DC analysis $I_G = 0$ and apply to the input circuit we get, $V_{GS} + V_{GG} = 0$
Therefore,

$$V_{GS} = -V_{GG}$$

Step 2: calculate I_{DQ}

$$I_{DQ} = I_{DSS} (1 - V_{GS}/V_P)^2$$

Step 3: Calculate V_{DS}

$$V_{DD} - I_D R_D - V_{DSQ} = 0$$

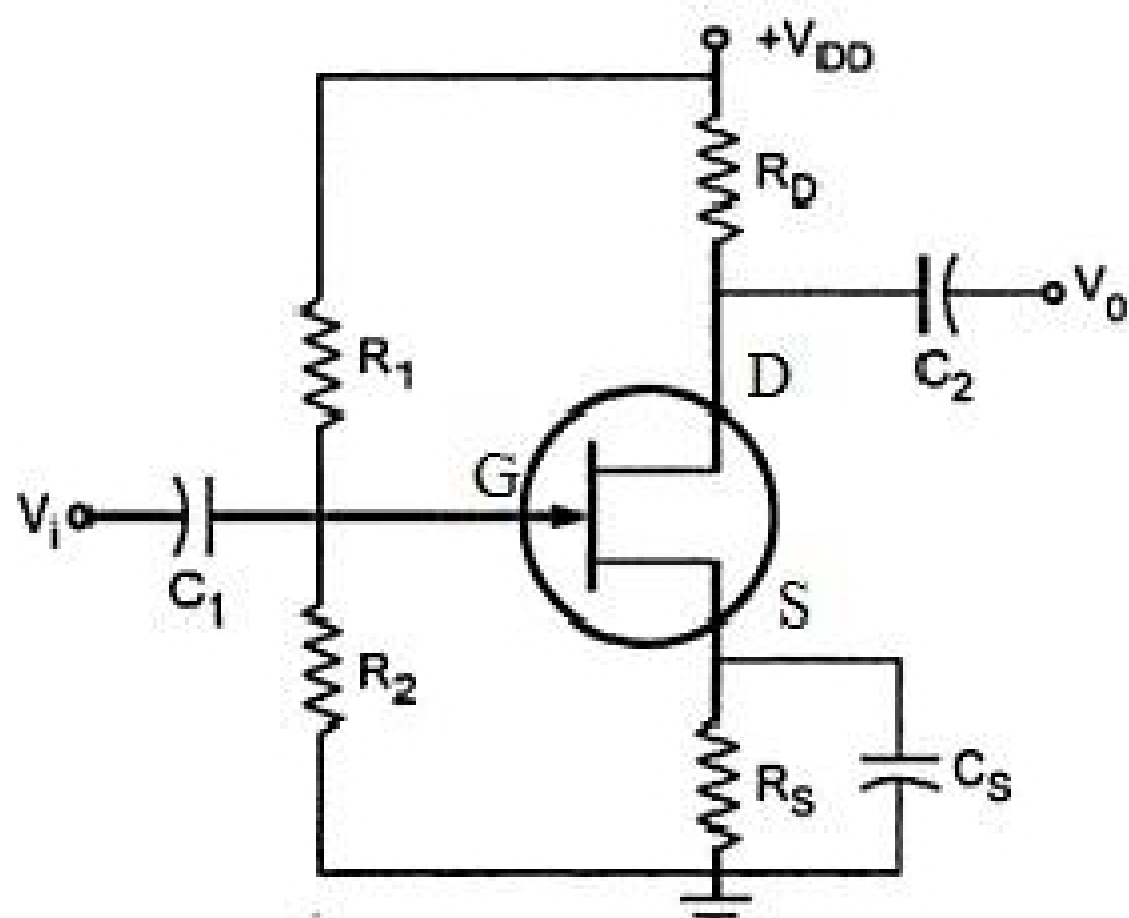
Therefore, $V_{DSQ} = V_{DD} - I_D R_D$

Note:

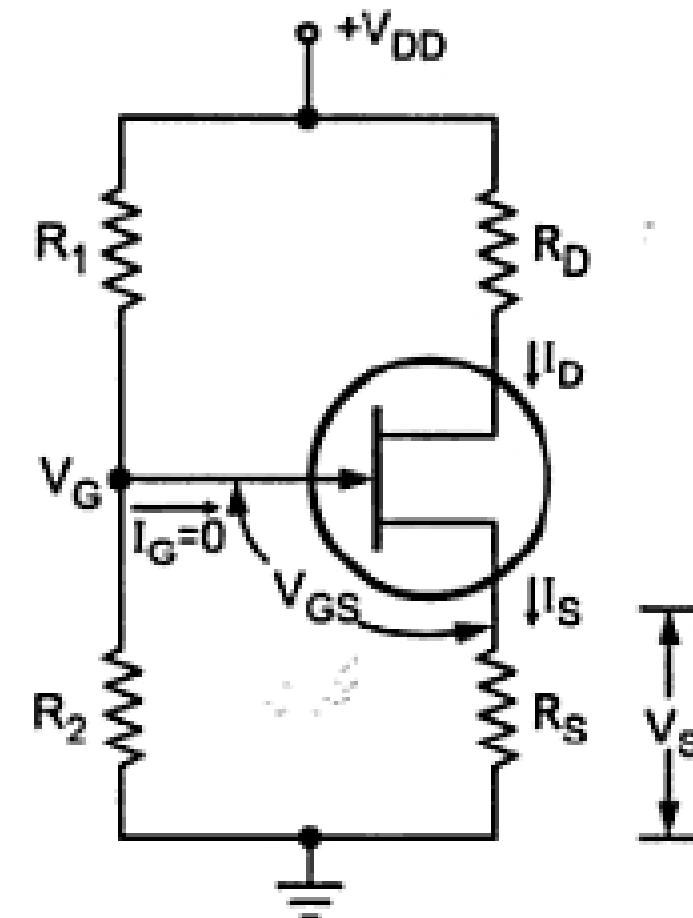
The main drawback of the fixed bias circuit of FET is that it requires two power supplies.

2.7.2 VOLTAGE DIVIDER BIAS:

- Figure shows n-channel JFET with voltage bias. The gate voltage is set by resistors R_1 and R_2 . Coupling capacitors C_1 and C_2 and source resistor bypass capacitor C_S are assumed to be open circuit for DC analysis.
- The source voltage is $V_S = I_D R_S$



Voltage divider bias



voltage divider bias simplified circuit

Dc analysis:Step1: Calculate Vg:

$$V_g = V_{DD} R_2 / (R_1 + R_2)$$

Step 2: Obtain expression for Vgs:

$$V_g - V_{gs} - I_d R_s = 0$$

Therefore, $V_{gs} = V_g - I_d R_s$

Step 3: Calculate Idq:

$$I_{dq} = I_{DSS} [1 - V_{gs} / V_p]^2$$

Step 4: Calculate Vds and Vgs:

Apply kvl to the output circuit we get;

$$V_{DD} - I_{DD} - V_{ds} - I_d R_s = 0$$

Therefore, $V_{ds} = V_{DD} - I_d R_d - I_d R_s$

$$V_{ds} = V_{DD} - I_d (R_d + R_s)$$

The point of a JFET amplifiers using

The voltage divider bias is given by

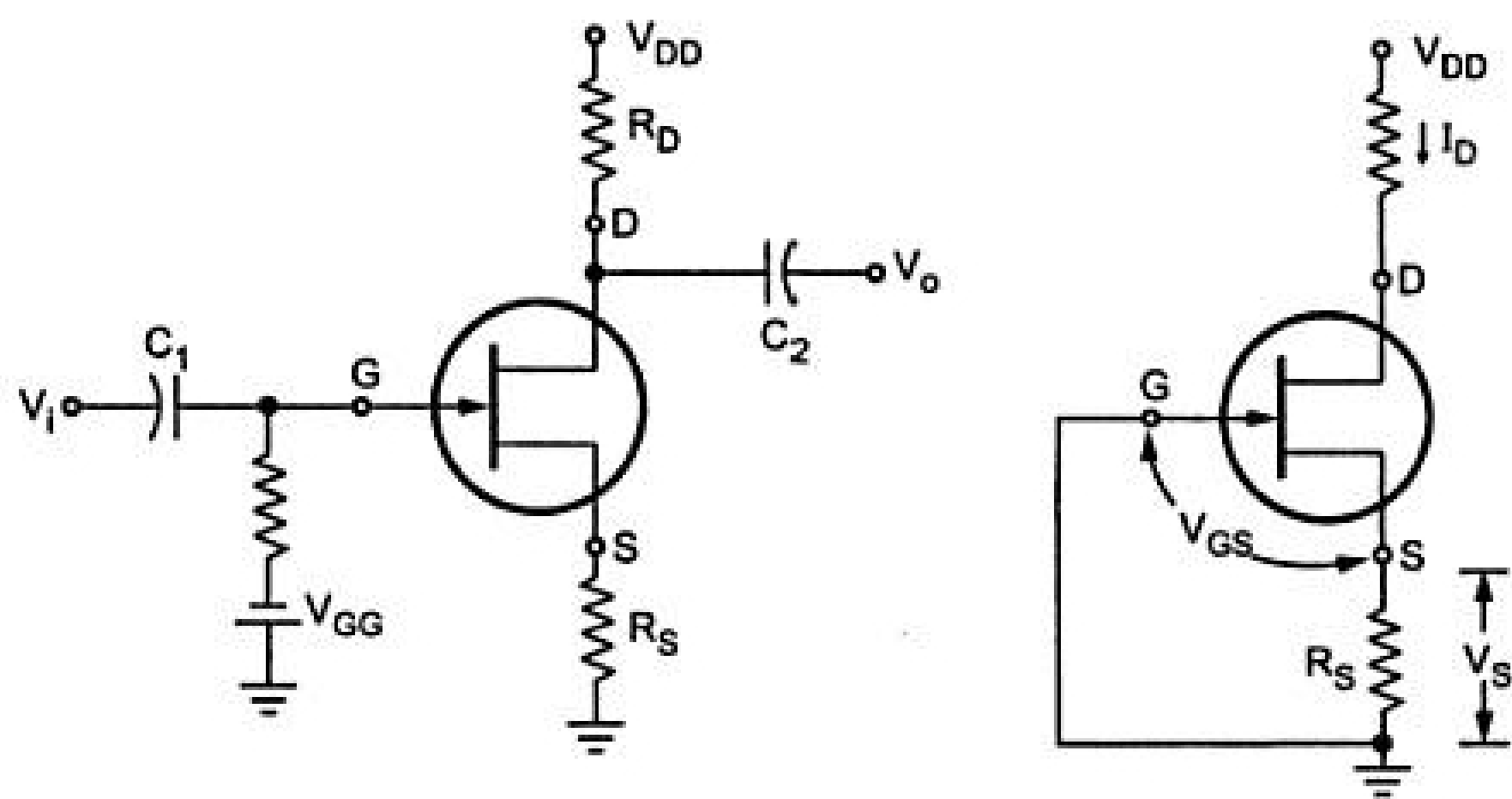
$$I_{dq} = I_{DSS} [1 - v_{gs} / V_p]$$

$$V_{dsq} = V_{DD} - I_d (R_d + R_s)$$

$$V_{gsq} = V_g - I_d R_s$$

2.7.3 SELF BIAS CIRCUIT:

- It is the most common type of JFET bias. Recall that a JFET must be operated such that the gate source junction is always reverse-biased. This condition requires a negative V_{GS} for an n-channel JFET this can be achieved using the self bias arrangement shows in figure .The gate resistor R_g does not affect the bias because it has essentially no voltage drop across it and therefore the gate remains at 0v, R_g is necessary only to isolate an ac signal from ground it analysis .the voltage drops across resistor , R_s makes gate source junction reverse biased .



N-channel JFET

Dc analysis:

Step 1: obtain expression fore V_{gs} :

- For the n-channel FET, it produces a voltage drop across R_s and makes the source positive with respect to ground, since $I_s = I_d$ and $V_g = 0$, then

$$V_s = I_s R_s = I_d R_s. \text{ the gate to source voltage is}$$

$$V_{gs} = V_g - V_s = 0 - I_d R_s = -I_d R_s$$

- For the p-channel FET, I_s produces a voltage drop across R_s and takes the source negative with respect to ground .since $I_s = I_d$ and $V_g = 0$ then

$$V_s = V_g - V_s = 0 - (-I_d R_s) = I_d R_s$$

- In the following dc analysis ,the n – channel a FET shows .for dc analysis we can replace coupling by open circuits and we can replace coupling capacitors by open circuit and we can also replace the resister Rg by a short circuit, since Ig=0.this is illustrated in figure.

Step2: calculate Idq:

$$I_d = I_{dss}[1 - V_{gs}/V_p]$$

Substitute value of Vgs we get,

$$I_d = I_{dss}[1 - (I_d R_s)/V_p] = I_{dss}[1 + I_d R_s/V_p]^2$$

Step 3: Calculate Vds:

Appllying KVL to the output circuit we get

$$V_s + V_{ds} + I_d R_d - V_{dd} = 0$$

$$V_{ds} = V_{dd} - V_s - I_d R_d$$

$$= V_{dd} - I_d R_s - I_d R_d$$

Therefore

$$V_{ds} = V_{dd} - I_d (R_s + R_d)$$

2.8 MOSFET:

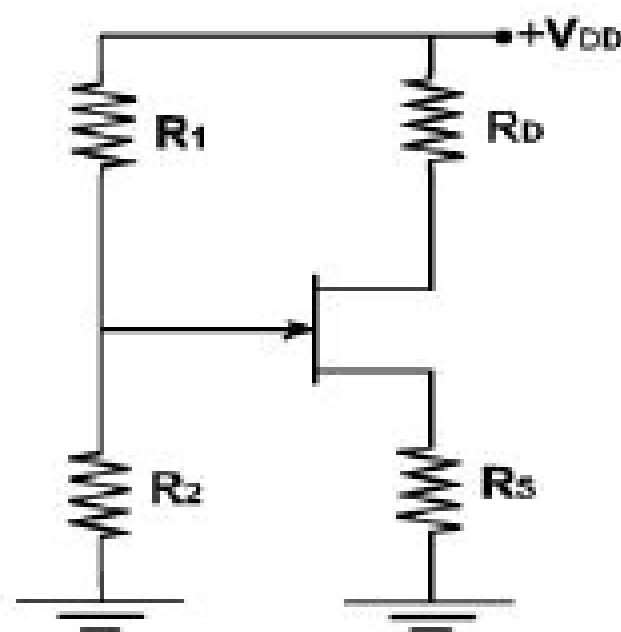
- To analyze and design the dc biasing of MOS transistor circuits. As with bipolar, the dc biasing of MOSFETs, is an importance part of the design of amplifiers. Here, resistors are used in junction with the MOS transistor. In real MOSFET integrated circuit, however, the resisters are generally replaced by other MOSFETs, so the circuit is composed entirely of MOS devices. In the dc analysis of MOSFET circuits, we can use the ideal current-voltage equations listed in table.

Si.no	nMOS	pMOS
1	Non saturation region $V_{ds} < V_{ds}(\text{sat})$ $I_p = k [\frac{1}{2} (V_{gs} - V_T) V_{ds} - \frac{1}{6} V_{ds}^3]$	Non saturation region $V_{sd} < V_{sd}(\text{sat})$ $I_d = k [\frac{1}{2} (V_{sg} - V_T) V_{sd} - \frac{1}{6} V_{sd}^3]$
2	Saturation region $V_{DS} > V_{DS}(\text{SAT})$ $I_p = k (V_{GS} - V_T)^2$	Saturation region $V_{SD} > V_{SD}(\text{SAT})$ $I_p = k (V_{SG} + V_T)^2$
3	Transition point $V_{DS}(\text{SAT}) = V_{GS} - V_T$	Transition point $V_{SD}(\text{SAT}) = (V_{SG} - V_T)$
4	Enhance ment mode	Enhance ment mode

	$V_T > 0$	$V_T < 0$
5	Depletion mode $V_T < 0$	Depletion mode $V_T > 0$

2.8.1 COMMON SOURCE CIRCUIT FOR EMOSFET WITH SOURCE RESISTOR

- Voltage divider bias circuit for W- Channel enhancement type MOSFET is shown in fig.



As $I_G = 0A$,

$$V_G = R_2 V_{DD} / (R_1 + R_2)$$

Applying KVL to the input circuit, we get

$$V_G - V_{GS} - V_S = 0$$

$$V_{GS} = V_G - I_S R_S = V_G - I_D R_D$$

$$V_{GS} = V_G - I_D R_S$$

Applying KVL to the output circuit, we get

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$V_{DS} = V_{DD} - I_D R_D - I_S R_S$$

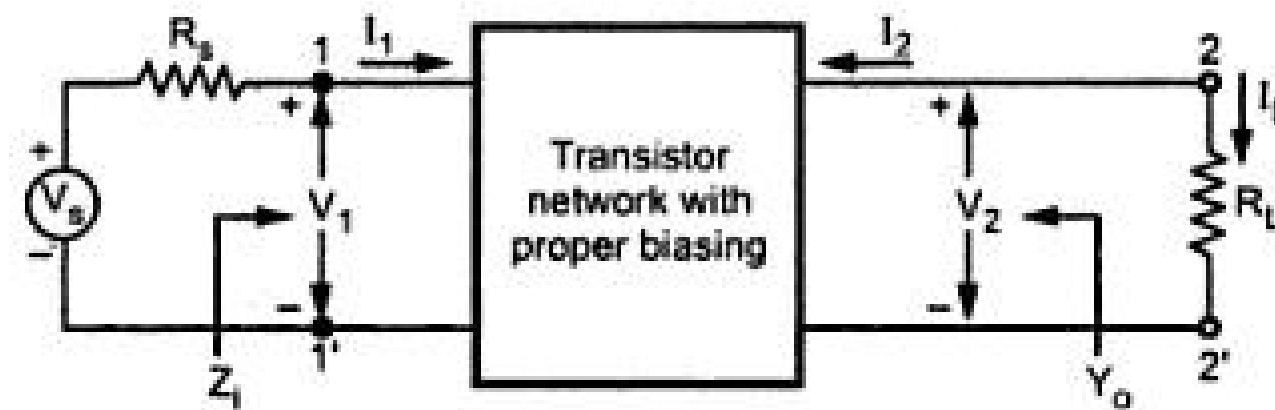
$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

Biasing Circuit for D – MOSFET:

- Biasing Circuit for depletion type MOSFET are quite similar to the circuits used for FET biasing
- The Primary difference between the two is the fact that depletion type MOSFET's also permit operating points with positive value of V_{GS} for n-channel and negative values of V_{GS} for p-channel MOSFET.
- To have positive value of V_{GS} for n-channel and negative value of V_{GS} for p-channel self-bias circuit is unsuitable.

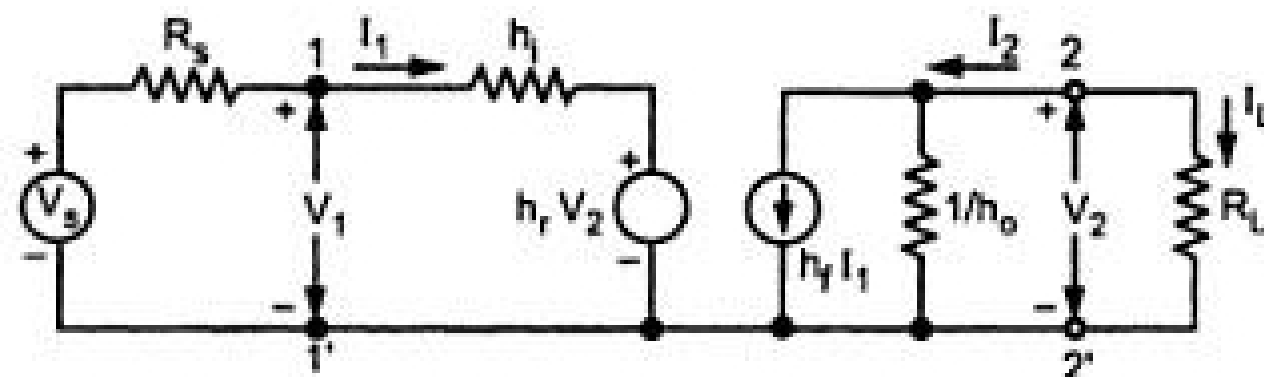
2.9 TRANSISTOR MODELLING:

- Figure shows the basic transistor amplifier only it is necessary to connect an external load and signal source along with proper biasing



Basic transistor amplifier

- Replace transmission circuit with its small signal hybrid model as shown in fig.



Transistor amplifier in its h-parameter model

Important Parameter of BJT:

- Let us analyze hybrid model to find the current gain, the input resistance, the voltage gain and the output resistance.

Current Gain:

- For transistor amplifier (A_i) is defined as the ratio of output to input currents. It is given by $A_i = I_2/I_1 = -I_2/I_1$
- Here I_1 and I_2 are equal in magnitude but opposite in sign (i.e) $I_1 = -I_2$

From the circuit, $I_2 = h_f I_1 + h_o V_2$

Substitute $V_2 = -I_2 R_L$ in the equation we have

$$I_2 = h_f I_1 + h_o V_2$$

Substitute $V_2 = -I_2 R_L$ in the equation we have

$$I_2 = h_f I_1 + h_o (-I_2 R_L)$$

$$I_2 + h_o I_2 R_L = h_f I_1$$

$$I_2 (1 + h_o R_L) = h_f I_1$$

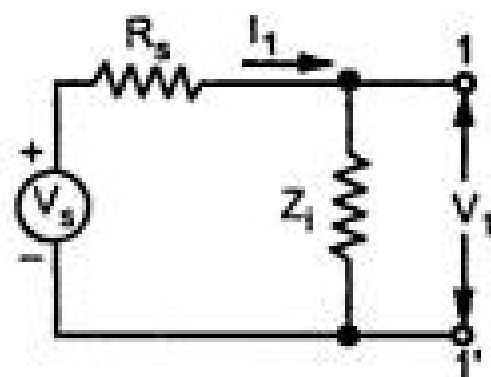
$$I_2/I_1 = h_f / 1 + h_o R_L$$

$$A_i = -I_2/I_1 = -h_f / 1 + h_o R_L$$

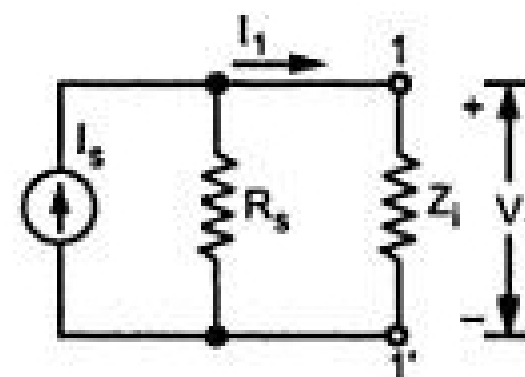
Current Gain (A_{I_s}):

- It is the current gain taking into account for the source resistance, R_s if the model is driven by the current source instead of voltage source. It is given by,

$$A_{I_s} = -I_2/I_s = -I_2/I_1 \cdot I_1/I_s = A_i \cdot I_1/I_s$$



(a) Input section of hybrid model



(b) Input section of hybrid model with current source instead of voltage source

Apply current divider rule for the figure.

$$I_1 = I_s R_s / Z_i + R_s$$

$$I_1/I_s = R_s / Z_i + R_s$$

$$A_{I_s} = A_i R_s / Z_i + R_s$$

Input Impedance (Z_i):

- From figure, R_i is the input resistance looking into the amplifier input terminals (1, 1'). It is given by

$$R_i = V_1 / I_1$$

From the input circuit,

$$V_1 = h_i I_1 + h_r V_2$$

$$Z_i = V_1 / I_1 = h_i I_1 + h_r V_2 / I_1$$

$$= h_i + h_r V_2 / I_1$$

Substitute $V_2 = -I_2 R_L = A_i R_L I_1$ then,

$$Z_i = h_i + h_r A_i R_L I_1 / I_1 = h_i + h_r A_i R_L$$

Substituting $A_i = -h_f / 1 + h_o R_L$

$$Z_i = h_i - h_r h_f R_L / 1 + h_o R_L$$

Divide by R_L in numerator and denominator,

$$Z_i = h_i - h_r h_f / Y_L + h_o$$

Note: Input Impedance is the function of load impedance

Voltage Gain (A_v):

- It is the ratio of output voltage to the input voltage. It is given by

$$A_v = V_2 / V_1$$

$$A_v = A_i I_1 R_L / V_1 = A_i R_L / Z_i$$

Voltage Gain (A_{vs}):

- It is voltage gain including the source. It is given by

$$A_{vs} = V_2 / V_s = V_2 / V_1 * V_1 / V_s = A_v V_1 / V_s$$

- From figure (a), apply voltage divider rule

$$V_1 = V_s \cdot Z_i / Z_i + R_s$$

$$V_1 / V_s = Z_i / Z_i + R_s$$

- Substitute V_1 / V_s in A_{vs} we get

$$A_{vs} = A_i R_L / R_s + Z_i$$

Output Admittance (Y_o):

- It is the ratio of output current I_2 to the output voltage V_2 . It is given by

$$Y_o = I_2 / V_2 \text{ with } V_i = 0$$

We know that,

$$I_2 = h_{f1} I_1 + h_{o2} V_2$$

$$I_2 / V_2 = h_{f1} I_1 / V_2 + h_{o2}$$

$$Y_o = h_{f1} I_1 / V_2 + h_{o2}$$

From fig $V_s = 0$ we can write,

$$R_s I_1 + h_{i1} I_1 + h_{r1} V_2 = 0$$

$$(R_s + h_{i1}) I_1 = -h_{r1} V_2$$

$$I_1 / V_2 = -h_{r1} / R_s + h_{i1}$$

Substitute I_1 / V_2 in Y_o we get

$$Y_o = h_{o2} - h_{f1} h_{r1} / h_{i1} + R_s$$

The output admittance is a function of the source resistance.

Power Gain (A_p):

- It is the ratio of average power delivered to the load R_L , to the output power. Output Power is given as

$$P_2 = V_2 I_2 = -V_2 I_L$$

Since the input power $P_1 = V_1 I_1$ the operating Power gain A_p of the transistor is defined as

$$A_p = P_2/P_1 = -V_2 I_2/V_1 I_1 = A_v A_i = A_{i2} R_L/Z_i$$

Relation between A_{vs} and A_{is} :

We know that,

$$A_{vs} = A_i R_L / Z_i + R_s \text{ and}$$

$$A_{is} = A_i R_s / Z_i + R_s$$

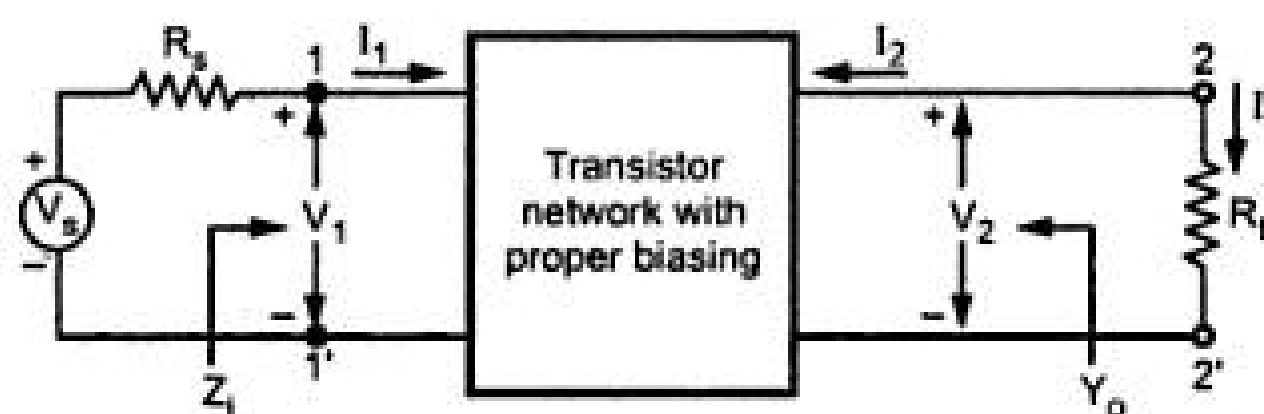
Taking ratio of A_{vs} , A_{is} we get

$$A_{vs}/A_{is} = R_L / R_s$$

$$A_{vs} = A_{is} R_L / R_s$$

2.10 H – PARAMETER MODEL:

- Let us consider transistor amplifier as shown in fig.



Basic transistor amplifier

- The transistor is a current operated device; input current is an independent variable. The input current and output voltage devices the input voltage as well as the output current. Hence input voltage and output current are the dependent variables whereas I_i and V_o are independent variables. Thus can write as

$$V_i = f_1 (I_i, V_o)$$

$$I_o = f_2 (I_i, V_o)$$

This can be written in the equation form as follows

$$V_i = h_{11} I_i + h_{12} V_o$$

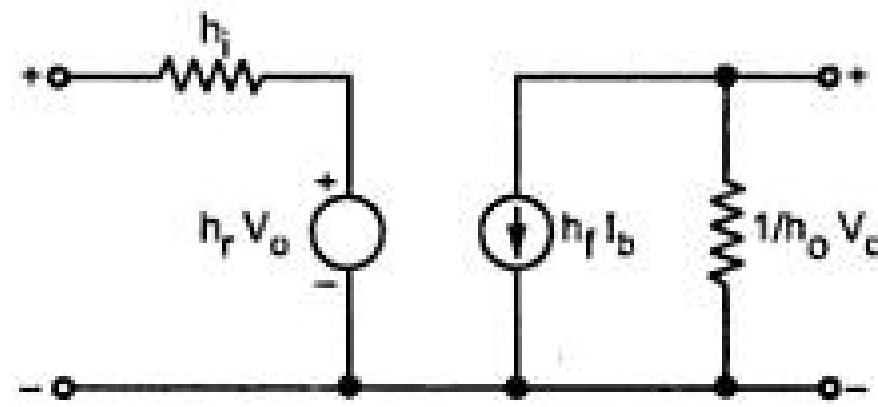
$$I_o = h_{21} I_i + h_{22} V_o$$

It also written in alphabet notations,

$$V_i = h_{i1} I_i + h_{r} V_o$$

$$I_o = h_{f1} I_i + h_o V_o$$

It is shown in fig



Hybrid model or equivalent circuit for transistor

Definition:

The Parameters are,

$H_{11} = V_i/I_i | V_o=0$ - Input resistance with output short circuits.

$H_{12} = V_i/V_o | I_i = 0$ - Fraction of output voltage at input with input open circuited. This parameter is ratio of similar quantities hence unit less

$H_{21} = I_o/I_i | V_o=0$ - Forward current transfer ratio or current gain with output short circuited. This parameter is a ratio of similar quantities, hence unit less.

$H_{22} = I_o/V_o | I_i=0$ - Output admittance with input open circuited.

These all are referred as hybrid in A.C analysis it is referred to as h - parameters.

a.) With output short circuited

$H_{11} = h_i$ - Input resistance

$H_{21} = h_f$ - Short circuit current gain

b.) With input open circuited:

$H_{12} = h_r$ - Reverse voltage transfer ratio

$H_{22} = h_o$ - Output admittance

Benefits:

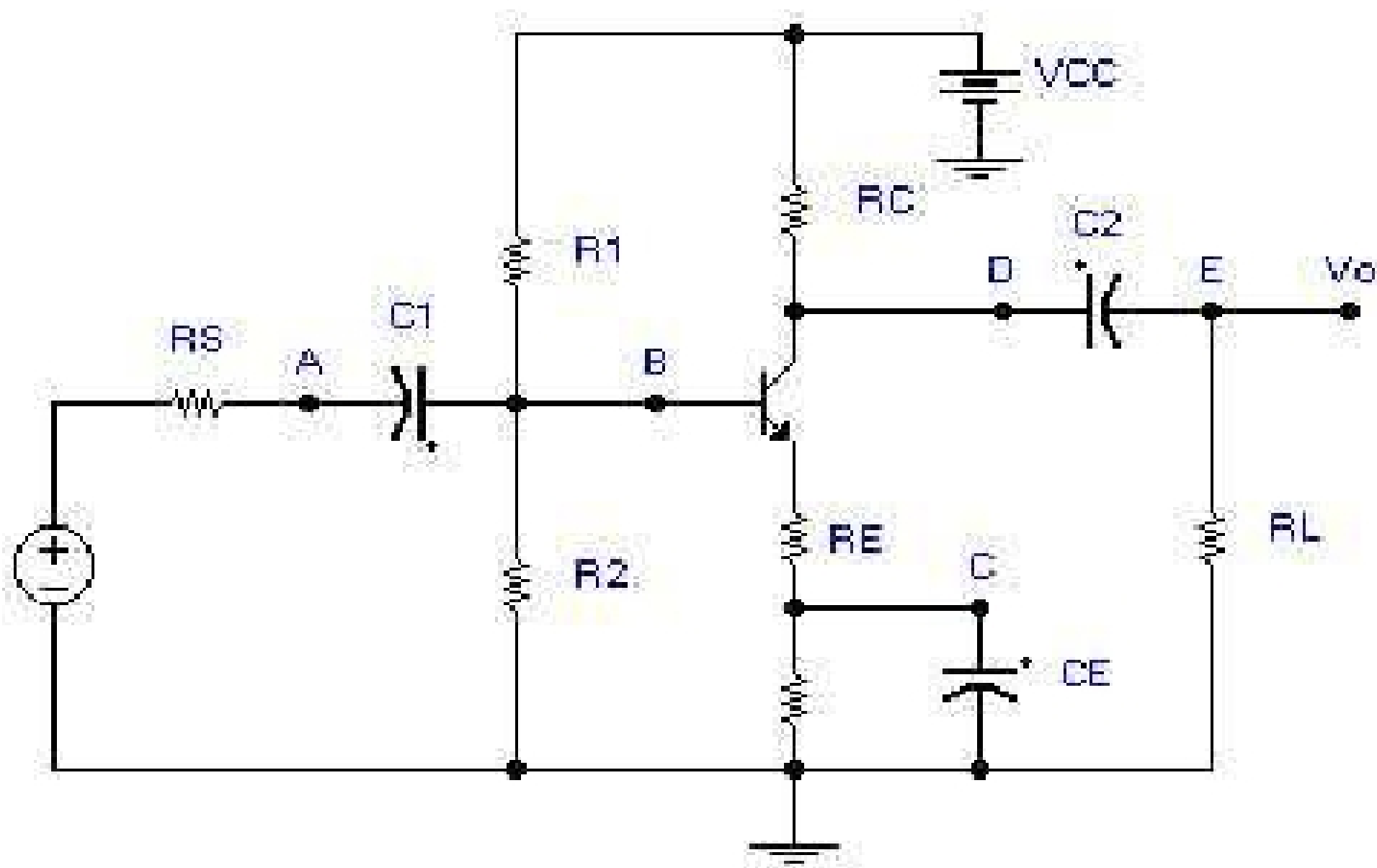
- ▶ Real numbers at audio frequencies
- ▶ Easy to measure
- ▶ Can be obtained from the transistor characteristics curves
- ▶ Convenient to use in circuit analysis and design
- ▶ Most of the transistor manufacturers specify the h - parameters

2.10.1 COMMON EMITTER AMPLIFIER:

- ▶ Figure shown the practical circuit of common emitter transistor amplifier. It consists of differential circuit components are

Biasing Circuit:

- The Resistances R_1 , R_2 and R_E form the voltage divider biasing circuit for the CE amplifier. It sets the proper operating point for the CE amplifier.

**Input Capacitor C_1 :**

- It couples the signal to the base of the transistor. It blocks any DC component present in the signal and passes only AC signal for amplification. Because of the biasing conditions are maintained constant.

Emitter Bypass Capacitor C_E :

- It is connected in parallel with the emitter resistance R_E to provide a low reactance path to the amplified AC signal. If it is not inserted, the amplified AC signal passed through R_E will cause a voltage drop across it. This will reduce the output voltage, reducing the gain of amplifier.

Output Coupling Capacitor C_2 :

- It couples the output of the amplifier to the load or to next stage of amplifier. It blocks DC and passes only AC part of the amplifier signal.

Need for C_1 , C_2 and C_E

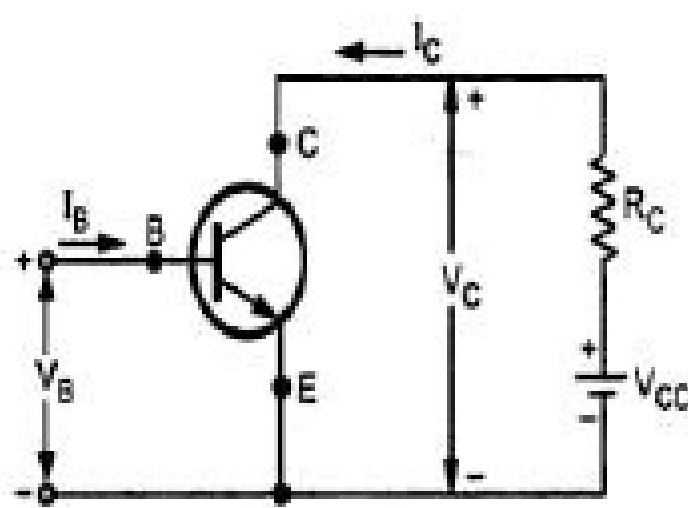
- We know that, the impedance of capacitor is given as

$$X_c = 1/2\pi f_c$$

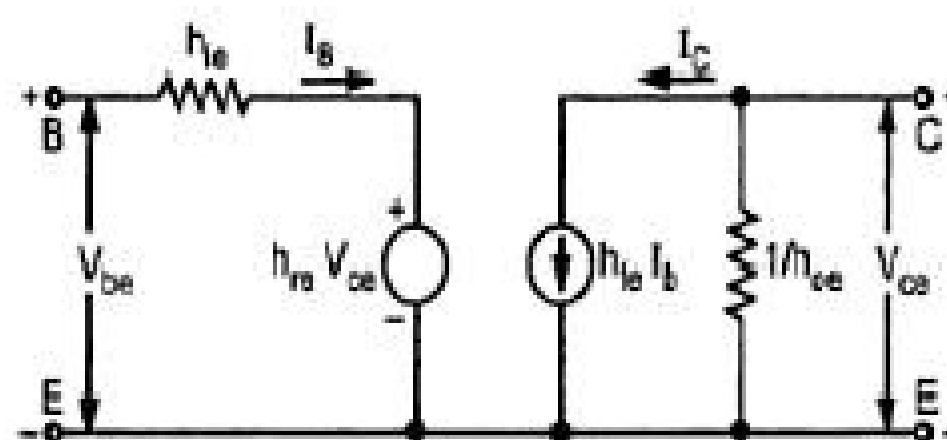
- Thus at signal frequencies all the capacitors have extremely small impedance and it can be treated as an AC short circuit. For bias on DC condition of the transistor all the capacitor acts as DC open circuit with this knowledge we have the C_1 , C_2 and C_E .

Hybrid Model for C_E :

- The Common emitter configuration and the h – parameters models are shown in fig. The variables I_b , I_c , V_{be} and V_{ce} represents input current, output current, input voltage and output voltage.



common emitter configuration



h-parameter equivalent circuit for the common emitter configuration

From the h – parameter model,

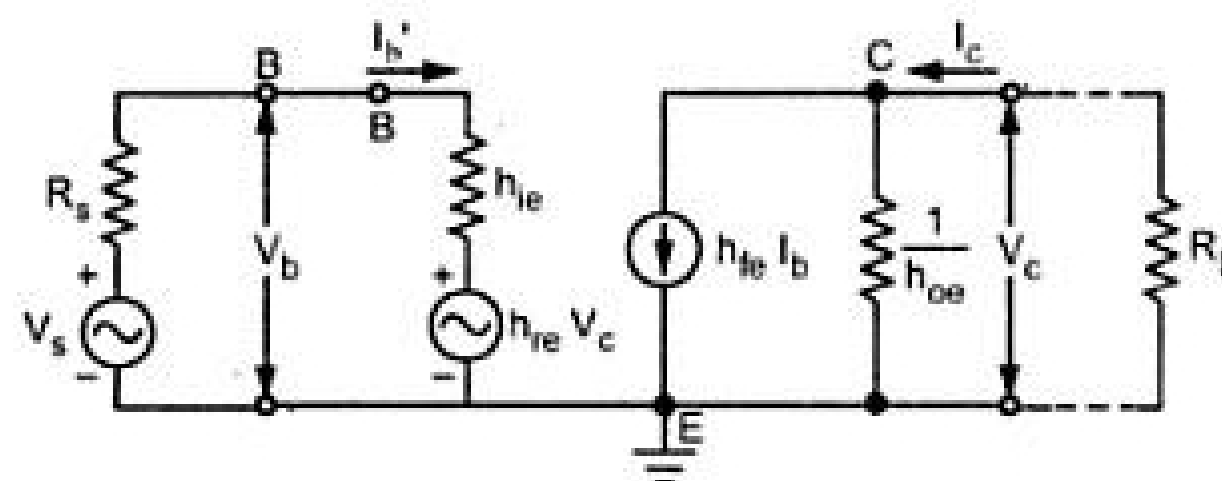
$$V_{be} = h_{ie} I_b + h_{re} V_{ce}$$

$$I_c = h_{fe} I_b + h_{oe} V_{ce}$$

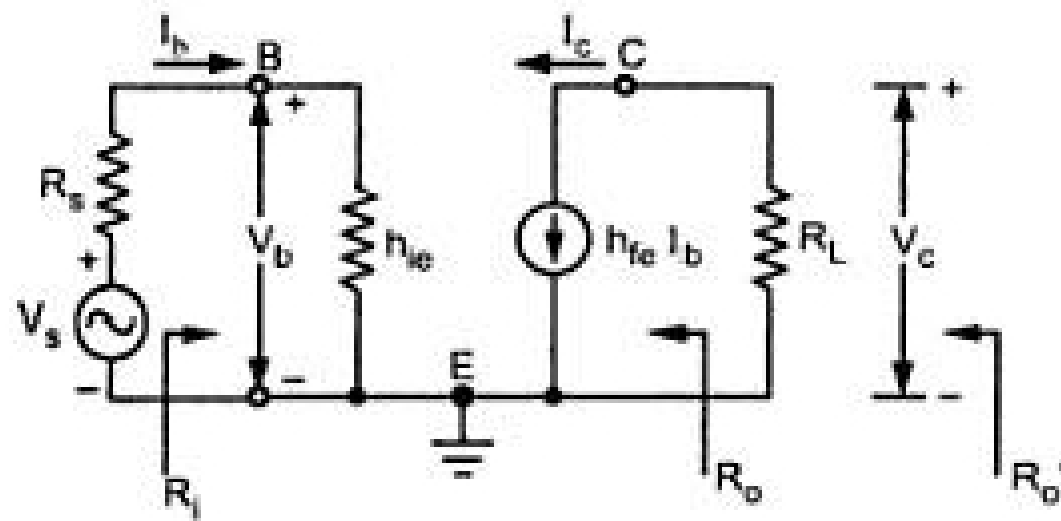
Where $h_{ie} = \Delta V_{be} / \Delta I_b | V_{ce} \text{ constant}$ $h_{re} = \Delta V_{be} / \Delta V_{ce} | I_b \text{ constant}$

$$h_{fe} = \Delta I_c / \Delta I_b | V_{ce} \text{ constant} \quad h_{oe} = \Delta I_c / \Delta V_{ce} | I_b \text{ constant}$$

Analysis of CE Amplifier:



- Let us consider the h – parameter model equivalent circuit for the amplifier as shown in fig.
- Now see how the modification of this model so as to make the analysis simple without greatly sacrificing accuracy.
- Since $1/h_{oe}$ is in parallel with R_L and R_C if $1/h_{oe} \gg R_L \parallel R_C$, then h_{oe} may be neglected



Approximate CE model

- Then h_{oe} , the collector current I_c is given by

$$I_c = h_{fe} I_b$$

- Under these conditions the magnitude of the voltage of the generator in the emitter circuit is

$$h_{re} |V_{ce}| = h_{re} I_c (R_L \parallel R_c) = h_{re} h_{fe} I_b (R_L \parallel R_c)$$

- Since $h_{re} h_{fe} = 0.01$, this voltage may be neglected in comparison with the $h_{ie} I_b$ drop across h_{ie} , provided that $R_L \parallel R_c$ is not too large.
- Therefore, if the load resistance $R_L \parallel R_c$ is small, it is possible to neglect the parameters h_{re} and h_{oe} is the h - parameter equivalent circuit. Fig shows the h parameter equivalent circuit.

Current Gain:

$$A_i = -I_c / I_b = -h_{fe} / 1 + h_{oe} R_L$$

By neglecting h_{oe} ,

$$A_i \approx -h_{fe}$$

Input Impedance:

$$R_i = h_{ie} + h_{re} A_i R_L$$

By neglecting h_{re} ,

$$R_i \approx h_{ie}$$

Voltage Gain:

$$A_v = A_i R_L / R_i = A_i R_L / h_{ie}$$

Output Admittance:

$$Y_o = h_{oe} - h_{fe} h_{re} / h_{ie} + R_s$$

By neglecting h_{oe} and h_{re}

$$Y_o = 0$$

$$R_o = 1/Y_o = \infty$$

$$R_o'' = R_o \parallel R_L = \infty \parallel R_L$$

$$R_o'' = R_L$$

2.11 IMPORTANT PARAMETERS OF JFET:

- ▶ The drain to source current of JFET is controlled by gate to source voltage. The change in the drain current due to change in gate to source voltage can be determined using the transconductance factor g_m . It is given as

$$\Delta I_d = g_m \Delta V_{GS}$$

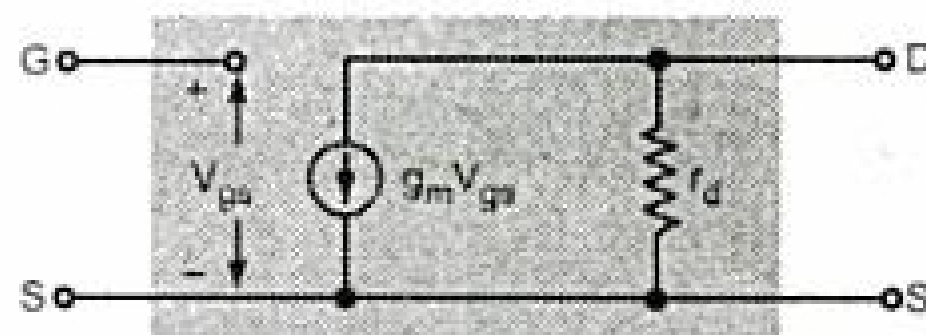
- ▶ In BJT the relation between an output and input quantity is given by amplification factor β , whereas in JFET this relation is given by transconductance factor g_m .
- ▶ The another important parameter of JFET is drain resistance R_d is given by

$$R_d = \Delta V_{ds} / \Delta I_d \mid V_{GS} \text{ constant}$$

- ▶ It determines the output impedance Z_o of the JFET amplifier.

Small Signal Model of FET:

- ▶ Figure shows the small signal low frequency AC equivalent circuit for n channel JFET. The relation of I_d by V_{GS} is included as a current source $g_m V_{GS}$ connected from drain to source. The input impedance is represented by the open circuit at its input terminals, since gate current I_g is 0. The output impedance by r_d from drain to source.



JFET low frequency ac equivalent circuit for n-channel JFET

JFET s mall signal model and Parameters:

- ▶ The JFET parameters are the major components of the low frequency small signal model for JFET. Hence before examining the low frequency small signal model we did JFET parameters.

JFET Parameters:

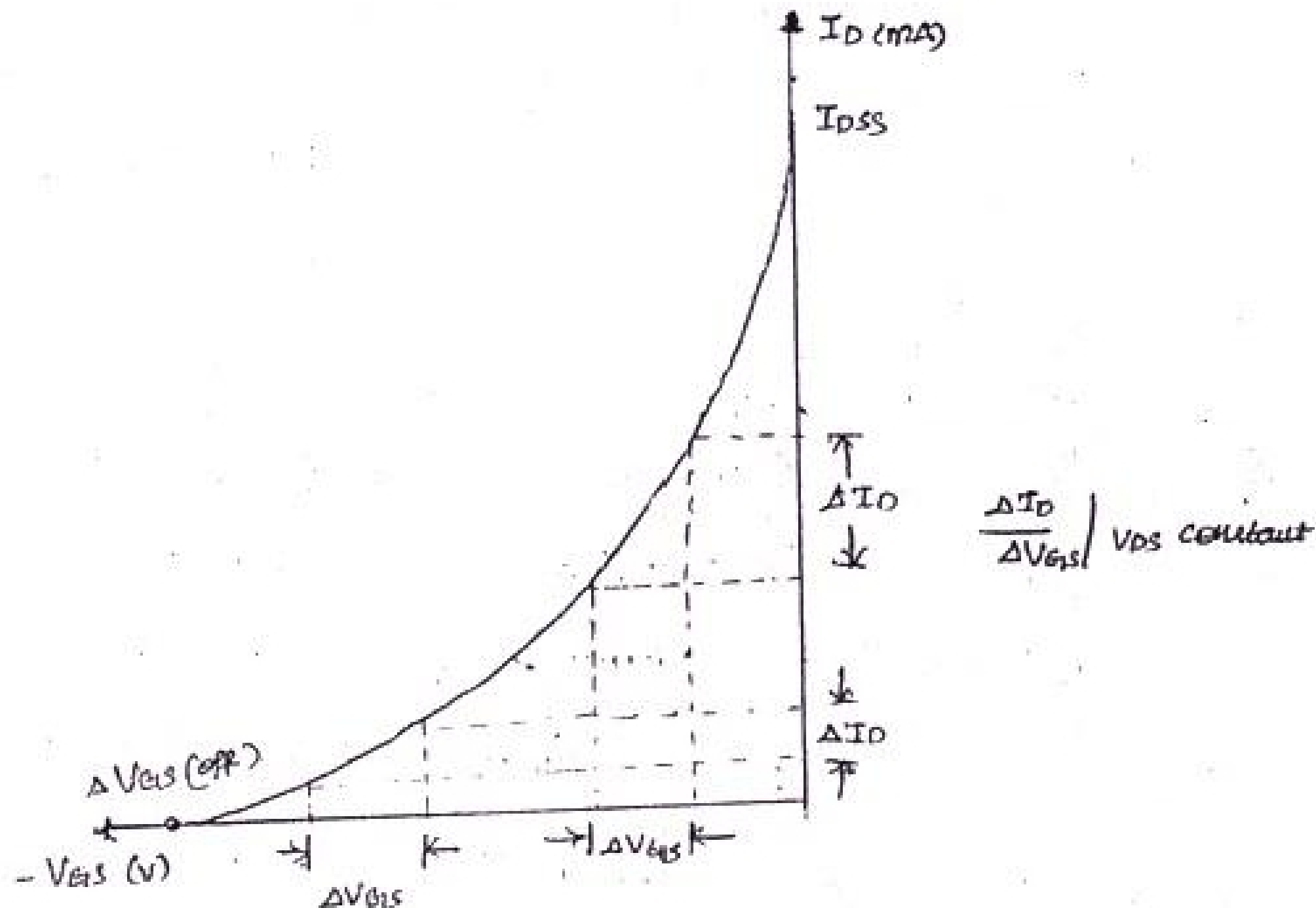
The important parameters of JFET are as follows.

- ▶ Transconductance
- ▶ Input resistance and capacitance
- ▶ Drain to source resistance (r_d)

- Amplification factor (μ)

Transconductance:

- The transconductance g_m is the change in the drain current for given change in gate to source voltage with the drain to source voltage constant as shown in fig.



Transconductance depends on V_{GS}

- From figure, we noted that it is the slope of the transfer characteristics. Since the slope varies, g_m also varies. G_m has greater value near the top of the curve than it does near the bottom. It is defined as

$$G_m = \Delta I_D / \Delta V_{GS} \mid V_{DS} \text{ constant}$$

- The transconductance (g_m) is also called mutual conductance. The practical unit for g_m is mS or mAN. For given g_m , we can calculate an approximate value for g_m at any point on the transfer characteristics of curve using the following equation.

$$G_m = g_{mo} [1 - V_{GS} / V_{GS(off)}]$$

- Where g_{mo} is the value of g_m for $V_{GS}=0$, and it is given by

$$G_{mo} = -2I_{DSS} / V_p$$

- This can be proved as given below, we know that

$$I_D = I_{DSS} [1 - V_{GS} / V_p]^2$$

Diff w.r.t V_{GS} we get

$$G_m = \Delta I_d / \Delta V_{GS} = -2I_{DSS} / V_p [1 - V_{GS}/V_p]$$

$$G_m = g_{mo} [1 - V_{GS} / V_p]$$

Input Resistance and Capacitance:

- ▶ A JFET operates with its gate-source junction reverse biased. Therefore the input resistance at the gate is very high. This high input resistance is one advantage of the JFET over the bipolar transistor. It often specifies the input resistance by giving a value of the gate reverse current (I_{GSS}), at a certain gate-to-source voltage (V_{GS}). The input resistance can then be determined using the following equation, where the vertical lines indicate an absolute value

$$R_{IN} = |V_{GS} / I_{GS}|$$

- ▶ The input resistance decreases with increase in temperature.
- ▶ The input capacitance C_{iss} is a result of JFET operating with a reverse biased PN junction. The reverse biased PN junction acts as capacitance whose capacitance depends on the amount of reverse voltage.

Drain to Source Resistance:

- ▶ From the drain characteristic, the important parameter of JFET, drain resistance r_d can be calculated. Fig shows drain characteristics of n-channel JFET.
- ▶ The drain resistance r_d is the AC resistance between drain and source terminals upon which the JFET is operating in the saturation region. It is reciprocal of the slope of the drain characteristics in the saturation region and is given by.

$$R_d = \Delta V_{DS} / \Delta I_D | V_{GS} \text{ constant}$$

UNIT-3

Topic	Page. No
3.1 RC Coupled Amplifier	59
3.1.1 Operation	60
3.1.2 Frequency Response	61
3.2 Power Amplifiers	61
3.2.1 Classification Of Large Signal Amplifier	62
3.2.2 CLASS A Amplifiers	62
3.2.2.1 Classification Of CLASS A Amplifier	63
3.2.3 CLASS B Amplifiers	68
3.2.3.1 Push Pull CLASS B Amplifier	69
3.2.3.2 Complementary Symmetry CLASS B Amplifier	71
3.2.4 Amplifier Distortion	73
3.2.5 CLASS C Amplifiers	74
3.2.6 CLASS D AMPLIFIERS	76
3.3 Concept Of Feedback	78
3.3.1 Sampling Network	79
3.3.2 Feedback Network	80
3.3.3 Mixer Network	80
3.3.4 Types Of Feedback	80
3.3.4.1 Negative Feedback	80
3.4 Barkhausen Criterion	82
3.5 Wein Bridge Oscillator	84
3.6 Hartley Oscillator	85
3.7 Colpitts Oscillator	87
3.8 Crystal Oscillators	88
3.8.1 Pierce Crystal Oscillator	89
3.9 Frequency Stability	90
3.9.1 Factors Affecting The Frequency Stability	90

Department of Computer Science and Engineering

Subject Name: ELECTRONIC DEVICES AND CIRCUITS

Prepared by:

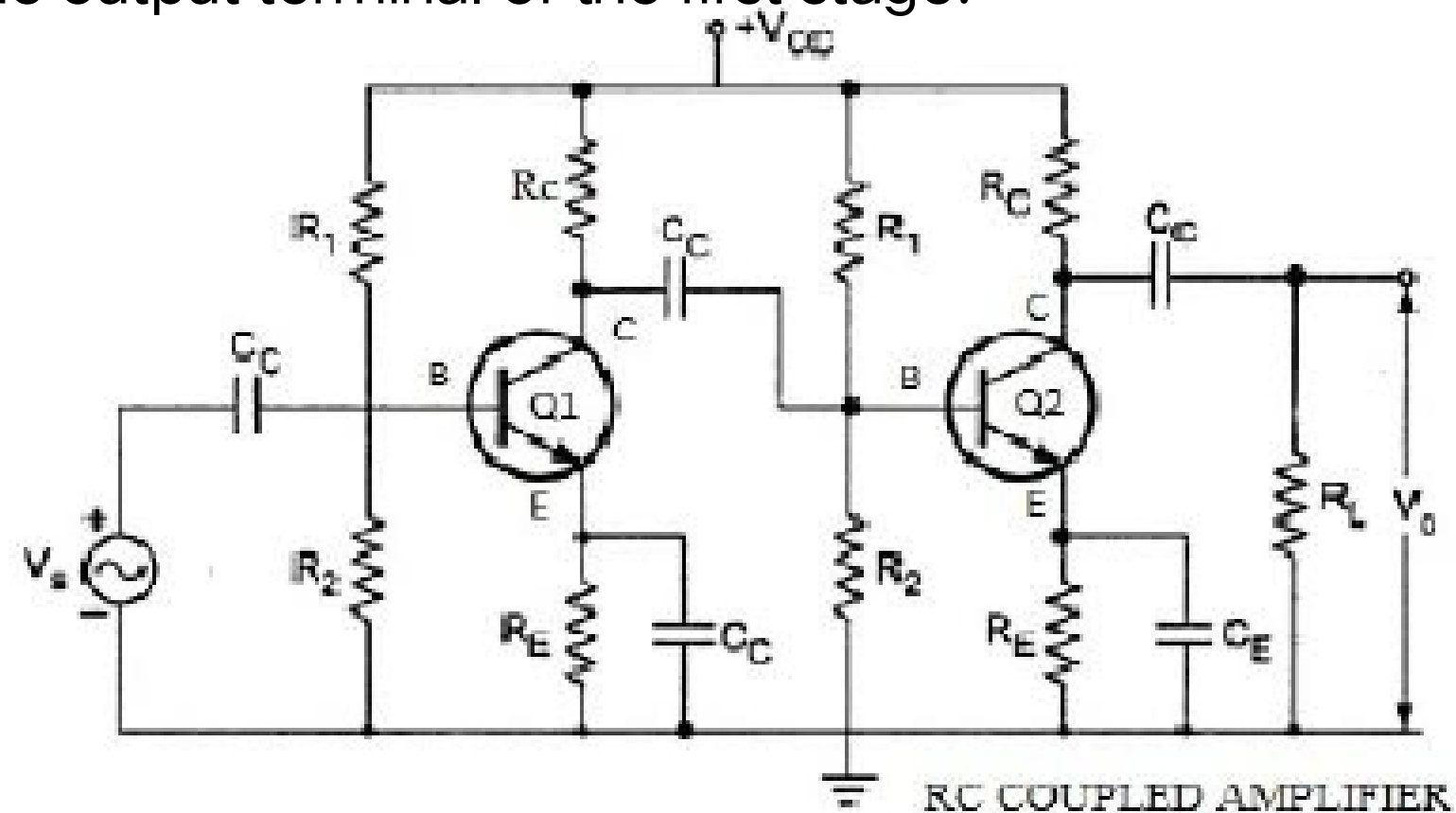
Mr. I.KAVIARASAN. M. Tech (Assistant Professor)

UNIT III

RC-coupled amplifier, Operation and Frequency response, Power amplifier – Series fed and transformer coupled Class A amplifiers, Class B amplifier, Circuit and Operation, conversion efficiency, amplifier distortion, Class C and D amplifiers. Concept of feedback- Negative and positive feedback, Barkhausen Criterion – Wien bridge oscillators, Hartley, Colpitts and crystal oscillator – Frequency stability.

3.1 RC COUPLED AMPLIFIER

- Figure shows RC amplifier using transistor. As seen from figure, the output signal of first stage is coupled to the input of the next stage through coupling capacitor and resistive load at the output terminal of the first stage.



3.1.1 OPERATION

- ▶ When input AC. signal is applied to the base of the transistor of the 1st stage of RC coupled amplifier, from the function generator, it is then amplified across the output of the 1st stage. This amplified voltage is applied to the base of next stage of the amplifier, through the coupling capacitor C_{out} where it is further amplified and reappears across the output of the second stage.
- ▶ Thus the successive stages amplify the signal and the overall gain is raised to the desired level. Much higher gain can be obtained by connecting a number of amplifier stages in succession.
- ▶ Resistance-capacitance (RC) coupling in amplifiers are most widely used to connect the output of first stage to the input (base) of the second stage and so on. This type of coupling is most popular because it is cheap and provides a constant amplification over a wide range of frequencies.

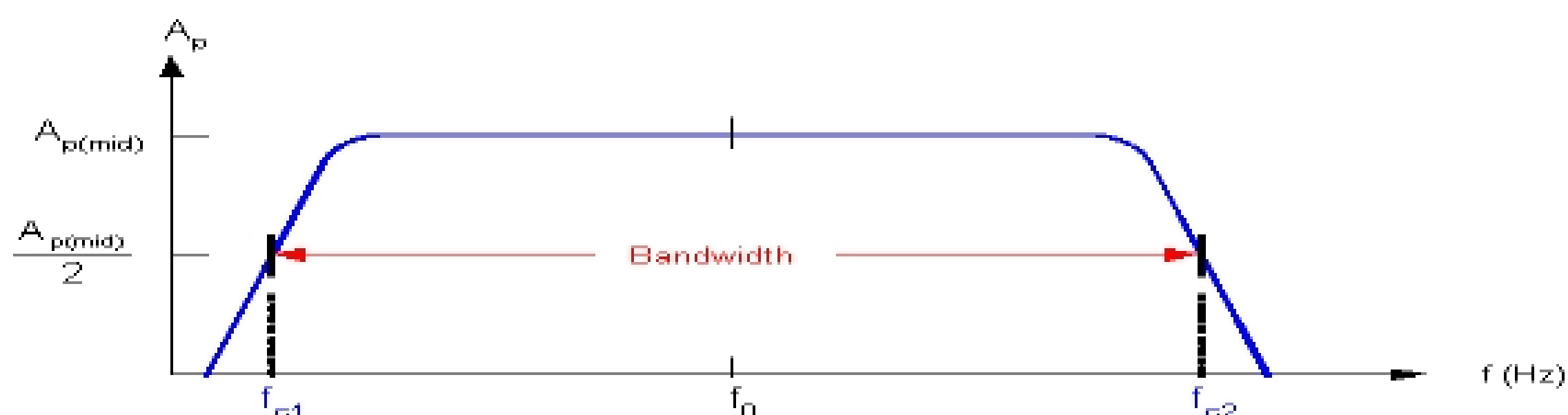
Transistor as Amplifiers

While knowing about different circuits for RC coupled amplifiers, it is important to know about transistors basics as amplifiers. The three configurations of the bipolar transistors that are commonly used are common base transistor (CB), common emitter transistor (CE) and common collector transistors (CE). Other than transistors, operational amplifiers can also be used for amplification purpose.

- Common emitter configuration is commonly used in the audio amplifier application because common emitter has a gain that is positive and also greater than unity. In this configuration emitter is connected to ground and has high input impedance. Output impedance will be medium. Most of these types of transistor amplifier applications are commonly used in RF communication and optical fiber communications (OFC).
- Common base configuration has a gain less than unity. In this configuration collector is connected to ground. We have low output impedance and high input impedance in common base configuration.
- Common collector configuration is also known as emitter follower because the input applied to the common emitter appears across the output of the common collector. In this configuration collector is connected to ground. It has low output impedance and high input impedance. It has a gain almost equal to the unity.

3.1.2 FREQUENCY RESPONSE

- ▶ The range of frequency that an amplifier circuit can amplify properly is known as the bandwidth of that particular amplifier. The curve below represents the frequency response of the single stage RC coupled amplifier.
- ▶ The curve which represents the variation of gain of an amplifier with frequency is called frequency response curve. The bandwidth is measured between the lower half power and upper half power points. P1 point is lower half power and P2 is upper half power respectively. A good audio amplifier must have bandwidth from 20 Hz to 20 kHz because that is the frequency range which is audible.



3.2 POWER AMPLIFIERS

- ▶ It is the last stage of multistage amplifiers. The previous stages develop sufficient gain and the input signal level and the amplitude of the power amplifier is large of the order of few volts.
- ▶ The output of power amplifier has large current and voltage swings. As it handles large signals called power amplifiers.
- ▶ The h-parameters analysis is appropriate to the small signal amplifiers and hence cannot be used for the analysis of power amplifiers. The analysis is carried out graphically by drawing a load line on the output characteristics of the transistors used in it.
- ▶ The power amplifiers (i.e.,) large signal amplifiers are used to feed the loads like loudspeakers having low impedance. So for maximum power transfer the impedance matching is important. Hence the power amplifiers must have low output impedance. Hence emitter follower circuit is very common in power amplifiers. The common emitter circuit with step down transformer for impedance matching is also commonly used.

- ▶ It develops an AC power of the order of few watts. Similarly large power gets dissipated in the form of heat at the junctions of the transistors used in the power amplifiers are of large size having large power dissipation rating called power transistors. Such transistors have heat sinks. A heat sink is a metal cap having bigger surface area, pressure fit on the body of the transistor to get more surface area to dissipate the heat to the surroundings. In general the power amplifiers have bulky components.
- ▶ A faithful reproduction of the signal after the conversion is important. Due to non-linear nature of the transistor characteristics, there exists a harmonic distortion in the signal. Ideally the signal should not be distorted in the case of power amplifier is important.
- ▶ In many cases the power amplifiers are used in public address systems and many audio circuits to supply large power to the loudspeakers. Hence power amplifiers are also called audio amplifiers or audio frequency (AF) power amplifier.

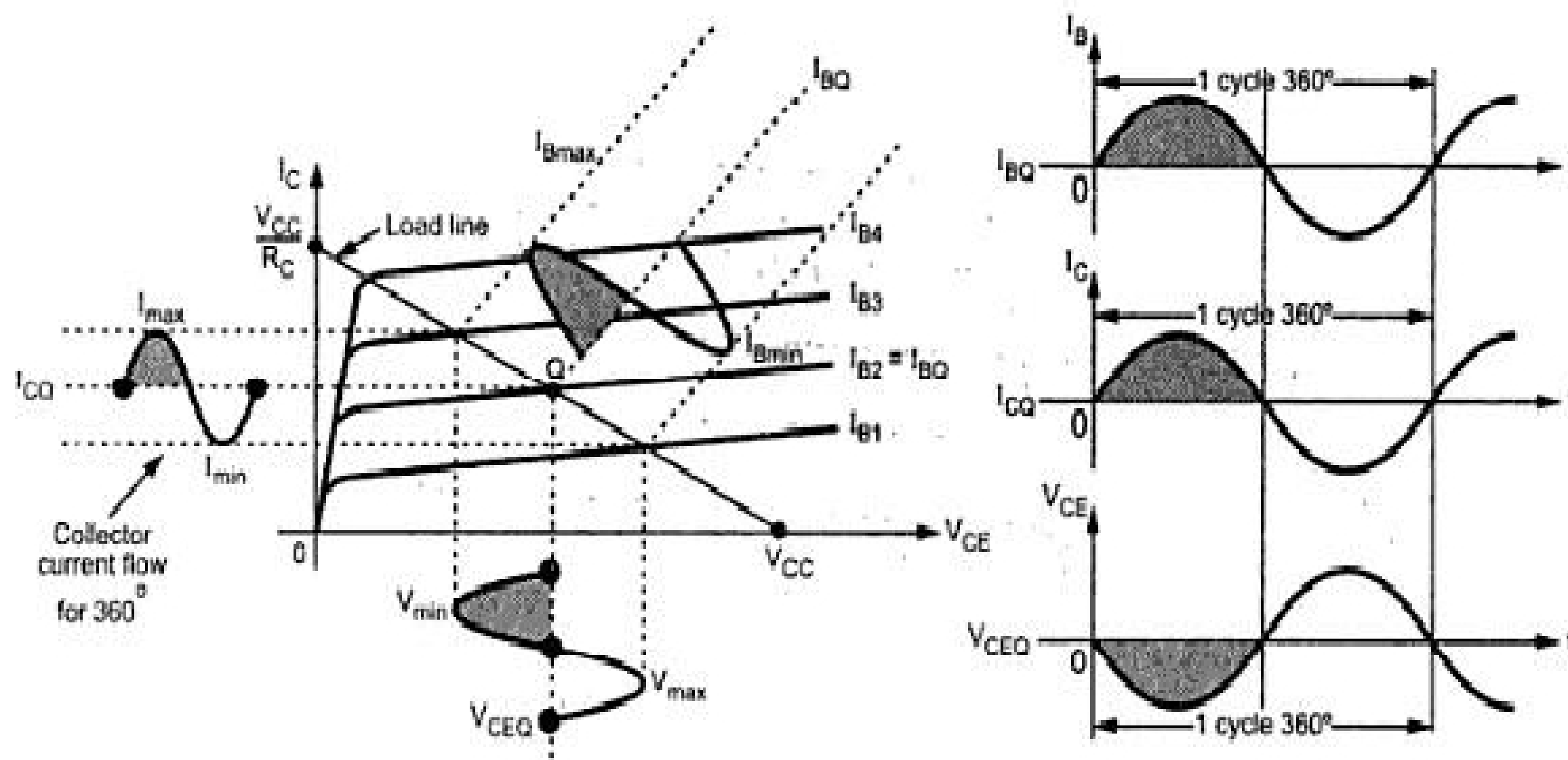
3.2.1 CLASSIFICATION OF LARGE SIGNAL AMPLIFIER

- ▶ For an amplifier, a quiescent operating point (Q-point) is fixed by selecting the proper DC biasing to the transistor used. The Q-point is shown in the load line which is plotted on the output characteristics of the transistor. The position of the Q-point on the load line decides the class of the operation of the power amplifier. The various classes of power amplifier are
 - ✓ Class A amplifier - Series fed, transformer coupled
 - ✓ Class B amplifier - Push pull, complementary symmetry
 - ✓ Class C amplifier
 - ✓ Class D amplifier

3.2.2 CLASS A AMPLIFIERS

- ▶ The power amplifier is said to be class A amplifier if the Q-point and the input signal are selected such that the output signal is obtained for full input cycle. For this class position of the point of the Q-point is approximately at the midpoint of the load line.
- ▶ For all values of the input signal, the transistor remains in the active region and never enters into the cut-off region or saturation region. When an AC input signal is applied, the collector voltage varies sinusoidal hence the collector current also varies sinusoidally. The collector current flows for 360° of the input signal. In other words, the angle of the collector current flows is 360°.

- The current and voltage waveforms for class A operators are shown with the help of the output characteristics and the load line is figure. From the figure, for all full input cycle, a full output cycle is obtained. Here signal is faithfully reproduced, at the output, without any distortion. This is an important feature of class A operation. The efficiency of class A operation is very small.

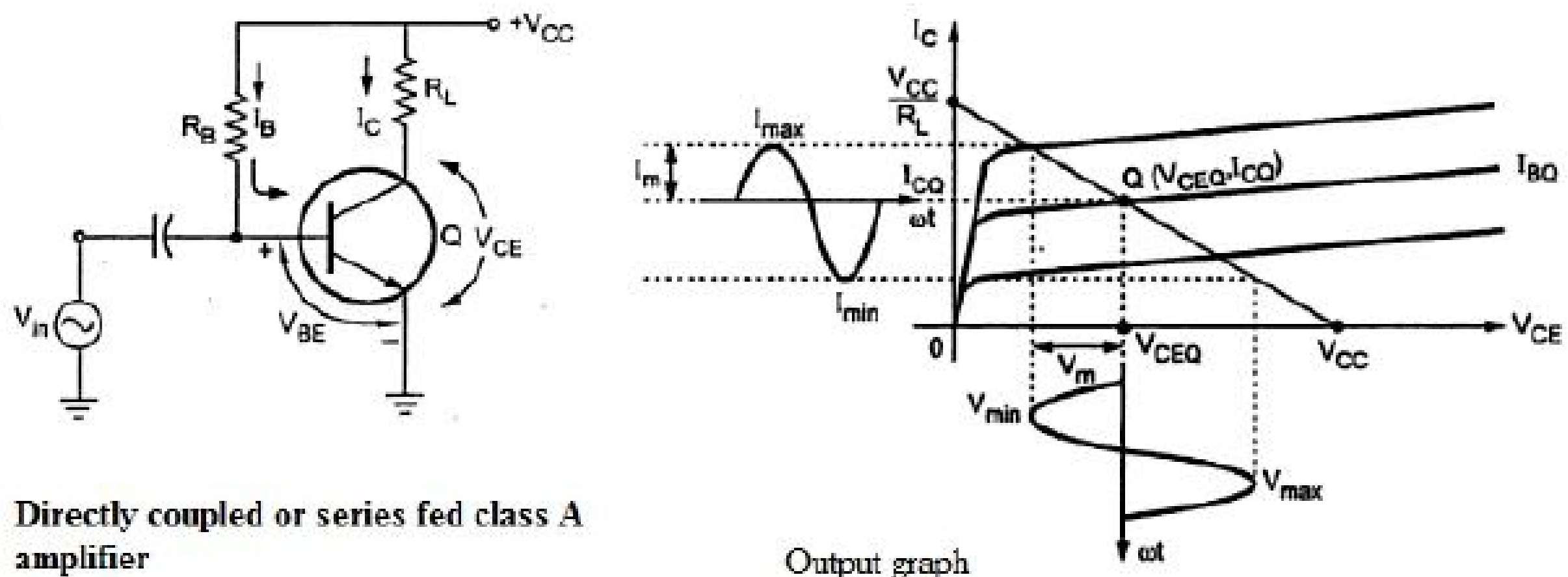


3.2.2.1 CLASSIFICATION OF CLASS A AMPLIFIER

- The class A amplifiers are further classified as **directly coupled or series fed and transformer coupled amplifiers**. In directly coupled type, the load is directly coupled to the collector circuit. While in the transformer coupled type, the load is coupled to the collector using a transformer called an output transformer.

SERIES FED, DIRECTLY COUPLED CLASS A AMPLIFIER

- The difference between small signal versions of the circuit is that the signals handled by this large signal circuit are of the order of few volts. Similarly the transistor used is a power transistor. The value of R_B is selected in such a way that the Q-point lies at the centre of the DC load line.
- The circuit represents the directly coupled class A amplifier as the load resistance is directly connected in the collector circuit. Most of the times the load amplifier is a loudspeaker, the impedance of which varies from $(3-4)\Omega$ to 16Ω . The beta of the transistor is less than 160. The overall circuit handles large power in the range of a few to tens of watts without providing much voltage gain.



Directly coupled or series fed class A amplifier

Output graph

Hence the RMS value of alternating output voltage and current can be obtained as

$$V_{rms} = V_m / \sqrt{2}$$

$$I_{rms} = I_m / \sqrt{2}$$

Hence we can write the equation as, $V_{rms} = I_{rms} R_L$

$$\text{i.e., } V_m = I_m R_L$$

- The AC power delivered by the amplifier to the load can be expressed by using RMS values, maximum (i.e.,) peak values and peak to peak values of the output voltage and current.

Maximum Efficiency

- Assume maximum swing of both the output voltage and the current to calculate the maximum efficiency the maximum efficiency as shown in figure.
- The efficiency of an amplifier represents the amount of power delivered or transported to the load, from the DC source (i.e.,) accepting the DC input. The generalized expression for efficiency of an amplifier is

$$\% = \{[(V_{max} - V_{min}) (I_{max} - I_{min}) / 8V_{CC}I_{CQ}] * 100\}$$

$$\text{i.e., } V_{max} = V_{CC} \text{ and } V_{min} = 0$$

$$I_{max} = 2I_{CQ} \text{ and } I_{min} = 0$$

$$\%_{max} = [(V_{CC} - 0)(2I_{CQ} - 0) / 8V_{CC}I_{CQ}] * 100$$

$$= [2V_{CC}I_{CQ} / 8V_{CC}I_{CQ}] * 100$$

$$\%_{max} = 25\%$$

Advantages

- ▶ The circuit is simple to design and to implement.
- ▶ The load is connected directly in the collector circuit hence the output transformer is not necessary. This makes the circuit cheaper.
- ▶ Less number of components required as load is directly coupled.

Disadvantages

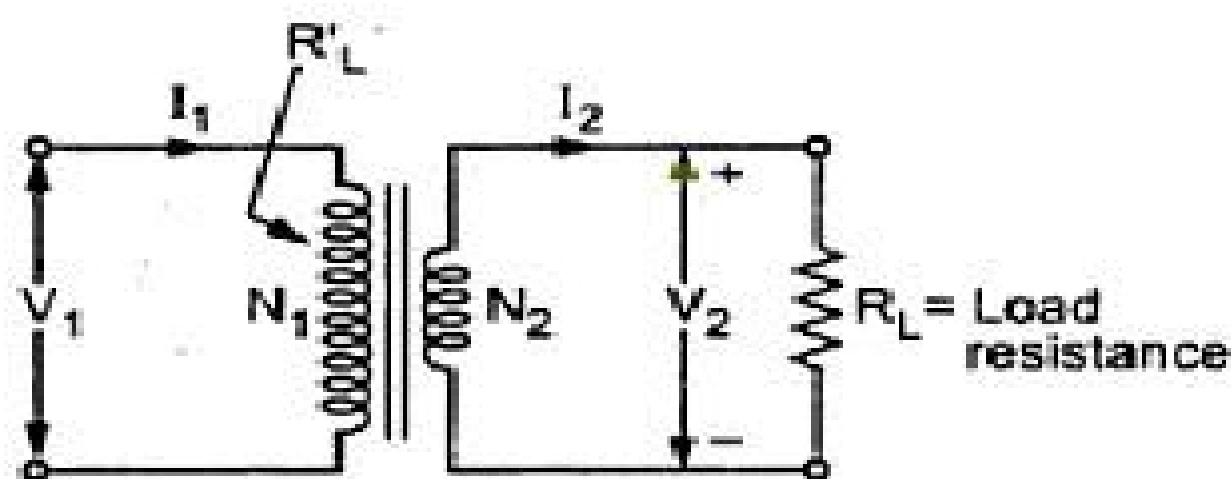
- ▶ The load resistance is directly connected to the collector and carries the quiescent collector current. This causes considerable wastage of power.
- ▶ Power dissipation is more. Hence power dissipation arrangements like heat sink are essential.
- ▶ The output impedance is high hence the circuit cannot be used for low impedance loads, such as loudspeakers.
- ▶ The efficiency is very poor, due to large power dissipation.

TRANSFORMER COUPLED CLASS A AMPLIFIER

- ▶ For maximum power transfer to the load, the impedance matching is necessary. For loads like loudspeaker, having low impedance values impedance matching is difficult using directly coupled amplifier circuits. This is because loudspeaker resistance is in the range of $(3-4)\Omega$ to 16Ω , while the output impedance of a class A amplifier is much more higher. This pattern can be eliminated by using transformer to deliver power to the load.
- ▶ Before studying the operation of the amplifier, let us revise few concepts regarding transformer.

Properties Of Transformer

- ▶ Consider a transformer as shown in the figure which is connected to a load of resistance R_L . While analyzing the transformer, it is assumed that the transformer is ideal and there are no losses in the transformer. Similarly the winding resistances are assumed to be zero.



Let us consider the following

N_1 – Number of turns in the Primary of the transformer

N_2 – Number of turns in the secondary of the transformer

V_1 – Voltage applied to the primary of the transformer

V_2 – Voltage applied to the secondary of the transformer

I_1 – Current applied to the primary coil

I_2 – Current applied to the secondary coil

Turns Ratio

- The ratio of the number of turns in the secondary to the number of turns in the primary is called turns ratio of the transformer.

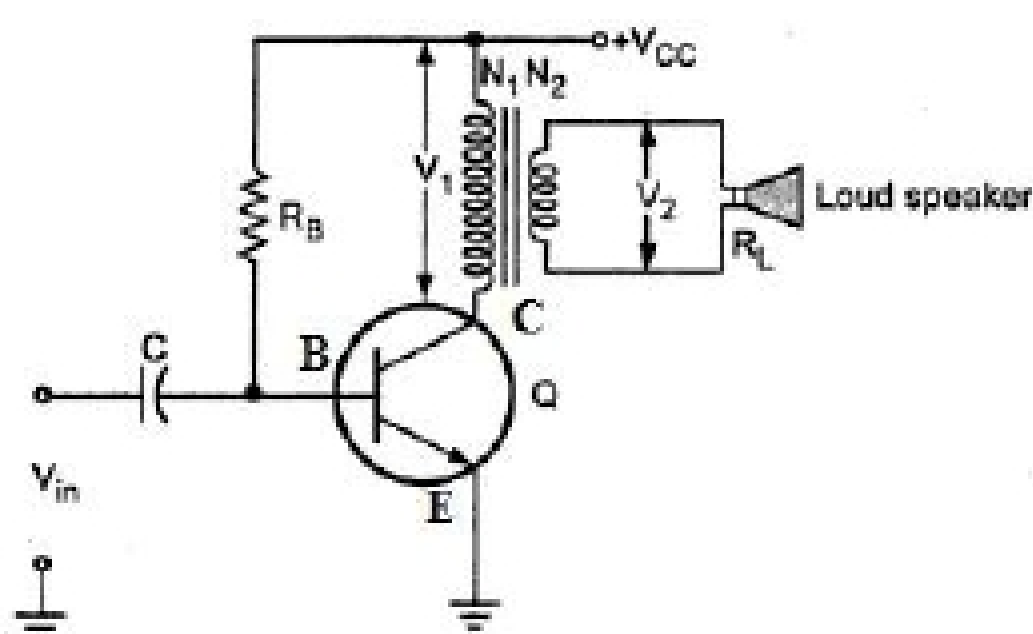
$$\text{Turns Ratio} = n = N_2/N_1$$

$$R_L' = R_L/n^2 = (N_1/N_2)^2 R_L$$

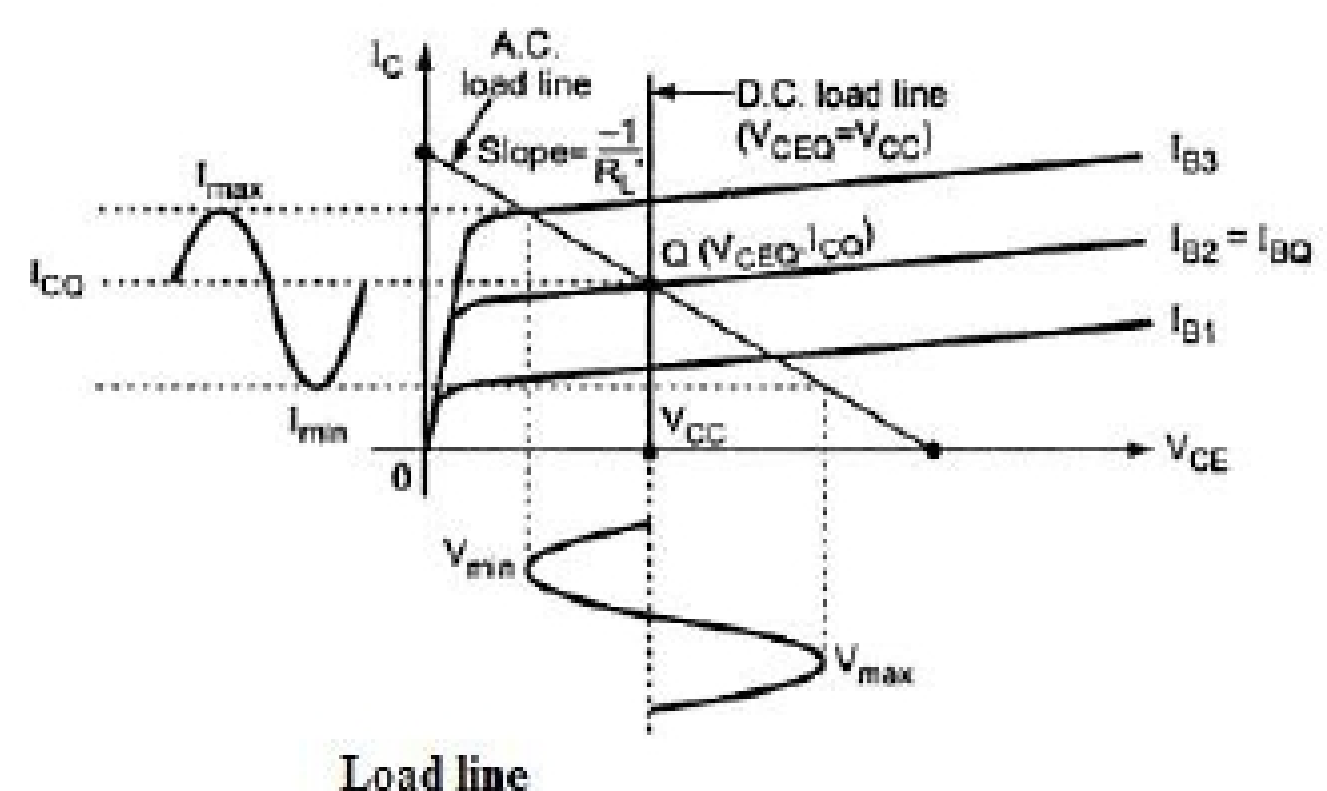
- The R_L' is repeated impedance and is related to the square of the turn's ratio of the transformer. Remember that for a step down transformer, the secondary voltage is less than the primary. And high voltage side is always high impedance side. Hence R_L' is always higher than R_L , for a step down transformer.

Circuit Diagram

- The basis circuit of a transformer coupled amplifier is shown in the figure. The loudspeaker connected to the secondary acts as a load having impedance of $R_L \Omega$. The transformer used is a step down transformer with the turns ratio of $n = N_2/N_1$.



Transformer coupled class A amplifiers



Similarly the AC power of the secondary is given by

V_{2m} – Peak value of secondary or load voltage

V_{2rms} – RMS value of secondary or load voltage

I_{2m} – Peak value of secondary or load current

I_{2rms} – RMS value of secondary or load current

$$P_{ac} = V_{2rms} I_{2rms} = I_{2rms}^2 / R_L = V_{2m} I_{2m} / 2 = I_{2m}^2 R_L / 2 = V_{2m}^2 / 2 R_L$$

- ▶ Power delivered to the primary is the same as power delivered to the secondary assuming ideal transformer. Primary and secondary values of voltages and currents are related to each other through turn's ratio of the transformer.
- ▶ The terms of peak to peak voltage/current are given as

$$V_{2m} = V_{pp}/2, I_{2m} = I_{pp}/2 \text{ and where } V_{pp} = (V_{\max} - V_{\min}) \text{ and } I_{pp} = (I_{\max} - I_{\min}).$$

Therefore AC output power is $P_{ac} = [(V_{\max} - V_{\min})(I_{\max} - I_{\min})]/8$

Efficiency

$$\% \text{Efficiency} = P_{ac} / P_{dc} * 100 = \{[(V_{\max} - V_{\min})(I_{\max} - I_{\min}) / 8 V_{cc} I_{CQ}]\} * 100$$

Maximum Efficiency

- ▶ The maximum efficiency calculation, assume maximum swings of both the output voltage and the output current. The maximum swings are shown in the figure. From the figure we can see that the minimum voltage possible is zero and the maximum voltage possible is V_{cc} , for a maximum swing, Similarly the minimum current is zero and the maximum current possible is $2I_{CQ}$ for maximum swing.

This maximum efficiency is an ideal value. For practical circuit, it is much less than 25% of the order of (10- 15)% negligible(DC) winding resistances and can be neglected.

From figure,

$$V_{\min} = 0, V_{\max} = 2V_{cc}$$

$$I_{\min} = 0, I_{\max} = 2I_{CQ}$$

$$\% \text{ max} = [(2V_{cc} - 0)(2I_{CQ} - 0) / 8 V_{cc} I_{CQ}] * 100$$

$$= [4V_{cc} I_{CQ} / 8 V_{cc} I_{CQ}] * 100$$

$$\% \text{ max} = 50\%$$

Advantages

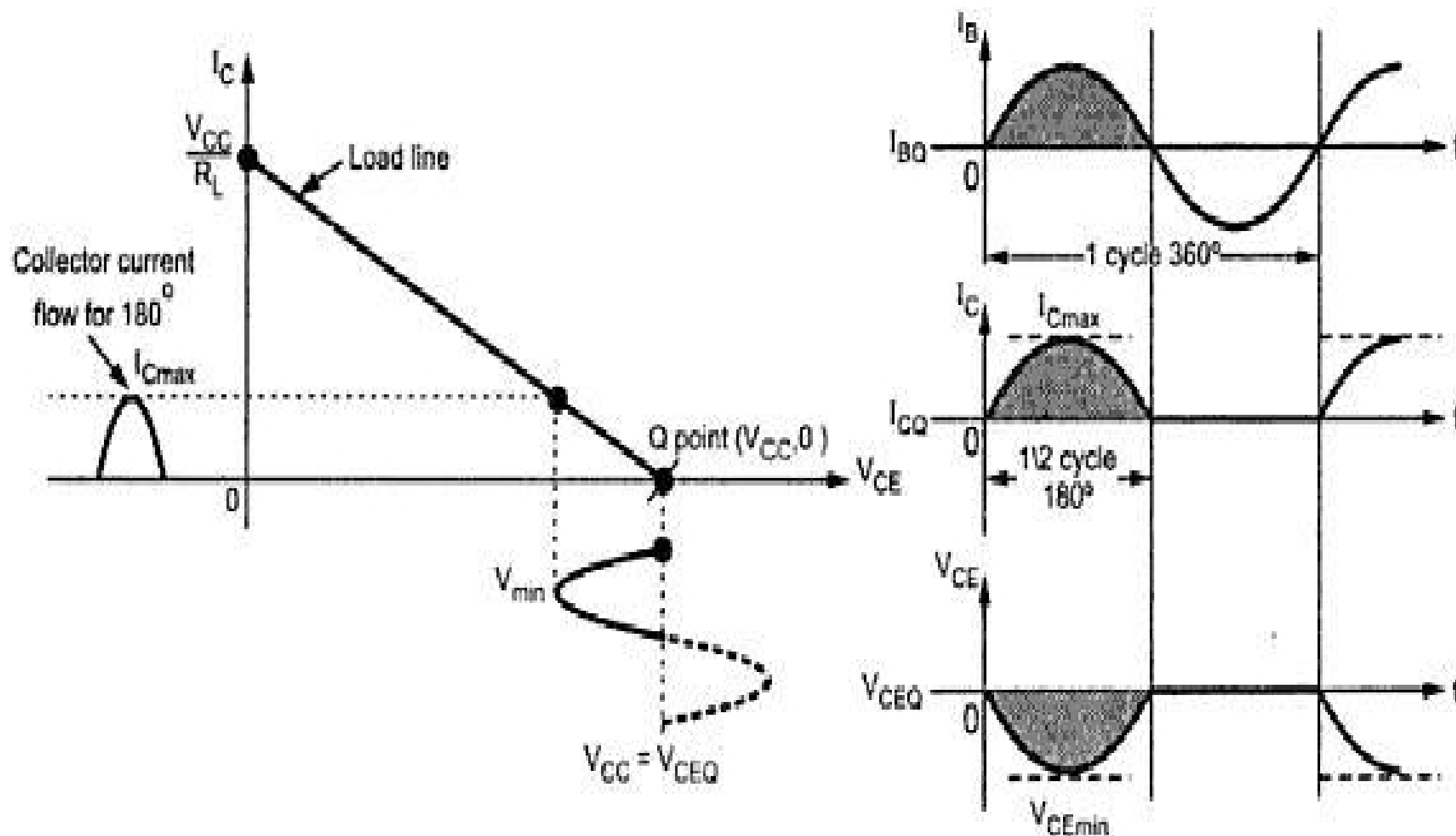
- ▶ The efficiency of operation is higher than directly coupled amplifier.
- ▶ The DC bias current flows through the load in case of directly coupled amplifier is stopped in case of transformer coupled.
- ▶ The impedance matching required for maximum power transfer is possible.

Disadvantages

- ▶ Due to transformer, the circuit becomes bulkier, heavier and costlier compared to directly coupled circuit.
- ▶ The circuit is complicated to design and implement compared to directly coupled circuit.
- ▶ The frequency response of the circuit is poor.

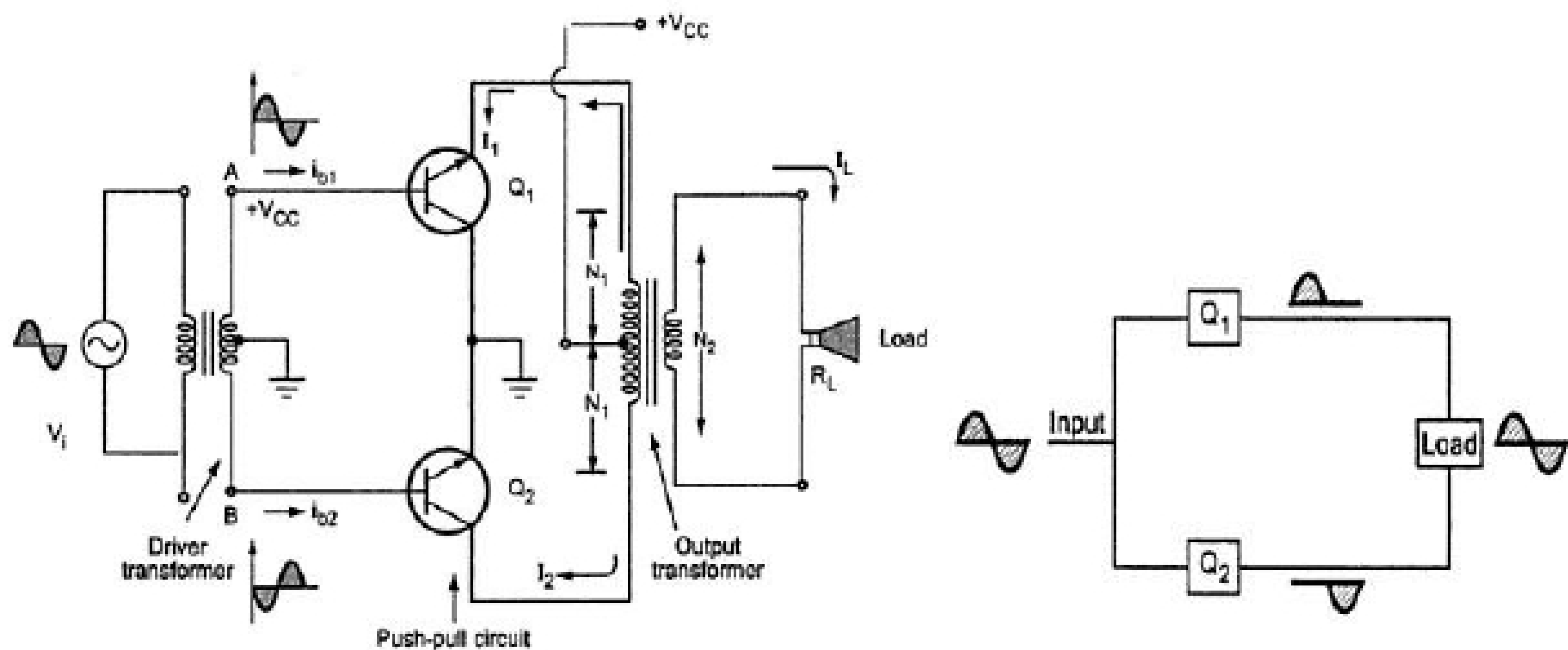
3.2.3 CLASS B AMPLIFIERS

- ▶ The power amplifier is said to be class B amplifier if the Q-point and the input signal are selected, such that the output signal is obtained only for one half cycle for the full input cycle.
- ▶ Due to the selection of Q-point on the x-axis the transistor remains in the active region only for positive half cycle of the input signal. Hence the half cycle is reproduced at the output. But in a negative half cycle of the input signal, the transistor enters into the cut-off region and no signal is produced in the output. The collector current flows only for 180(half cycle) of the input signal. In other words, the cycle of the collector current flow is 180(i.e.,) one half cycle.
- ▶ The current and voltage waveforms for a class B operation are shown in figure. As only a half cycle is obtained at the output, for full input cycle, the output signal is distorted in the mode of operation. To eliminate the distortion, practically two transistors are used in alternate half cycles of the input signal. The overall a full cycle of output signal is obtained across the load. Each transistor conducts only for a half cycle of the input signal.
- ▶ The efficiency of class B operation is much higher than the class A operation



3.2.3.1 PUSH PULL CLASS B AMPLIFIER

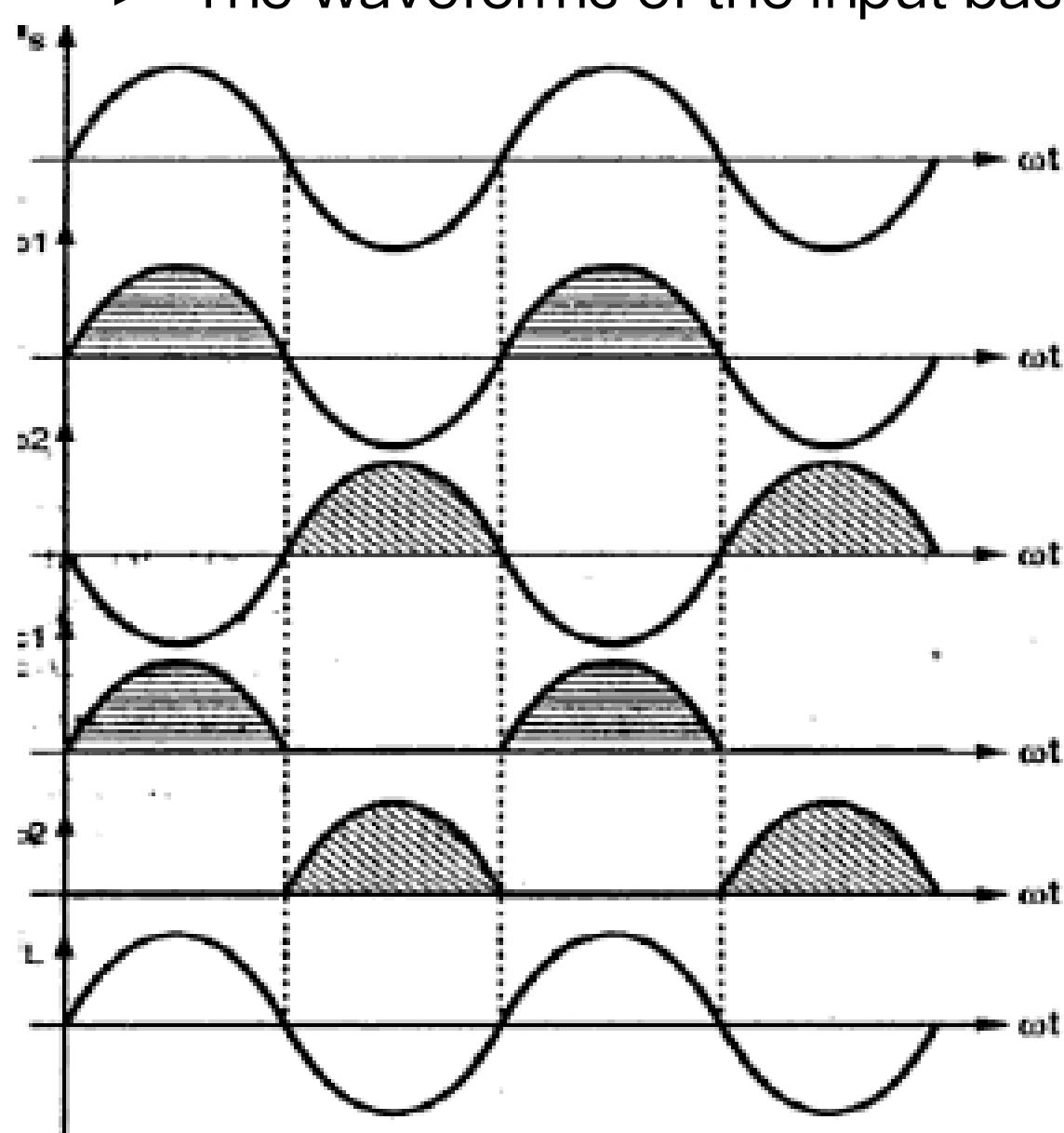
- It requires two transformers, one as input transformer called driver transformer and the other to connect the load called output transformer. The input signal is applied to the primary of the driver transformer. Both the transformers are center tapped transformers. It is as shown in the figure.



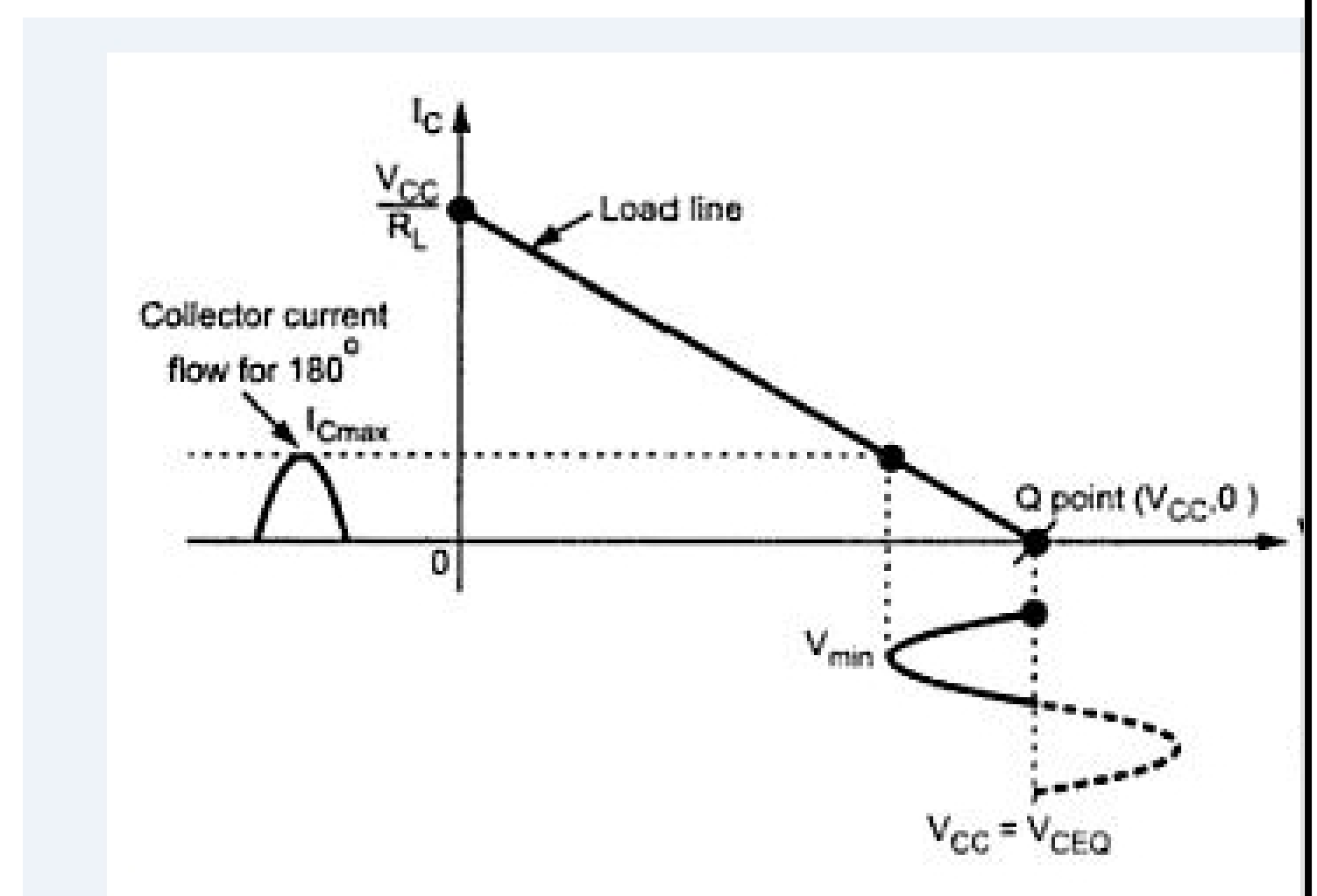
Operation

- In the circuit, both Q₁ and Q₂ transistors are of NPN type of circuit can use both Q₁ and Q₂ of PNP type. In such a case, the only change is that the supply voltage must be $-V_{cc}$, the basic circuit remains the same. Generally the circuit using NPN transistors.

- ▶ The driver transformer derives the circuit signal is applied to the primary of the driver transformer. The centre tap on the secondary of the driver transformer is provided. The centre tap on the primary of the output transformer is connected to the supply voltage $+V_{CC}$.
- ▶ With respect to the centre tap, for a positive half cycle of the input signal, the point A shown on the secondary of the driver transformer will be positive. While the point B will be negative. Thus the voltages is the two halves of the secondary of the driver transformer will be equal but with opposite polarity. Hence the input signals applied in the base of the transistor Q_1 and Q_2 will be 180 out of phase.
- ▶ The transistor Q_1 conducts for the positive half cycle of the input producing positive half cycle across the load. While the transistor Q_2 conducts for the negative half cycle producing negative half cycle across the load, we get a full cycle for a full input cycle. It is shown in figure.
- ▶ When the point A is positive, the transistor Q_1 gets driven into the active region while the transistor Q_2 is negative, the point B is positive, hence the transistor Q_2 gets driven into the active region while the transistor Q_1 is in the cut-off region.
- ▶ The waveforms of the input base and collector, load accounts are shown in figure.



Input and Output waveform



Load line

Efficiency

$$\begin{aligned} \% &= P_{ac}/P_{dc} * 100 \\ &= [(V_m I_m / 2) / (2/\pi * V_{CC} I_m)] * 100 \\ \% &= [\pi/4 (V_m/V_{CC})] * 100 \end{aligned}$$

Maximum Efficiency

From load line curve $V_{min} = V_m = V_{CC}$

$$\begin{aligned} \%_{max} &= [\pi/4 (V_{CC}/V_{CC})] * 100 \\ \%_{max} &= 78.5\% \end{aligned}$$

Thus practical circuits have efficiency of (65-70)%

Advantages

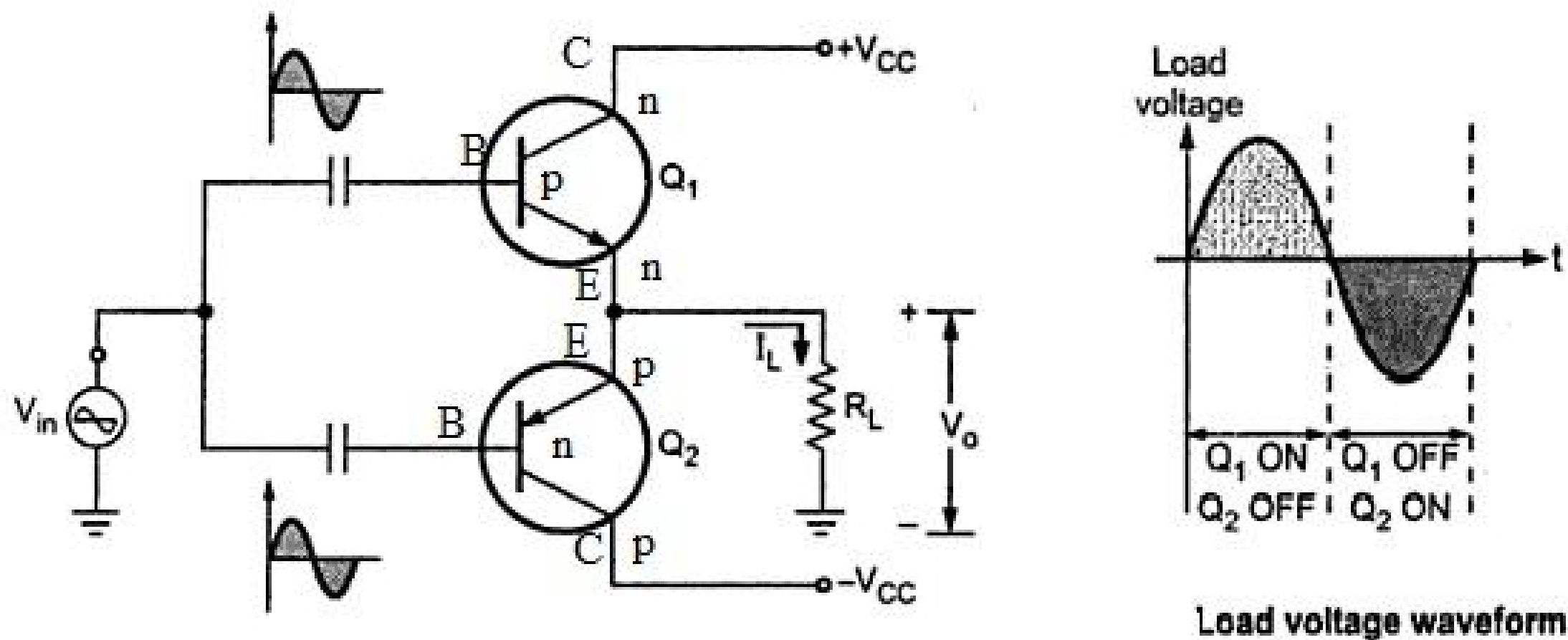
- ▶ The efficiency is much higher than class A operation.
- ▶ When there is no input signal, power dissipation is zero.
- ▶ The even harmonic get cancelled. This reduces harmonic distortion.
- ▶ As the DC current components flow in opposite direction through the primary winding, there is no possibility in DC saturation of the core.
- ▶ Ripples present in the supply voltage also get eliminated.
- ▶ Due to transformer, impedance matching is positive.

Disadvantages

- ▶ Two centre tap transformer are necessary.
- ▶ The transformer, make the circuit bulky and hence costlier.
- ▶ Frequency response is poor.

3.2.3.2 COMPLEMENTARY SYMMETRY CLASS B AMPLIFIER

- ▶ It is a transformer less circuit. But with common emitter configuration, it becomes difficult to match the output impedance for maximum power transfer without output transformers. Hence the matched pair of complementary transistors are used in common collector configuration in this circuit.
- ▶ In addition, voltage feedback can be used to reduce the output impedance for matching. It is shown in figure.



Operation

- ▶ The circuit driver from a dual supply of $\pm V_{CC}$. The transistor Q_1 gets driven into the active region and start conducting. The same signal gets applied to the base of the Q_1 but as it is complementary type, remains in OFF condition, during positive half cycle. This result into positive half cycle across the load.
- ▶ During the negative half cycle, the transistor Q_2 being PNP gets biased into conduction. While the transistor Q_1 gets driven into the cut-off region. Hence only Q_2 conducts during the negative half cycle of the input, producing the negative half cycle across the load R_L .
- ▶ Thus for a complete cycle of input, a complete cycle of output signal is developed across the load as shown in the waveform.

Analysis

- ▶ All the results derived for push pull transformer coupled applicable to the complementary class B amplifier, the only change is that as the output transformer is not present, hence in the expression R_L value must be used as it is instead of R_L'' .

Advantages

- ▶ As the circuit is transformer, its weight size and cost are less.
- ▶ Due to common collector configuration, impedance matching is possible.
- ▶ The frequency response improves due to transformer class B circuit

Disadvantages

- ▶ The circuit needs two separate voltage supplies.
- ▶ The output is distorted to cross-over distortion.

3.2.4 AMPLIFIER DISTORTION

- ▶ The transistor is a non-linear device. As power amplifiers handle large signals, hence there is always a harmonic distortion. Harmonic distortion increases as we go from class A to class C. Even in class A operation some distortion occurs. Because of non-linearity, the signal is distorted in wave slope. The waveform of output signal is distorted in wave slope. The waveform of output signal can be expressed as

$$i_o = I_0 + I_1 \sin \omega t + I_2 \sin 2\omega t + \dots\dots\dots$$

Where,

I_0 – peak value of first harmonic or fundamental

I_1 – peak value of first harmonic or fundamental

I_2 – peak value of second harmonic and so on

- ▶ Due to distortion, the output power calculated on the basis due to harmonic component of distorted signal.

Output power due to first harmonic $P_1 = (I_1/\sqrt{2})^2 R_L = I_1^2 R_L / 2$

Output power due to second harmonic $P_2 = I_2^2 R_L / 2$

Total power due to harmonic components is

$$\begin{aligned} P &= (I_1^2 + I_2^2 + I_3^2 + \dots\dots\dots) R_L / 2 \\ &= [(1 + (I_2/I_1)^2 + \dots\dots\dots)] I_1^2 R_L / 2 \\ P &= [1 + D_2^2 + D_3^2 + \dots\dots\dots] P_1 \end{aligned}$$

Where D_2 , D_3 are second and third harmonic distortions respectively

The distortion factor $D = \sqrt{D_2^2 + D_3^2 + \dots\dots\dots}$

Total power = $[1 + D^2] P_1$

- ▶ If the distortion is 10% (i.e.,) 0.1, the percentage of increase in power is given by $P = [1 + (0.1)^2] P_1 = 1.01 P_1$

$$\% \text{ Increase in power} = [(1.01 P_1 - P_1) / P_1] \times 100$$

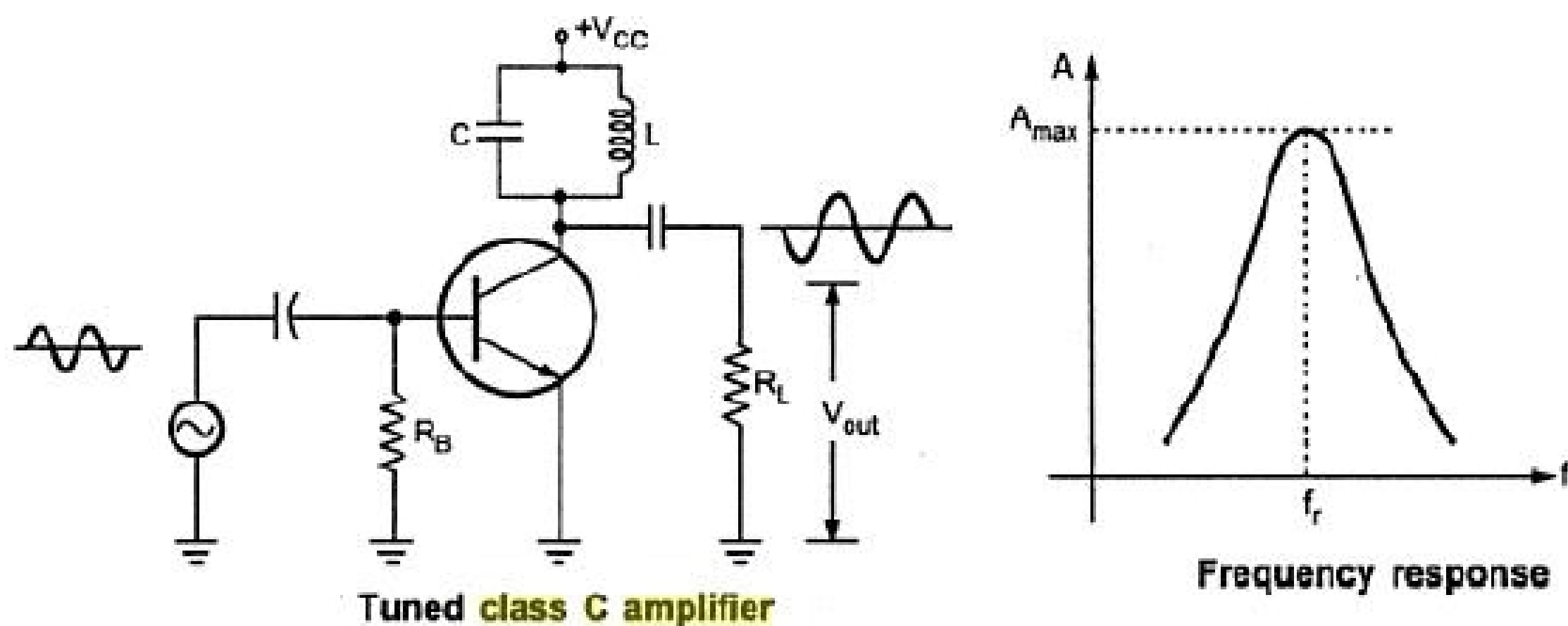
$$= 1\%$$

Comparison of push pull and complementary symmetry circuit

	Push Pull Class B	Complementary Symmetry Class B
1.	Both the transistors are similar either pnp or npn.	Transistors are complementary type i.e. one npn other pnp.
2.	The transformer is used to connect the load as well as input.	The circuit is transformerless.
3.	The impedance matching is possible due to the output transformer.	The impedance matching is possible due to common collector circuit.
4.	Frequency response is poor.	Frequency response is improved.
5.	Due to transformers, the circuit is bulky, costly and heavier.	As transformerless, the circuit is not bulky and costly.
6.	Dual power supply is not required.	Dual power supply is required.
7.	Efficiency is higher than class A.	The efficiency is higher than the push pull.

3.2.5 CLASS C AMPLIFIERS

- ▶ The power amplifiers is said to be class C amplifier if the Q-point and the input signal are collected such that the output signal is obtained for less than a half cycle, for a full input cycle.
- ▶ Due to such selection of the Q=point, transistor remains active for less than a half cycle. Hence only that much peak is represented at the output. For remaining cycle of the input cycles the transistor remains cut-off and no signal is produced at the output. The angle of the collector current flow is less than 180 degrees
- ▶ The current and voltage waveforms for a class C amplifier operation are as shown in the figure



- ▶ A parallel resonant circuit acts as load impedance. The collector current flows for less than half a cycle hence it consists of a series of pulses with the harmonics of the input signal. A parallel tuned circuit acting as a load is tuned to the input frequency. Thus it filters the harmonic frequencies and produces a sine wave output. Voltage consisting of fundamental component of the input signal.
- ▶ The AC input voltage drives the base and amplified output voltage is available at the collector. The amplified and inverted collector voltage is connected to the load resistance R_L through coupling capacitor.
- ▶ As class C amplifier is used with parallel tuned circuit, the output voltage is maximum at the resonant frequency. It is given by

$$f_r = 1/2\pi\sqrt{LC}$$
- ▶ The gain drops on either side of the resonant frequency. Thus the response of class C amplifier is shown in figure. As the gain is maximum at the resonant frequency, these amplifiers are used to amplify only narrow of frequencies.

Bandwidth

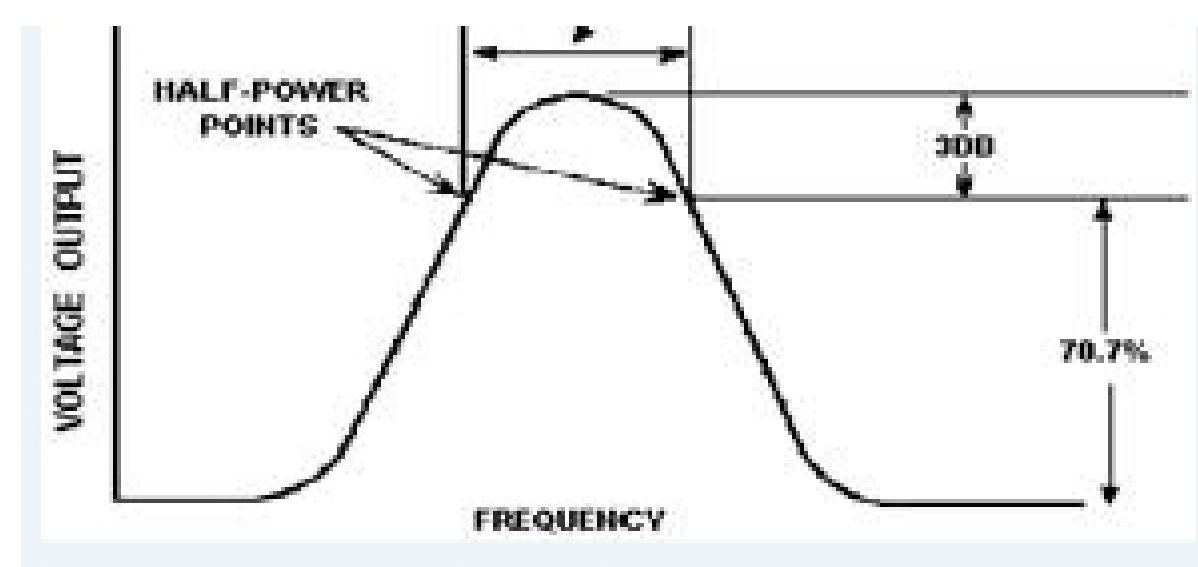
Bandwidth of resonant circuit is defined as $BW = (f_2 - f_1)$

Where f_1 – lower half power (3dB) frequency

f_2 – upper half power (3dB) frequency

- ▶ The half power frequencies are identical to the frequencies at which the voltage gain is equal 0.707 times the maximum gain. It is shown in the figure.
- ▶ The bandwidth of class C tuned amplifier is given by $BW = f_r / Q$

Where Q – Quality factor of the circuit



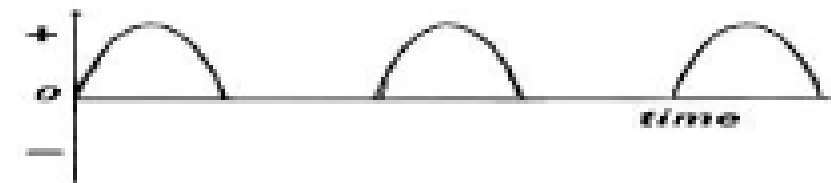
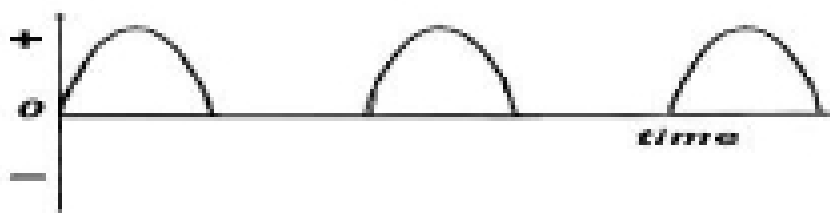
Duty Cycle

- ▶ It is the ratio of ON period of the transistor to the total period of the pulses. The width of current pulse represents on period of the transistor.
- ▶ Thus W – Width of the pulse

T – Period of the pulses

Their duty cycle D is given by

$$D = W/T$$



Efficiency

- ▶ It is given by the ratio of AC output to the DC input power.

$$\% = P_{out}/P_{DC} * 100 = P_{out}/V_{CC}I_{DC} * 100$$

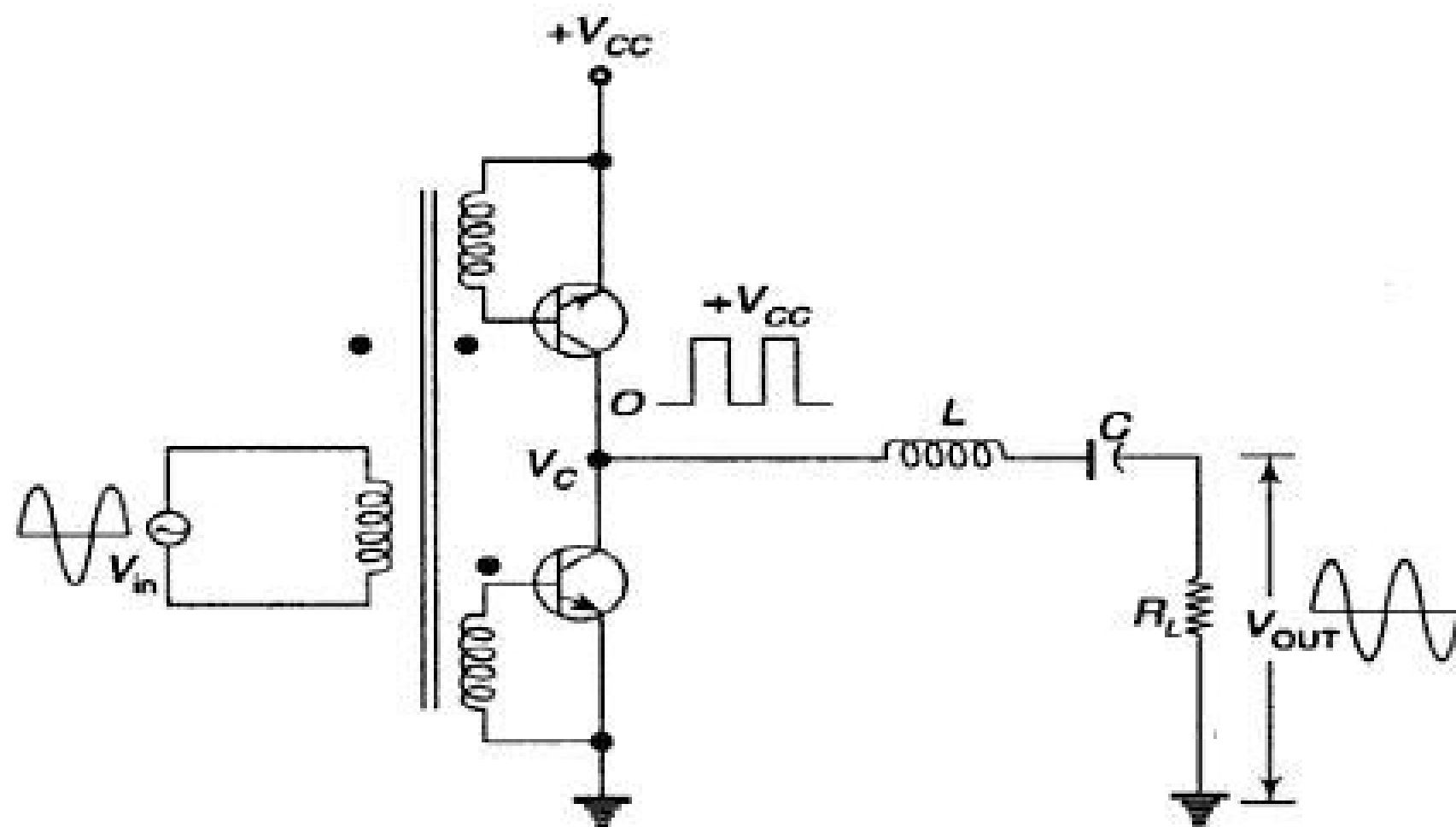
- ▶ The efficiency depends on the conduction angle $\emptyset$. The figure shows the graph of efficiency against the conduction angle $\emptyset$.
- ▶ In a class C amplifier, most of the DC input power is connected into AC load power because the transistor and coil losses are small. When the conduction angle is 180° , the efficiency is 78.5%. The efficiency increases when the conduction angle decreases. It has maximum efficiency of 100% approached at very small conduction angles $\emptyset$.

Applications

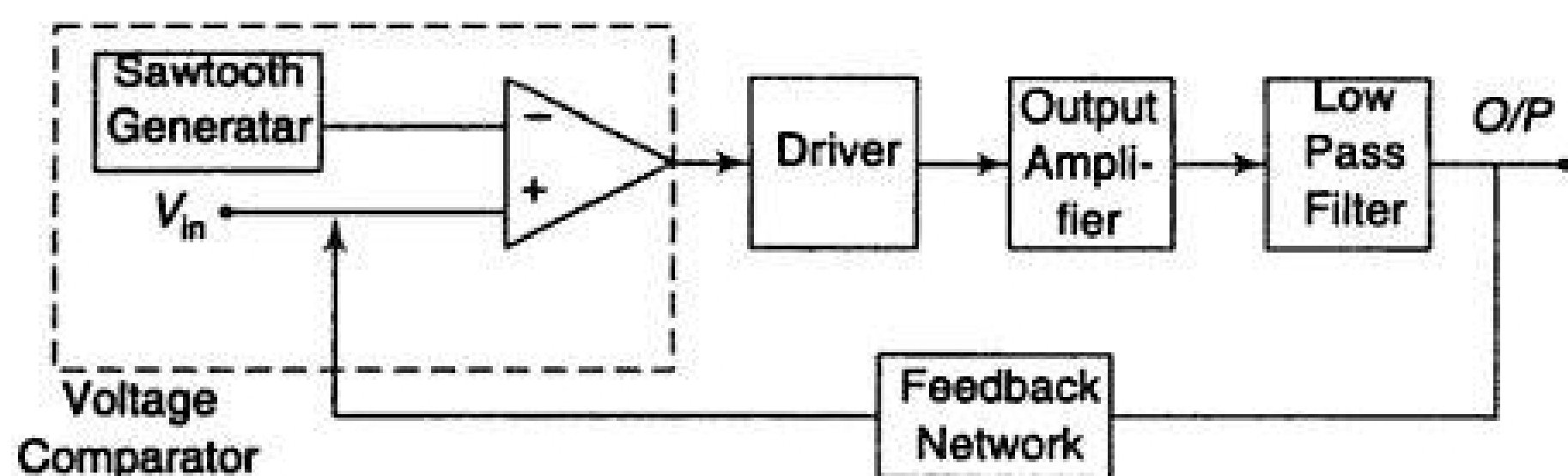
- ▶ The class C operation is not suitable for audio frequency power amplifiers.
- ▶ The class C amplifiers are used in tuned circuits used in communication areas and in radio frequency (RF) circuits with tuned RLC loads.
- ▶ It is used in tuned circuits; class C amplifiers are called tuned amplifiers.
- ▶ Used in mixer convertor circuits used in radio receivers and wireless communication systems.
- ▶ Used as final amplifier stage in communication circuits.

3.2.6 CLASS D AMPLIFIERS

- ▶ Figure shows the basic concept of class D amplifier. The amplifier consists of two complementary symmetry transistors driving a load R_L . This means one transistor is PNP and other is NPN.



- The transistors are biased in such a way that they behave as ideal switches. When transistor is ON, it is biased to saturation so that the voltage across it is zero while the current is high. When the transistor is OFF, it is biased so that the current through it is zero while the voltage is high. Thus when the input goes positive Q1 conducts heavily acting as closed switch while Q2 is OFF. While the input goes negative, Q2 conducts heavily acting as closed switch while Q1 is OFF. Thus the load voltage V_0 across R_L has one of two possible values which are V_{supply} or $0V$. This is a type of digital output having two levels high and low.



Block diagram of class D amplifier

- The transistors dissipate almost zero power as in any of the states, either voltage is zero or current is zero for the transistors. Thus entire power input, available to the load. Here efficiency of class D amplifiers is almost 100%. The figure of merit which is the ratio of the maximum power dissipated in transistor to that delivered to the load is zero. The facts make the class D amplifier as an ideal amplifier.

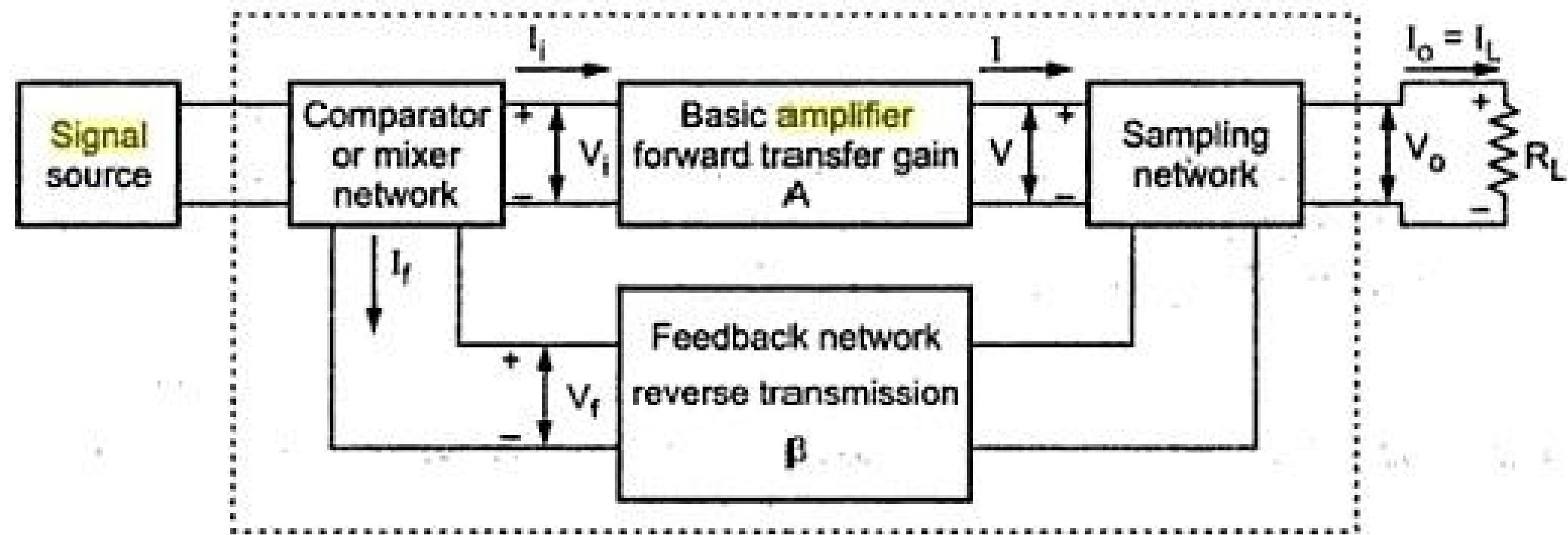
- ▶ Practically class D amplifiers are designed to operate with digital or pulse type of signals. The basic block diagram of unit used along with class D amplifier in the application circuits is shown in the figure
- ✓ The op-amp comparator is used for which one input is saw tooth type while other is sinusoidal. The comparator converts sinusoidal signal to digital pulse type signal with the help of saw tooth waveform. This is called chopping of sinusoidal signal to produce digital signal.
- ✓ The signal drives the class D amplifier. When pulse signal is high the transistors are ON and when it is low, the transistors are OFF.
- ✓ The most of power supplied is delivered to the load by producing high power output signal.
- ✓ The digital signal is connected back to the original sinusoidal signal using low pass filter.
- ✓ Practically instead of power BJT, power MOSFET devices are used as driver devices for class D amplifier.

Applications

- ▶ It is mainly used in pulse and digital circuits.

3.3 CONCEPT OF FEEDBACK

- ▶ Feedback plays an important role in almost all electronic circuits. It is almost invariably used in amplifier to improve the performance and to make it more ideal. In the process of feedback, a part of the output is sampled and fed back to the input of the amplifier. Therefore at input we have two signals input signal and part of the output which is fed back to the input.
- ▶ It can sample the output voltage or current by means of a suitable sampling network and apply this signal to the input through a feedback two port network, as shown in the figure. At the input the feedback signal is combined with the input signal through a mixer network and is fed into the amplifier.

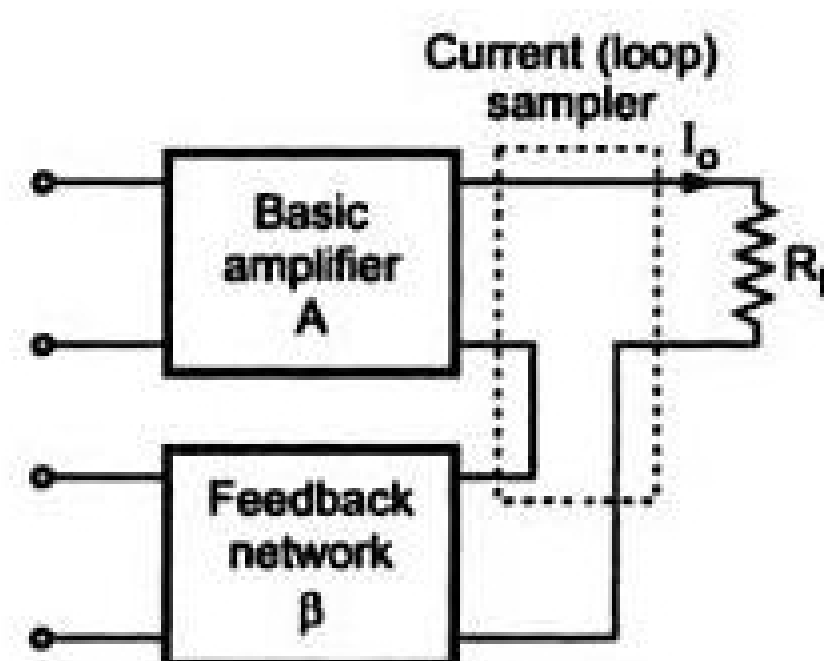
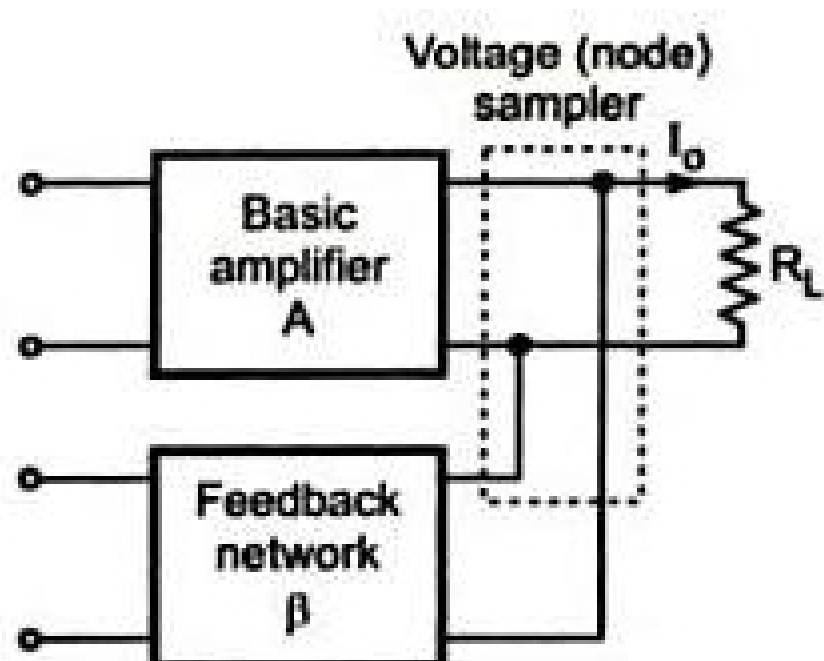


From the figure we have three following networks,

- ✓ Sampling network
- ✓ Feedback network
- ✓ Mixer network

3.3.1 SAMPLING NETWORK

- There are two ways to sample the output, according to the sampling parameter, either voltage or current. The output voltage is sampled by connecting the feedback network in shunt across the output as shown across the output as shown in figure. This type of connection is referred to as voltage or node sampling. The output current is sampled by connecting the feedback network in series with the output as shown in the figure. This type of connection is referred as current or loop sampling.

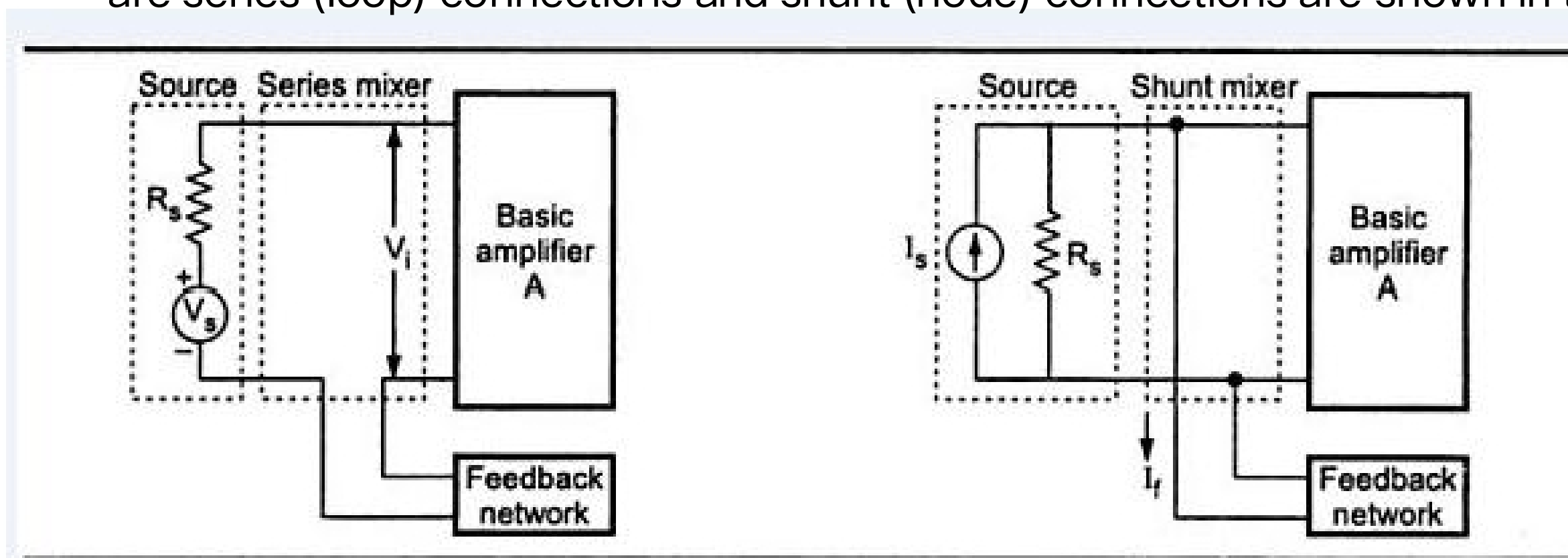


3.3.2 FEEDBACK NETWORK

- ▶ It may consist of resistors, capacitors and inductors. Most often it is simply a resistive configuration. It provides reduced portion of their output as feedback signal to the input mixer network. It is given $V_f = \beta V_o$ as
- ▶ Where β – feedback factor or feedback ratio. The symbol β used in feedback circuits which always lies between 0 and 1. It is totally different from symbol β used to represent current gain in common emitter amplifier, which is greater than 1.

3.3.3 MIXER NETWORK

- ▶ Like sampling, there are two ways of mixing feedback signal with the input signal. There are series (loop) connections and shunt (node) connections are shown in the figure.



3.3.4 TYPES OF FEEDBACK

Depends on phase of the signals. It can be divided into two types

- ✓ Negative feedback
- ✓ Positive feedback

3.3.4.1 NEGATIVE FEEDBACK

- ▶ If the feedback signal (i.e.,) voltage or current is applied in such a way, that it is out of phase (180°) with the input signal and thus decreases it, then it is called negative feedback amplifier or degenerative or inverse feedback amplifier.
- ▶ The negative feedback reduces the gain of the amplifier. However it improves the amplifiers performance in many other respects. Thus a negative feedback is frequently used in small signal as well as large signal amplifier circuits.
- ▶ The voltage or open loop gain (A_v) is the ratio of output voltage (V_o) to the input voltage (V_{in}), mathematically $A_v = V_o / V_{in}$

Let V_f – feedback voltage and β – feedback factor, A_{vf} – gain without feedback then,

$$A_{vf} = A_v / (1 + \beta A_v)$$

- It may be noted that voltage gain expression of negative feedback that as the term $(1 + \beta A_v) > 1$. The value of A_{vf} is smaller than A_v . In other words a negative feedback amplifier (A_{vf}) is smaller than the open loop gain (A_v). Thus the negative feedback reduces the gain of an amplifier

If $(\beta A_v) \gg 1$, therefore $1 + \beta A_v = \beta A_v$

Thus, $A_{vf} = A_v / \beta A_v = 1 / \beta$

- The voltage gain of negative feedback amplifier (A_{vf}) depends only upon the feedback ratio (β). Though the value of voltage gain of a negative feedback amplifier (A_{vf}) is quite low as compared to A_v , yet it is found to be very stable. It means that the voltage gain (A_{vf}) is not altered.
- It may be noted that the stability of voltage gain (A_{vf}) comes only if the feedback network is made up of stable elements such as resistors, inductors and capacitors. If the feedback network has an active device, the gain will not remain stable.

Advantages

- Though there are numerous advantages of employing negative feedback in an amplifier, yet only some of the advantages are given below
 - ✓ Increased Stability
 - ✓ Increased bandwidth
 - ✓ Less amplitude distortion
 - ✓ Less harmonic distortion
 - ✓ Decreased noise
 - ✓ Less frequency distortion
 - ✓ Less phase distortion
 - ✓ Input and output resistances can be modified as required

Disadvantages

- The only disadvantage of negative feedback amplifier is that it reduces the gain of the circuit.
- However the disadvantage is outweighed due to the presence of numerous advantages.

3.4 BARKHAUSEN CRITERION

Now if V_f has to be equal to V_i , then

$$V_i = -A\beta V_i$$

$$-A\beta = 1$$

This satisfies only when,

The $-A\beta = 1$ condition is called Barkhausen criterion

$$A\beta = -1+j0$$

Taking magnitude on both sides

$$|A\beta| = 1$$

- ▶ The phase of V_f , must be same as V_i (i.e.,) feedback network should introduce 180° phase shift in addition to 180° introduced by inverting amplifier. This ensures a positive feedback and hence the total phase shift is 360° .

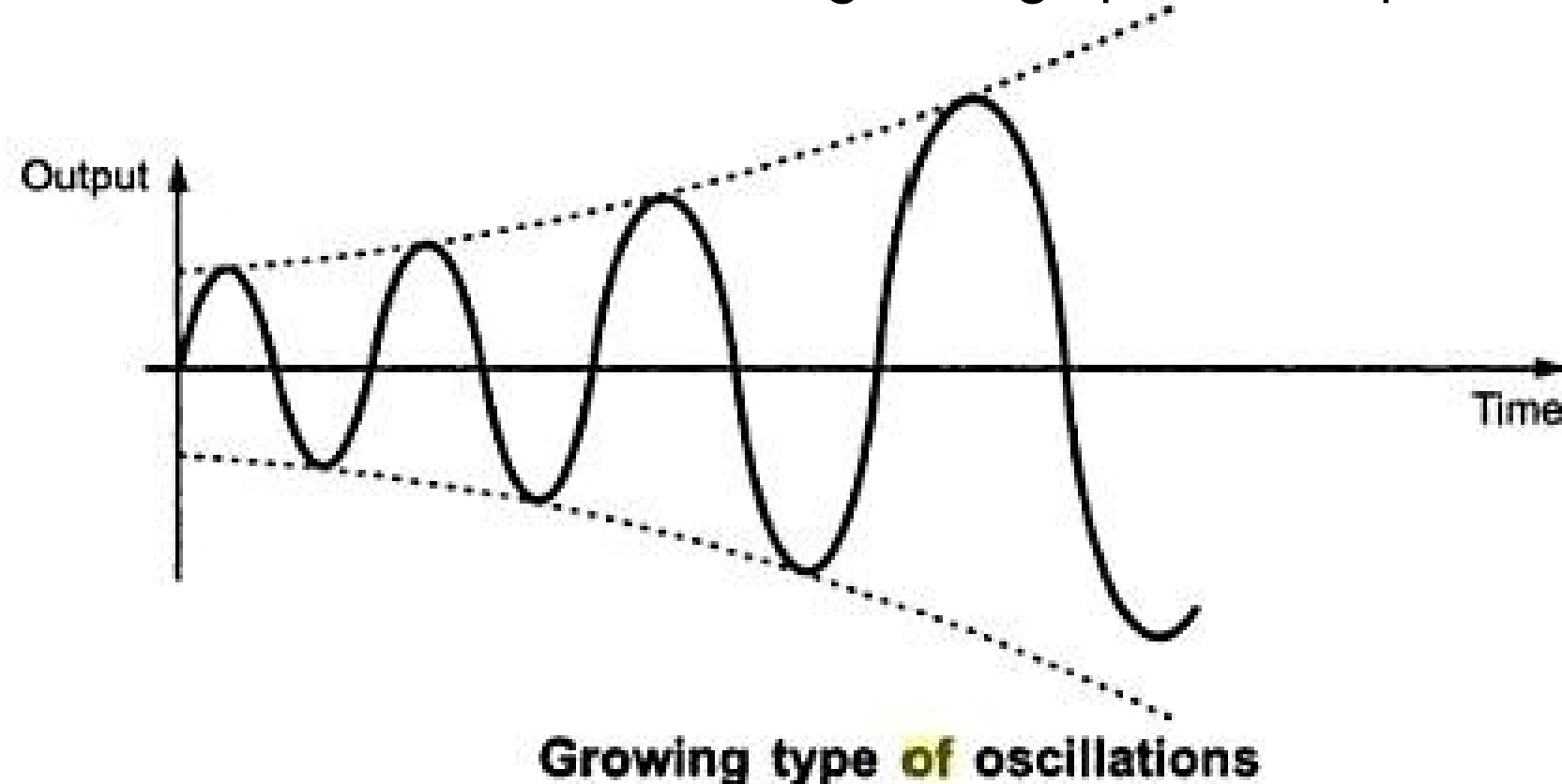
The Barkhausen criteria states that

- ▶ The total phase shift around a loop, as the signal proceeds from input through amplifier feedback network back to input again, completing a loop, is precisely 0° to 360° or an integral multiple of 2π radians (i.e.,) Angle of $A\beta = 0^\circ$ or 360° .
- ▶ The magnitude of the product of the open loop gain of the amplifier (A) and the feedback factor is unity (i.e.,) $|A\beta| = 1$.

Let us see the effect of magnitude of the product $A\beta$ on the nature of oscillations.

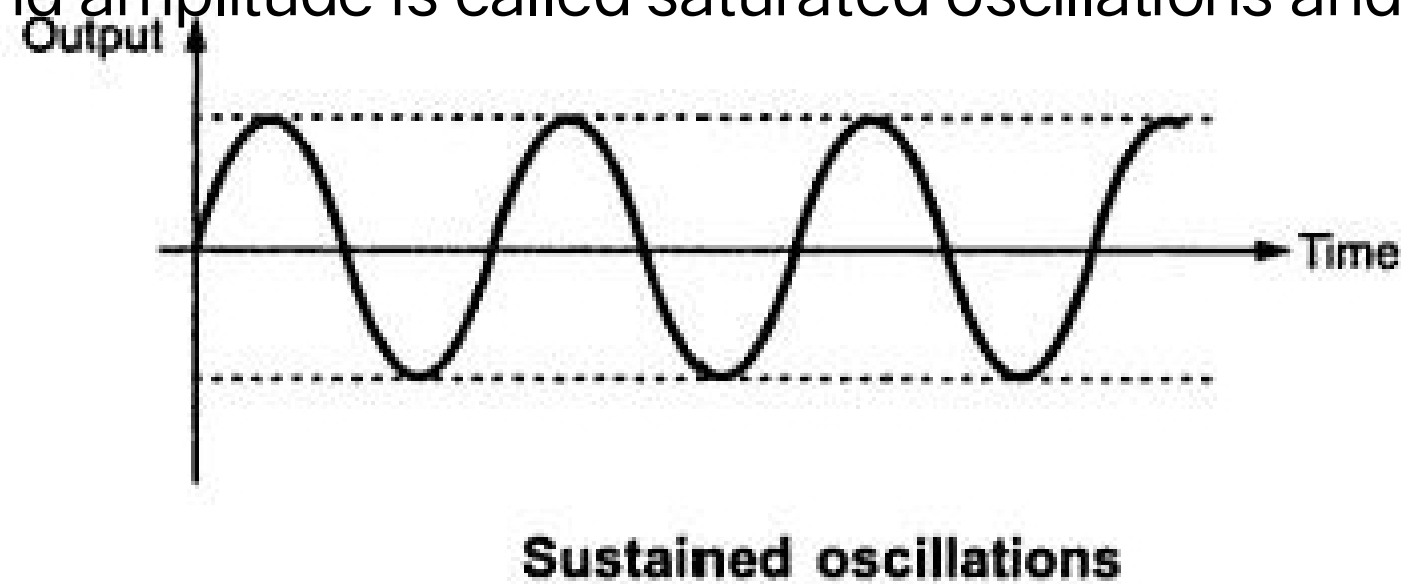
CASE 1: $|A\beta| > 1$

- ▶ When the total phase shift occurred at loop 0° or 360° and $|A\beta| = 1$, then the input oscillates but the oscillations are growing up. The amplitude of the oscillations

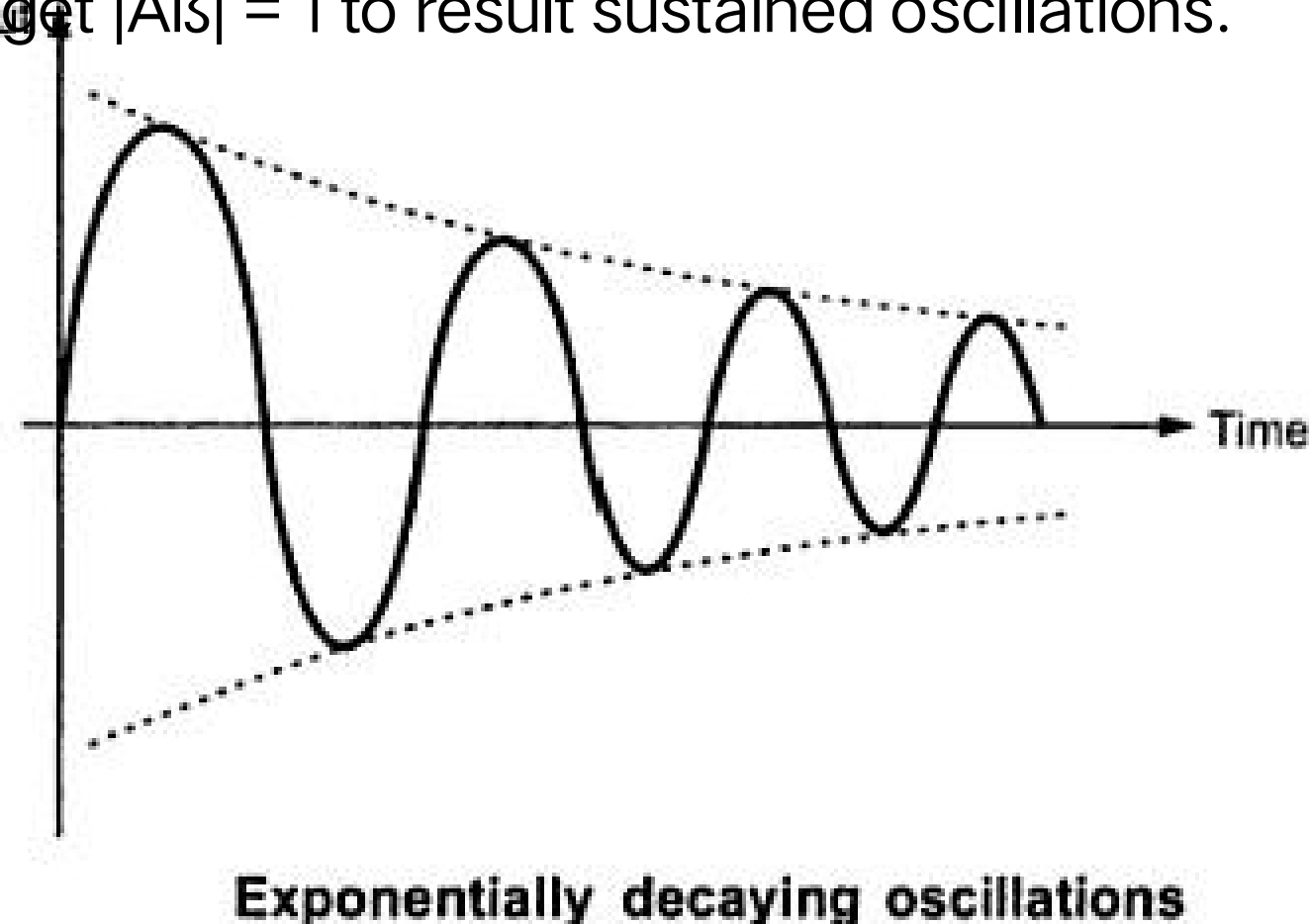


CASE 2: $|A\beta| = 1$

- As stated by Barkhausen criterion, when total phase shift around the loop is 0° or 360° ensuring that the positive feedback $|A\beta| = 1$, then the oscillation with constant frequency and amplitude is called sustained oscillations and shown on figure.

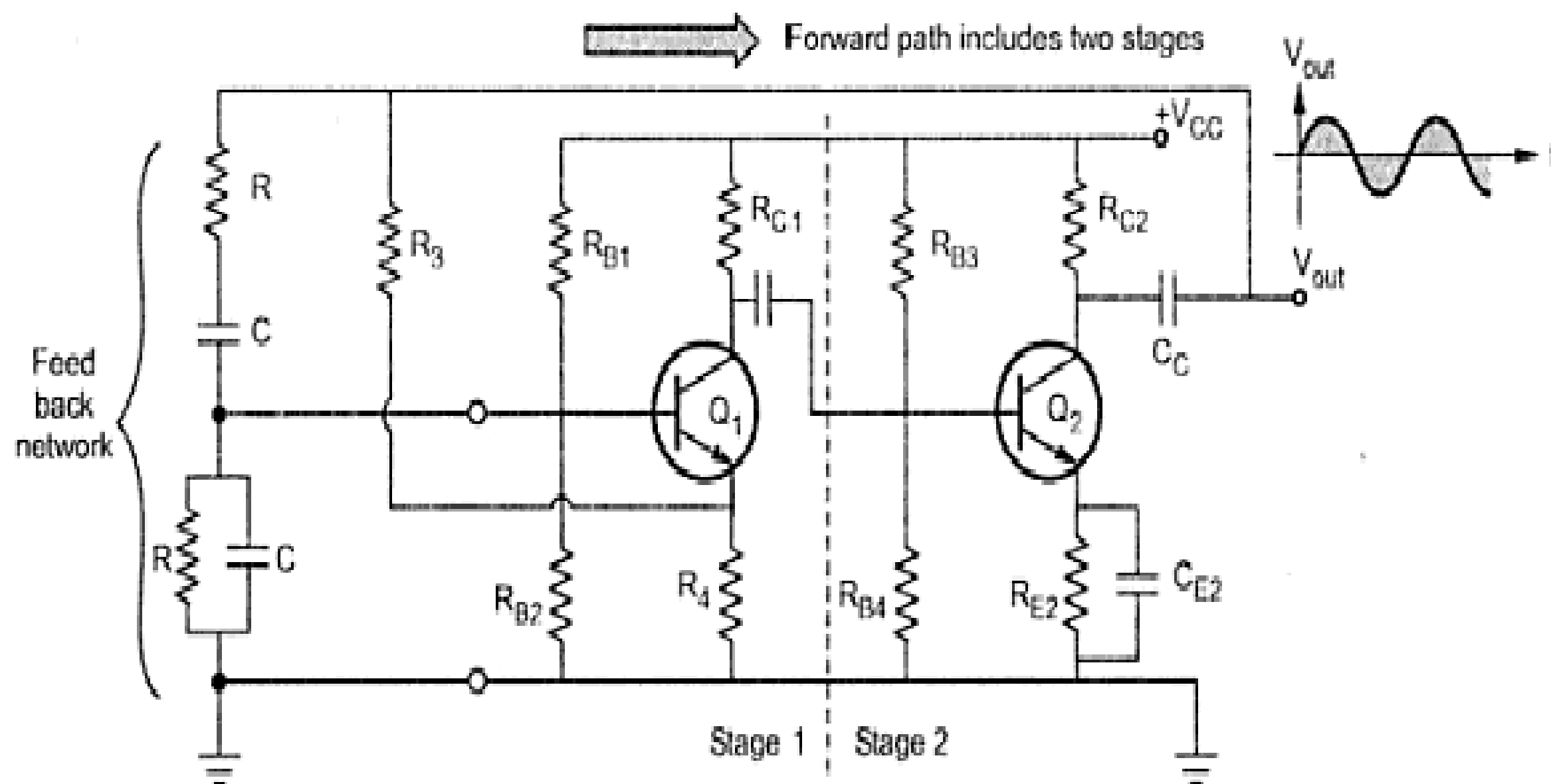
**CASE 3: $|A\beta| < 1$**

- When the total phase shift around the loop is 0° or 360° and $|A\beta| < 1$, then the oscillations are of decaying type (i.e.,) in such an oscillation the amplitude decreases exponentially and the oscillations finally cease. Thus the circuit works as an amplifier without oscillations as shown in the figure.
- So to start the oscillations without input $|A\beta|$ is kept higher than unity and then the circuit adjusts itself to get $|A\beta| = 1$ to result sustained oscillations.



3.5 WEIN BRIDGE OSCILLATOR

- It consists of two stages common emitter transistor amplifier used. Each stage contributes 180° phase shift hence the total phase shift due to the amplifier stage becomes 360° or 0° which is necessary as per the oscillator conditions. It is shown in the figure.



- The bridge consists of R and C that are in series and parallel, R_3 and R_4 . The feedback is applied from the collector of Q_2 through the coupling capacitor, to the bridge circuit. The resistance R_4 serves the dual purpose of emitter resistance of the transistor Q_1 and also the element of the Wein Bridge.
- The two stage amplifier provides a gain much more than 3 and it is necessary to reduce it. To reduce the gain, the negative feedback is used without bypassing the resistance R_4 . The negative feedback can accomplish the gain stability and can control the output magnitude. The negative feedback also reduces the distortion and the output is a pure sinusoidal in nature. The amplitude stability can be improved by using a non-linear resistor R_4 . Due to this, the loop gain depends on the oscillations. Increase in amplitude of oscillations, increases the current through the value of non-linear resistance, which results to an increase in the value of non-linear resistance R_4 . When this value increases, a greater amount of negative feedback is applied.

- The two arms of the bridge are namely R_1, C_1 are in series and the R_2, C_2 are in parallel are called frequency sensitive arms. This is because the components of the two arms decide the frequency of the oscillator. Let us find the gain of the feedback network. As seen earlier input V_{in} in the feedback network is called lead-lag network. This is because at very low frequencies it acts like a lead while at very high frequencies it acts like lag network.

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

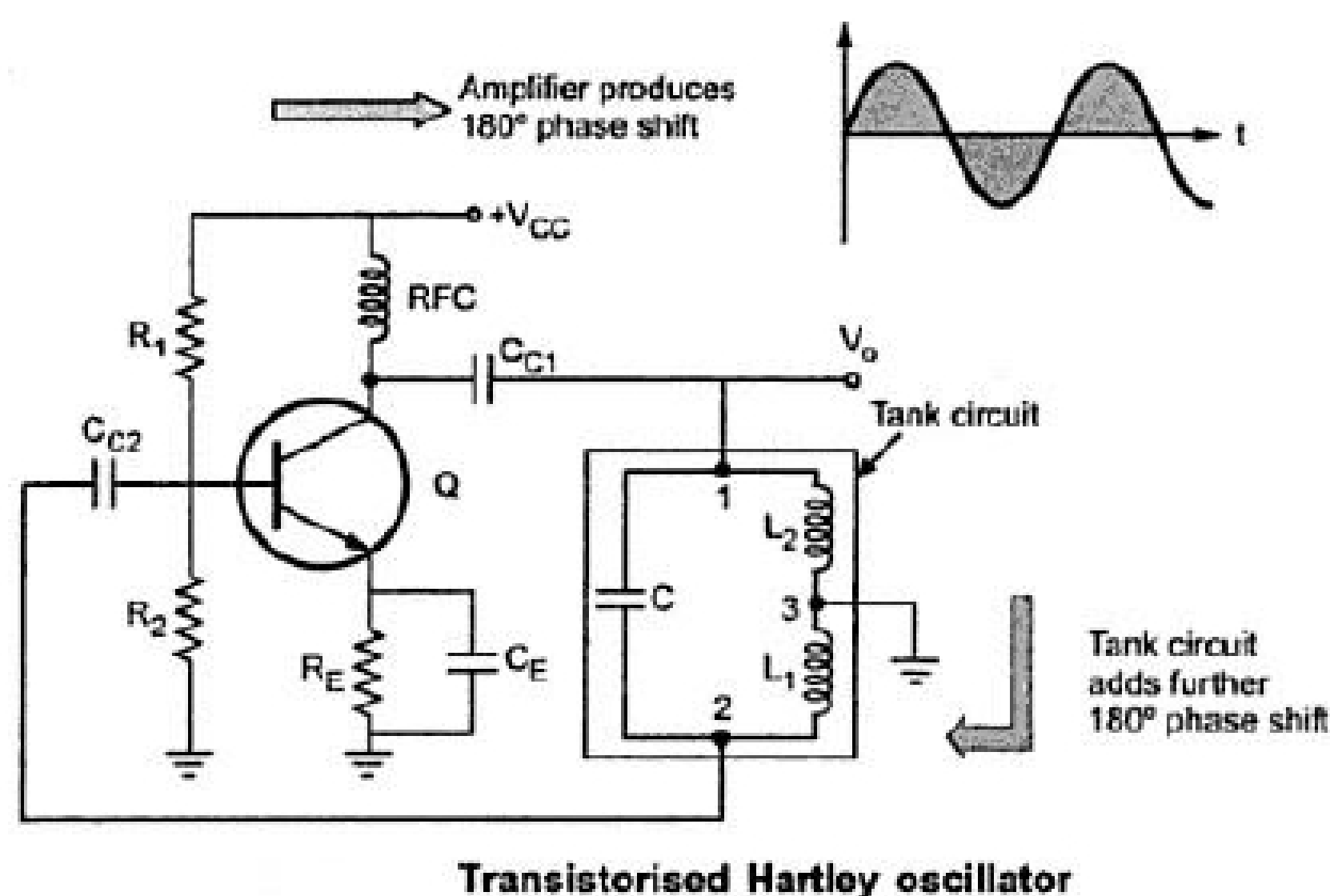
$$A = \frac{R_1 C_1 + R_2 C_2 + C_1 R_2}{C_1 R_1}$$

Advantages

- By varying the two capacitors values simultaneously difference in frequency ranges can be provided.
- Good frequency stability, pure sine output and highly stabilized output.
- It is used between 20Hz to 1MHz.

3.6 HARTLEY OSCILLATOR

- A LC oscillator uses two inductive reactance and one capacitive reactance in its feedback network is called Hartley Oscillator. The amplitude stage uses an active device as a transistor in common emitter configuration. It is shown in the figure.



- ▶ The resistance R_1 and R_2 are biasing resistances. The RFC is the radio frequency choke. Its reactance value is very high for high frequencies, hence it can be treated as an open circuit. While for DC conditions the reactance is zero hence causes no problem for DC capacitors.
- ▶ Hence due to RFC, the isolation between AC and DC operation is achieved. R_E is also a biasing circuit resistance and C_E is the emitter bypass capacitor. C_{c1} and C_{c2} are coupling capacitors.
- ▶ The common emitter amplifier provides a phase shift of 180° . AS the emitter is grounded, the base and the collector voltages are out of phase by 180° . As the centre of L_1 and L_2 is grounded, when the upper end becomes positive, the lower becomes negative and vice versa. So the LC feedback network gives an additional phase shift of 180° , necessary for satisfaction of oscillation conditions.
- ▶ This is the frequency of oscillations. At the frequency, the restriction of the value of h_{fe} can be obtained, by equating the magnitudes of both sides,

the frequency of oscillations is given by,

$$f = \frac{1}{2\pi\sqrt{C L_{eq}}}$$

NOTE

- ▶ So if the capacitor C is kept variable, frequency can be varied over a large range as per requirement.
 - ▶ In practice L_1 and L_2 may be wound on a single core so that there exists a mutual inductance between them and is denoted as M
 - ▶ In such a case the mutual inductance is considered while determining the equivalent inductance (L_{eq}).
- Hence, $L_{eq} = L_1 + L_2 + 2M$
- ▶ If L_1 and L_2 are assisting each other, then sign of $2M$ is positive while L_1 and L_2 are in series opposite then sign of $2M$ is negative.
 - ▶ The expression for the frequency of oscillation is

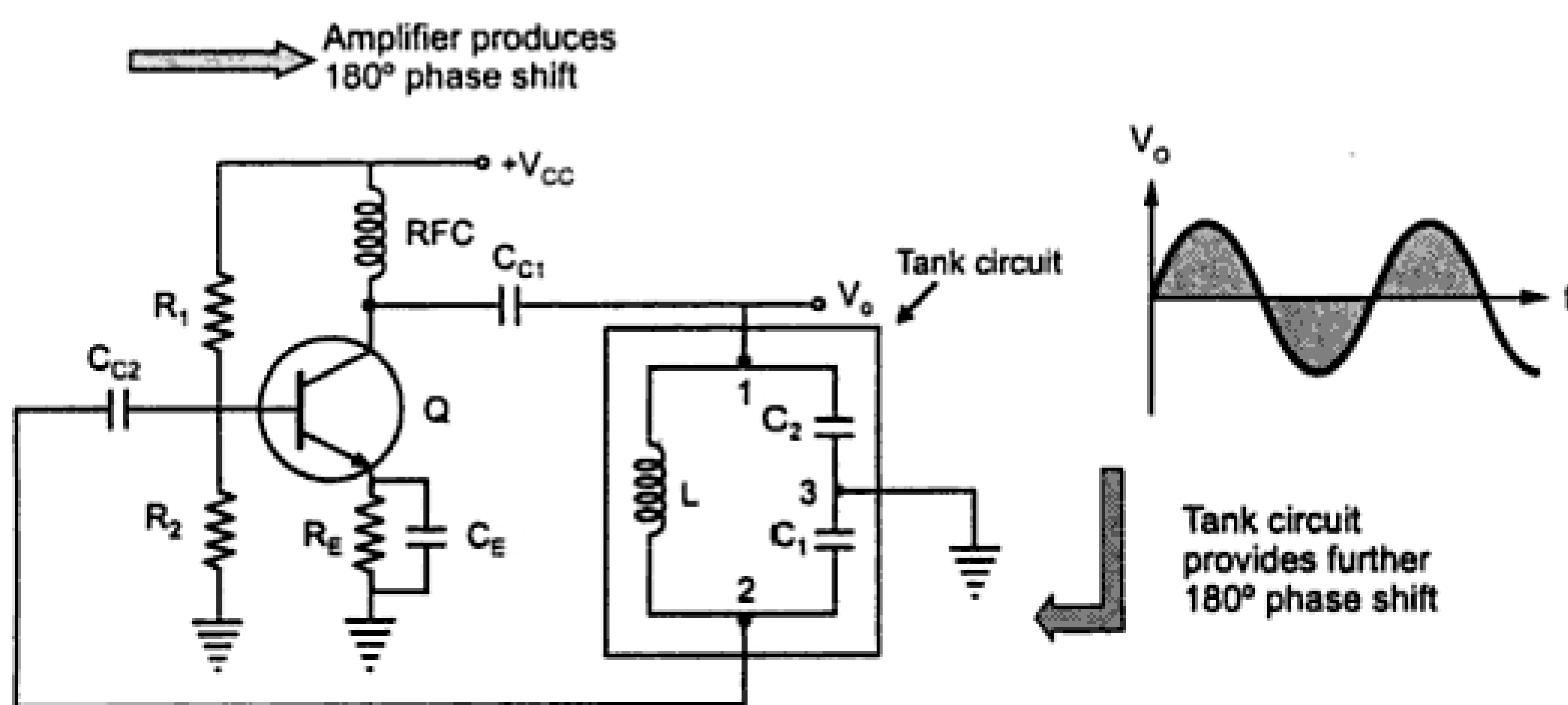
$$f = 1 / 2\pi\sqrt{C L_{eq}}$$

where, $L_{eq} = L_1 + L_2 + 2M$ (positive) same

$$L_{eq} = L_1 + L_2 - 2M \quad (\text{negative}) \text{ opposite}$$

3.7 COLPITTS OSCILLATOR

- An LC oscillator which uses two capacitive reactance and one inductive reactance in the feedback network is called colpitts oscillator. The amplifier stage uses active devices such as transistor in common emitter configuration. It is shown in the figure.
- The basic circuit is same as transistorized Hartley oscillator, except the tank circuit. The common emitter amplifier causes a phase shift of 180° phase shift, to satisfy the oscillating conditions.



DERIVATION OF FREQUENCY OF OSCILLATIONS

- As seen earlier, the output current I_c which is $h_{fe}I_b$ acts as input to the feedback network. While the base current I_b acts as the output current of the tank circuit, flowing through the input impedance of the amplifier h_{ie} . The equivalent circuit is shown in the figure. Converting the current source into the voltage source, we get the equivalent circuit in simplified form are also shown in the figure.

$$f = 1 / 2\pi\sqrt{LC_{eq}}$$

This is the frequency of oscillator in the colpitts oscillator.

Substitute $\omega^2 = 1 / L[(C_1C_2) / (C_1+C_2)]$ in magnitude part

$$h_{fe} = [1 - (C_1 + C_2) / C_1] = (C_1 - C_1 + C_2) / C_1$$

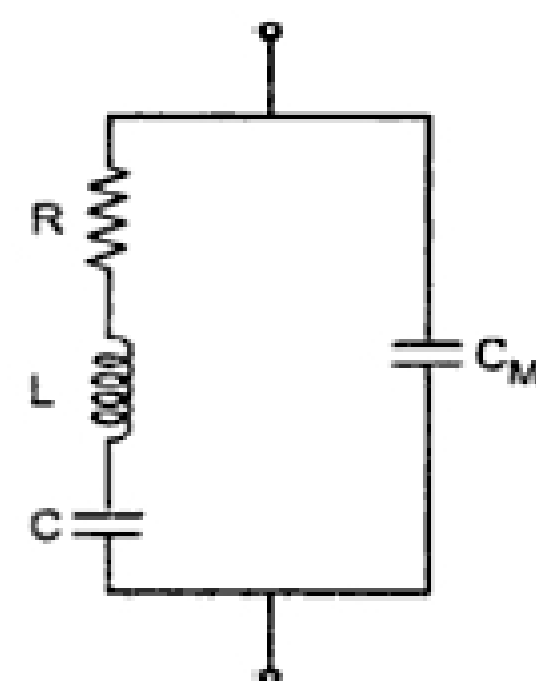
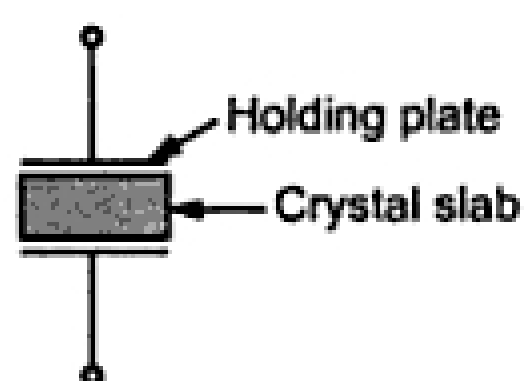
$$h_{fe} = C_2 / C_1$$

3.8 CRYSTAL OSCILLATORS

- ▶ The crystal oscillators are either naturally occurring or synthetically manufactures exhibiting the piezo-electric effect. It means under the influence of the mechanical pressure, the voltage gets generated across the opposite face of the crystal to vibrate, the AC voltage; it vibrates causing mechanical distortion in the crystal shape. Every crystal has its own resonating frequency depending on it. So under the influence of the mechanical vibrations, the crystal generates an electrical signal of very constant frequency. The crystal oscillator is a tuned circuit oscillator using a piezo-electric crystal as its resonant tank circuit. The crystal oscillator are preferred when greater frequency stability is required. Hence the crystals are used in watches, communication transmitters and receivers etc.,.
- ▶ The main substance exhibiting the piezo-electric effect is quartz, Rochelle salt and tourmaline. Rochelle salts have greatest piezo-electric effects. For a given AC voltage, they vibrate more than quartz or tourmaline. Hence these are prepared in using microphones associated with portable tape recorders, headsets and loudspeakers. Rochelle salt is mechanically weakest of the three and can break easily. Tourmaline shows the least piezo-electric effect but is mechanically the strongest. Tourmaline is more expensive and hence it is used rarely in practice. Quartz is more commonly used since it is easily obtained and is less expensive when compare to Rochelle salt and tourmaline.

Construction Details

- ▶ The natural shape of quartz crystal is hexagonal prism. But for its practical use it is cut into rectangular slab. This is mounted between the two metal plates.

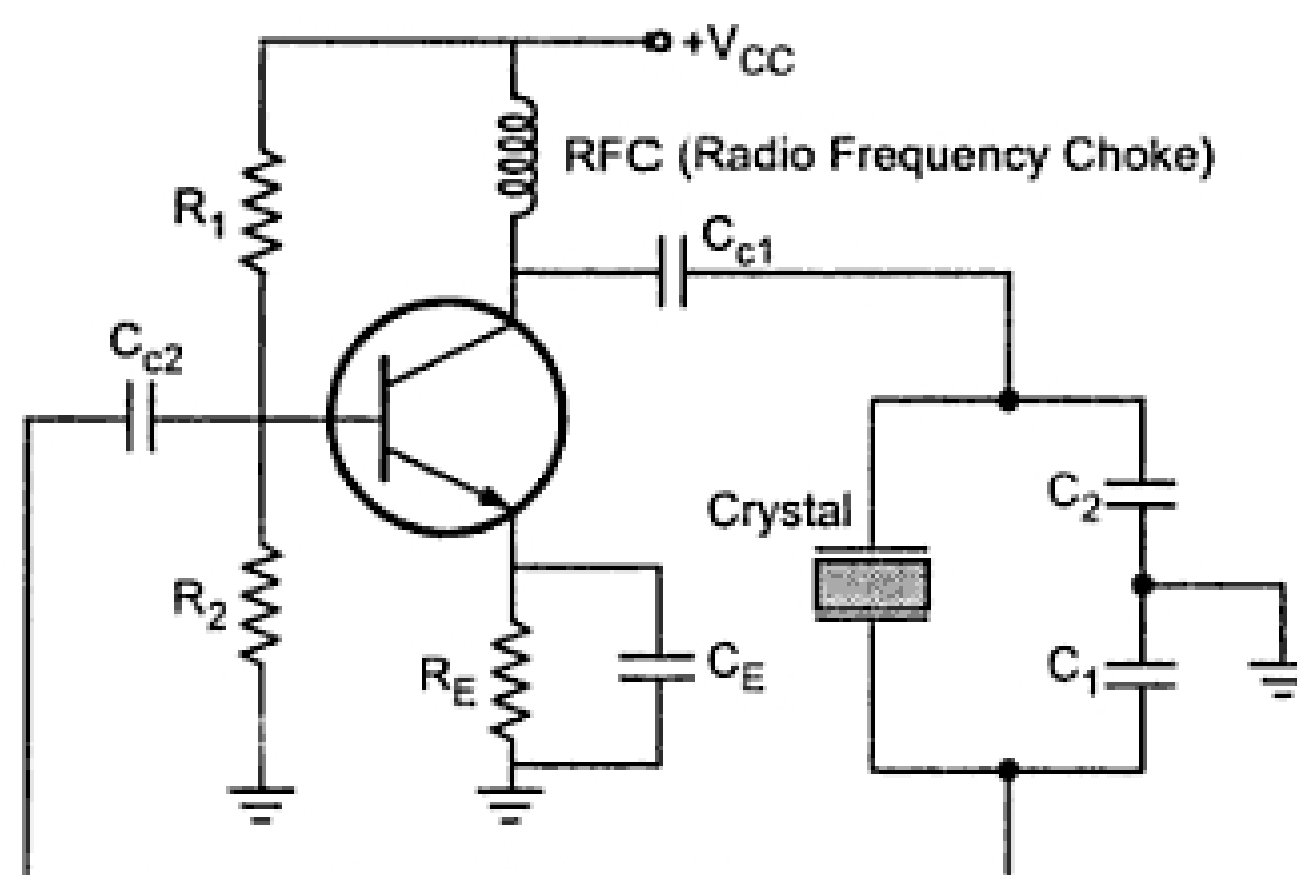


- The symbol of crystal is shown in the figure. The metal plates are called holding plates, as they hold the crystal slab between them

$$f = 1 / 2\pi\sqrt{LC}$$

3.8.1 PIERCE CRYSTAL OSCILLATOR

- The colpitts oscillator can be modified by using the crystal to behave like an inductor. The circuit is called Pierce Crystal Oscillator. The crystal behaves as an inductor for a frequency stability higher than the series resonance frequency f_r . The capacitors C_1, C_2 are required in the tank circuit along with an inductor are used as they are used in colpitt's oscillator. As the only inductor gets placed by the crystal, which behaves as an inductor, the basic working principle is the same as that of the colpitt's oscillator. It is shown in the figure.



- The resistances R_1, R_2, R_E provide DC bias while the capacitor C_E is emitter bypass capacitor. RFC (radio frequency choke) provides isolation between AC and DC operation. C_{c1} and C_{c2} are coupling capacitors.
- The resulting circuit frequency is set by the series resonant frequency of the crystal. Change in the supply voltages, temperature, transistor parameters have no effect on the circuit operating conditions and hence good frequency stability is obtained.

3.9 FREQUENCY STABILITY

- ▶ For an oscillator, the frequency of the oscillators must remain constant. The analysis of dependence of the oscillating frequency on the various factors like stray capacitance temperature etc is called or the frequency stability analysis. The measure of the stability of an oscillator to measure the desired frequency as precisely as possible for a long time as possible is called frequency stability of an oscillator.
- ▶ In transistorized colpitts oscillator or Hartley oscillator, the base collector junction is reverse biased and there exists internal capacitors which is dominant at high frequencies. This capacitance affects the value of capacitance in the tank circuit and hence the oscillating frequency.
- ▶ Similarly the transistor parameters are temperature sensitive. Hence as temperature changes, the oscillatory frequency also changes and no longer remains stable. Hence practically the circuit cannot provide stable frequency.

3.9.1 FACTORS AFFECTING THE FREQUENCY STABILITY

In general following are the factors which affect the frequency stability of an oscillator.

- ▶ Due to changes in temperature, the values of the components of tank circuit get affected. So the changes in the values of the inductors and capacitors due to the changes in temperature are the main cause due to which frequency does not remain stable.
- ▶ Due to changes in temperature, the parameters of the active device used like BJT, FET get affected which in turn affect the frequency.
- ▶ The variation in power supply is another factor affecting the frequency.
- ▶ The changes in temperature in the atmosphere conditions, aging and unstable transistor parameters affect the frequency.
- ▶ The changes in the load connected, affect the effective resistance of the tank circuit.
- ▶ The capacitive effect in the transistor and stray capacitance of the tank circuit and hence the frequency.

The variation of frequency with the temperature is given by the factor denoted by „ S“

$$S_{\omega,T} = \frac{\Delta\omega / \omega_r}{\Delta T / T_r} \text{ parts per million per } ^\circ\text{C}$$

where

- ω_r = desired frequency
- T_r = operating temperature
- $\Delta\omega$ = change in frequency
- ΔT = change in temperature

The frequency stability is defined as,

$$S_{\omega} = \frac{d\theta}{d\omega}$$

where $d\theta$ = phase shift introduced for a small frequency change

It can be improved by the following modifications

- ▶ Enclosing the circuit in a constant temperature chamber.
- ▶ Maintaining constant voltage by using the zener diode.
- ▶ The load effect is reduced by coupling the oscillator to the load loosely or with the help of a circuit having high impedance and low output impedance.

UNIT-4

Title	Page. No
4.1 INTRODUCTION	93
4.1.1 CIRCUIT SYMBOL OF IC 741	93
4.1.2 BLOCK DIAGRAM OF OP-AMP	94
4.1.3 EQUIVALENT CIRCUIT	94
4.1.4 INPUT-OUTPUT CHARACTERSTICS	95
4.1.5 CHARACTERSTICS OF AN IDEAL OP-AMP	96
4.1.6 PARAMETERS	97
4.2 OPEN LOOP OP-AMP CONFIGURATIONS	101
4.2.1 INVERTING AMPLIFIER	101
4.2.2 NON INVERTING AMPLIFIER	103
4.3 SUMMER	105
4.4 DIFFERENTIAL CONGIGURATION	106
4.5 SUBTRACTOR	107
4.6 DIFFERENTIATOR	108
4.7 INTEGRATOR	109
4.8 COMPARATOR	110
4.9 ANALOG MULTIPLIERS	111
4.9.1 ANALOG VOLTAGE DIVIDER CIRCUIT	113
4.9.2 MULTIPLIER INTEGRATED CIRCUIT	113
4.10 PERFORMANCE PARAMETERS	115
4.11 APPLIATIONS	116
4.12 SECOND ORDER LOW PASS FILTER	117
4.13 SECOND ORDER HIGH PASS FILTER	118

Department of Computer Science and Engineering

Subject Name: ELECTRONIC DEVICES AND CIRCUITS

Prepared by:

Mr. I.KAVIARASAN. M.Tech (Assistant Professor)

:

UNIT IV

Operational Amplifier: Introduction to op-amp, Characteristics of op-amp. - Differential and common mode operation, op-amp parameters - Applications: Inverting and non-inverting amplifier, summer, subtractor, differentiator, integrator, comparator, analog multiplier, second order low pass and high pass active filters.

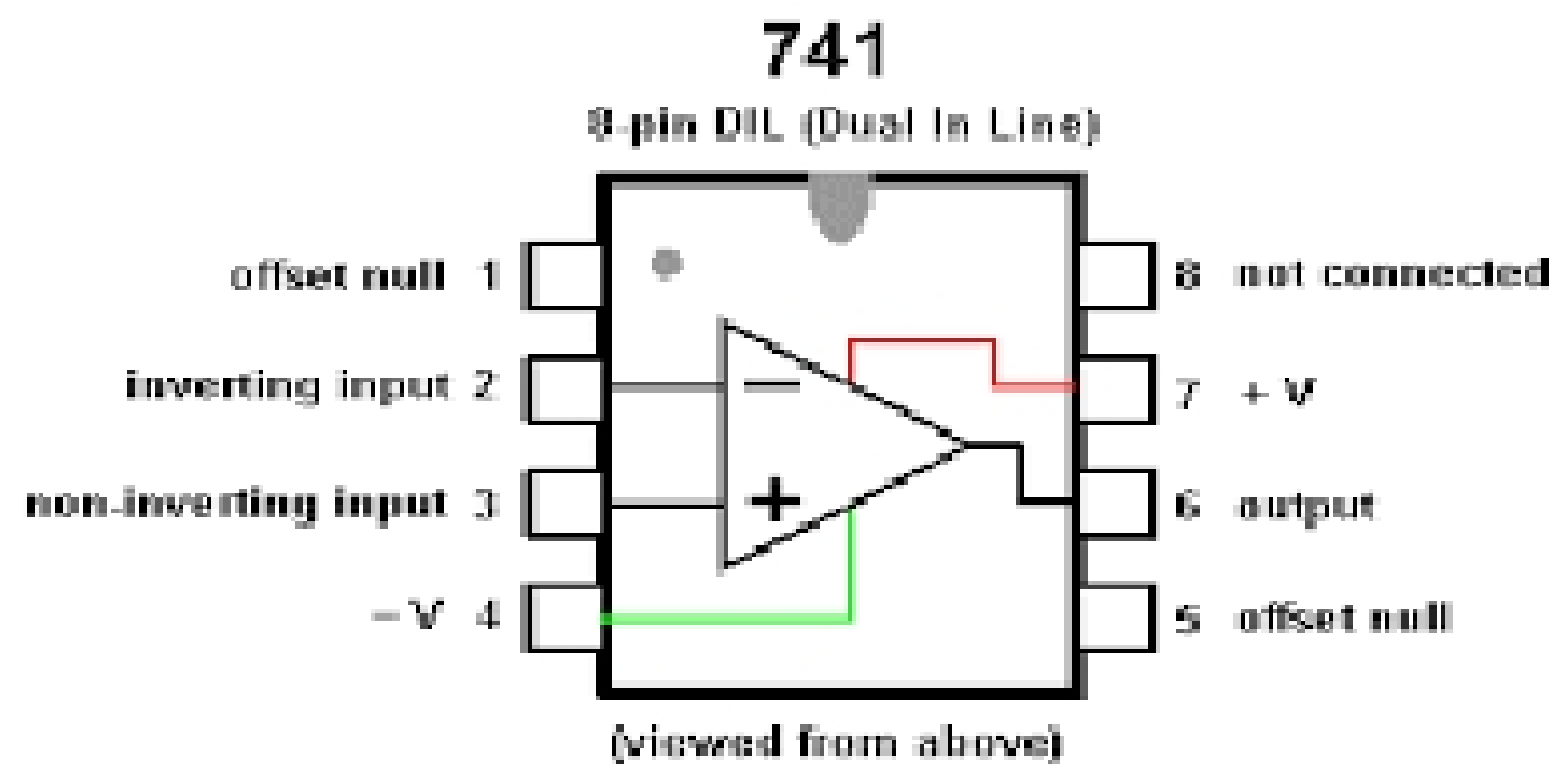
4.1 INTRODUCTION:

- ▶ The operational amplifier is a direct coupled, high gain, negative feedback amplifier. The term was originally used for DC amplifier which performs mathematical operations as summation, subtraction, integration and differentiation in analog computers. It is also used in voltage regulators, instrumentation and control systems, phase shift and oscillator, pulse generators, square wave generators, analog to digital and digital to analog convertors, scale changer etc. It is otherwise known as OP-AMP
- ▶ It can amplify signals having frequency ranging from (0-1) MHz (i.e) it can be used to amplify dc signals (0Hz) as well as AC input signals (High frequency).

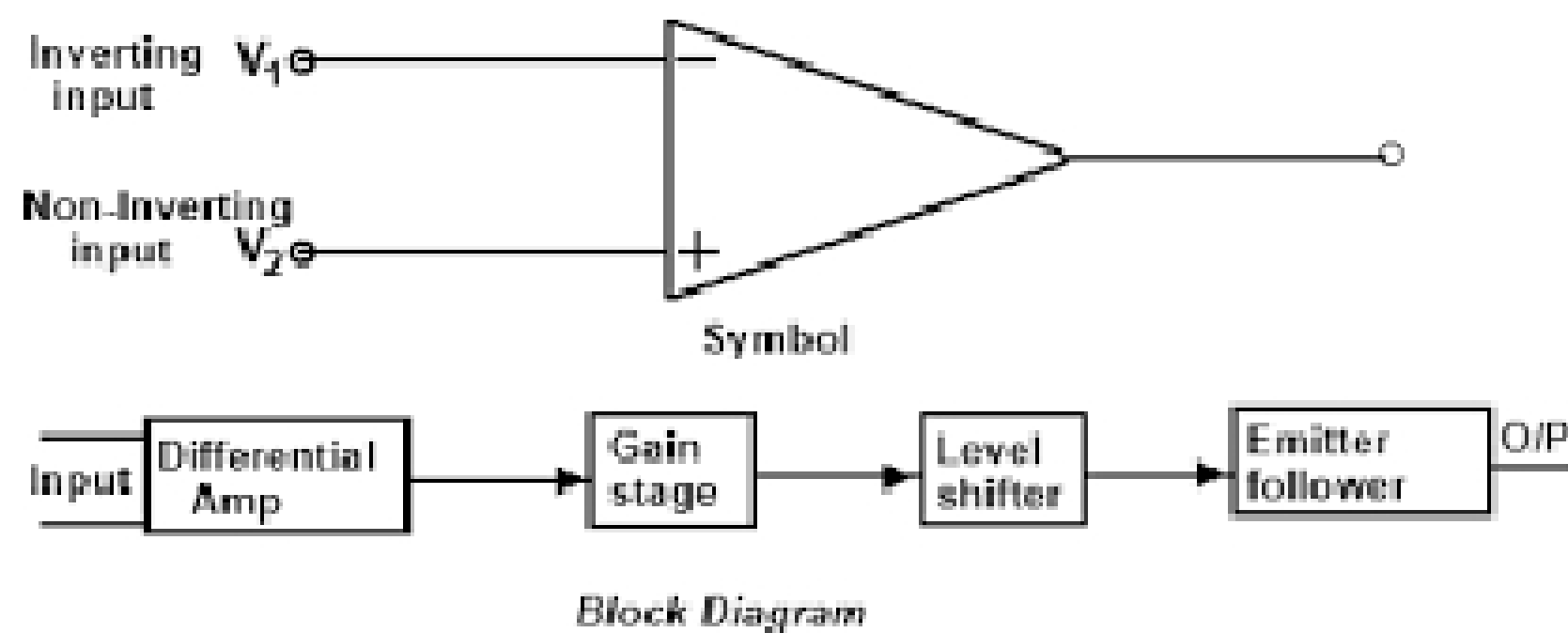
4.1.1 CIRCUIT SYMBOL OF IC 741:

- ▶ Figure shows the circuit symbols of IC 741 OP_AMP. The evident frequency figure, we see that OP-AMP has two input terminals and one output terminals. The difference between OP-AMP as circuit device and other circuit elements such as resistor, inductor, and capacitor or voltage source is that OP-AMP is a two port element whereas all the

elements are one port. In one port element there is a eight pair of terminals into or out of which a current flows and across which a voltage appears.

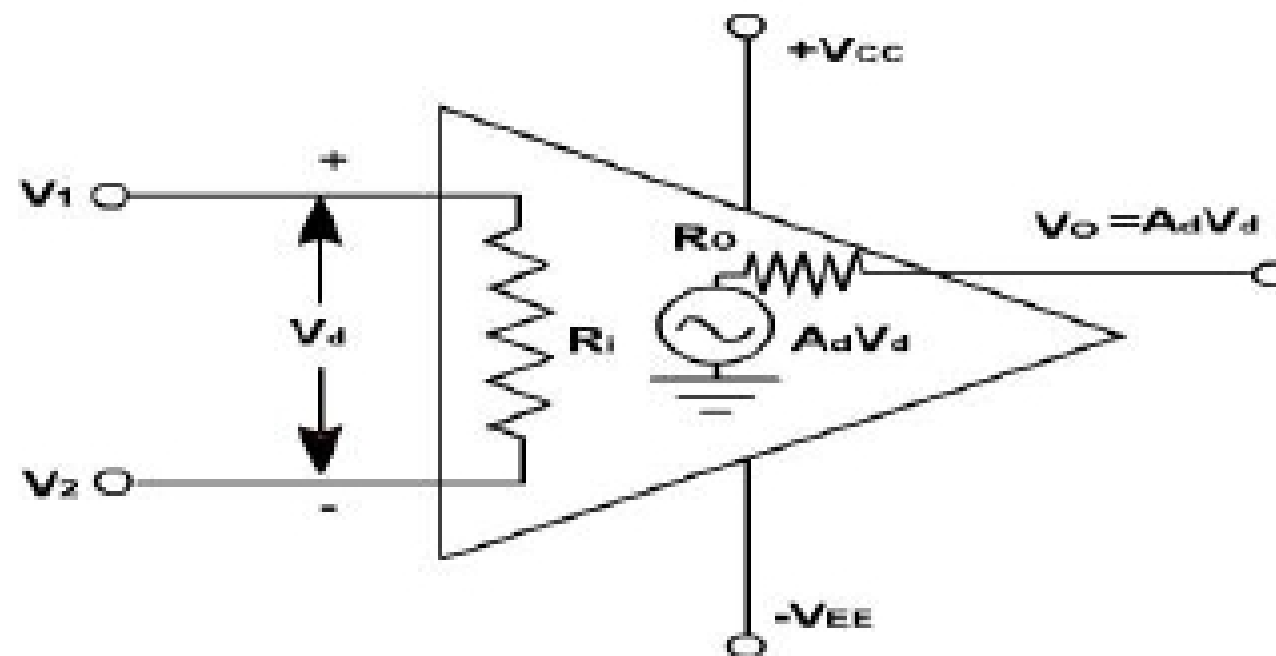


4.1.2 BLOCK DIAGRAM OF OP-AMP:



4.1.3 EQUIVALENT CIRCUIT:

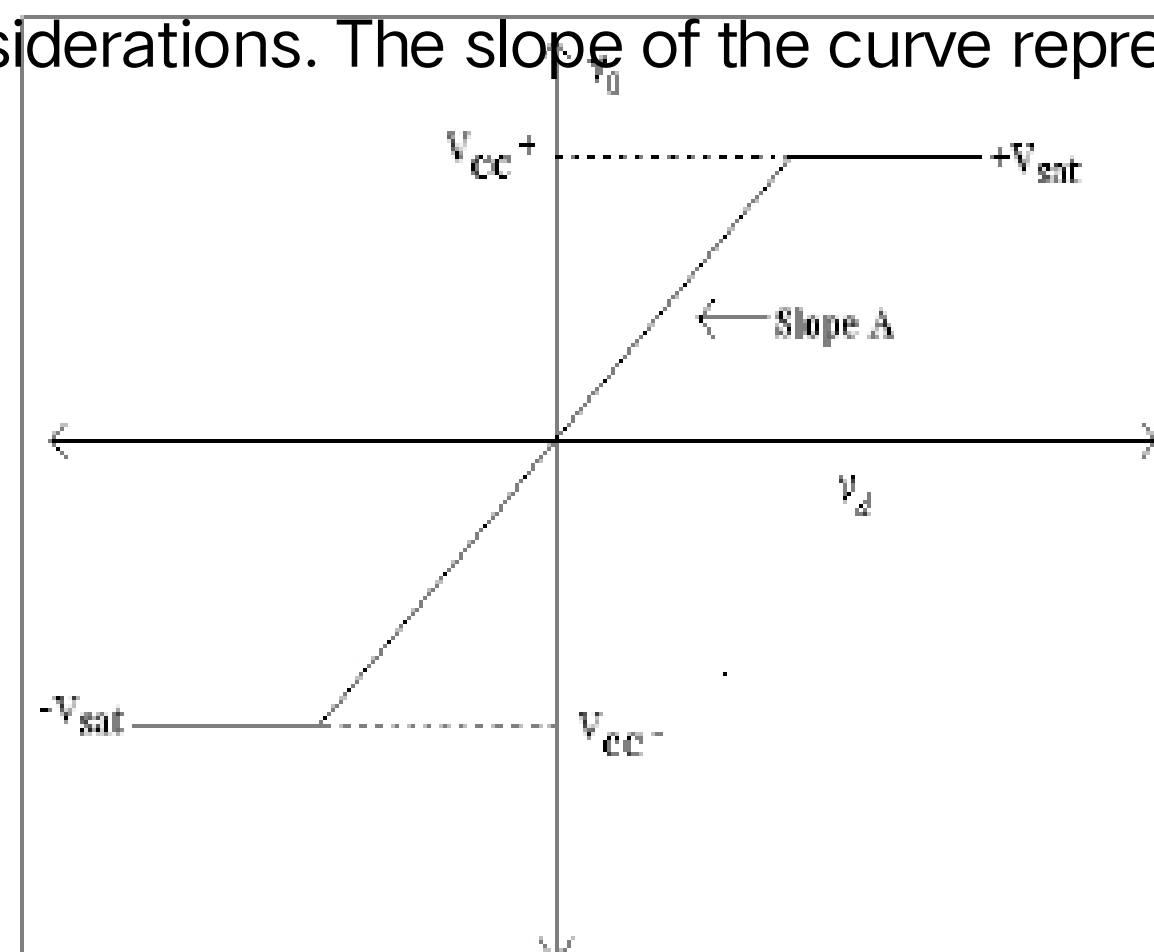
- The equivalent circuit of an OP-AMP is helpful in analyzing the operating principles of an OP-AMP and in observing the effects of feedback. The equivalent circuit of an OP-AMP and an ideal OP-AMP is shown in figure. V_1 is the voltage at the inverting terminal and V_2 is the voltage at the non-inverting terminal. V_{id} is the difference of two input voltages (i.e) $(V_2 - V_1)$. R_i is the input resistance which appears between the inverting and non-inverting input terminals. R_o is the output. The terminal of OP-AMP, the voltage of thevenin equivalent resistance looking back into the output terminal of OP-AMP. This voltage source (AV_{id}) is called Thevenin voltage source. The output voltage is directly proportional to the algebraic difference between the two input voltages.



- ▶ The ideal OP-AMP has infinite input resistance ($R_i = \infty$), zero output resistance ($R_o = 0$) and an infinite voltage gain ($A = \infty$). The difference between the two equivalents is well evident. R_i is infinite for open circuit in case of ideal OP-AMP equivalent circuit also R_o is replaced by a short circuit.

4.1.4 INPUT-OUTPUT CHARACTERISTICS:

- ▶ Voltage transfer curve is drawn between input and output voltages. In case of OP-AMP output voltage (V_o) is given $V_o = A V_{id}$ by
- ▶ Graphical representation of the above equation is called transfer characteristics curve of OP-AMP such a curve is shown in figure above the curve it is observed that saturations of the output voltage takes place once the differential input signal reaches a transfer value (V_{th}). This saturation value is approximately equal to the supply voltage for all practical considerations. The slope of the curve represents the gain A .



Transfer characteristics of op amp

4.1.5 CHARACTERISTICS OF AN IDEAL OP-AMP:

The ideal OP-AMP has the following characteristics

An infinite voltage gain:

- ▶ It is the open loop voltage gain (i.e) ($A = \infty$)

An infinite bandwidth:

- ▶ It is any frequency signal ranging from (0 to ∞) Hz may be amplified without any distortions. In other words, we can say that it provides a constant gain for all frequency range.

An infinite input resistance:

- ▶ It is the resistance measured between inverting and non-inverting terminals. By infinite input resistance, the resistance offered to a source applied at either of the two input terminals of the OP-AMP is infinite. Hence input current is zero and there is no loading effect. The OP-AMP does not change the behavior of the signal source. Another inference that can be drawn is that the OP-AMP is a voltage controlled device.

Zero output resistance:

- ▶ It is like the internal resistance of the battery and hence should be very small ($= 0$). It means that V_o is independent of the load resistance connected across the output. In other words, the output voltage is independent of the current drawn from the output.

Perfect balance:

- ▶ The output is zero ($V_o = 0$) when equal voltages ($V_1 = V_2$) are applied at the two input terminals.

Zero offset voltage:

- ▶ When input signal voltage is zero, the output signal will also be zero

Infinite CMRR:

- ▶ Common Mode Rejection Ratio (CMRR) is infinite (i.e) common mode noise voltage is zero.

Infinite slew rate:

- ▶ Changes in the output voltage simultaneously with the input voltage changes

Zero drift:

- ▶ The characteristics do not (drift) with the temperature changes. The term CMRR, slew rate and drift have been explained later in this unit. It is important to note that a practical

OP-AMP is however non-ideal. The comparison table of ideal and non-ideal OP-AMP characteristics is shown.

Characteristics of OPAMP	Ideal OP-AMP	PRACTICAL OP-AMP
Open loop voltage gain(A)	∞	2×10^5
Input resistance	∞	$2M\Omega$
Output resistance	0	75Ω
Input offset voltage	0	2mV
Input offset current	0	200nA
Bandwidth	∞	1MHz
CMRR	∞	90db
Slew rate	∞	$0.5V/\mu s$

4.1.6 PARAMETERS:

- There are few constantly used electrical parameters of OP-AMP are defined as follows:

Input offset voltage:

- When the input is 0v, the output of OP-AMP should be zero. But in actual operation, there is some offset voltage at the output. The input offset voltage is defined as the voltage that must be applied between the two input terminals of OP-AMP to nullify the output. Typically it ranges from (1-5)mv. The smaller the value of input offset voltage; the better is the input terminals matched.

Input offset current:

- An output offset voltage will also result due to any difference in DC bias current at both inputs. The reason is that the two input transistors are never exactly matched. Each has a slightly different current. The input offset current is the difference between the two input currents. It typically ranges from (20-60)nA. The smaller the input offset current; the better is the OP-AMP's performance.

Input bias current:

- It is the average of the current that flow into the inverting and non-inverting input terminals of an OP-AMP. The smaller the input bias current, the smaller is the possible unbalance.

Input offset voltage drift:

- It is the ratio of the change of input offset voltage to the change in temperature.

Input offset current drift:

- It is the ratio of the change of input offset voltage to the change in temperature.

Input resistance:

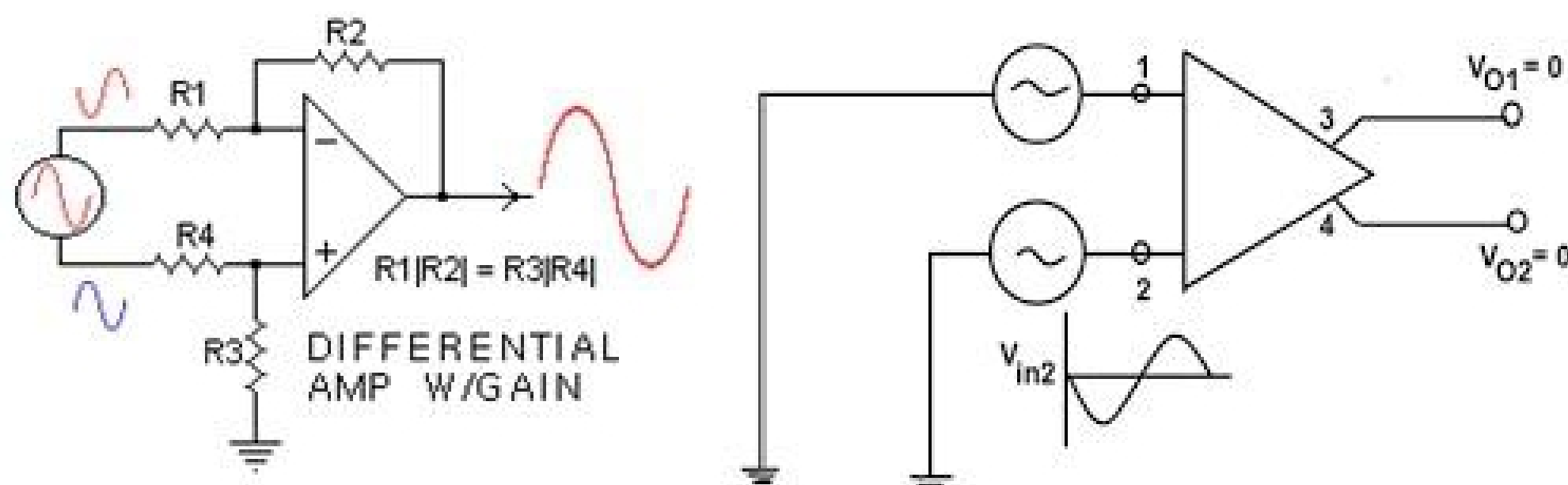
- It is the differential input resistance that can be measured at either the input terminal with the other terminals connected to ground. The OP-AMP having bipolar input stage has an input resistance in the range of $100\text{K } \Omega$ to $1\text{M } \Omega$. Usually the voltage gain is large enough. Hence the input resistance has little effect on the circuit performance.

Output resistance:

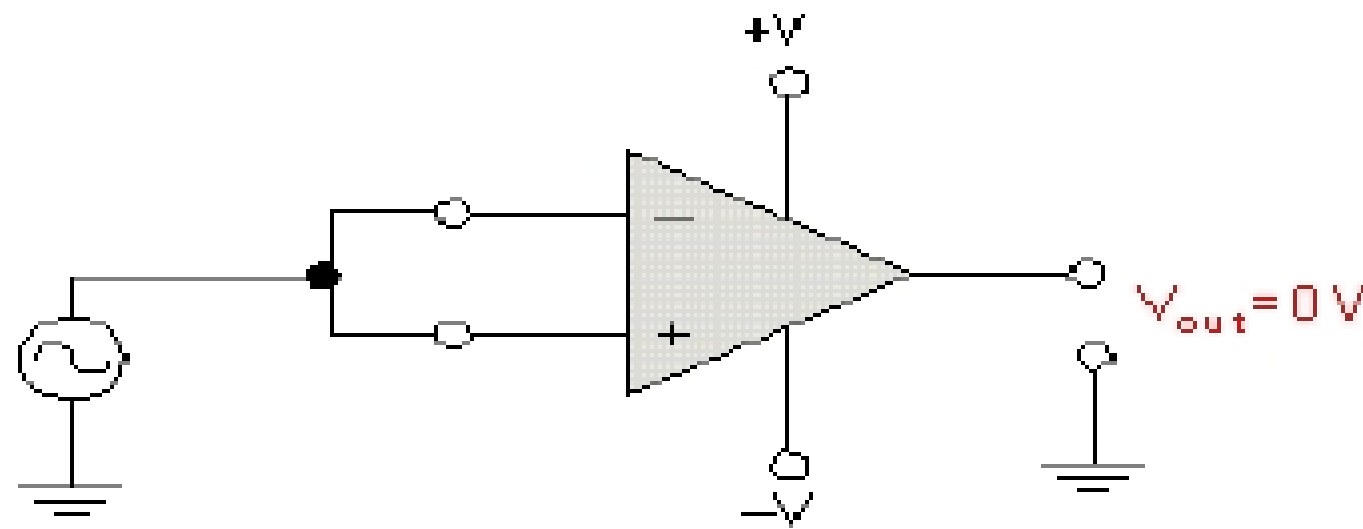
- It is the resistance measured between the output terminal of the OP-AMP and the ground. This is of the order $(40-100)\Omega$. this resistance does not significantly effect the closed loop performance of the OP-AMP.

Differential mode operation:

- Instead of using only one input, it is possible to apply signals at each input of OP-AMP. The operation is known as double-ended or differential mode operation. It is shown in figure an input V_d applied between three input terminals. The output is in phase with the applied signal between plus and minus inputs. The other figure shows the same action resulting when two separate signals are applied to the inputs. The difference signal being $(V_{i1} - V_{i2})$.

Common mode operation:

- When the same input signals are applied to both the inputs of an OP-AMP, then the operation is known as Common Mode Operation. It is shown in figure at output terminal giving 0v output. Practically, a small output signal will result.



- So, a differential amplifier, amplifying signals, that are opposite at the two inputs to a large extent while a very small amplification to those signals that are common to both inputs. Since amplification of opposite input signal is much greater than that of the common input signals, the circuits provide a common mode rejection. The ability of a differential amplifier to reject a common mode signal is expressed by its Common Mode Rejection Ratio (CMRR).

Common mode rejection ratio:

It is defined as the ratio of the differential voltage gain A_d to the common mode voltage gain A_c

$$CMRR = \frac{A_d}{A_c}$$

Where A_d = gain of OP-AMP when two differential voltage are applied at the two inputs.

Where A_c = common mode voltage gain of OP-AMP when the two terminals of OP-AMP are applied with the same voltage from the same source

- For commercial OP-AMP the CMRR lies in the range of (60-100) db. When two separate inputs are applied to the OP-AMP, the resulting difference signal is difference between two inputs.

$$V_d = V_1 - V_2$$

- When both the input signals are equal a common signal element due to two inputs is defined as the average of the sum of the two signals.

$$V_c = \frac{V_1 + V_2}{2}$$

- When any signal is applied to an OP-AMP, the output is general have both in phase and out of phase components. The resulting output is

$$V_o = A_d V_d + A_c V_c$$

- Where V_d and V_c are difference and common voltages respectively. Further A_d and A_c are differential and common- mode gains of OP-AMP respectively

We know that $A_c = \frac{A_d}{CMRR}$

Substitute in V_o , we get

$$V_o = A_d V_d + (A_d / CMRR) V_c$$

Slew rate(SR):

- It is defined as the maximum rate of change of output voltage per unit of time and is expressed in volts/microsecond (i.e)

$$SR = \max \left(\left| \frac{dv_{out}(t)}{dt} \right| \right)$$

- Slew rate indicates how rapidly the output of OP-AMP can change in response to change in input frequency.

Power supply rejection ratio (or) supply voltage rejection ratio:

- This is defined as the change in OP-AMP input offset voltage V_{io} caused by variation in supply voltage (i.e)

$$SVRR = PSRR = \frac{dV_{io}}{dV}$$

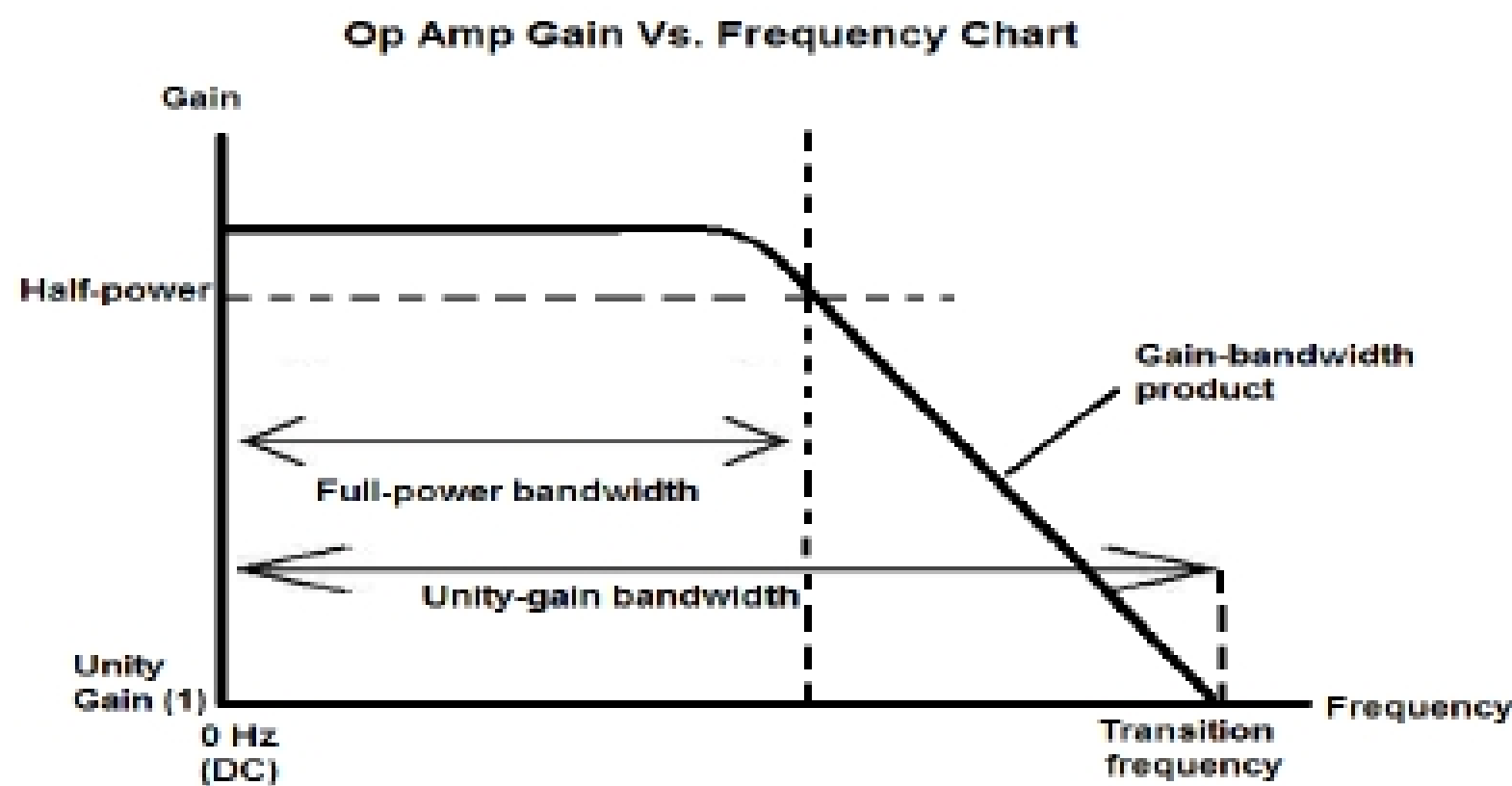
Where dV_{io} – change in input offset voltage

dV – change in supply voltage.

Typically lies in the range $(10^{-5} \text{ to } 7 \times 10^{-5})$ V

Gain bandwidth product:

- Due to parasite junction capacitance and minority charge storage in devices making up to the circuit, the voltage gain of OP-AMP decreases at high frequencies. It is shown in figure, a plot of gain versus frequency for a typical OP-AMP. At low frequency down to DC operation, the gain is A_{VD} (Voltage difference gain). As the frequency of input signal increases, the open loop gain drops off until it finally reaches the value 1. The frequency at which the gain reduces to 1 is defined as unity gain frequency (f_1) or unity-gain bandwidth (B_1).



4.2 OPEN LOOP OP-AMP CONFIGURATIONS:

- ▶ Where there exists no connection between the output and input terminals of OP-AMP (i.e) No feedback, then it is known as open loop OP-AMP. In this case, the OP-AMP functions as high gain amplifier. There are two main open loop configurations are
 - ✓ Inverting amplifier
 - ✓ Non-inverting amplifier
- ▶ These configurations are based on the number of inputs used and the terminal to which the input is applied. It is important to mention here that OP-AMP amplifier both AC and DC input signals.

4.2.1 INVERTING AMPLIFIER

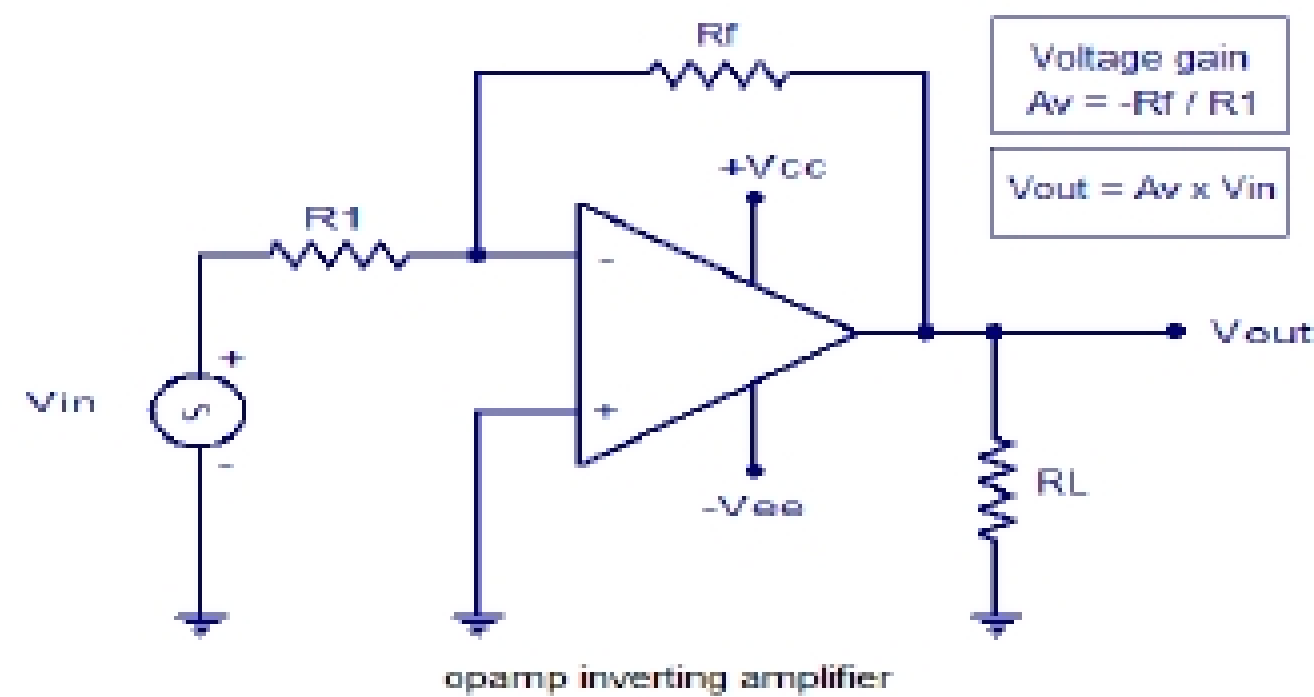
- ▶ In this configuration, the input signal is applied to the inverting input terminal as shown in figure. The non-inverting terminal is grounded (i.e) $V_2=0$

The output is given by

$$V_O = AV_{id} = A(V_2 - V_1)$$

$$V_O = -AV_1 \quad \text{i.e } (V_2=0)$$

- ▶ The negative sign indicates that the output voltage is out of phase with respect to input (i.e) output voltage is 180° out of phase with respect to input. Thus in inverting amplifier, the input is amplified by gain A with change in polarity. So it is known as inverting amplifier.



- Figure shows the basic inverting amplifier with an input resistance R_1 and feedback resistance R_f . In this mode of operation, the positive input terminal of the amplifier is grounded and the input signal is
- Note that V_1 may be a DC or an AC signal within the bandwidth of the amplifier. The output signal is feedback to the inverting input of OP-AMP through feedback resistor R_f . It is obvious from the figure that the feedback currents are algebraically added at point G, so this point is called as summing point. When input voltage V_i is applied at the input terminal, the point G attains same positive potential. Now there exists output voltage V_o . Due to negative feedback, fractions of the output Voltage with phase inverted is feedback to point G. The algebraic sum of the two voltages at G is almost zero. Thus the point G, although not connected to ground, but is held virtually at ground potential respective of the magnitudes of the positive V_1 and V_o . The voltage at point G will become exactly zero when negative feedback voltage is exactly equal to positive voltage produced by V_1

Gain:

The current i_1 flowing through point G is given by

$$i_1 = \frac{V_1 - V_i}{R_1} = \frac{V_i}{R_1} \quad (G - \text{ground potential})$$

$$\text{Similarly } i_2 = \frac{V_i - V_o}{R_f} = \frac{-V_o}{R_f}$$

$$\text{At point G } \frac{V_i}{R_1} = \frac{-V_o}{R_f}$$

- The ratio of output voltage V_o and input voltage V_i is known as the closed loop gain of the output of the amplifier. Hence the gain of inverting amplifier is given by

$$A_v = \frac{V_o}{V_i} = \frac{-R_f}{R_1}$$

- ▶ Thus the voltage gain is equal to the ratio of feedback resistance R_f of the input Resistance R_1 . The negative sign indicates that the output voltage is inverted with respect to the input voltage. It also indicates that closed-loop gain of the inverting OP-AMP can be changed by simply changing the values of two resistance ($-R_f$) and R_1 . Hence depending on a particular application any desired gain can be attained.

Input resistance:

- ▶ It is defined as the ratio of voltage V_1 to the input current.

$$R_{in} = V_1 / (V_1 - V_i) / R_1 \approx R_1.$$

- ▶ It has infinite input resistance.

Negative scalar:

- ▶ Here we shall show that OP-AMP works as a negative scalar. We denote $\frac{V_o}{V_i}$ by the ratio of

K , a real constant then

$$V_o = -KV_i$$

- ▶ This shows that the input voltage has been multiplied by a factor $-K$ to give the output voltage scale. Thus the circuit can act as scalar changer. In scalar changer the precision resistors are used to get the accurate value of scalar factor K .

Unity gain amplifier:

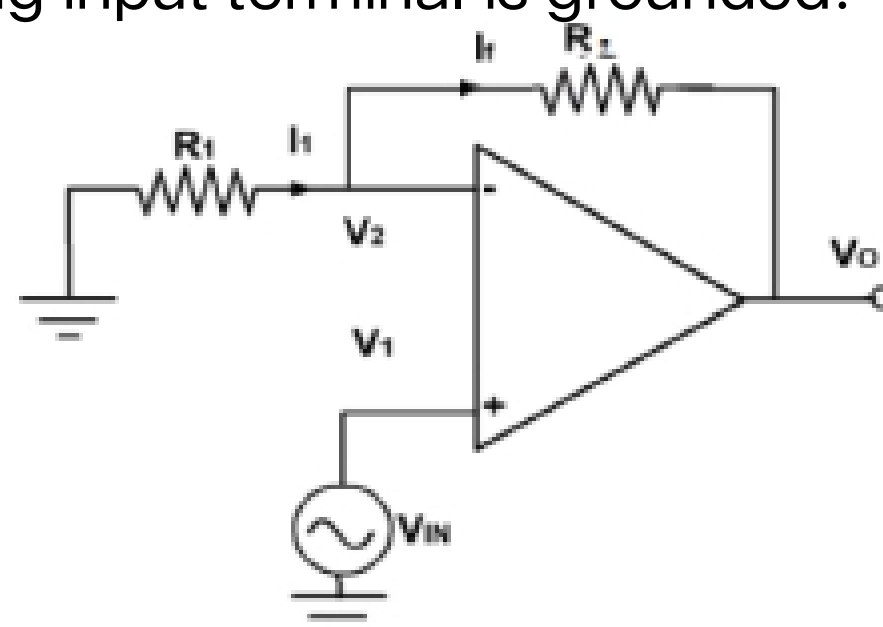
- ▶ The output of basic OP-AMP is given by

$$\text{If } R_f = R_1 \quad A_v = \frac{V_o}{V_i} = \frac{-R_f}{R_1}.$$

- ▶ The circuit provides a unity voltage gain with 180° phase shift inversion. It is used in inverter, paraphrase amplifier.

4.2.2 NON INVERTING AMPLIFIER

- ▶ Here, the input signal is applied to the non-inverting input terminal of OP-AMP as shown in figure. The inverting input terminal is grounded.



The output voltage is given by

$$\begin{aligned} V_O &= AV_{id} \\ &= A (V_2 - V_1) \\ V_O &= AV_2 \end{aligned}$$

- ▶ This shows that the output voltage gain A times the input voltage V_2 and is in phase with the input. As in non-inverting amplifier, the input signal is amplified by gain A without change in polarity and hence named as non-inverting amplifier.
- ▶ Here the input voltage V_2 is applied to the non-inverting terminal and hence the name non-inverting amplifier. The potential of point G also V_2 since the gain of OP-AMP is infinite. The polarity of V_O is the same as that of V_2 . The voltage across R_1 is V_2 and across R_f is $(V_O - V_2)$

Gain:

- ▶ The values of current i_1 and i_2 are given as

$$i_1 = \frac{V_2}{R_1} \quad \text{and} \quad i_2 = \frac{V_O - V_2}{R_f}$$

- ▶ Applying Kirchhoff's first law at point G, we get

$$-i_1 + i_2 = 0$$

$$\frac{-V_2}{R_1} + \frac{V_O - V_2}{R_f} = 0$$

$$\frac{V_O}{V_2} = 1 + \frac{R_f}{R_1}$$

$$A_V = 1 + \frac{R_f}{R_1}$$

- ▶ So the gain is one plus the ratio of two resistances R_f and R_1 . Here the output voltage is in phase with the input voltage. This circuit offers high input and low output impedance.

Positive scalar:

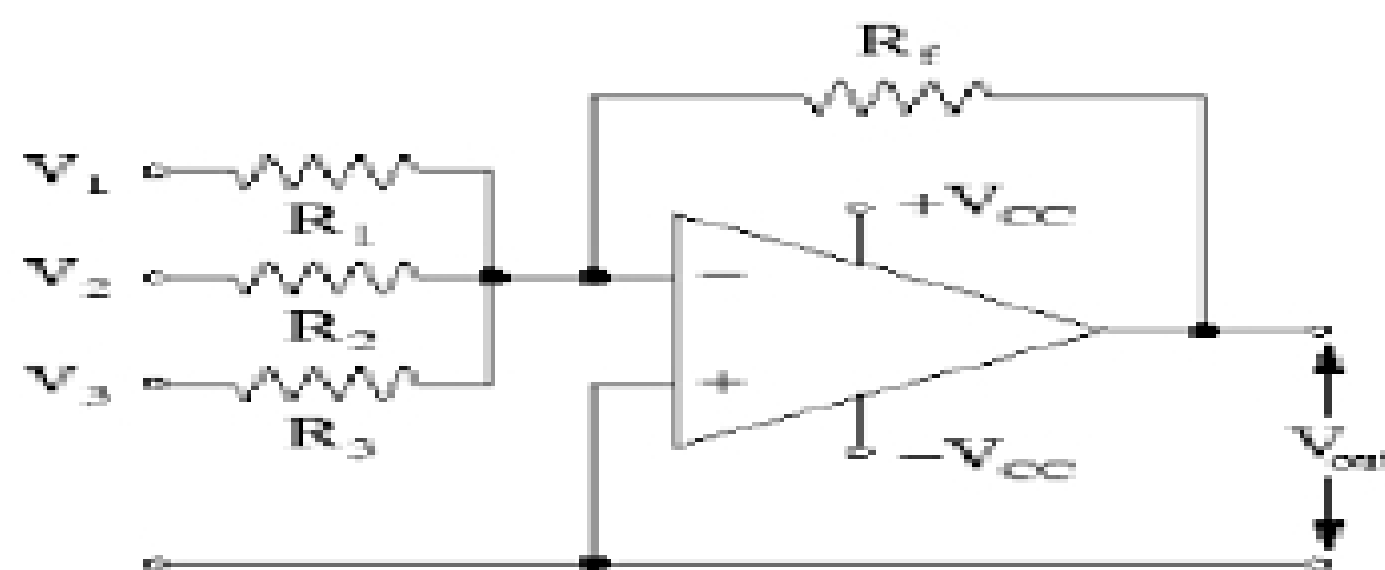
$$\text{Here } \frac{V_O}{V_2} = 1 + \frac{R_f}{R_1} = k \quad V_O = K V_2$$

- So it acts as a positive scalar. The amplifier is used, when we require an output which is equal to input multiplied by a positive current. As the circuit has high input and low output impedance, this is generally used as impedance matching devices between high impedance source and low.

4.3 SUMMER

- The most useful of the OP-AMP circuit used in analog computers is the summing amplifier circuit. Figure shows a three input summing circuit. The circuit provides a means of algebraically summing three input voltages, each multiplied by a current gain factor. In the circuit, G is a virtually ground and the output voltage is phase reverted.
- As G is virtually, the different currents are given by

$$i_1 = \frac{v_1}{R_1} \quad i_2 = \frac{v_2}{R_2} \quad i_3 = \frac{v_3}{R_3} \quad \text{and} \quad i_f = \frac{-v_o}{R_f}$$



- Applying Kirchhoff's law current distribution law at point G is

$$i_1 + i_2 + i_3 - i_f = 0$$

$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \frac{v_o}{R_f} = 0$$

$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

- Thus the output voltage is proportional to the algebraic sum of three input voltages. Again if $R_f = R$ we have,

$$V_o = -[V_1 + V_2 + V_3]$$

- In this case the output voltage V_o is numerically equal to the algebraic sum of the input voltages. Hence the name summing or adder amplifier.

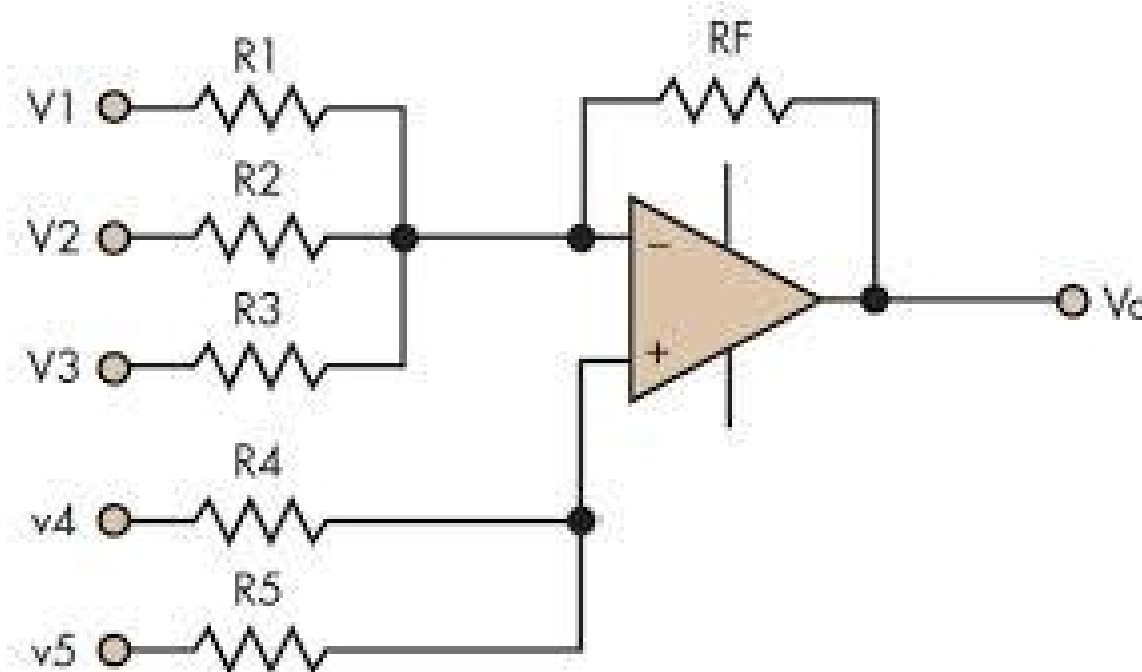
If $R_f = R/3$ then,

$$V_o = -(1/3)[V_1 + V_2 + V_3]$$

In this case the output voltage is equal to the average of the three inputs.

4.4 DIFFERENTIAL CONFIGURATION

- A summing amplifier in differential configuration is shown in figure. As seen from figure it consists of four inputs such that two input resources are connected of four inputs such that two input resources are connected each to non-inverting and inverting terminal of the OP-AMP.



For simplicity in calculation, we have taken all resistances to be equal to R

- The circuit output is now determined using superposition theorem. First consider only one input, making $V_b, V_c, V_d = 0$ (Connecting to earth) and calculate the output due to V_a alone. The output due to V_a is given by

$$v_{od} = \frac{R}{R} v_o = -v_o$$

Similarly $V_{ob} = -V_b$

- Now V_a, V_b and V_d are made equal to zero leading the circuit to be non-inverting amplifier whose output is given by

$$v_{oc} = 1 + \frac{R}{R} v_2 = 3 \frac{v_c}{3} v_c$$

Similarly $V_{od} = V_d$

- The net output voltage is now calculated

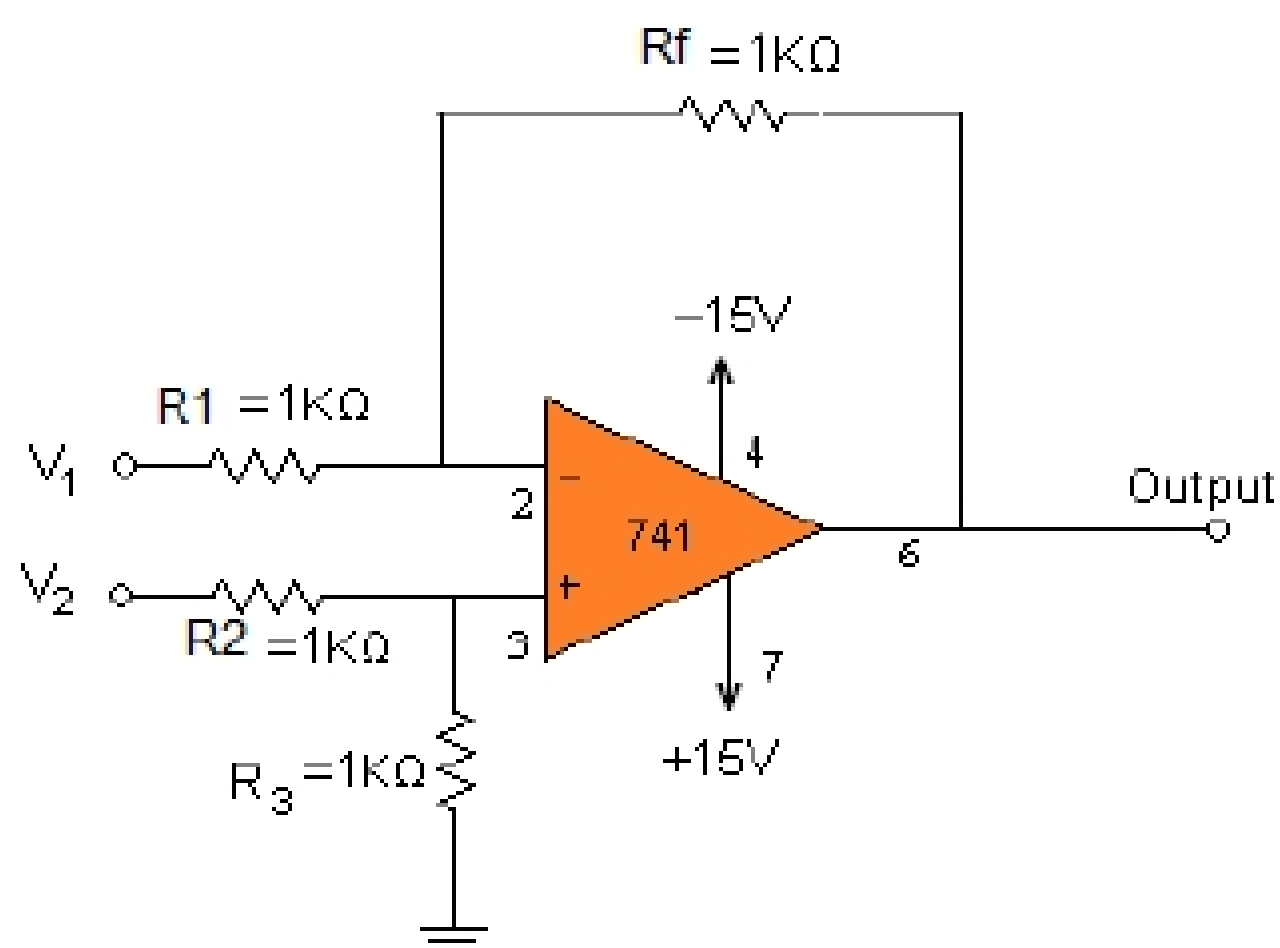
$$v_o = v_{oa} + v_{ob} + v_{oc} + v_{od}$$

$$v_o = -v_a - v_b + v_c + v_d$$

- From above equation it is observed that output voltage is equal to the sum of input voltages applied in the non-inverting terminal plus the negative sum of input voltages applied to non-inverting terminal.

4.5 SUBTRACTOR

- Figure show the circuit of the subtractor. The function of the subtractor is to provide an output which is equal to the difference of two input signals or proportional to the difference of two input signals. Here one output signal is applied at the non-inverting terminal of an OP-AMP.



- Let $(V_o)_1$ and $(V_o)_2$ be the output voltages produced by input voltages V_1 and V_2 respectively. Now,

$$v_{o1} = \frac{-R_f}{R_1} v_1$$

$$v_{o2} = 1 + \frac{R_f}{R_1}$$

$$v_o = v_{o1} + v_{o2}$$

$$= \frac{-R_f}{R_1} v_1 + \frac{-R_f}{R_1} v_2$$

$$= \frac{R_f}{R_1} v_2 - v_1$$

$$v_o = K (v_2 - v_1)$$

This equation shows that output voltage is proportional to the difference of two input voltages

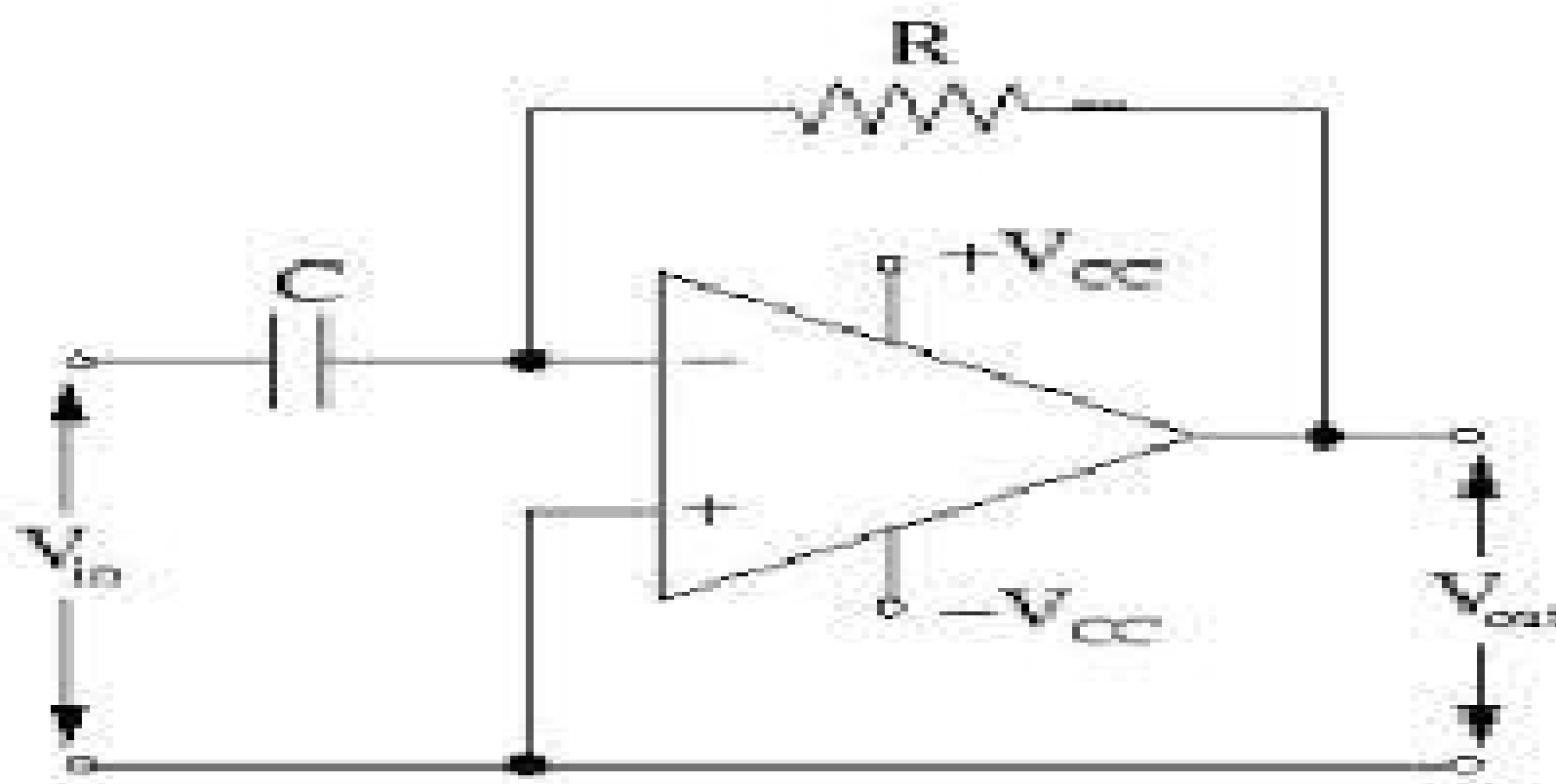
If $R_f=R_1$, then from above equation , We get,

$$v_o = (v_2 - v_1)$$

Thus the output voltage is equal to difference of two input voltages.

4.6 DIFFERENTIATOR:

- The function of the differentiator is to give an output voltage which is proportional to the rate of change of the input voltage. The circuit of the differentiator is same as the inverting OP-AMP except that the input resistance R_1 is replaced by a capacitor. When we feed linearly increasing voltage to the differentiator, we get a current DC output. So it is an increasing mathematical operation to that of an integrator.



- Assuming that G is at ground Potential, we can write for capacitor

$$v_1 = \frac{q}{c}$$

$$\frac{dv_1}{dt} = \frac{1}{c} \frac{dq}{dt} = \frac{i}{c}$$

Again $V_o = -iR$

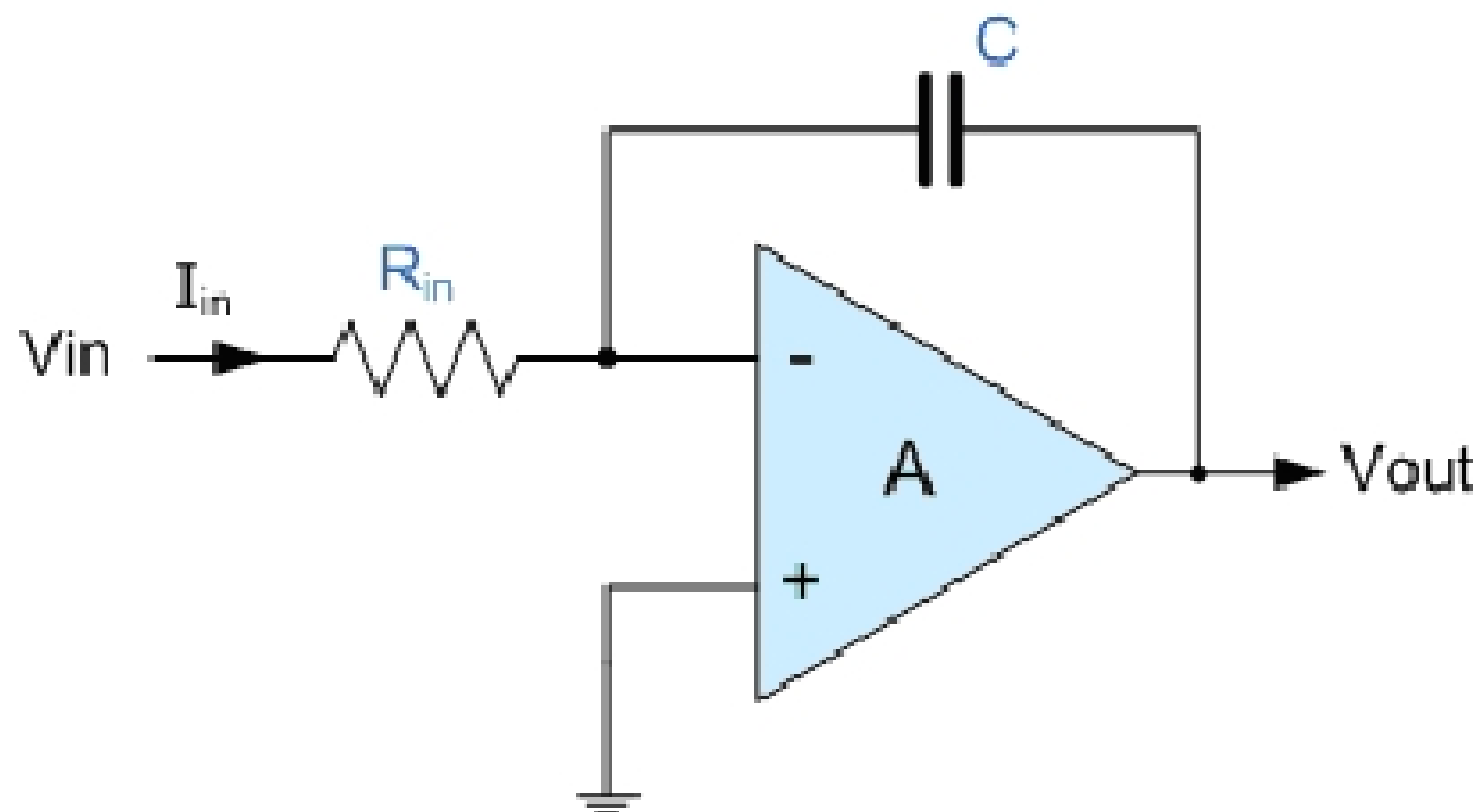
Substitute the value of i , we get

$$v_o = -RC \frac{dv_1}{dt}$$

- This relation shows that the output voltage V_o is equal to a constant ($-RC$) times the derivative of the input voltage V_1

4.7 INTEGRATOR

- The circuit of integrator is shown in figure. This circuit provides an output voltage which is proportional to the time signal integrator of the input voltage. Due to this reason this is known as integrator. The integrator is an inverting OP-AMP in which feedback resistor R_f has been replaced by a capacitor C . Feedback through the capacitor forces a virtual ground to exist at the inverting input terminal. Now the voltage across the condenser is simply the output voltage V_o



The capacitance impedance X_c can be expressed as

$$X_c = \frac{1}{j\omega C} = \frac{1}{sC}$$

Where $s=j\omega$ for Laplace notation

From the figure

$$i_1 = \frac{V_1}{R_1}$$

$$i_2 = \frac{-V_o}{X_c} = -sC V_o$$

At point G,

$$i_1 = i_2$$

$$\frac{V_1}{R_1} = \frac{-1}{sC} V_o$$

In time domain the above equations can be rewritten as

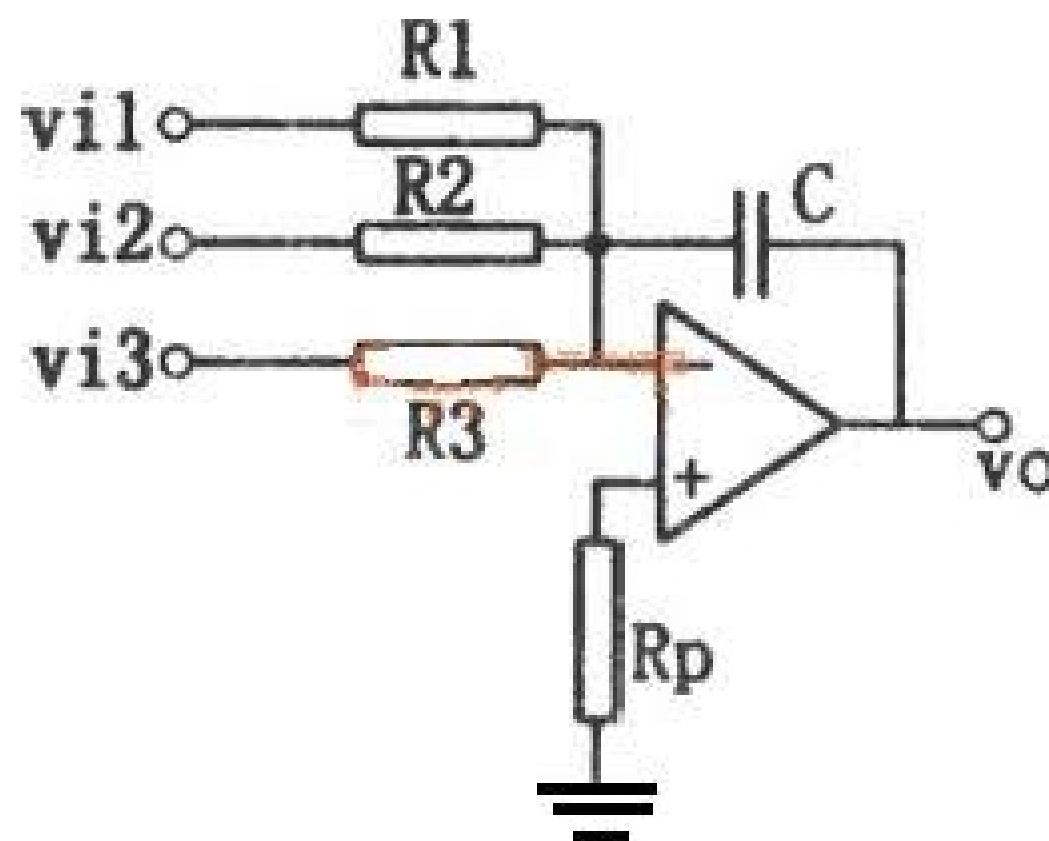
$$V_o(t) = \frac{-1}{RC} \int V_1(t) dt$$

- This equation shows that the output voltage is the integral of the input with all inversion and scale multiplier of $1/RC$. When a step voltage is applied at the input terminal, the

output voltage is a ramp or linearly changing voltage. Integrators are used in ramp or sweep generators, in filters, analog computers etc.

SUMMING INTEGRATOR

- The circuit of summing integrator is shown in figure. The output of such circuit is given by

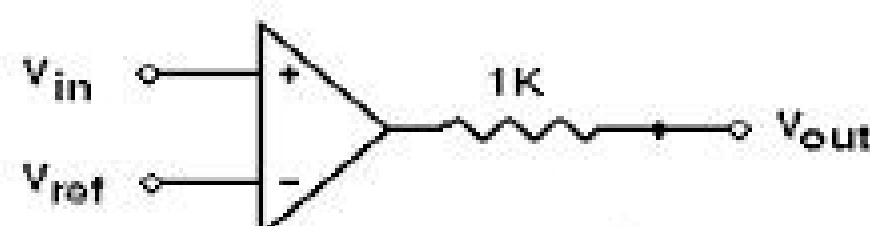


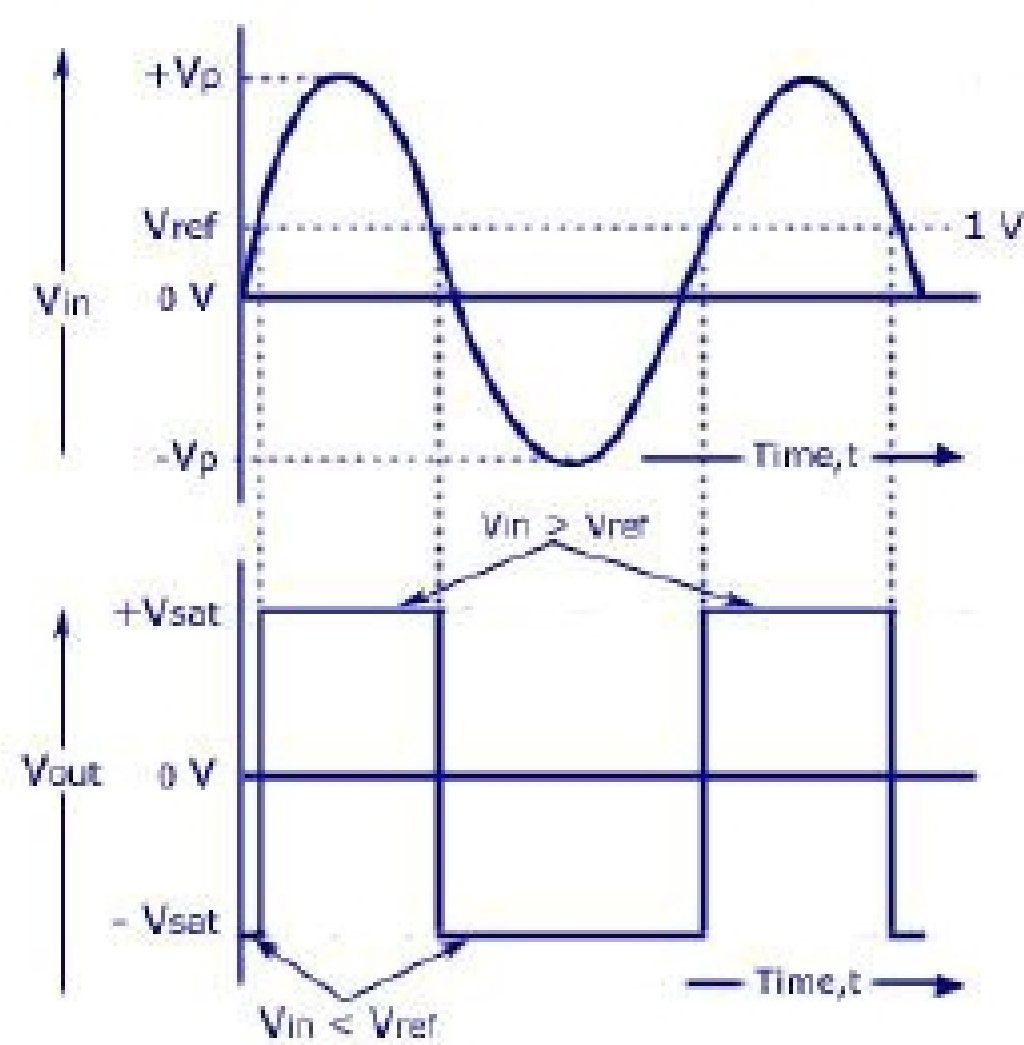
$$v_{O(t)} = -\frac{1}{R_1 C} \int v_1(t) dt - \frac{1}{R_2 C} \int v_2(t) dt - \frac{1}{R_3 C} \int v_3(t) dt$$

It is used in analog computers.

4.8 COMPARATOR

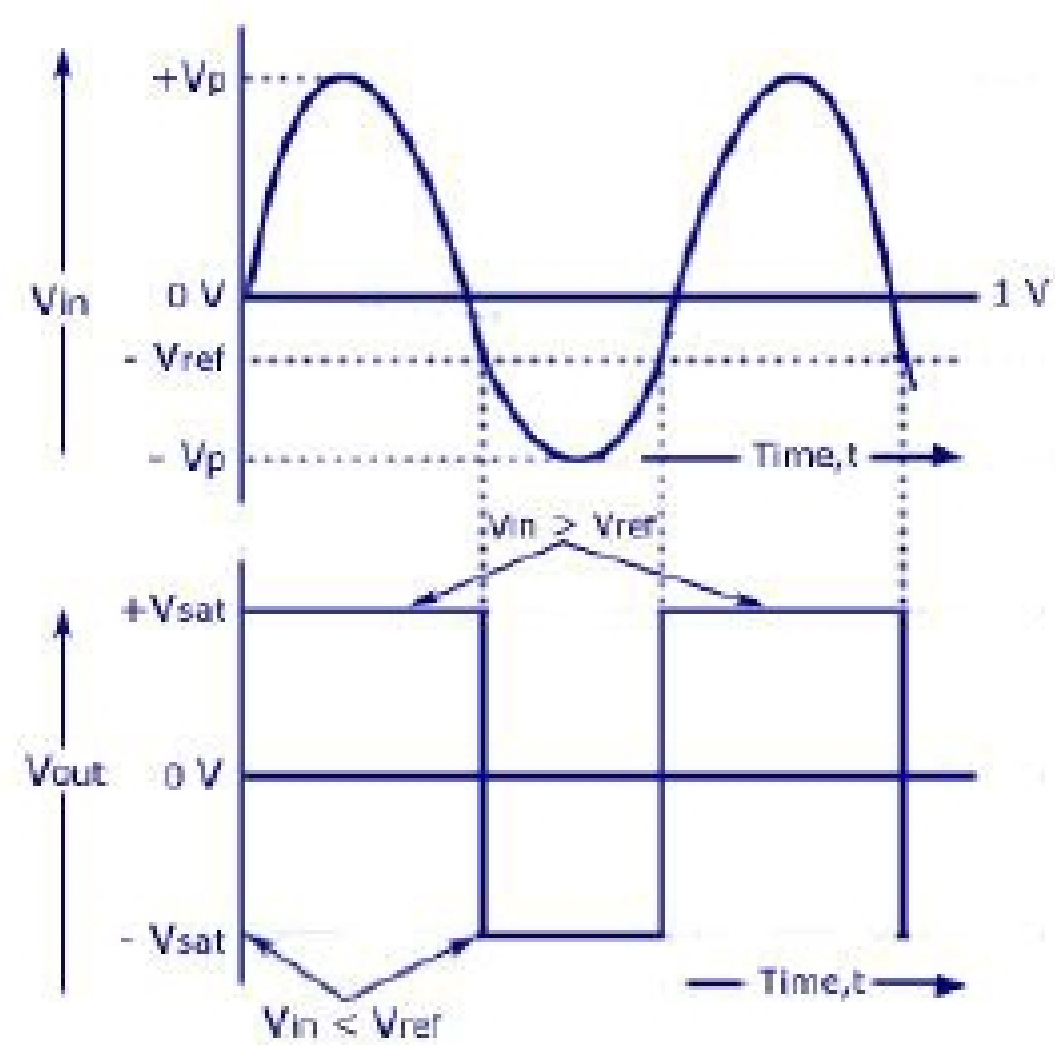
- Comparator is a circuit which compares two voltages and provides an output that indicates the relationship between them. Hence comparator may be used to compare.
 - ✓ Two changing voltages like two sine waves
 - ✓ A changing voltage with a set of DC reference voltage
- The circuit for a comparator along with its input and output waveforms is shown in figure. As evident from circuit diagram, OP-AMP is operated in open-loop configuration. The input voltage is applied to that non-inverting input terminal and a set DC reference voltage V_{ref} is applied to the inverting terminal of the OP-AMP. As the open loop gain of OP-AMP is very high, it goes into saturation even with small input voltage. Thus the two possible output levels of this comparator are $+V_{sat}(+V_{max})$ and $-V_{sat}(-V_{max})$.





Input and Output Waveforms
For Positive V_{ref}

www.CircuitsToday.com



Input and Output Waveforms
For Negative V_{ref}

The transition of V_o from $+V_{max}$ to $-V_{max}$ or vice versa takes place when the input voltage, V_i crosses the V_{ref} level.

ZERO CROSSING DETECTORS:

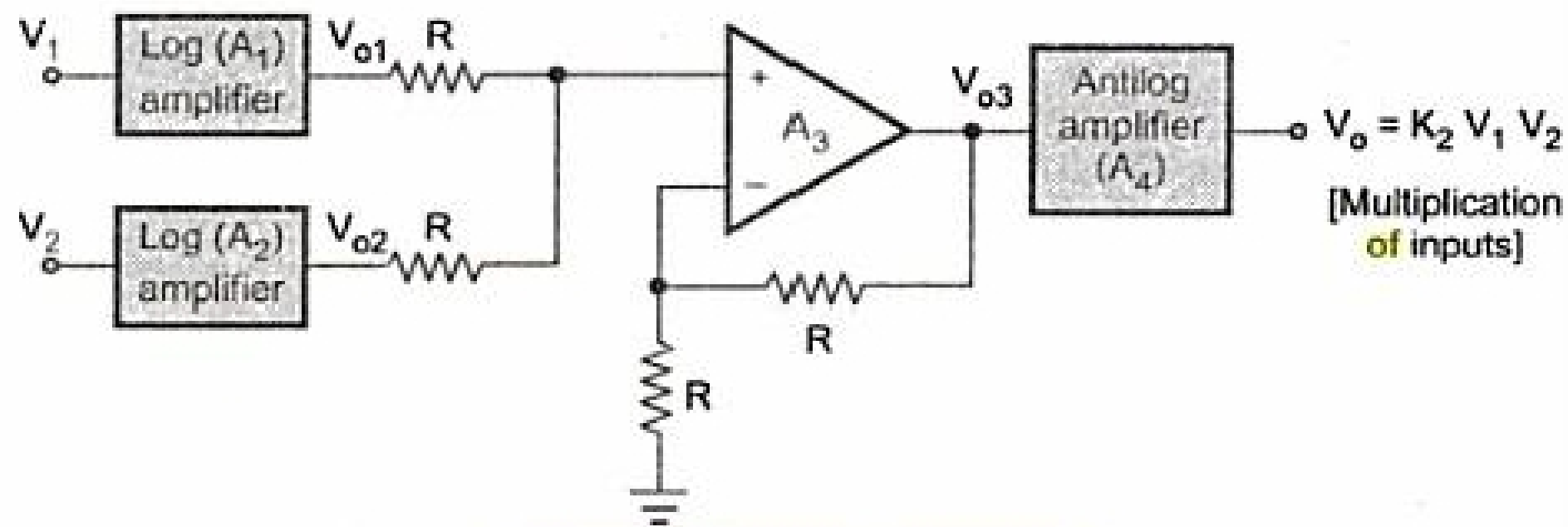
- To use the above circuit as zero crossing detector, V_{ref} is chosen as zero potential (earthened).

4.9 ANALOG MULTIPLIERS:

- The circuit performs the multiplication of the two input voltage is called multiplier circuit. Such multiplier circuits are used in squarer, square extractor, frequency doubler etc.

Analog voltage multiplier:

- Using log and antilog amplifiers, the circuit can be built to obtain the output proportional to the product of the two input voltages. It is called analog voltage multipliers. It is shown in figure.



Analog voltage multiplier circuit

- The log amplifier circuits with A_1, A_2 give the output as V_{01} and V_{02} given by

$$V_{01} = k_1 \ln k_2 V_1$$

$$V_{02} = k_1 \ln k_2 V_2$$

Where $k_1 = \frac{-V_T (R_2 + R)}{R_T}$

$$k_2 = \frac{1}{V_{ref}}$$

Then $V_0 = -k_1 \ln \frac{V_{in}}{R I_{ref}}$

OP-AMP A_3 is non-inverting amplifier with unity gain

$$V_{03} = V_{02} + V_{01}$$

$$= k_1 \ln k_2 V_2 + k_1 \ln k_2 V_1$$

$$V_{03} = k_1 \ln k_2^2 V_1 V_2$$

Now V_{03} is applied as input to antilog amplifier A_4

$$V_0 = k_2^{-1} e^{\frac{k_1 \ln k_2^2 V_1 V_2}{k_1}}$$

For this equation, refer to equations

$$V_0 = I_{ref} R_f \ln^{-1} \frac{-V_{in} R_2}{n V_T + R_1 + R_2}$$

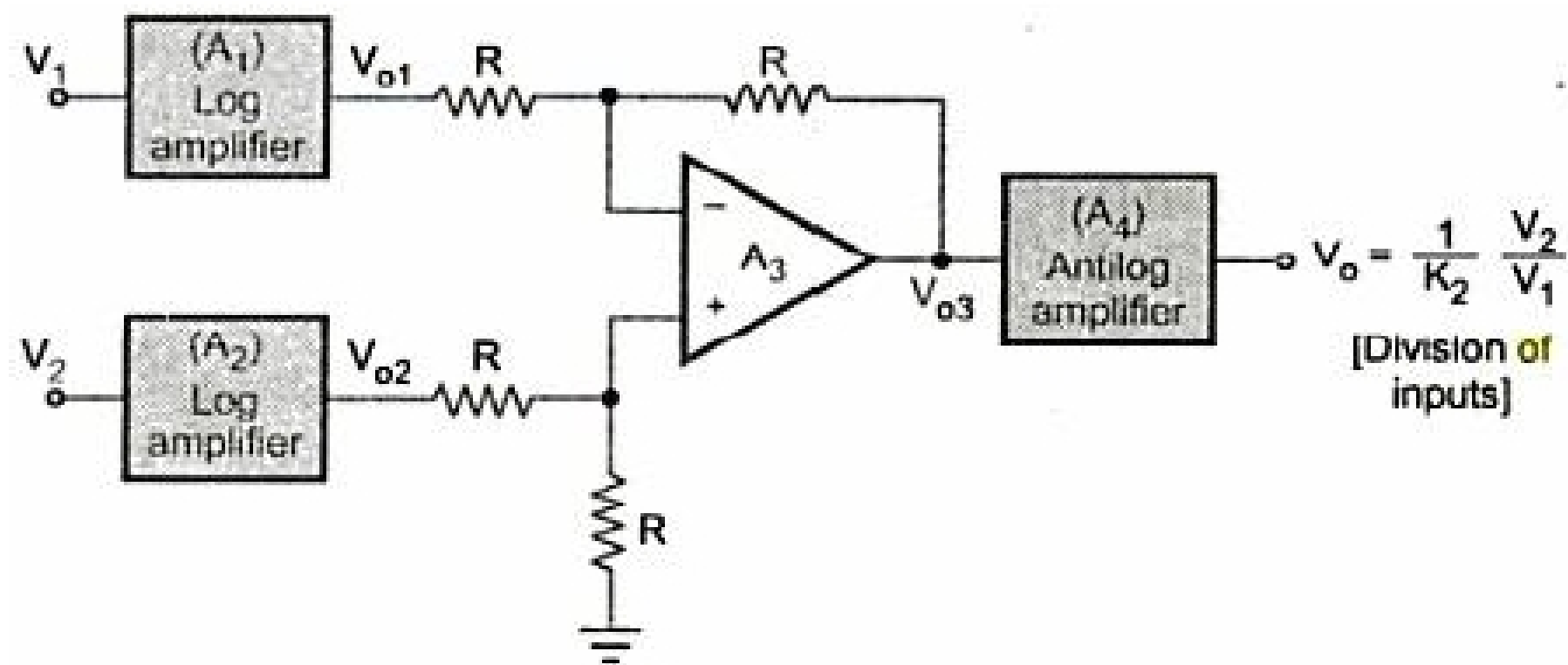
$$V_0 = k_2^{-1} k_2^2 V_1 V_2$$

$$V_0 = k_2 V_1 V_2$$

Thus the output is proportional to the product of two analog inputs V_1 and V_2

4.9.1 ANALOG VOLTAGE DIVIDER CIRCUIT

- Similar to analog voltage multiplier analog voltage divider can be obtained using log and antilog amplifiers. The circuit gives output which is proportional to the division of two input signals. The circuit for the voltage divider is shown in figure.



$$V_{01} = k_1 \ln k_2 V_1$$

$$V_{02} = k_1 \ln k_2 V_2$$

Now the OP-AMP A3 is difference amplifier hence we get

$$V_{03} = V_{02} - V_{01}$$

$$= k_1 \ln k_2 V_2 - k_1 \ln k_2 V_1$$

$$V_{03} = k_1 \ln \frac{V_2}{V_1}$$

Now A4 is again an antilog amplifier which gives

$$V_0 = k \frac{1}{2} n - k_1 \frac{V_1}{V_2}$$

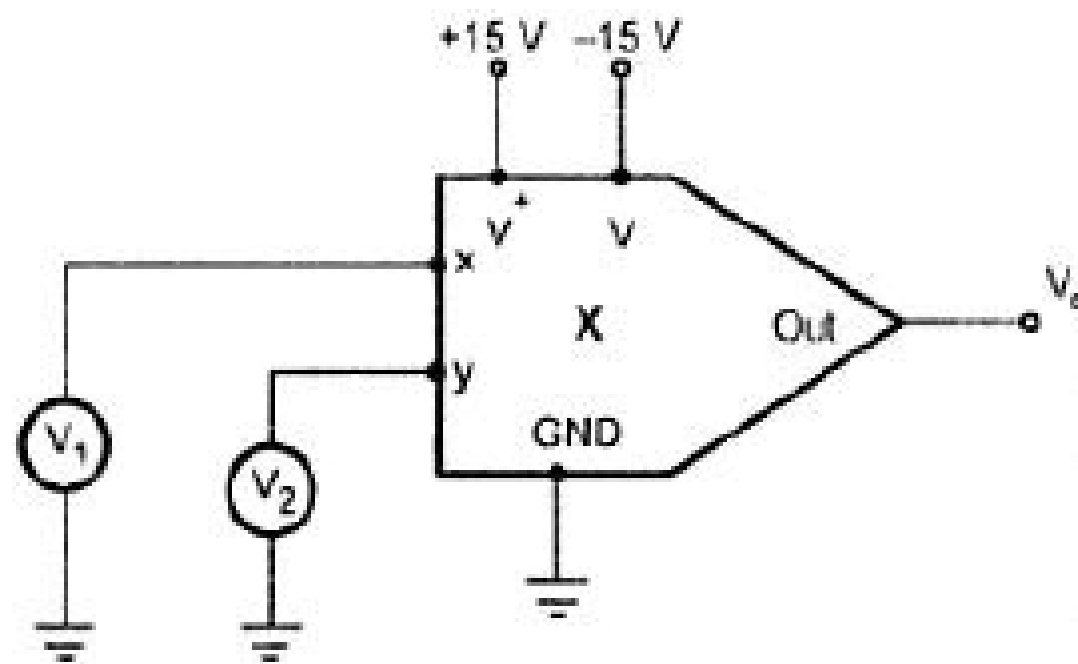
Thus the output is proportional to the division of two analog inputs V1 and V2

4.9.2 MULTIPLIER INTEGRATED CIRCUIT:

- It is commonly used in practical. Monolithic integrator has lowered cost of multiplier. Its considerably, such an it is not only useful as multiplier but can be configured to use in many applications as signal multiplications in process instrumentation, chemical

analyzers, servo mechanic control systems, frequency doublers, phase angle detectors, true RMS convertor, controlling of oscillator frequency and so on. The IC's can be used to improve the data acquisition through rationing two signals. $v_{01} = k_{1} \ln R_{2} v_{1}$.

- The basic multiplier is an active circuit in which the output voltage is proportional to the product of the two input signals. A schematic symbol of such basic amplifier it is shown in figure.

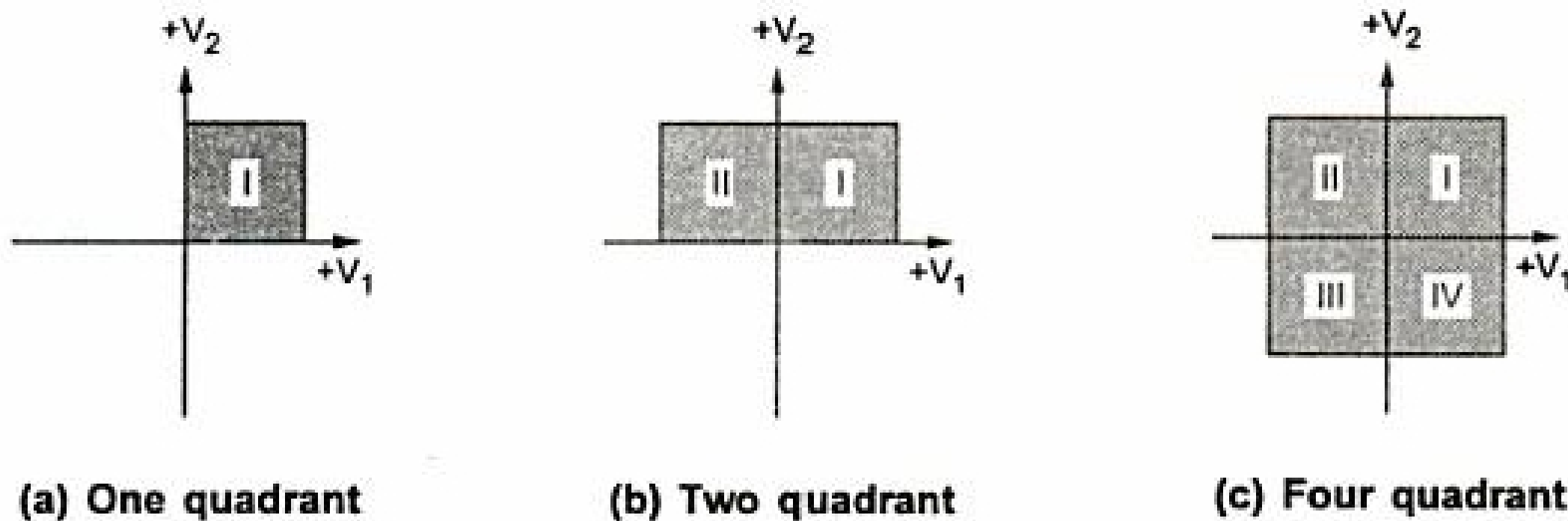


- The terminals V+ and V- are supply terminals for IC where dual supply is to be connected generally $\pm 15V$ as shown. The x and y are the two input terminals where inputs V1 and V2 are connected.
- The output of such basic multiplier is $V_o = KV_1V_2$
- Where K is connected and is equal to $1/V_{ref}$. Usually V_{ref} is set to 10V internally and hence

$$v_0 = \frac{v_1 v_2}{10}$$

- As long as it is ensured that both the input voltage are below the reference voltage ($V_1, V_2 < V_{ref}$), the output of the basic amplifier (multiplier) will not saturate.
- Depending on the use of the basic amplifier (multiplier), it is necessary to retrieve the polarity of one or both the inputs. Depending upon the polarity restriction, the IC operations is called as
 - ✓ **One quadrant multiplier:** In such operation, the polarities of both the inputs must always be positive.
 - ✓ **Two quadrant multiplier:** It functions properly if one input is held positive and the other is allowed to ensure in both positive and negative.

- ✓ **Four quadrant multiplier**: If both the inputs are allowed to ensure in both positive and negative directions, the operation is four quadrant multiplier operations are shown in figure.



4.10 PERFORMANCE PARAMETERS

- Let us define the performance parameters of a multiplier

Accuracy:

- It is maximum deviation of the actual output level from the ideal one given by the $V_o = KV_1V_2$, for any choice of V_1 and V_2 , within the dynamic range of the multipliers. This deviation is called as total error. It is generally specified in terms of percentage of full scale output.

Linearity:

- It is the maximum output deviation from the best fit straight line at the output, where one input is varied from end to end while the other is kept fixed, usually at $+10V$ or $-10V$. It is also expressed in terms of percentage of full scale output.

Bandwidth:

- It is the range up to the frequency where the output is 3db below its low frequency value. This is also called as small signal bandwidth. It depends on the active device used.

1% absolute error bandwidth:

- It represents the frequency where the output magnitude starts to deviate from its low frequency value by one percent.

Feed through voltage:

- It is the peak to peak voltage at the output when one of the two input is grounded. As the output is the multiplication of the two inputs, if one of the input is made zero, the output

of ideal multiplier also must be zero. But in practice, there exists a small voltage at the output. This voltage is different for the two input terminals.

Zero trim:

- It is the ability of the multiplier to set the feed through voltage at the output to zero.

Scale factor:

- It is the proportionality constant (K) relating the output voltage and the product of the two input voltages.

$$K = \frac{V_0}{V_1 V_2}$$

Quadrant

- This indicates the restriction on the polarities of the two input voltages. For one quadrant, both the inputs must be positive, for two quadrants one must be positive while the other can be bipolar. And for the four quadrant operations, both the inputs are bipolar.

4.11 APPLICATIONS

The multiplier is used in many practical applications. Some of their applications are:

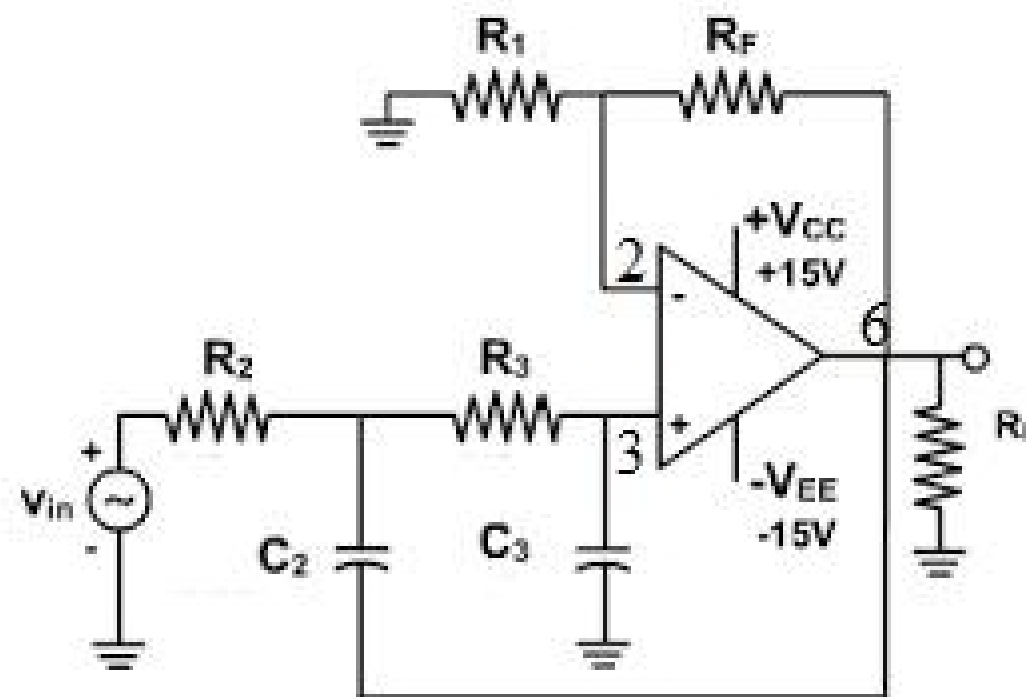
- ✓ In communication it is used in amplitude modulation, phase modulation, frequency modulation, phase detection, suppressed carrier modulation etc.
- ✓ In instrumentation and control it is used to measure velocity, acceleration, instantaneous power, automatic gain control etc.
- ✓ For voltage control attenuators and for voltage controlled amplification.
- ✓ It is used for voltage divider, true RMS calculation, rectifier phase shift detection etc.
- ✓ It is used for frequency convertors, frequency doubling and frequency shifting etc.
- ✓ It is used for squaring and square root calculations.
- ✓ It is used to solve non-linear equations.
- ✓ It is used in oscillators to generate the waveform and also used for square wave generators etc. Such oscillators are Wein Bridge, Hartley, Colpitt and other oscillators.

4.12 SECOND ORDER LOW PASS FILTER:

- A stop-band response having a 40-dB/decade at the cut-off frequency is obtained with the

second-order low-pass filter. A first order low-pass filter can be converted into a second-order low-pass filter by using an additional RC network as shown in figure.

- The gain of the second order filter is set by R_1 and R_F , while the high cut-ff frequency f_H is determined by R_2 , C_2 , R_3 and C_3 as follows:



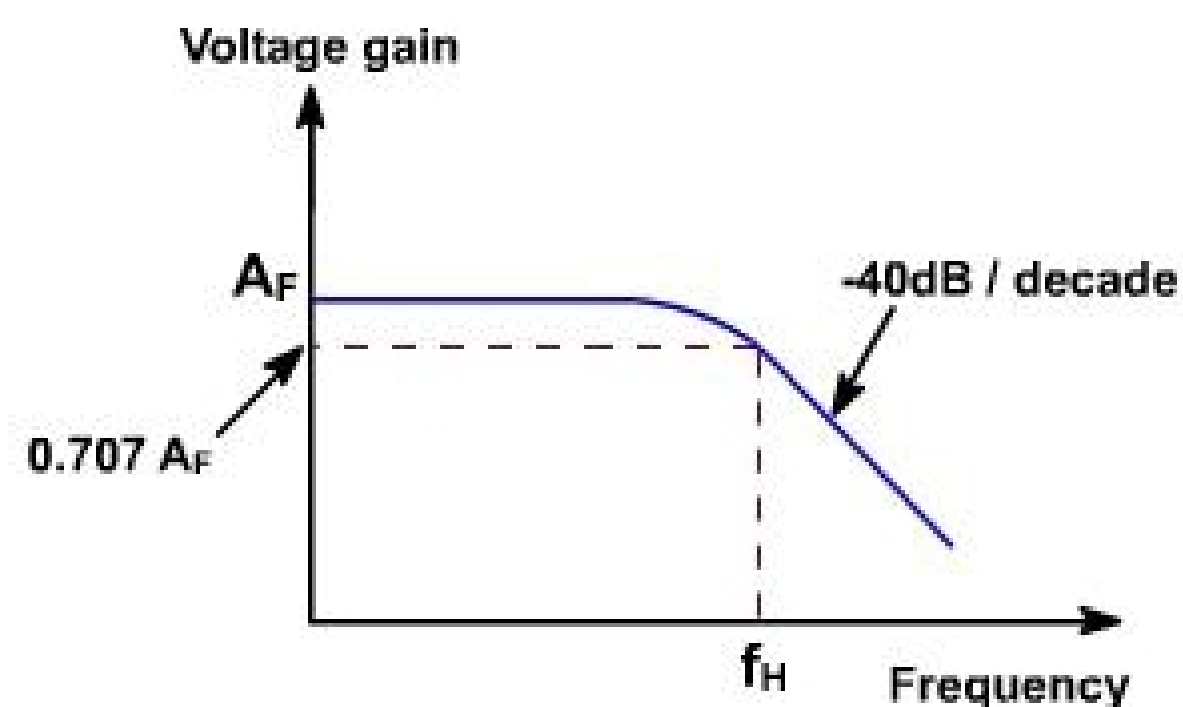
$$f_H = \frac{1}{2\pi\sqrt{R_2 R_3 C_2 C_3}}$$

- Furthermore, for a second-order low pass Butterworth response, the voltage gain magnitude is given by

$$\left| \frac{v_o}{v_{in}} \right| = \frac{A_F}{\sqrt{1 + (f/f_H)^4}}$$

$$A_F = 1 + \frac{R_F}{R_1} = \text{passband gain of the filter}$$

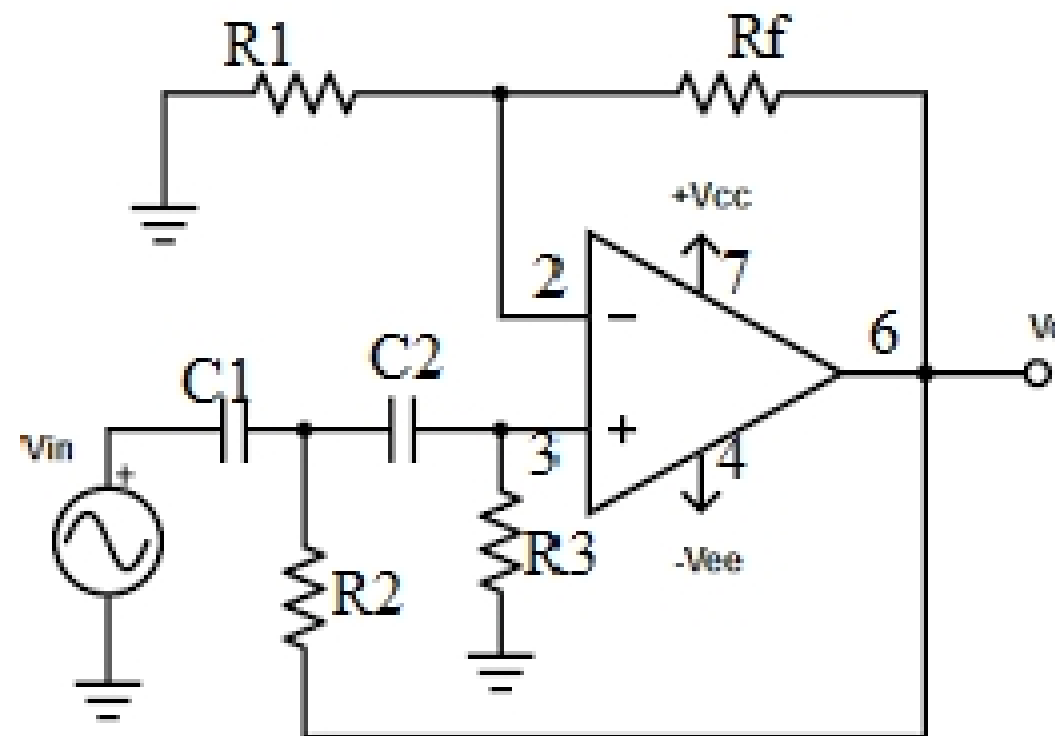
f = frequency of the input signal



- Except for having the different cut off frequency, the frequency response of the second order low pass filter is identical to that of the first order type as shown in figure

4.13 SECOND ORDER HIGH PASS FILTER:

- ▶ The basic operation of an active high pass filter is exactly the same as that for its equivalent RC passive high pass filter circuit except this time the circuit has an operational amplifier included within its filter design providing amplification and gain control.



- ▶ As its name implies, attenuates low frequencies and passes high frequencies signals. It consists simply of a passive filter section followed by a non-inverting operational amplifier. The frequency response of the circuit is the same as that of the passive filter except that the amplitude of the signal is increased by the gain of the amplifier and for a non-inverting amplifier the above value of the pass band voltage is given as $1 + R_2/R_1$.
- ▶ As with the passive filter, a first order high pass active filter can convert into a second order high pass filter simply by using an additional RC network in the input path. The frequency response of the second order high pass filter is identical to that of the first order type except that the stop band roll-off will be twice the first order filters at 40 dB / Decade.
- ▶ Active filters are widely used in modern communication systems, because they have the following advantages.
- ▶ Because the transfer function with inductive characteristic can be achieved by particular circuit design, resistors can be used instead of inductors.
- ▶ The high input impedance and low output impedance of the operational amplifier means that the filter circuit is excellent in isolation characteristic and suitable for cascade.
- ▶ Because active components provide amplification, therefore active filters have gain.

UNIT-5

Title	Page. No
5.1 VARACTOR DIODE	121
5.1.1 SPECIFICATIONS	122
5.1.2 APPLICATIONS	123
5.2 TUNNEL DIODE	123
5.2.1 PARAMETERS:	125
5.2.2 APPLICATIONS	126
5.3 PIN DIODE	126
5.3.1 OPERATION	127
5.3.2 ADVANTAGES	127
5.3.3 APPLICATIONS	127
5.4 LIGHT EMITTING DIODE	128
5.4.1 OPERATION	128
5.4.2 MATERIALS AND COLOURS	129
5.4.3 CONSTRUCTION	129
5.4.4 BIASING	129
5.4.5 ADVANTAGES	130
5.4.6 DISVANTAGES	131
5.4.7 APPLICATIONS	131
5.5 LIQUID CRYSTAL DISPLAY	131
5.5.1 ADVANTAGES	132
5.5.2 DISADVANTAGES	133
5.5.3 APPLICATIONS	133
5.6 SEVEN SEGMENT DISPLAY:	133
5.6.1 TYPES OF SEVEN SEGMENTS DISPLAY:	133
5.6.1.1 COMMON ANODE TYPE:	134
5.6.1.2 COMMON CATHODE TYPE	134
5.7 OPTO- ISOLATORS	134
5.8 UJT (UNI-JUNCTION TRANSISTOR):	135

5.8.1 CONSTRUCTION:	135
5.8.2 EQUIVALENT CIRCUIT OF UJT	136
5.8.3 INTRINSIC STAND OFF RATIO (η)	137
5.8.4 OPERATION	137
5.8.5 UJT CHARACTERISTICS	138
5.8.6 UJT RELAXATION OSCILLATOR	139
5.8.6.1 OPERATION	139
5.9 SILICON CONTROLLED RECTIFIER (SCR)	141
5.9.1 CONSTRUCTION	141
5.9.2 OPERATION	141
5.9.3 CHARACTERISTICS OF SCR	143
5.9.3.1 FORWARD CHARACTERISTICS	143
5.9.3.2 REVERSE CHARACTERISTICS	143
5.9.4 TWO TRANSISTOR MODEL	144
5.9.5 TURNING ON (TRIGGERING) SCR	145
5.9.6 TURNING OFF SCR	145
5.9.7 SCR RATINGS	146
5.9.8 REVERSE BREAKDOWN VOLTAGE	146
5.9.9 APPLICATION OF SCR	147
5.10 TRIAC	148
5.10.1 TRIAC OPERATION	149
5.10.2 VI CHARACTERISTICS OF A TRIAC	150
5.10.3 TRIAC RATINGS	151
5.10.4 MAXIMUM PEAK GATE POWER ($P_{GAV(max)}$):	151
5.10.5 APPLICATIONS OF TRIAC	152
5.11 DIAC	153
5.11.1 OPERATION:	153
5.11.2 VI CHARACTERISTICS OF DIAC	154
5.11.3 APPLICATIONS	155

Department of Computer Science and Engineering

Subject Name: ELECTRONIC DEVICES AND CIRCUITS

Prepared by:

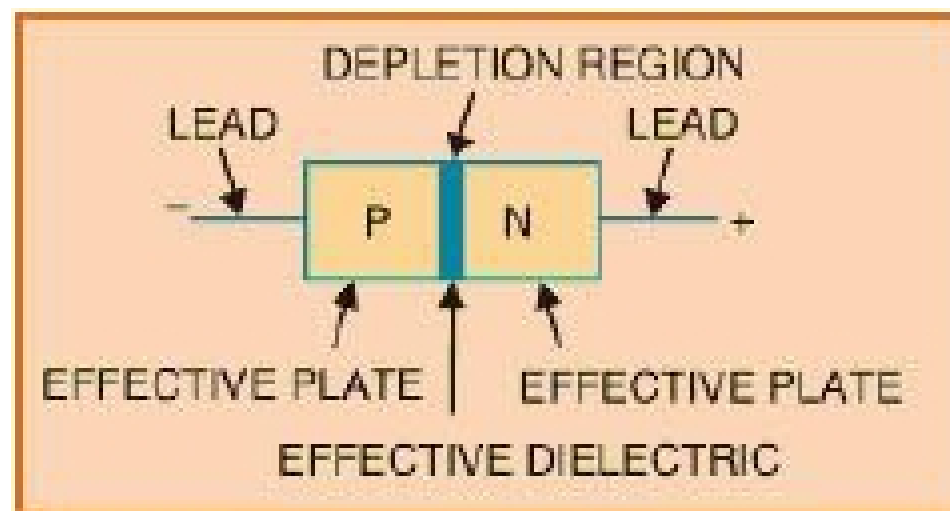
Mr. T.KAVIARASAN. M.Tech (Assistant Professor)

UNIT V

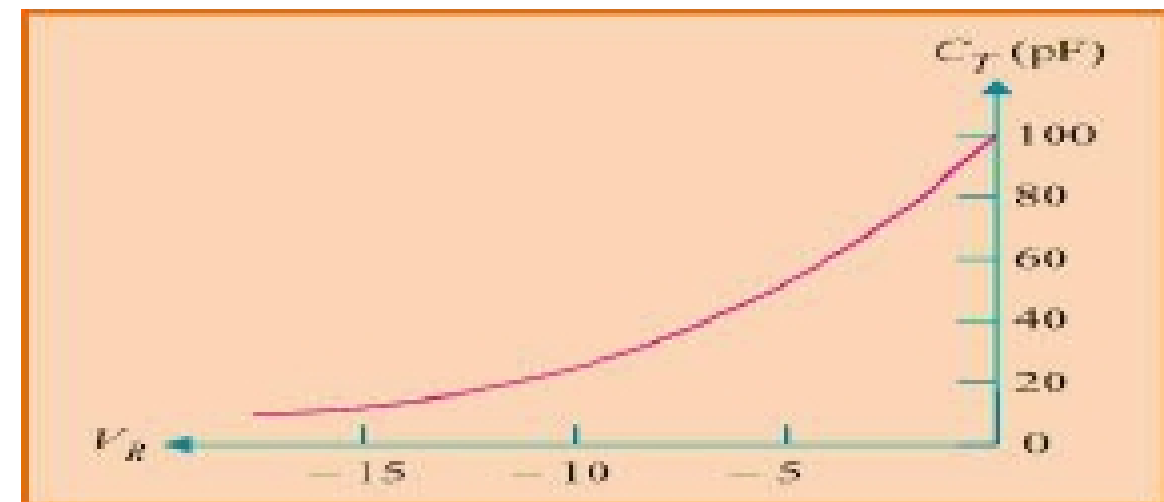
Special Devices: Varactor diode, Tunnel diode, PIN diode, LED, LCD, Seven segment displays, Opto-isolator. UJT - Characteristics and equivalent circuit – intrinsic standoff ratio – UJT relaxation oscillator, SCR - Two transistor model, DIAC and TRIAC - Operation, Characteristics and their applications.

5.1 VARACTOR DIODE:

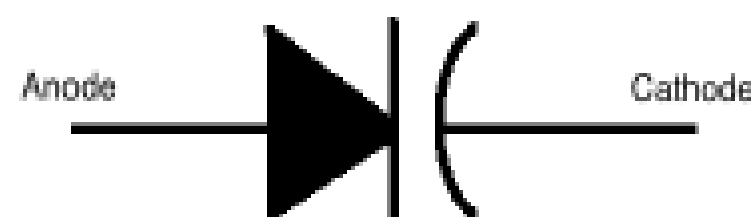
- ▶ It is basically a reverse biased PN junction, which utilizes the inherent capacitance of the depletion layer. It is also known as varicap, voltcap or tuning diode. It is used as voltage variable capacitor. In PN junction, the depletion layer created by the reverse bias acts as a capacitor dielectric, whereas the P and N regions acts as the capacitor plates as shown in figure.
- ▶ When the reverse bias voltage increases the dielectric thickness which in turn, reduces the capacitance. When the reverse bias voltage decreases, the depletion layer narrows down. This decreases the dielectric thickness, which in turn increases the capacitance. Figure shows the variation of capacitance with reverse voltage. (i.e) capacitance is maximum, when the reverse voltage is equal to zero. It reduces in a non linear manner, as the value of reverse voltage is increased.



Reverse biased PN junction



C vs VR



Symbol

- ▶ In this device, the capacitance parameters are controlled by the method of doping in the depletion layer, or the size and geometry of the diode construction. Figure shows the doping profile for an abrupt junction diode (ie) normal PN junction. In this type, the doping is uniform on both sides of the junction. The range of capacitance variation (turning angle) of an abrupt junction diode is 4:1. It means that if its maximum transition capacitance is 100pf and the minimum is 25pf, and then is turning angle is 4:1.
- ▶ The higher turning angle is obtained for varactor, which have hyper abrupt junction. The doping profile of such a junction is shown in figure. In this type, the doping level increases as approach the junction. The heavy doping is at junction results in a narrow depletion layer and hence a larger capacitance. Moreover, any change in reverse voltage produces a larger variation in capacitance than that of abrupt junction. A hyper abrupt varactor diode has a broadcast receiver (500-1650) khz range.
- ▶ The figure shows the symbol and equivalent circuit for varactor diode. Rs- Reverse series resistance, Ct- variable capacitor (2-100)pf

5.1.1 SPECIFICATIONS :

The diode capacitance temperature co-efficient (Tcc)rating:

It gives change in the components capacitance for each 1 c rise in temperature above 25°C. For MV209 series, Tcc=300ppm/°c (ie) capacitance will increase by 300 parts per million for each 1°C.

$$\Delta C = (0.0003) * 29Pf = 0.0087Pf$$

Is the increase in capacitance /c rise in temperature.

- ▶ The MV209 series would have to experience a temperature increase of nearly 115°C as increase in capacitance of 1pf. This leads us to conclude that the capacitance of a varactor diode is relatively independent of operating temperature.

Diode capacitance rating:

The rating for MV209 series is (26-32)pf when V_r is 3VDC

Capacitance ratio(C_r):

It gives us the varieties of the junction capacitance over the given range of voltages for MV209 series, $C_r = (5-6.5)\text{pf}$ for $V_r = (3-25)\text{Vdc}$

Figure of merit (Q):

It is the quality factor of a junction capacitance. We know that 'Q' of a capacitor is the ratio of energy stored in the capacitor to the energy lost through leakage current or it is the ratio of power returned to the circuit by the capacitor to the power dissipated by the capacitor. For MV209 series $Q = 200$ (minimum).

5.1.2 APPLICATIONS :

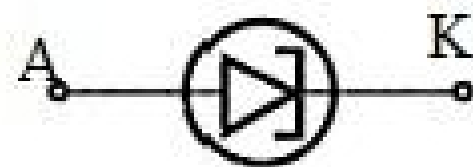
- ▶ Tuning circuits
- ▶ Electronic tuner in radio, television and other commercial receivers
- ▶ Automatic frequency controlled device
- ▶ Adjustable band pass filter
- ▶ Parametric amplifier
- ▶ When it is used in a tuning circuit it allows a resonant frequency to be adjusted by a variable voltage level

5.2 TUNNEL DIODE:

- ▶ If the concentration of impurity atoms is greatly increase in a normal PN junction (by 1000 times or more) characteristics are completely changed it is known as tunnel diode. It is invented in 1958 by Dr. Leo Esaki. So it is also known Esaki diode.
- ▶ In PN junction diodes, an impurity concentration is of one part in 10^9 atoms. There is a width of depletion layer is of order of one micron (10^{-6}m). This constitutes a potential barrier at the junction, which control the forces a charge carrier across the junction. The charge carrier cannot cross-over the potential barrier, unless they acquire sufficient energy to overcome it. However, when impurity concentration is increased one part in 10^{10}

atoms, the width of depletion layer reduces to about 10 nanometer. Under each condition, the charge carrier will penetrate through the junction at the speed of light, even though they do not have enough energy to overcome the potential barrier. As a result of this, a large forward current is produced, even if the applied forward voltage is much less than 0.3V.

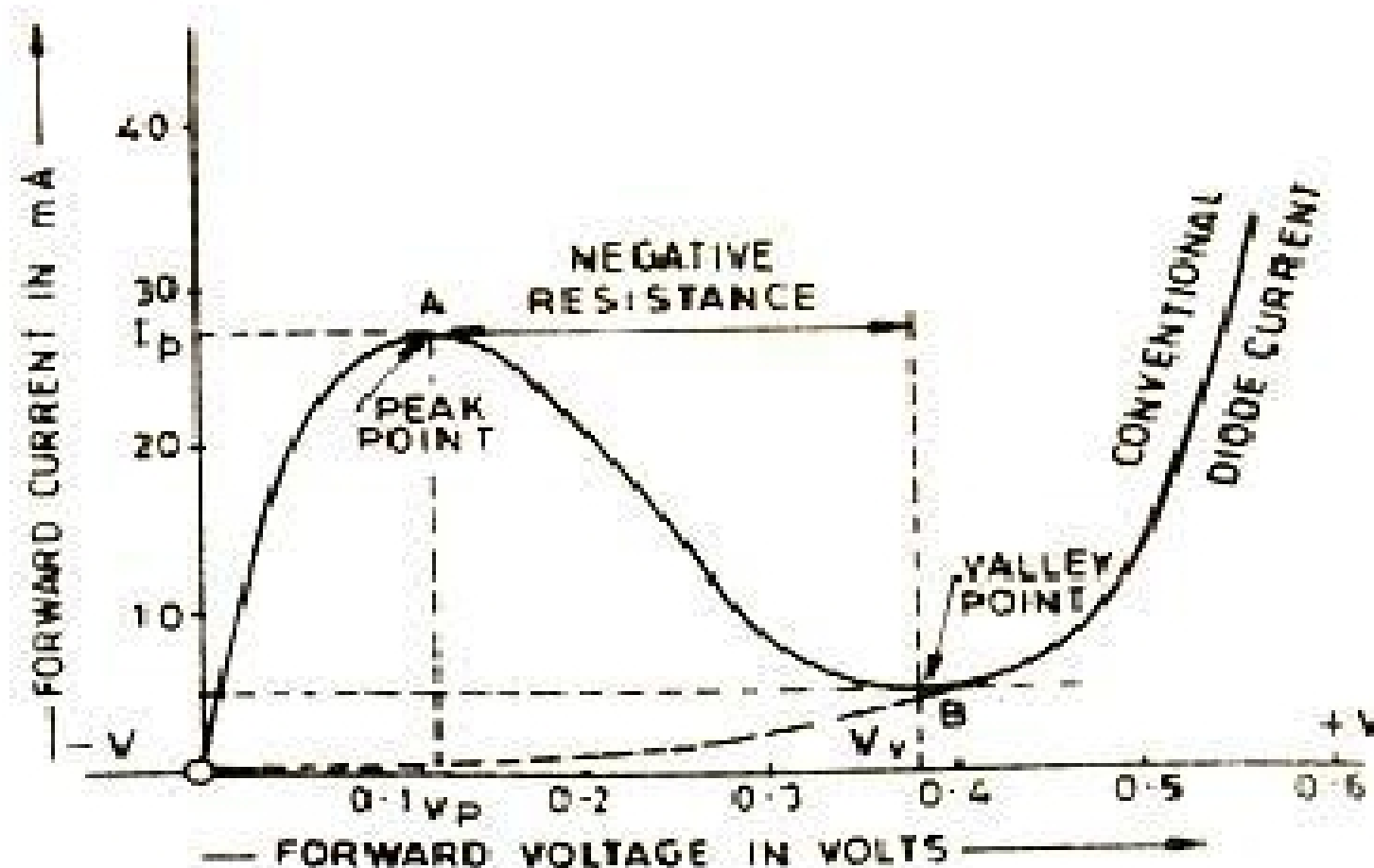
- The phenomenon of penetrating the charge carriers, directly through the potential barrier, instead of climbing over it is called tunneling. So highly doped PN junction diodes are called tunnel diodes. It is usually made of germanium or gallium arsenide. Figure shows two commonly used symbols for a tunnel diode. It should be handled with great care because these are low power devices and can be easily damaged by heat and static electricity.



Tunnel diode symbol

Characteristics :

- Characteristic curve we see that as the applied forward voltage is increased from low, the current increases very rapidly. Till it reaches its maximum value known as peak current (I_p) as indicated by the point A. The corresponding value of the forward voltage is indicated by peak voltage V_p . The value of this voltage is typically 65mV for germanium and 160 mV for GaAs.
- It will be interesting to know that if the forward voltage is further increased ($>V_p$), the current decreases, till it reaches its minimum value known as valley current (I_v) as indicated by the point B. As the voltage is further increased the current increases in a usual manner as in a normal PN junction diode. It has been observed that the current again reaches its peak value. The corresponding value of the voltages, the current increases beyond these values.



- ▶ If the tunnel diode is reverse biased, it acts like an excellent conductor (ie) the reverse current increase with the increase in reverse voltage. It is indicated by OD in the figures.
- ✓ Between A and B, useful in high frequency oscillators.
- ✓ Between B and C used as forward characteristic of PN junction

5.2.1 PARAMETERS :

Negative resistance :

- ▶ It is the resistance of a tunnel diode which it offers when operated in a negative resistance region
- ▶ It is given by

$$R_n = -\Delta V_f / \Delta I_f.$$

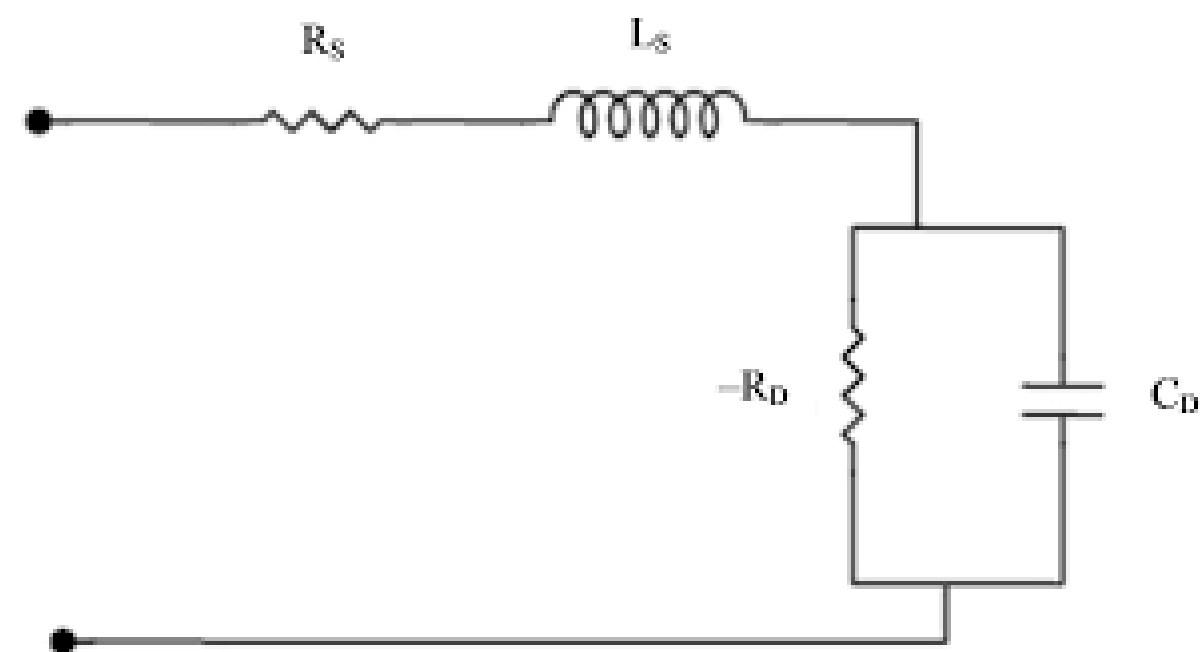
Where

- ✓ ΔV_f -change in forward voltage
- ✓ ΔI_f -change in forward current.

It also be expressed in terms of slop of the characteristic in the negative resistance region. Its value depends on the semi conductor materials used. It varies from (10 – 200).

Ratio of peak to valley current:

It is important for high speed switching circuits which are used in computers. The ratio I_p/I_v of Ge is 6 and GaAs is 10. There is silicon type tunnel diode as a low I_p/I_v ratio which is about 3. So it is not used in actual practice.

Equivalent circuit:

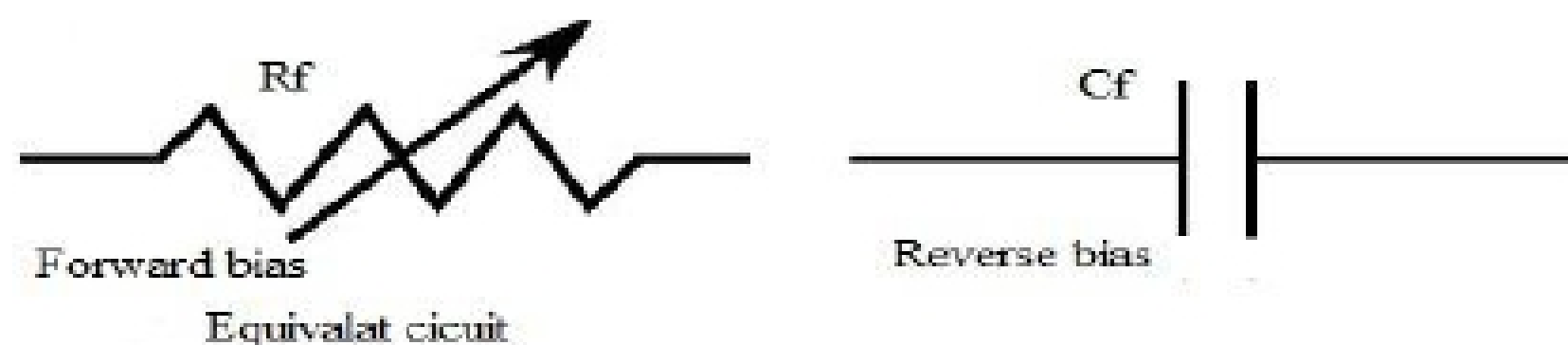
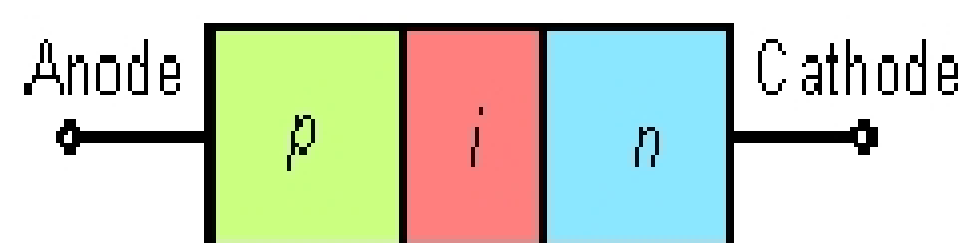
- ✓ R_s - series resistance, L_s - series inductance
- ✓ C -junction capacitance
- ✓ R_n -negative resistance

5.2.2 APPLICATIONS:

- ✓ At ultra high speed (10) switching device.
- ✓ Logic memory storage device.
- ✓ Frequency of 10 GHz as a microwave oscillator.
- ✓ Used in relaxation oscillator due to negative resistance of the device.

5.3 PIN DIODE:

- It is made up of three semiconductor materials two heavily doped P and one N type material separated by an intrinsic semiconductor (I).as shown in figure. This region offers high resistance to the current flowing through it.



5.3.1 OPERATION:

- ▶ When a PIN diode is unbiased, there is a diffusion of electron and holes across the junction due to the different concentration of atoms in the P and I and N regions. The diffusion of electrons and holes produce a depletion layer across the PI and IN junction as shown in figure. It penetrates the little distance in I region. It has a high value of resistance.
- ▶ When it is forward biased, the width of depletion layer decreases. As a result of this, more carrier are injected into the I- region. This reduces the resistance of the I-region. If the depletion layers are not thick, then the I- region becomes flooded with the carriers at a suitable bias. Thus, when a PN diode is forward biased, it acts as a suitable bias. Thus, when a PN diode is forward biased, it acts like a variable resistance as shown in figure. It decreases with increasing current.
- ▶ When the PIN diode is reverse biased, the depletion layer becomes thicker. As the reverse bias is increased, the thickness of the depletion layers increases till the I- region becomes free of mobile carriers. The reverse bias, at which happens, is called sweep out voltage. At this stage it acts as an almost constant capacitance as shown in figure.

5.3.2 ADVANTAGES

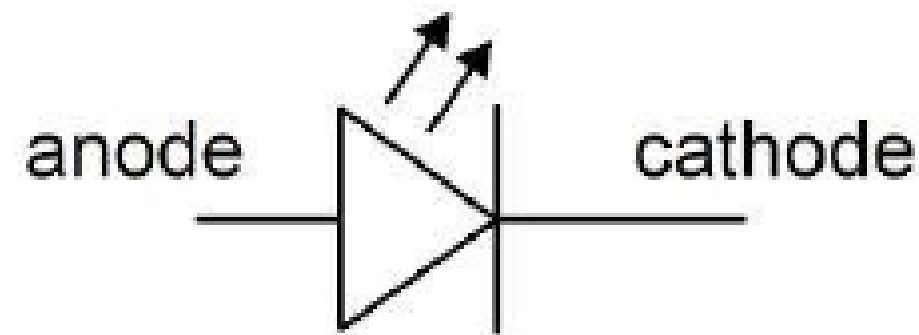
- ▶ The capacitance between the P and N region decreases because of the increased separation between the P and N regions. This advantage allows the PIN diode to have fast response time. Hence, these diodes are useful at very high frequencies above 300MHz.
- ▶ There is greater electron hole pair generation because of the increased electric field between the P and N regions. This advantage allows the PIN diode to process even weak signals.

5.3.3 APPLICATIONS :

- ▶ As a DC controlled microwave switch operated by rapid changes in bias modulating device.
- ▶ This application takes advantages of the variable forward resistance characteristics.
- ▶ Modulators are used to combine two signals of different frequencies into a signal.
- ▶ In attenuator applications because its resistance can be controlled by the current.

5.4 LIGHT EMITTING DIODE:

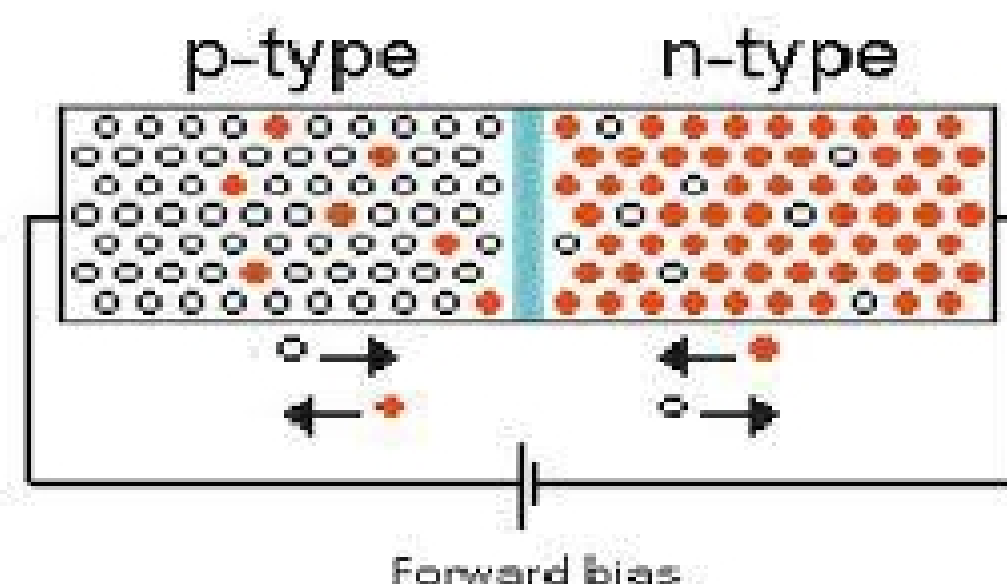
It is a optical diode, which emits light when forward biased. It is shown in figure , similar to PN junction diode apart from the two arrows indicating that the device emits light energy.



LED symbol

5.4.1 OPERATION :

- ▶ Whenever a PN junction is forward biased, the electrons cross the PN junction from the N type semiconductor materials and recombine with the holes in the p type semiconductor material. The free electrons are in the conduction band while the holes are present in the valance band. Thus the free electrons are at higher energy level associated with it changes from higher to lower value. The energy corresponding to the difference between higher and lower level is released is in the form of heat. But LED is made up some special material which release this energy is the form of photons which emit the light energy. Hence such diodes are called light emitting diode. This process is called electro luminance.
- ▶ Figure shows the basic principle of this process the energy released in the form of light depends on the energy corresponding to the forbidden gap. This determines the wavelength of the emitted the forbidden gap. The wavelength determines the colour of the light and also determines whether the light is visible or invisible (infra red).



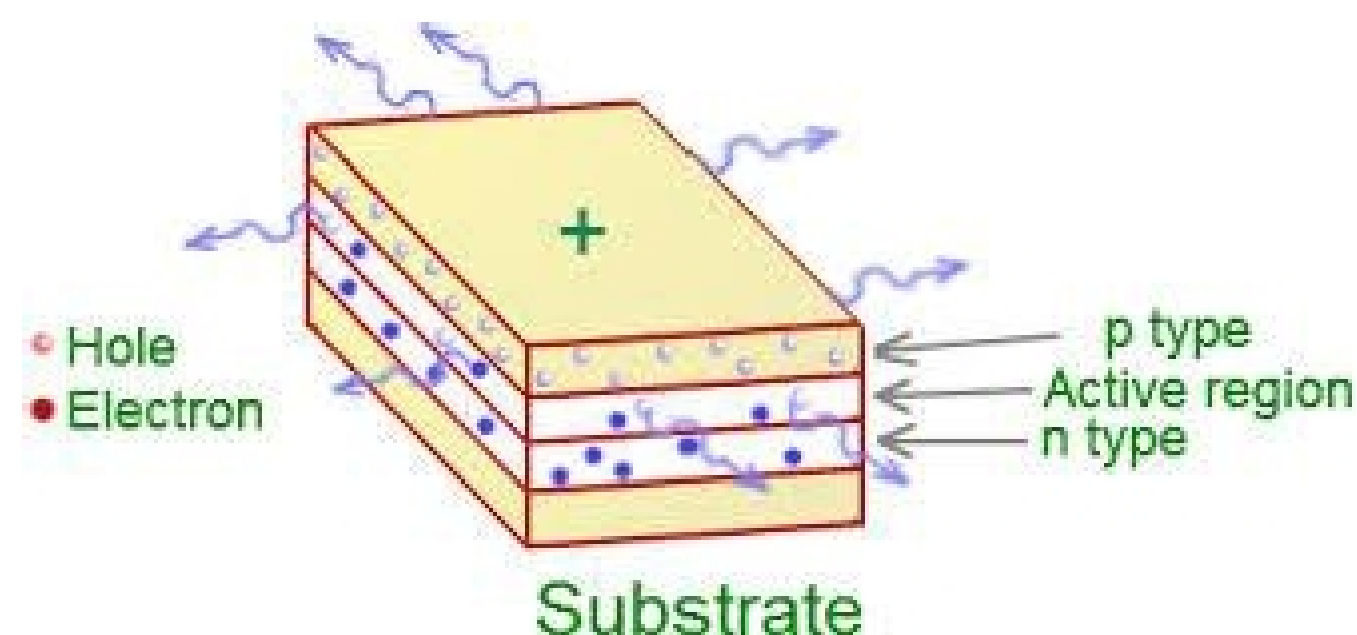
- ▶ The various impurities are added during the doping process to control the wavelength and color of the emitted light for normal silicon diode , the forbidden energy gap is 1.1 eV and wavelength of the spectrum hence in normal diodes the light is not visible . The

infrared light is not visible.

5.4.2 MATERIALS AND COLOURS:

Mixture used	Symbol	colour
gallium arsenide	GaAs	infrared, invisible
gallium phosphide	GaP	red or green
gallium arsenide	GaAsP	red or yellow

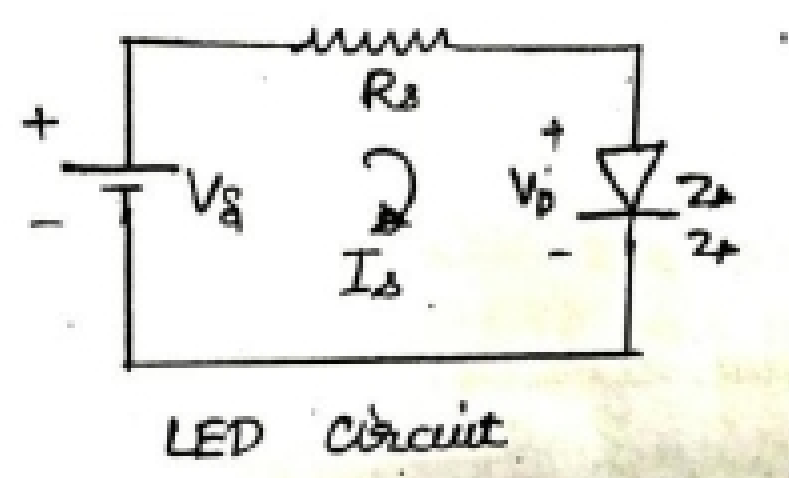
5.4.3 CONSTRUCTION :

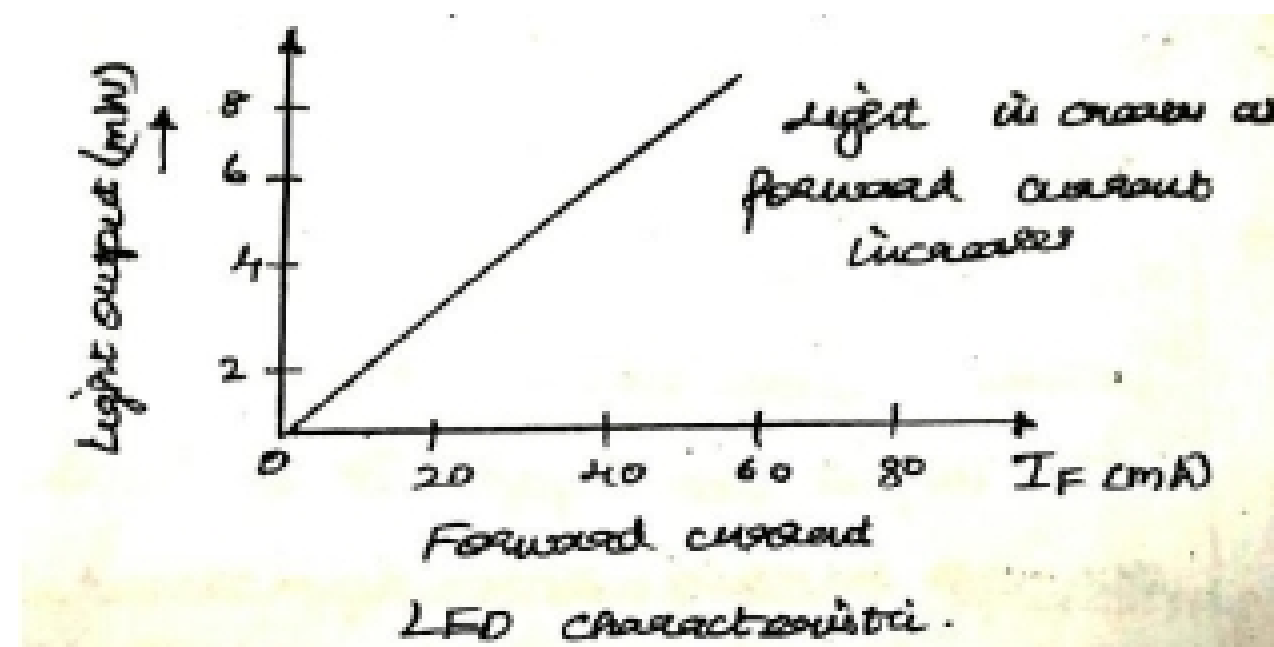


- It is shown in figure with three semiconductor layers. In between P and N type, there is an active region. It emits light, when an electron and hole recombine. When the diode is forward biased, holes from p-type and electrons and hole recombine. When the diode is forward biased, holes from p-type and electrons from N-type, both get driven into the active region. And when recombine the light is emitted.
- In this particular structure the LED emit light all the way around the layered structure. Thus the bias layered structure is placed in a tiny reflective cup so that the light from the active layer will be reflected towards the desired exit direction.

5.4.4 BIASING:

- Consider a source connected to LED and a resistor is shown in figure. The outward arrow associated with a diode indicates that it is a led. The resistor R_s is the current limiting resistor. Due to this resistor, the current through the circuit is limited and prevented from exceeding the maximum current rating of the diode.





- ▶ When a forward biased, as forward current increases, the light output also increases. The graph of output light in mW against forward current I_F in (mA) is shown in the figure.
- ▶ When forward biased, the voltage drop conducting LED is about (2-3)V which is considerably greater than that across a normal silicon or germanium diode. The current range of commercially available LED's is (10-80) mA .unless and otherwise specified , while analyzing the led circuits the drop across LED ions considered as $V_d=2v$.the reverse breakdown voltage of LED is much less than normal diode , which is about (3-10)v.

5.4.5 ADVANTAGES :

- ▶ Small in size and hence can be regarded as point source of light. Because of their small size, several thousand LED can be packed in one square area.
- ▶ The brightness of LED depends on the current flowing through LED. Hence the brightness of light of light can be smoothly controlled by varying the current, this makes possible to operate LED displays under different ambient lighting conditions.
- ▶ LEDs are fast operating devices. They can be turned on and off in time less than 1 microseconds
- ▶ Light in weight, available in variable colours have long life.
- ▶ Cheap and readily available, easy to interface with various electronic circuits.
- ▶ It radiates IR light which is invisible but still useful in some application like burglar alarm systems.
- ▶ It useful for applications which are subjected to frequent on-off cycling. The fluroscents layers turn out more quickly when cycled.
- ▶ It can be easily dimmed using pulse width modulation or by controlling the forward current.
- ▶ Shock resistance and difficult to damage due to shocks
- ▶ It don't contain toxic material like mercury.

5.4.6 DISVANTAGES :

- ▶ It draws considerable current requiring frequent replacement of battery in low power battery operated devices.
- ▶ Luminous efficiency of LEDs is low (1.5 Lumiere/Watt).
- ▶ The characteristic are affected by temperature
- ▶ Need large power for the operation compared to PN junction diode.

5.4.7 APPLICATIONS:

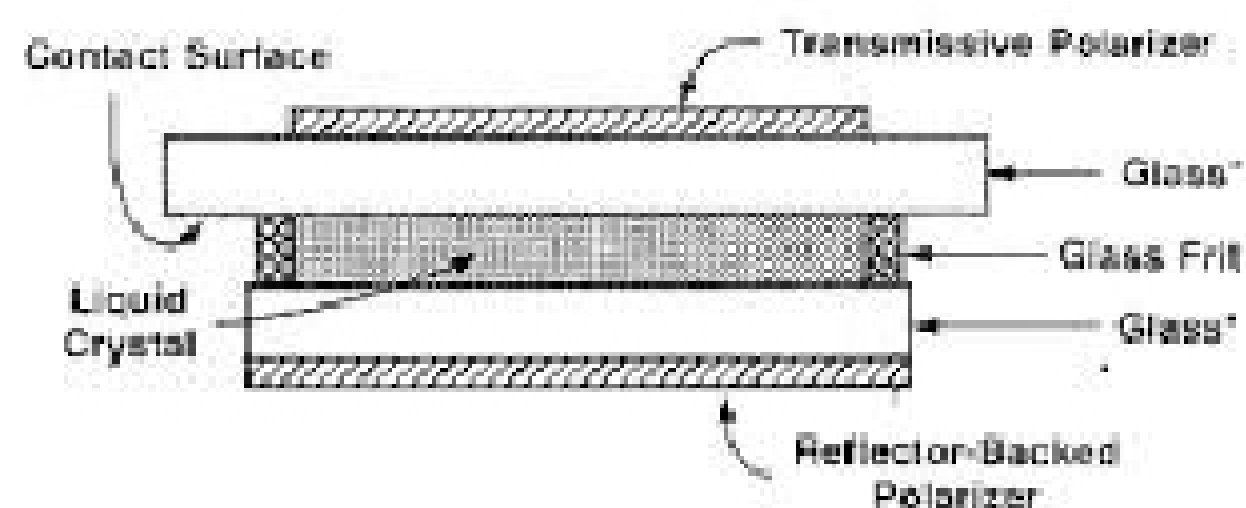
- ▶ In 7-segment, 16-segment and dot matrix displays. It's used to indicate alphanumeric characters and symbol in various systems such as digital clocks, microwave ovens, stereo tuners. Calculators etc...
- ▶ For indicating power ON /OFF conditions, power level indicators or stereo amps.
- ▶ In optical switching applications
- ▶ In the field of optical communication, where LEDs are used to transfer energy from one circuit to another. They are also used to send light energy thorough fiber optical cable, which transmits energy by means of total interval reflection. The fiber optical cable is of light weight, feerible, other transparent and as small as 0.043 mm in diameter.
- ▶ For image sensing circuits in picturephon.
- ▶ In burglar alarm systems in such applications, LEDs radiating infrared light are preferred.
- ▶ While LEDs are well established in the market these days. They are ideal for backlighting of automobiles dashboards, backlighting of keyboards on cell phones and path marking lights.
- ▶ Widely used for traffic signal management. Many countries have replaced traffic control signals using normal incandescent lamps with the LEDs
- ▶ Even most of the car manufacturers have placed LED lights in place of incandescent lamps
- ▶ These days IC's are available which can drive 10 LEDs or more .for example BCT3211S chip drive up to 10 LEDs with each LED consuming upto 20mA current.

5.5 LIQUID CRYSTAL DISPLAY:

- ▶ It is a material (usually an organic compound) which flows like a liquid at room temperature but whose molecular structure has some properties normally associated with solids (eg: cholestery 1 nonanoate and p-azoxyanisole). As is well known, the molecules in

ordinary liquids have random oriented in a definite crystal pattern. Normally, a thin layer of liquid crystal is transparent to incident light but when an electric field is applied across it, its molecular arrangement is distributed causing changes in falls on activated layers of a liquid crystal. It is either absorbed or scattered by the disoriented molecules.

- ▶ As shown in figure, a liquid crystal 'cell' consists of a thin layer about 10mm of a liquid crystal sandwiched between two glass sheets with transparent electrodes deposited on their mode . With both glass sheets transparent, the cell is known as transmitter type cell. When one glass is transparent and the other has a reflective coating, the cell is called reflective type. The LCD does not produce any illuminative of its own. In fact, it depends entirely on illumination falling on it from an external source for its visual effect.



- ▶ Dynamic scattering display. When field effect display is energized, the energized areas of the LCD absorb the incident light and hence give localized black display. When dynamic scattering display is energized, the molecules of energized area of the display become turbulent and scatter light in all directions, consequently, the display of course, the UN-energized areas remain translucent.
- ▶ As shown in figure, a digit on an LCD has a segment appearance. For example, if number 5 is required, the terminals 8, 2, 3, 6 and 5 would be energized so that only these regions would be activated while the other areas remain clear.

5.5.1 ADVANTAGES :

- ▶ Less power consumption
- ▶ Low cost
- ▶ Uniform brightness with good contrast
- ▶ Low operating voltage and current

5.5.2 DISADVANTAGES :

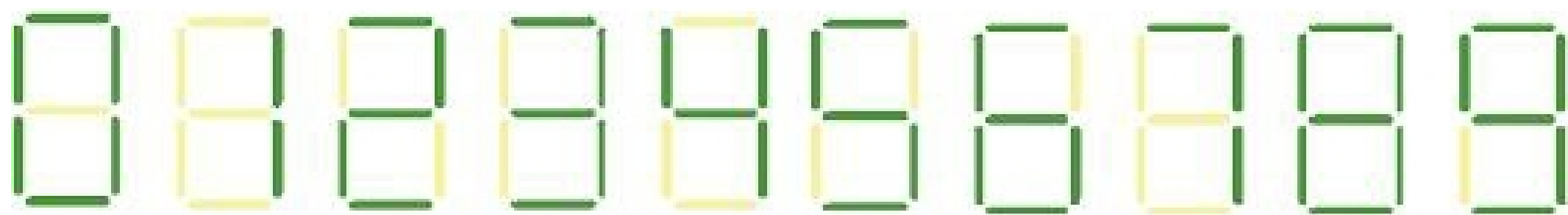
- ▶ Poor reliability
- ▶ Limited temperature range
- ▶ Poor visibility in low ambient temperature
- ▶ Slow speed
- ▶ Requires an AC drive.

5.5.3 APPLICATIONS :

- ▶ Watches, portable instruments such as digital thermometers, blood pressure, blood sugar, pressure monitoring etc..
- ▶ Recent desk top LCD monitors
- ▶ Notebook computer display.
- ▶ Cellular phone display.
- ▶ To display data on personal digital assistant (pDAs) such as palm Vx and pocket PCs etc...

5.6 SEVEN SEGMENT DISPLAY:

- ▶ A display consisting of seven LEDs arranged in seven segments is called seven segment displays. It is shown in figure. The seven LEDs are arranged in a rectangular fashion and are labeled A through G. Each LED is called a segment because it forms a part of digital displayed. An additional LED is used for the indication of decimal point (DP)
- ▶ By forwarding biasing different LEDs can display the digits 0 through 9. For example to display a zero, the LEDs A, B, C, D, E and F are forward biased. To light up a 5, we used to forward bias segment A, F, G, C, D. Thus in a seven segment display depending upon the digit to be displayed, the particular set of LEDs is forward biased. The various digits are shown in figure. Microprocessor kits often use such seven segment displays.



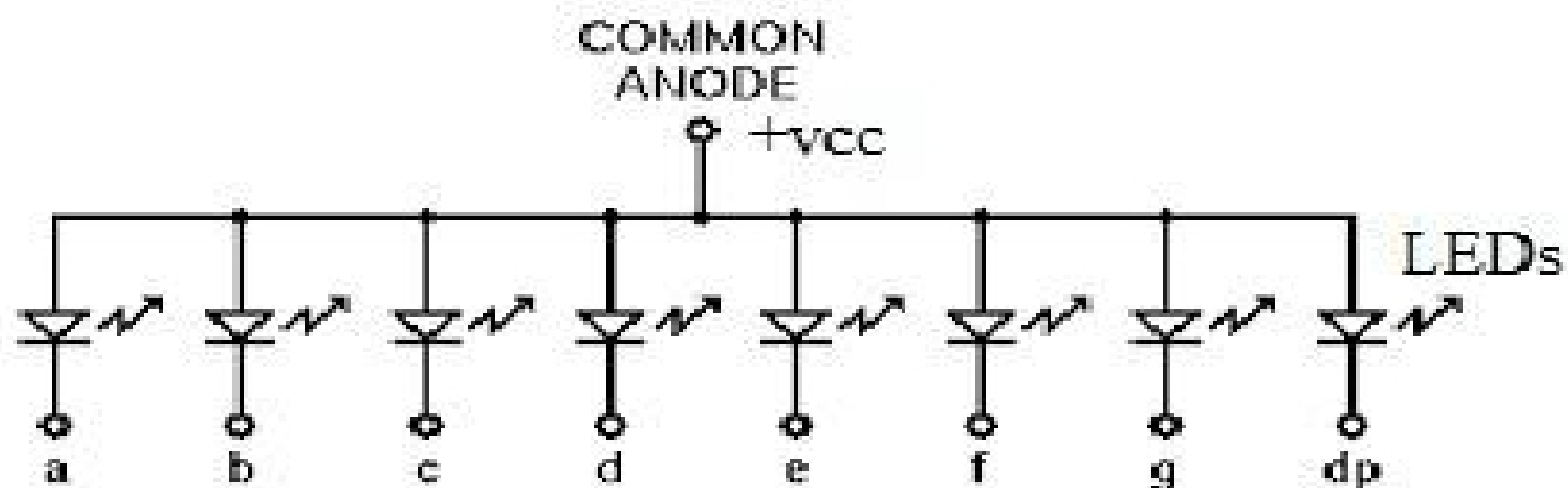
Seven segment displays

5.6.1 TYPES OF SEVEN SEGMENTS DISPLAY:

- ▶ Common anode type
- ▶ Common cathode type

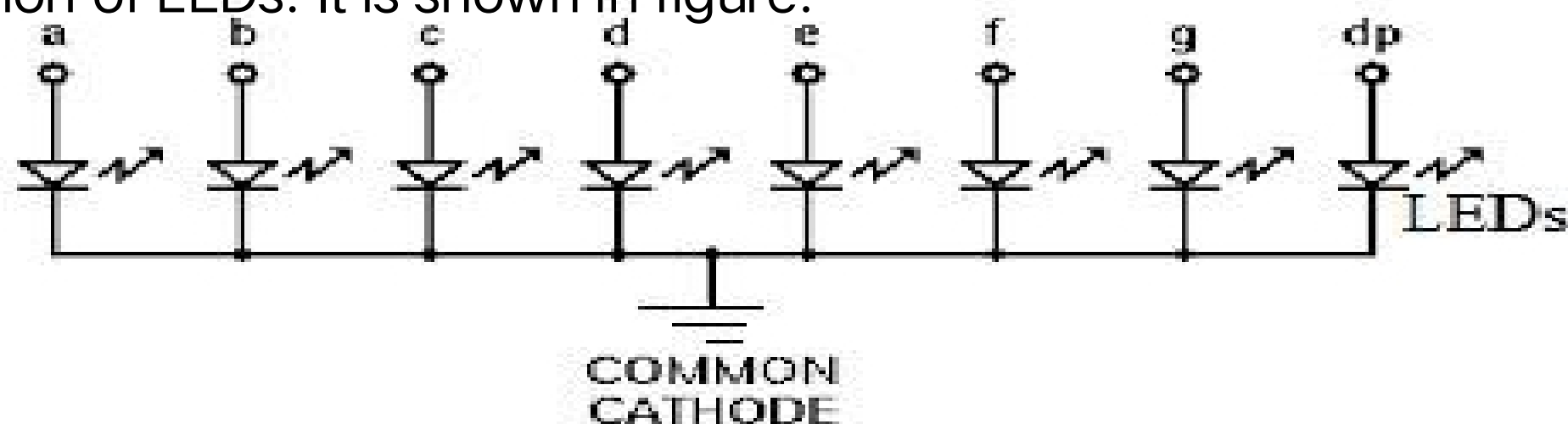
5.6.1.1 COMMON ANODE TYPE:

- ▶ In this type all anodes of LEDs are connected together and common point is connected to +Vcc which is positive supply voltage. A current limiting resistor is required to be connected between each LED and ground. It is shown in figure.



5.6.1.2 COMMON CATHODE TYPE:

- ▶ In this type, all cathodes of LEDs are connected together and common point is connected to the ground, a current limiting resistor is connected between each LED and the supply +Vcc. The anodes of the respective segments are to be connected to +Vcc for the required operation of LEDs. It is shown in figure.

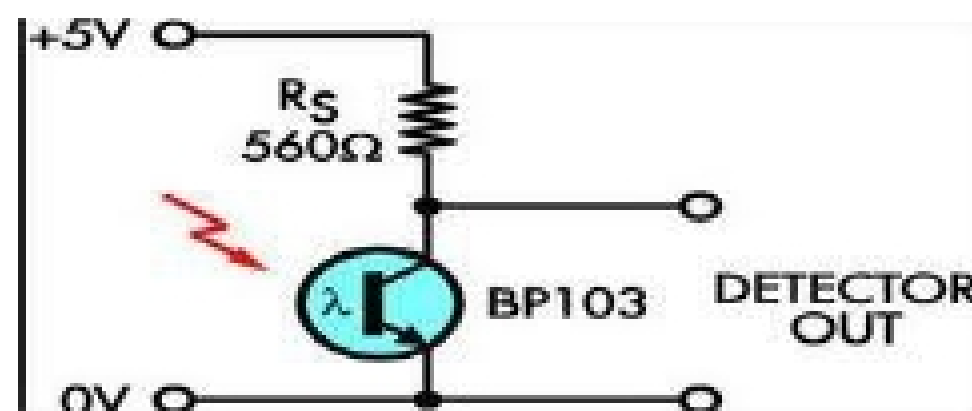


5.7 OPTO- ISOLATORS:

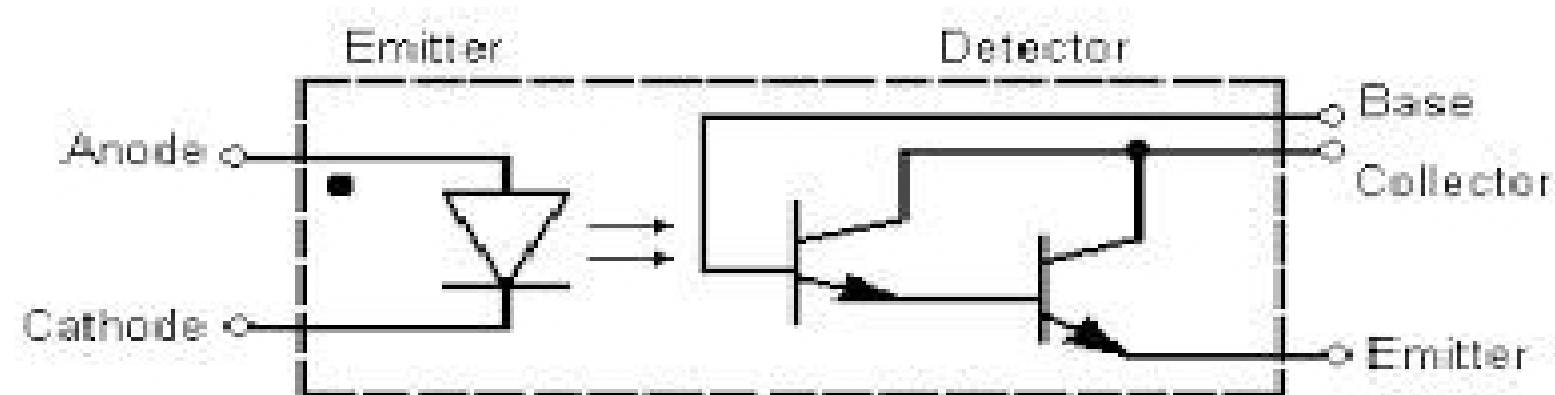
- ▶ Optical isolators are designed to electrically isolate one circuit from another allowing one circuit to control the other. The usual purpose of isolation is to provide protection from high voltage transients surge voltages and low level electrical noise that could possibly result in an error output or damage to the device. Such isolators allow interfacing of circuits with different voltage levels and different grounds etc...
- ▶ An optical isolator (or coupler) consists of a light source such as LED and a photo detector such as photo transistor as shown in figure and is available in a standard IC package. A photo transistor is a 3 terminal device which responds to its input in the same way as photo-diode. In addition it produces larger output.

- ▶ When LED is forward biased, the light produced by it is transferred to the photo transistor

which is turned ON thereby producing current through the external load. The second figure shows a Darlington transistor coupler which is used when increased output current capability is needed beyond that provided by the phototransistor output. The LASER output coupler of figure can be used in application where a low – level input voltage is required to latch a high voltage relay for activating some kind of elect- mechanical device.



a) Photo detector



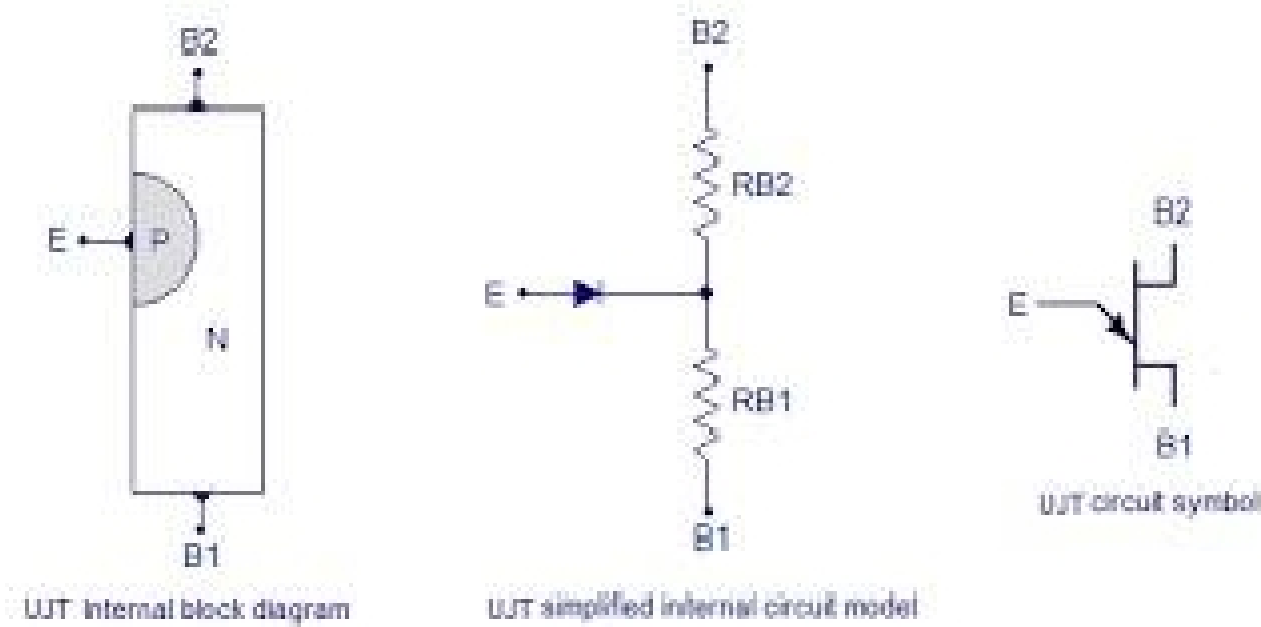
b) Darlington transistor coupler

5.8 UJT (UNI-JUNCTION TRANSISTOR):

- Uni-junction transistor (UJT) is a device which does not belong to thyristor but is used to turn on SCRs.

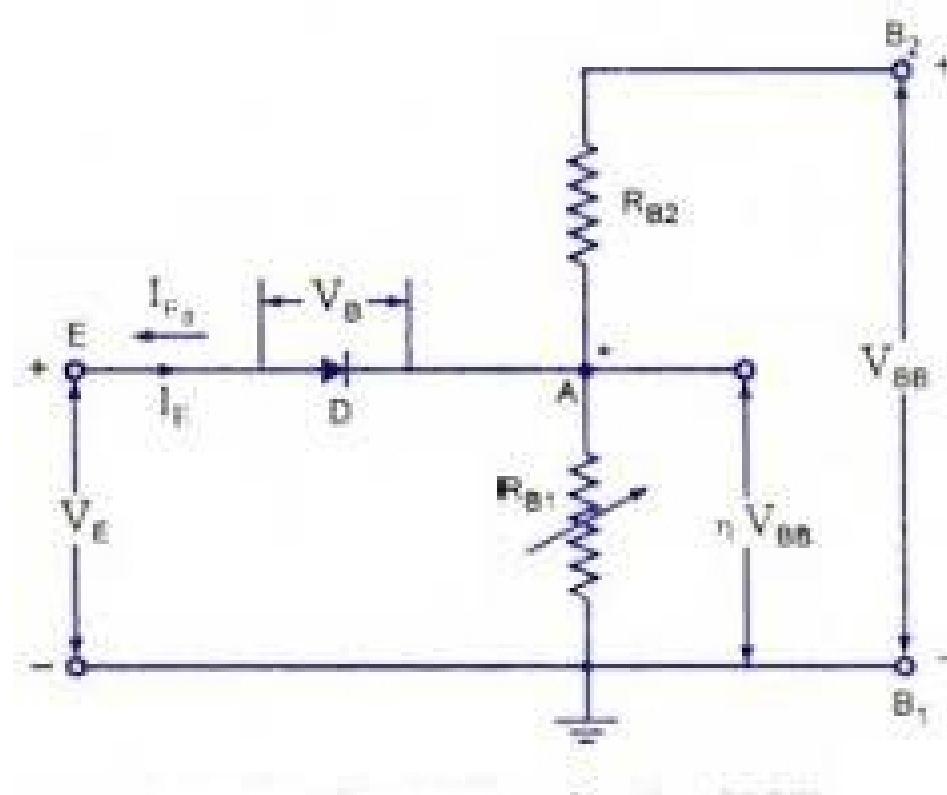
5.8.1 CONSTRUCTION:

- It is a three terminal device having two layers. It consists of a slab of lightly doped N-type silicon material. The two base contacts are attached to both the ends of this N type surface. These are denoted as B1 and B2 respectively. A p-type material is used to form a PN junction at the boundary of the aluminum rod and n-type silicon slab. The third terminal called emitter (E) is taken out from this P-type material. The N-type is lightly doped while P-type is heavily doped. It is shown in figure.



- As N-type is lightly doped, it provides high resistivity and P-type as heavily doped, provides low resistivity. The symbolic representation of UJT is shown in figure. The emitter is shown by an arrow which is at an angle to the vertical line representing N-type material. This arrow indicates the direction of flow of conventional current when the UJT is forward biased.

5.8.2 EQUIVALENT CIRCUIT OF UJT:



- Figure shows the basic structure and equivalent circuit of UJT. The internal resistances of the two bases are represented as R_{B1} and R_{B2} . In the actual construction the terminal E is closer to B_2 , as compared to B_1 . Hence resistance R_{B1} is more than R_{B2} . The PN junction is represented as normal diode with V_D as the drop across it.
- When the emitter diode is not conducting then the resistance between the two bases B_1 and B_2 is called interbase resistance denoted as R_{BB} . Its value ranges from $(4-12)\Omega$.

$$R_{BB} = R_{B1} + R_{B2}$$

5.8.3 INTRINSIC STAND OFF RATIO (η)

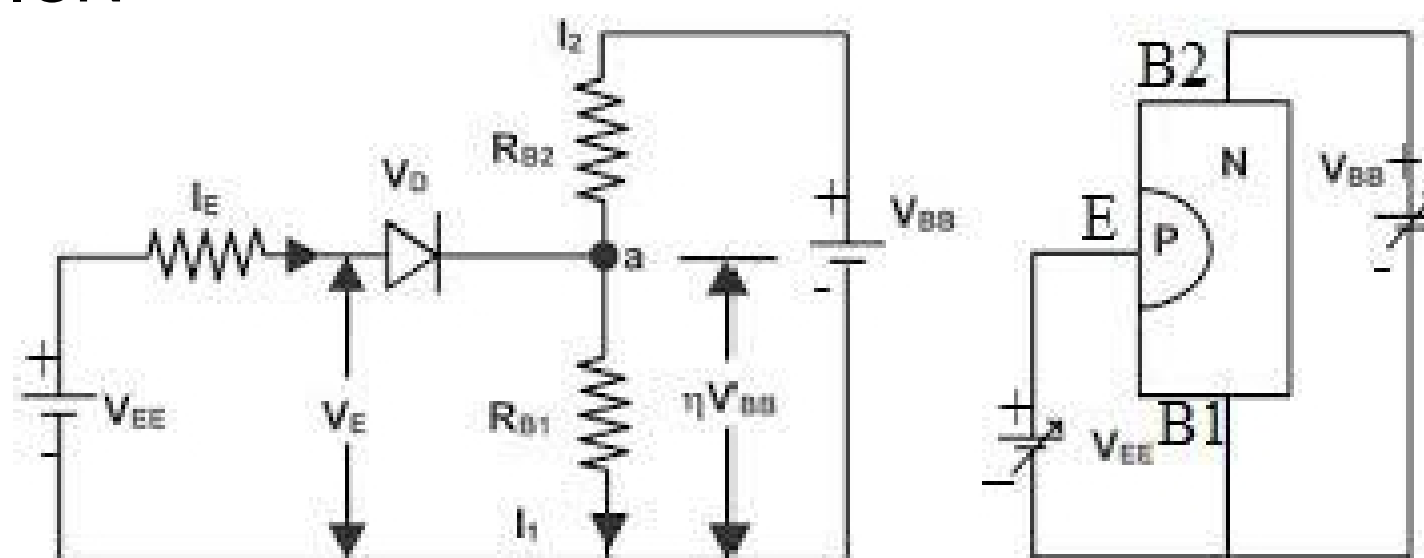
- Consider UJT as shown in the figure to which supply V_{BB} is connected with $I_E = 0$ (i.e.,) emitter diode is not conducting. The voltage drop across R_{B1} can be obtained by using the potential divider rule.

$$V_{RB1} = R_{B1} V_{BB} / (R_{B1} + R_{B2}) = \eta V_{BB} \quad I_E = 0$$

Thus η = Intrinsic stand-off ratio = R_{B1} / R_{BB}

- The typical range of η is from 0.5 to 0.8. The voltage V_{RB1} is called intrinsic stand-off voltage because it keeps the emitter diode reverse biased for the emitter voltages less than V_{RB1} .

5.8.4 OPERATION



- While operating an UJT, the supply V_{BB} is applied between B2 and B1 while the variable emitter voltage is applied between the emitter terminals. It is shown in the figure. Let us see the effect of change in V_E . The potential is decided by η and is equal to V_{RB1} .

CASE 1:

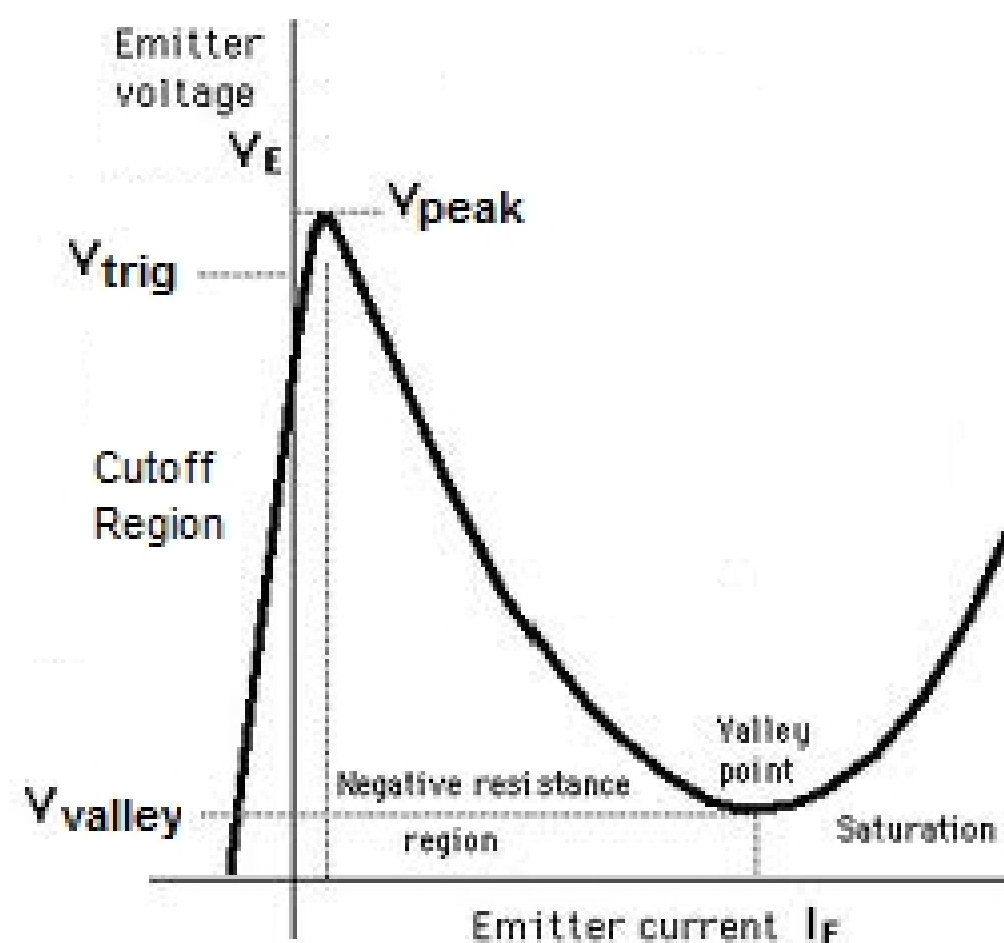
$V_E < V_A$ – as long as the PN junction is reverse biased. Hence the emitter current I_E will not flow. Thus the UJT is said to be in OFF.

CASE 2:

$V_E > V_P$ – The diode drop V_D is generally between (0.3 – 0.7) V. Hence we can write $V_P = V_A + V_D = \eta V_{BB} + V_D$

When V_E becomes equal to or greater than V_P , then the PN junction becomes forward biased and current I_E flows. The UJT is said to be ON.

5.8.5 UJT CHARACTERISTICS



- ▶ The graph of emitter current against emitter voltage is plotted for a particular value of V_{BB} is called the characteristics of UJT. For a particular fixed value of V_{BB} such that the characteristics is shown in the figure. This is divided into three main regions

Cut-off region

- ▶ The emitter voltage V_E is less than V_P and the PN junction is reverse biased. A small amount of reverse saturation current I_{EO} flows through the device, which is negligible small of the order of μA . This condition remains till the peak point.

Negative resistance region

- ▶ When the emitter voltage V_E becomes equal to V_P and the PN junction becomes forward biased and I_E starts flowing. The voltage across the device decreases in this region, though the current through the device increases. Hence the region is called negative resistance region. This decreases the resistance region R_{B1} . This region is stable and used in many of the applications. This region continues till valley point.

Saturation region

- ▶ Increase in I_E further valley point I_E drives the device into the saturation region. The voltage corresponding to the valley point is called valley point voltage denoted as V_v . In this region, further decrease in voltage does not take place. The characteristic is similar to that of a semi-conductor diode, in this region. The active region (i.e.,) negative resistance region the notes which are large in number on P-side, get injected into N-side.

This causes increase in free electrons in the N-type slab. This increases the conductivity that increases the resistivity. Hence the resistance R_{B1} decreases in this region.

As the V_{BB} increases the potential V_P corresponding to peak point will increase

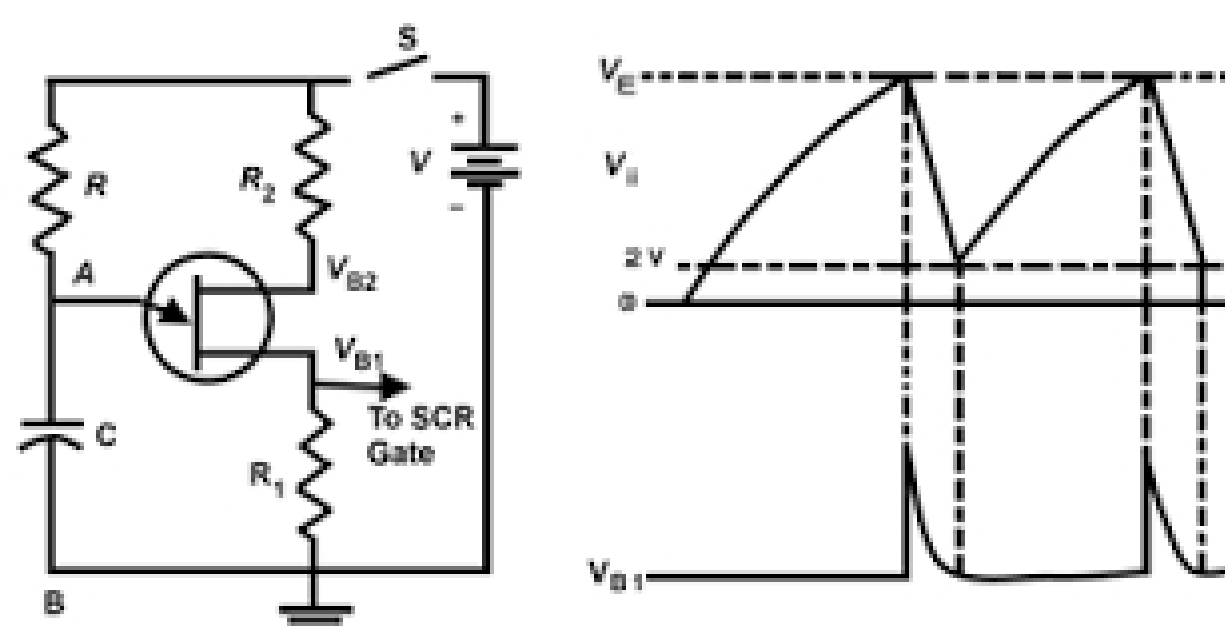
Applications

- The UJT is mainly used in triggering of other devices such as SCR. It is also used in sawtooth wave generators and timing circuits. The most popular application of UJT is that it can be used as a relaxation oscillator to obtain short pulses for triggering of SCR's.

5.8.6 UJT RELAXATION OSCILLATOR

- The pulse signal required to drive the digital circuits can be obtained from a single stage oscillators circuits used are uni junction transistor. Such oscillator which uses UJT is
- The R_1 and R_2 are biasing resistances which are selected such that they are lower than inter base resistances R_{B1} and R_{B2} . They resistance R_T and the capacitance C_T decide the oscillating rate. The value of R_T is so selected that the operating point of UJT remains in the negative resistance region. The UJT characteristics and the negative resistance is already shown. It varies between emitter voltage and emitter current.

5.8.6.1 OPERATION



The capacitor C_T gets charged through resistance R_T , towards the supply voltage V_{BB} . As long as the capacitor voltage is less than the peak voltage V_P , the emitter appears as an open circuit.

$$V_P = \eta V_{BB} + V_P$$

Where η - stand-off ratio of UJT

V_P - cut-in voltage of the diode

- When the capacitor voltage V_C exceeds the voltage V_P , the UJT fires. The capacitor starts discharging through $R_1 + R_{B1}$ internal base resistance.

- ▶ Due to the design of R_1 , this discharge is very fast, and it produces a pulse across R_1 . Where the capacitor voltage falls below V_V (i.e) $V_C = V_V = V_E = V_V$, the UJT gets turned OFF. The capacitor starts charging again.
- ▶ The discharge time of the pulse is controlled by the time $C_T R_1$ while the charging time constant by $R_T C_T$. It is shown in waveform.
- ▶ There is voltage drop across R_2 and voltage rise across R_1 , when UJT fires. The charging equation of the capacitor is given

$$V_C(t) = V_V + V_{BB}[1 - e^{-t/R_T C_T}]$$

But $V_C(t) = V_P$ at $t = T$ then

$$V_P = V_V + V_{BB} [1 - e^{-T/R_T C_T}]$$

Neglecting V_D and V_V to get approximate relation for T

$$T = R_T C_T \ln [1 / 1 - \frac{V_P - V_V}{V_{BB}}]$$

$$f_0 = 1 / R_T C_T \ln [1 / 1 - \frac{V_P - V_V}{V_{BB}}]$$

where f_0 – oscillating frequency

Condition for turn on and turn off

- ▶ To ensure turn ON of UJT, I_E should not be less than the peak point. To achieve this

$$V_{BB} - V_P > I_P R_T$$

$$R_T < [(V_{BB} - V_P) / I_P] \quad \text{ON condition}$$

- ▶ To turn OFF the device I_E at the valley point must be less than I_V specified. Thus voltage across R_T must be less than $I_V R_T$.

$$V_{BB} - V_P > I_P R_T \text{ and } V_{BB} - V_V < I_V R_T$$

$$R_T > [(V_{BB} - V_V) / I_V] \quad \text{OFF condition}$$

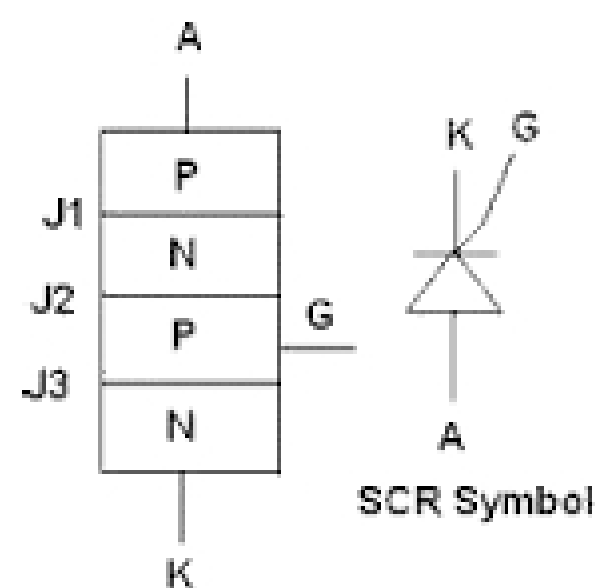
Hence the range of R_t for the proper turn ON and OFF is

$$[(V_{BB} - V_P) / I_P] > R_T > [(V_{BB} - V_V) / I_V]$$

5.9 SILICON CONTROLLED RECTIFIER (SCR)

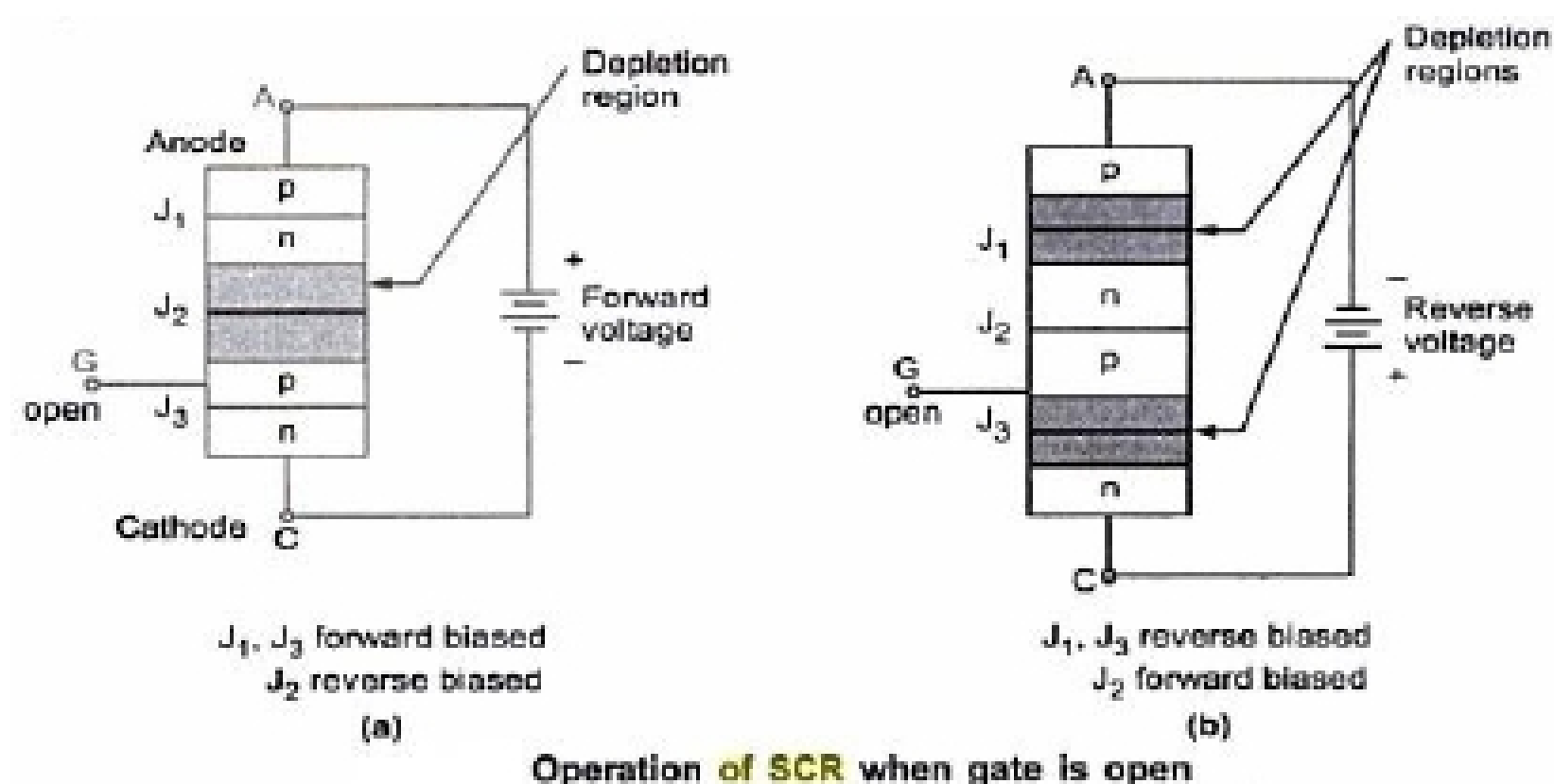
- The SCR is an unidirectional device and like diode, it allows to flow current in only one direction. But unlike diode, it has a built in feature to switch „ ON“ and „ OFF“. The switching of SCR is controlled by the additional part of input called gate and biasing conditions. The switching property of SCR allows to control the „ ON“ periods of the SCR thus controlling average power delivered to the load. It can be used as a rectifier element like diode to convert AC signals to DC signals.

5.9.1 CONSTRUCTION

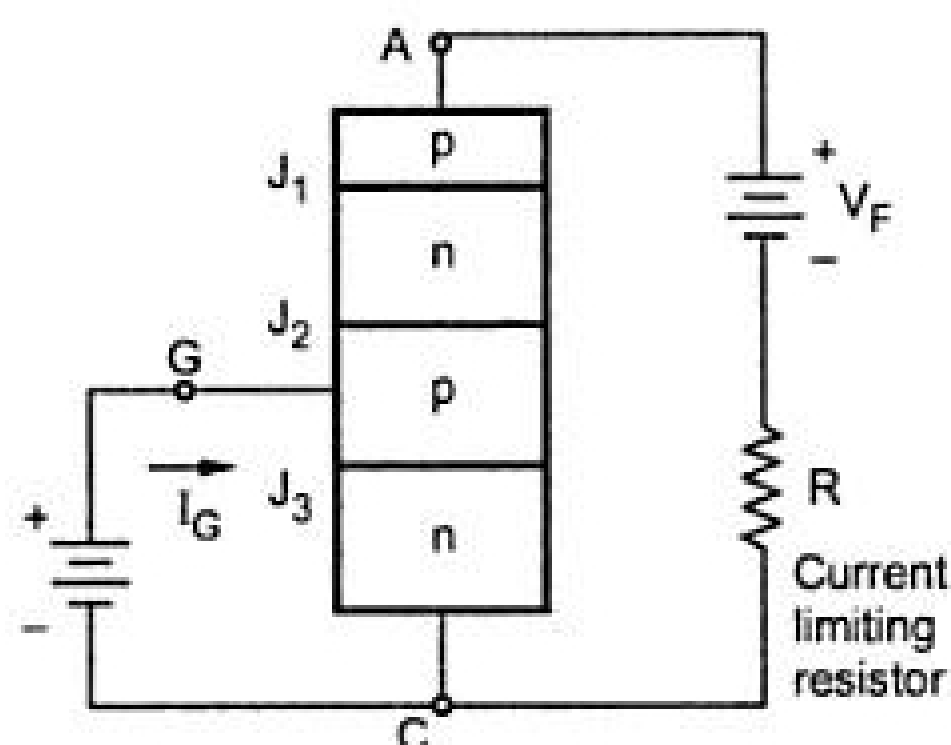


- SCR is a four layer PN-PN device where P and μ layers are alternately arranged. The outer layers are heavily doped. There is a PN junction called J_1 , J_2 and J_3 . The outer p-layer is called anode and outer μ - layer is called cathode. Middle p- layer is called gate. The three terminals are taken out respectively from the three layers as shown in the figure.
- Anode must be positive with respect to cathode to forward with the SCR. But this is not sufficient criterion to turn SCR ON. To make it ON a current is to be passed through the gate terminal denoted as I_{GT} . Thus it is a current operated device.

5.9.2 OPERATION



- ▶ It is divided into two categories
- ▶ **When the gate is open:** Consider that the anode is positive with respect to cathode and gate is open. The junctions J_1 and J_3 are forward biased and junction J_2 is reverse biased. There is depletion region around J_2 and only leakage current flows which is negligibly small. Practically SCR is said to be OFF. This is called forward blocking state of the SCR and voltage applied to the anode and cathode with anode positive is called forward voltage. This is shown in the figure.
- ▶ With the gate open, if the cathode is made positive with respect to anode, the junctions J_1 , J_3 become reverse biased and the J_2 is forward biased. Still the current flows is leakage current, which can be neglected as it is very small. The voltage applied to make the cathode positive is called reverse voltage and SCR is said to be reverse blocking state. This is as shown in the figure.
- ▶ In the forward blocking state, if the forward voltage is increased, the current remains almost zero upto certain limit. At a particular value, the reverse junction J_2 breaks down and SCR conducts heavily. This voltage is called forward breakdown voltage V_{BO} of the SCR. In such condition, SCR is said to be triggered or ON.

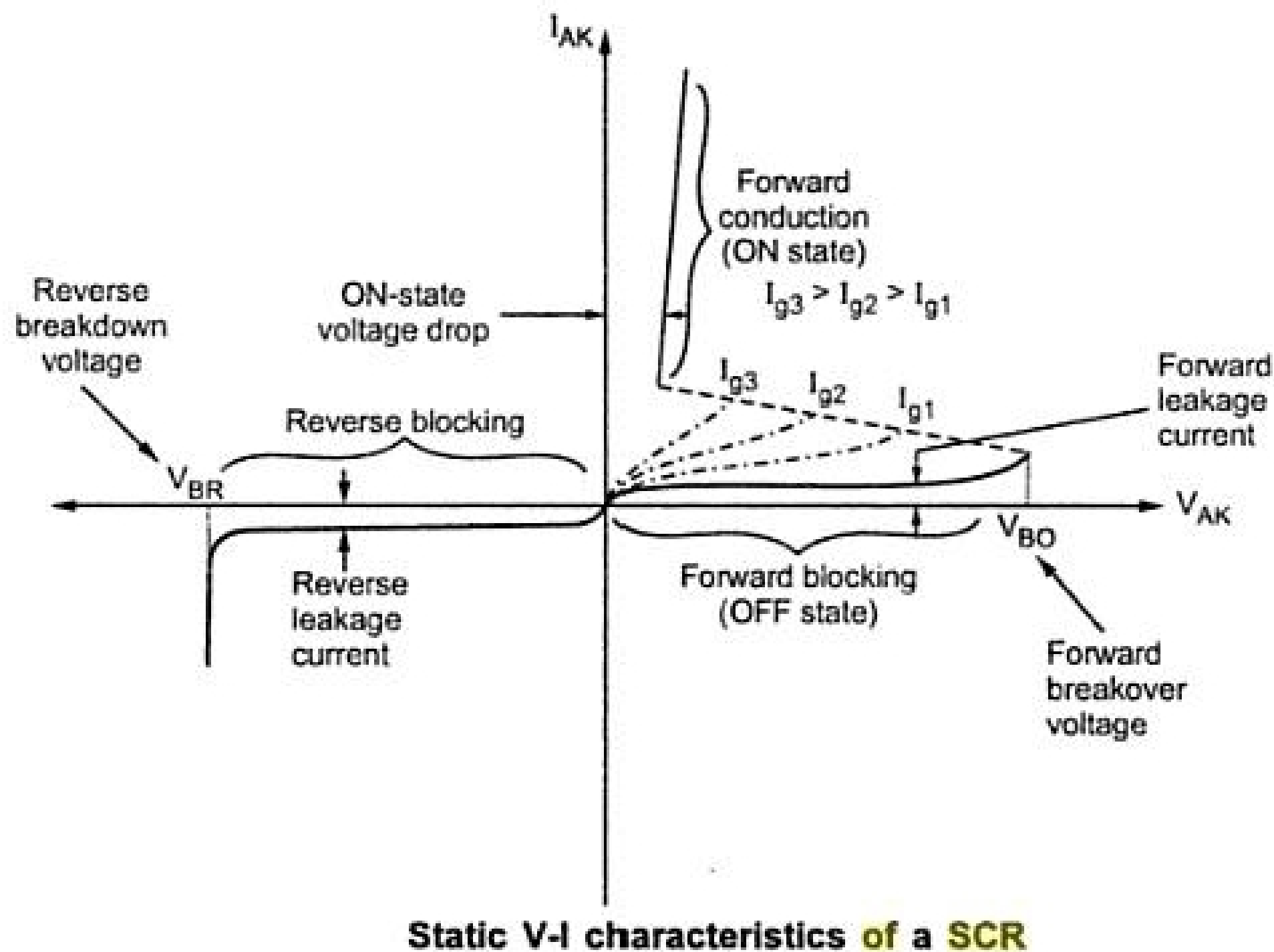


Operation of SCR when gate is closed

- ▶ **When the gate is closed:** Consider the Voltage is applied between the gate and the cathode when the SCR is in forward blocking state. The gate is made positive with respect to the cathode. The electrons from the junction J_3 to reach to positive of battery.
- ▶ When holes from the p-type move towards the negative of the battery. This constitutes the gate current. This current increases more electrons across the junction J_2 and anode

current further increases. Due to regeneration action, within short time, the junction J_2 breaks and the SCR conducts, the gate loses its control.

5.9.3 CHARACTERISTICS OF SCR



5.9.3.1 FORWARD CHARACTERISTICS

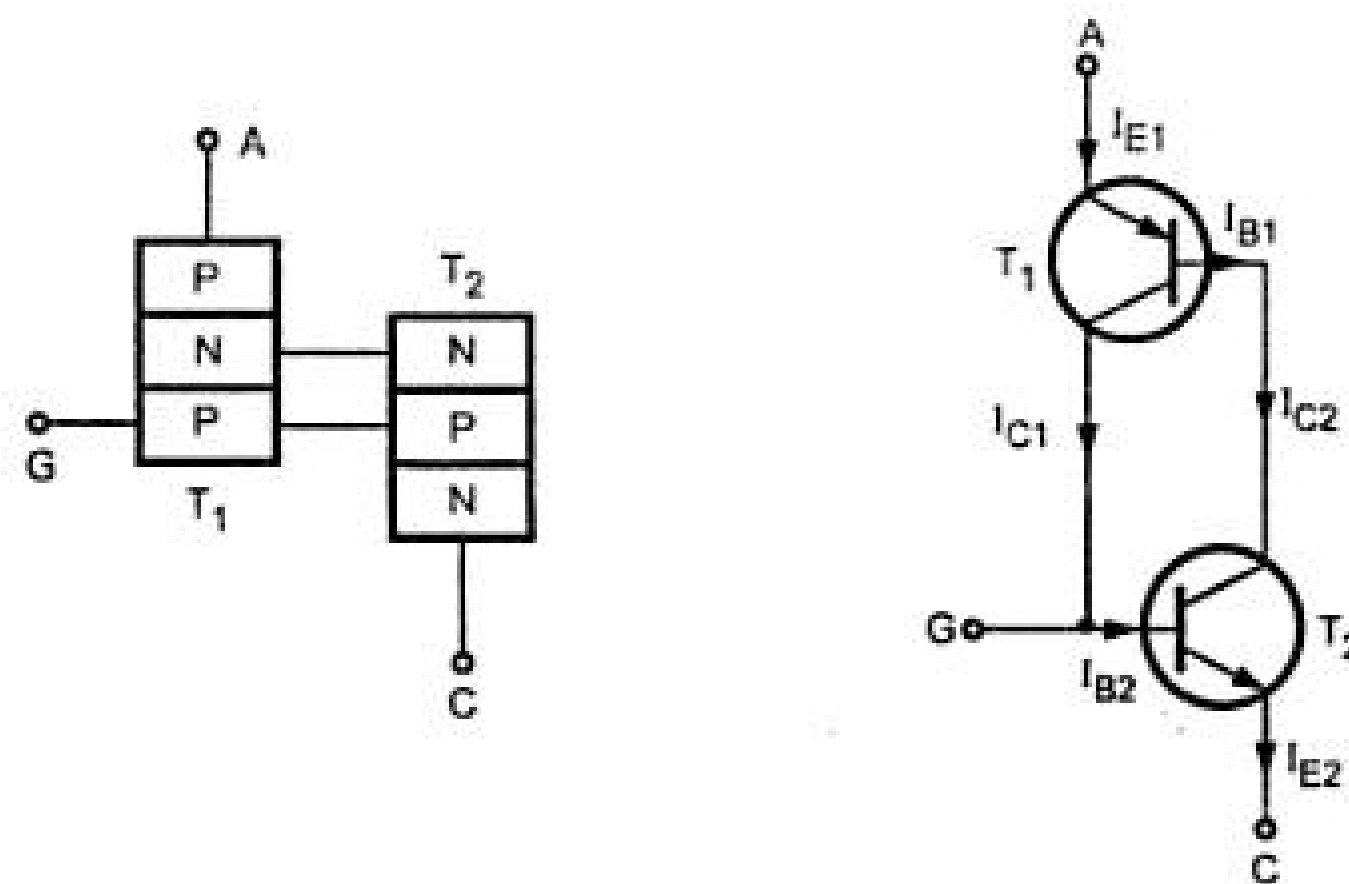
- It shows a forward blocking region, when $I_g = 0$. It also shows that when forward voltage increases up to V_{BO} , the SCR turns ON and a high current results. The drop across the SCR reduces suddenly which is now ohmic drop in the four layers. The current must be limited only by the external resistances in series with the device.
- It also shows that, if gate bias is used, as gate current increases, less voltage is required to turn ON the SCR. If the forward biased current falls below the level of the holding current I_H , then a depletion region begins to develop around J_2 and the device goes into the forward blocking region.
- When SCR is turned ON from OFF state, the resulting forward current is called latching current I_L . It is slightly higher than I_H .

5.9.3.2 REVERSE CHARACTERISTICS

- If the anode to cathode voltage is reversed, then the device enters into the reverse blocking region. The current is negligibly small and practically neglected.

- ▶ If the reverse voltage is increased, similar to the diode at a particular value avalanche breakdown occurs and a large current flows through the device. This is called reverse breakdown and the voltage which happens is called reverse breakdown voltage V_{BR} .
- ▶ It is shown in the figure.

5.9.4 TWO TRANSISTOR MODEL



Two transistor model of SCR

- ▶ The basic structure of SCR can be divided into three layer structure as shown in the figure. It may be noted that the upper 3- layer structure is a PNP transistor whereas the lower is an NPN transistor. Thus SCR also known as two transistor model is shown in the figure say Q_1 – PNP and Q_2 – NPN interconnected. Sometimes this circuit is also called as the two transistor analogy or an ideal latch of a SCR.
- ▶ From the figure the collector of each transistor is connected to the base of each transistor. Therefore the collector current of the transistor Q_1 is the base current of the transistor Q_2 and the collector current of the transistor Q_2 is the base current of the transistor Q_1 . For this circuit we have an action of positive feedback or regeneration. It means that if there is a change in current at any point in the loop it is amplified and returned to the starting point with the same phase. For example I_{B2} increases I_{C2} also increases. It causes I_{B1} increases and I_{C1} also to increase. It continues till both the transistors are driven into the saturation region. In this case SCR will act as an ON switch and will pass the current from anode to cathode.
- ▶ On the other hand, I_{B2} decreases, I_{C2} will also decrease. It causes reduction in I_{B1} and hence I_{C1} also gets reduced. This process will continue till both the transistors are driven

into the cut-off region. In this case SCR will act as an OFF switch and hence block the current flowing from anode to cathode.

5.9.5 TURNING ON (TRIGGERING) SCR

The SCR can be turned ON from its OFF position by several methods.

Forward Breakdown Voltage

- ▶ If the voltage across the SCR exceeds the forward breakdown voltage, the SCR will start conducting due to avalanche breakdown.

Gate Triggering

- ▶ This is the most commonly used method to trigger the SCR. In this method the SCR is operated in the anode voltage slightly less than the rated forward breakdown voltage and is triggered into conduction region by low power gate pulse. It may be noted that once the SCR is switched ON the gate current has no control over the device current. The gate pulse signals can be applied either from DC source or from AC source.
- ▶ Figure shows the SCR connected to the DC source through a load this case, the gate signal is generated by push button switch. When the switch S is pressed momentarily a positive voltage is applied at the gate. AS a result of this the SCR is turned ON and the current flows through the load. It will remain in its ON position until the supply voltage is removed or reversed.
- ▶ Other figure shows a SCR connected to an AC source through a load. In this circuit the signal is provided by the timing pulses. Such pulses can be generated by a number of devices called triggering devices. Such as UJT, DIAC etc.,.

Rate – Effect Or dV/dt Triggering

- ▶ Here the SCR is turned ON by rapidly increasing the anode to cathode voltage. It provides a charging current which triggers the SCR to conduction.

Light Triggering

- ▶ Here the SCR is triggered by irradiating its light. When the light falls in the middle junction (J_2) of the SCR, the device turns ON. Such a device is called light activated silicon controlled rectifier (LASER).

5.9.6 TURNING OFF SCR

- ▶ The SCR is turned ON it continues to conduct even when the gate signal is removed. The ability of the SCR to keep on conducting even when the SCR is turned ON is called

latching. It means the SCR cannot be turned OFF simply by removing the gate signal. A number of methods are used for turning OFF the SCR turned below.

- ▶ Reverse the polarity of anode to cathode voltage.
- ▶ Interrupting anode current by means of momentarily series parallel switching arrangement. This method is called anode current interruption.
- ▶ Reducing the current of SCR below the holding current. This method is known as forced commutation.

5.9.7 SCR RATINGS

Forward breakdown voltage

- ▶ It is the voltage at which the SCR is switched from OFF to ON position. It is maximum for $I_G=0$. Its range is from (50-1200)V.

Holding current

- ▶ It is the value of anode current below which the SCR is turned ON or OFF. It is the minimum value of current required to handle the conduction.

Gate trigger current

- ▶ It is the anode current when the SCR switches from OFF to ON position used for specified conditions.

Average forward current

- ▶ It is the maximum value of anode current which SCR can handle in its ON position. It varies from (1-1800)A.

5.9.8 REVERSE BREAKDOWN VOLTAGE

- ▶ It is value of reverse current from cathode to anode at which the device breaks into avalanche region and starts conducting heavily.
- ▶ Other parameters such as turn ON time, turn OFF time and junction temperature and case temperature, circuit fusing rating, gate non-triggering voltage and critical rise rating. The circuit fusing rating shows the maximum forward surge current capability of the SCR device. For example I_{t2} of 2N682-92 series is 103A2s. If the product of the square of the surge current times the duration of surge current exceeds 103A2s, the SCR is destroyed by excessive power dissipation.
- ▶ When I_{t2} is known, we can determine the maximum allowable duration of surge current is given by the amount of surge current it follows

$$t_{\max} = I_{\text{st}}(\text{rated}) / I_{\text{r2}}$$

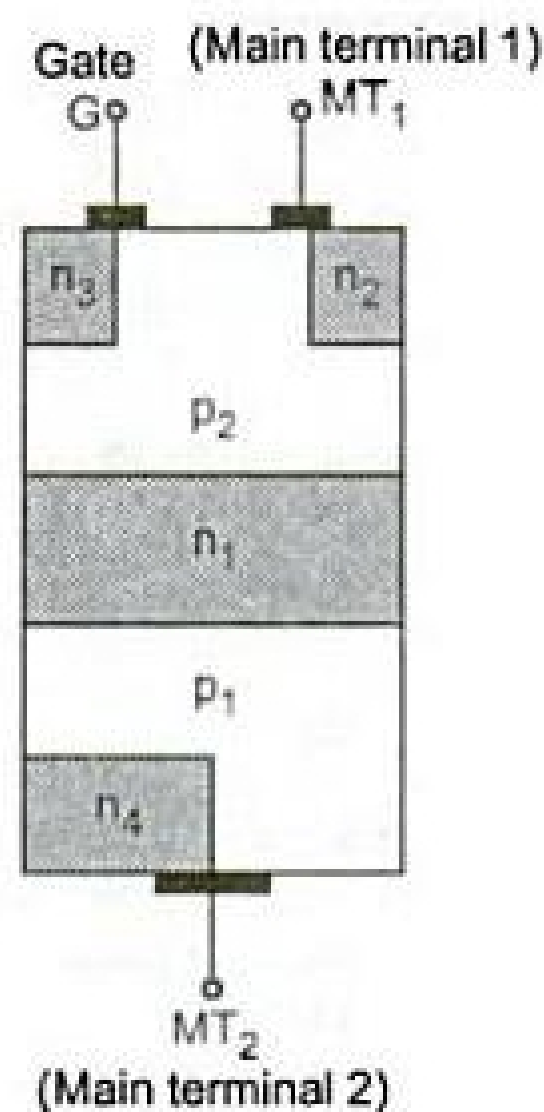
where I_{s} – value of surge current

- ▶ The I_{st} rating becomes critical in applications which SCR is used as a surge protection device. The gate triggering voltage (V_{GD}) rating indicates the maximum gate voltage that can be applied without triggering the SCR into conduction. If V_{G} exceeds the rating, SCR will be triggered into the ON state. The V_{GD} is important because it points out of the potential reason of the false triggering of SCR. False triggering is a situation where SCR is accidentally triggered into conduction, usually by some type of electrical noise. For example 2N682-92 series, $V_{\text{GD}} = 250\text{mV}$ appears at the gate the device may be triggered into conduction.
- ▶ Another common reason for false triggering is a rise in anode voltage that exceeds the critical rise (dV/dt) rating of the SCR. This rating indicates the maximum rate of increase in anode to cathode voltage that the SCR can handle without false triggering occurring. The (dV/dt) rating of 2N682-92 series is 30v/ms . This means that a noise signal is V_{AK} with a rate of rise equal to 30 v/m may cause false triggering.

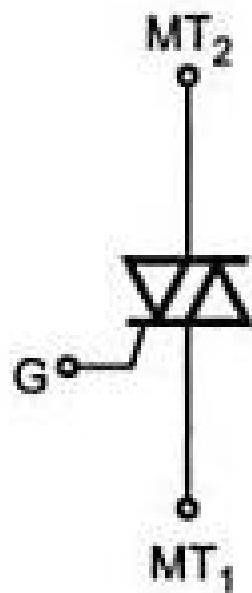
5.9.9 APPLICATION OF SCR

- ✓ Motor speed control
- ✓ Light dimming control
- ✓ Heater control
- ✓ Phase control
- ✓ Battery charger
- ✓ Invertors
- ✓ Static switches
- ✓ Rectifier power amplifier
- ✓ Relay control.

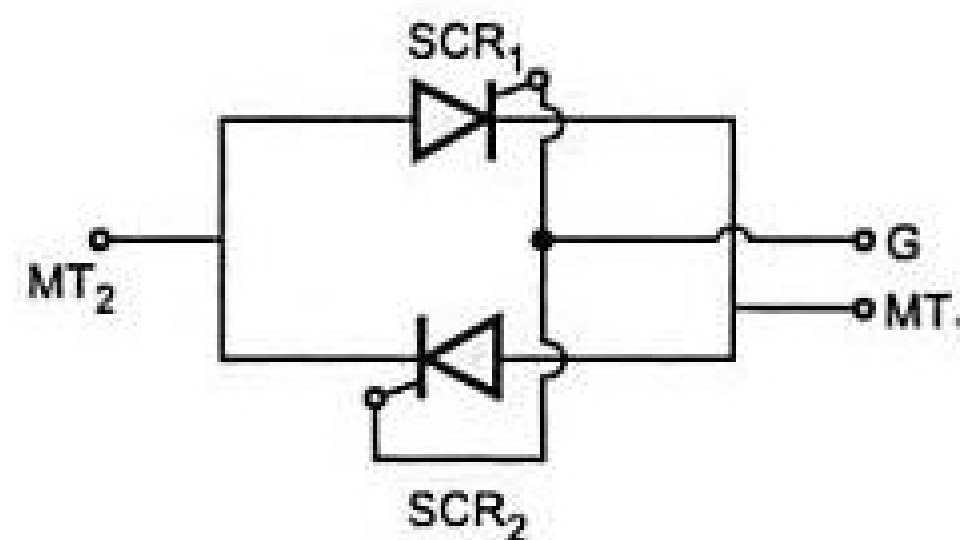
5.10 TRIAC:



(a) Construction



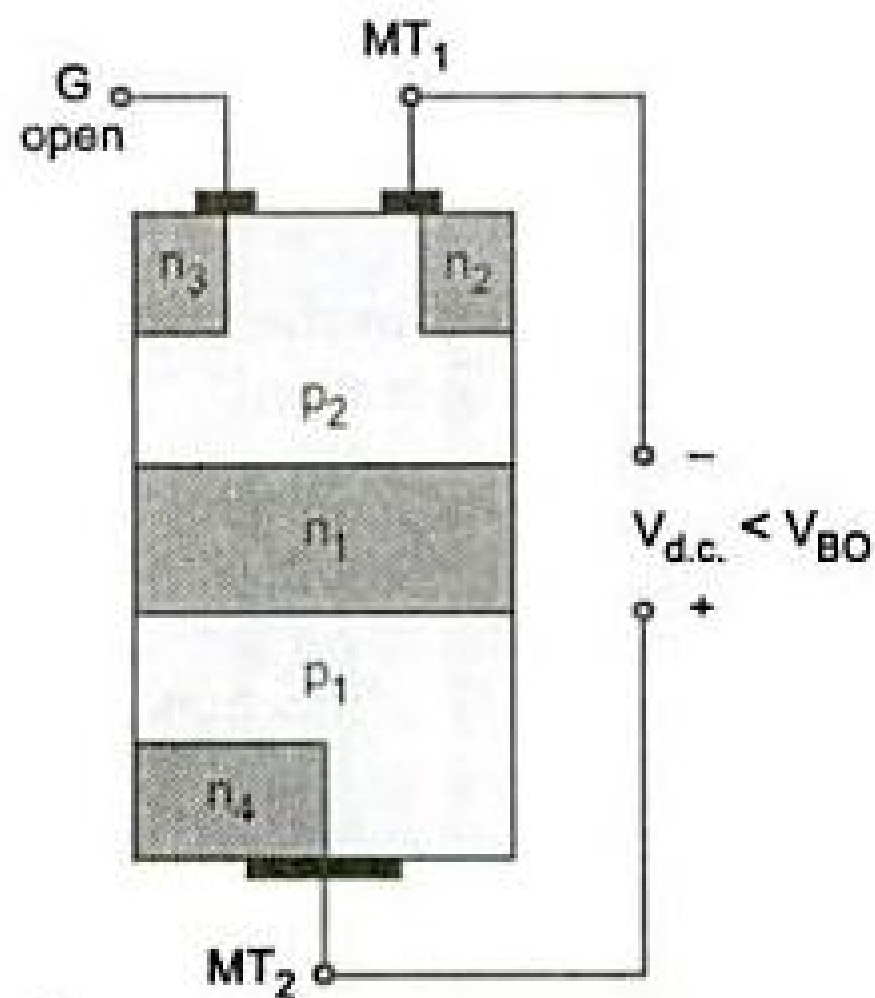
(b) Symbol



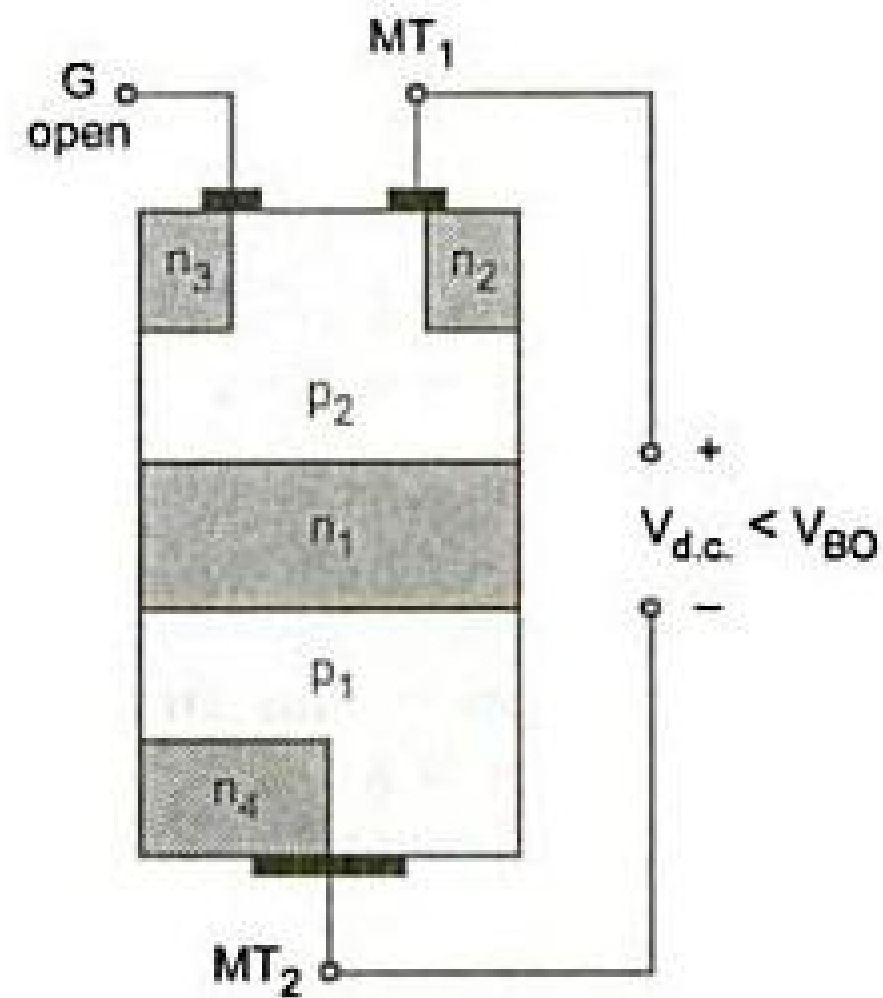
(c) Equivalent circuit

- It is a three terminal device which can conduct in either directions when triggered either by a positive or a negative pulse irrespective of the polarity of the voltage across its main terminals. The triac behaves like two SCR"s connected in parallel but in opposite directions, with a common gate terminal i.e., anode of one SCR is connected to the cathode of the other and their gates are connected together. Thus anode and gate voltage applied in either direction will trigger the triac, it is due to the fact that the applied voltage will trigger at least one of the SCR"s connected in opposite direction. The triac are available with current rating upto 25A and voltage rating upto 500V.
- The basic connection of a triac is shown in figure. It may be noted from this figure that the triac consist of two four layer switches in parallel. These switches are P1 N1 P2 N2 and P2 N3 P1 N4 as shown by the broken lines in the figure. Since the triac conducts in both directions therefore the terminals are designated by numbers instead of anode and cathode as in SCR. The triac has two main terminals namely main terminal 1(MT 1) and one gate terminal (G). The equivalent circuit of a triac is shown which consist of two SCR"s connected in parallel but in opposite directions with a common gate terminal. Figure shows also the schematic symbol of a triac.

5.10.1 TRIAC OPERATION:

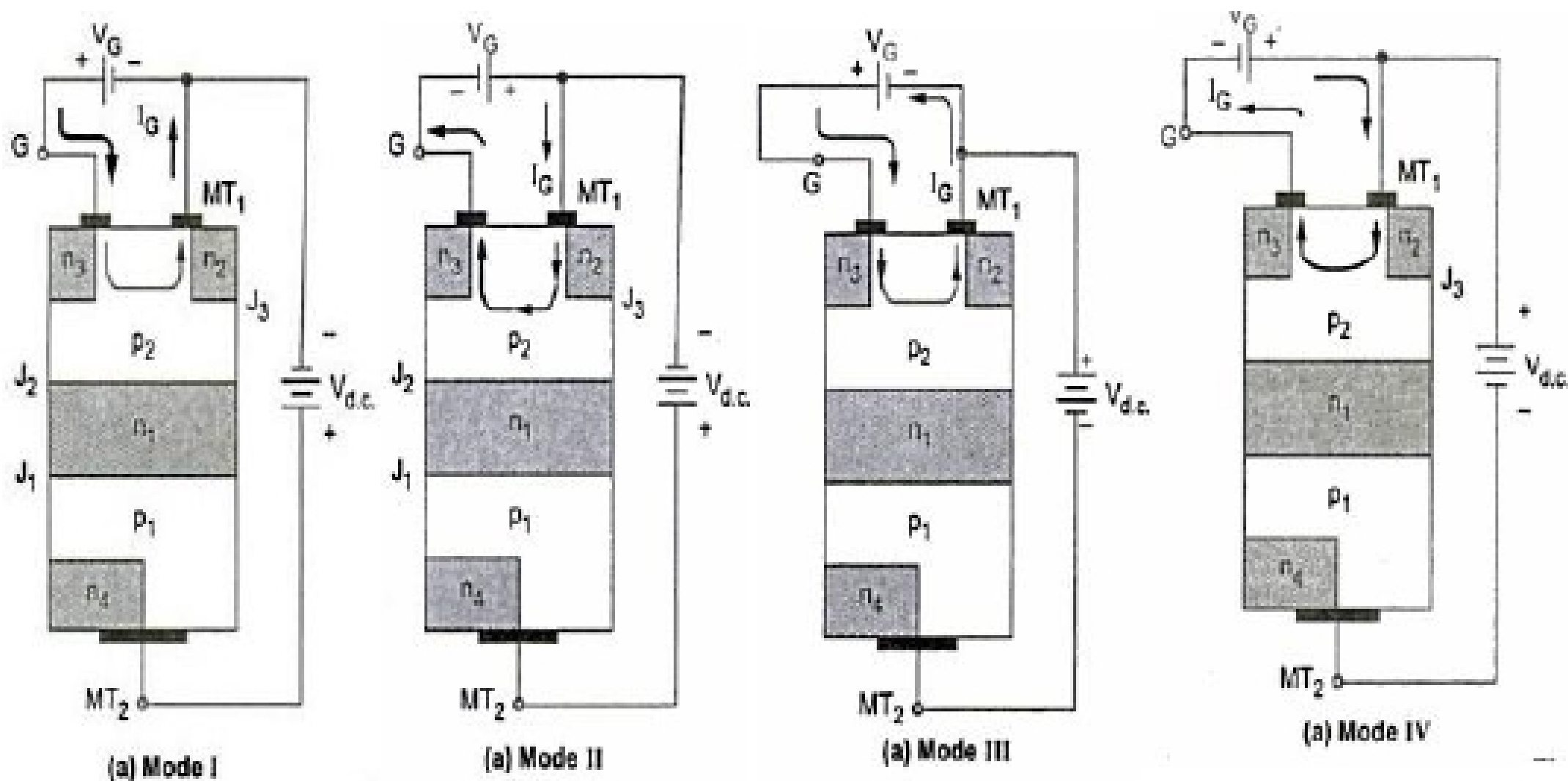


(a) Forward blocking



(b) Reverse blocking

- ▶ There are four number of triac operation, depending upon the polarity of voltage across its main terminal and gate terminal. These modes are described as below.
- ▶ **MT 2 is positive and G is positive** : The triac is identical to SCR. The current flows through the switch P1 N1 P2 N2 from MT2 to MT1.
- ▶ **MT 2 is negative and G is positive** : The current flows through the switch P2 N1 P1 N4 from MT1 to MT2. It is insufficient than mode and must be avoided.
- ▶ **MT 2 is positive and G IS negative** : The current flows through the switch P1 N1 P2 N2 from MT 2 to MT 1. It is less efficient than mode 1 but not as poor as mode 2.



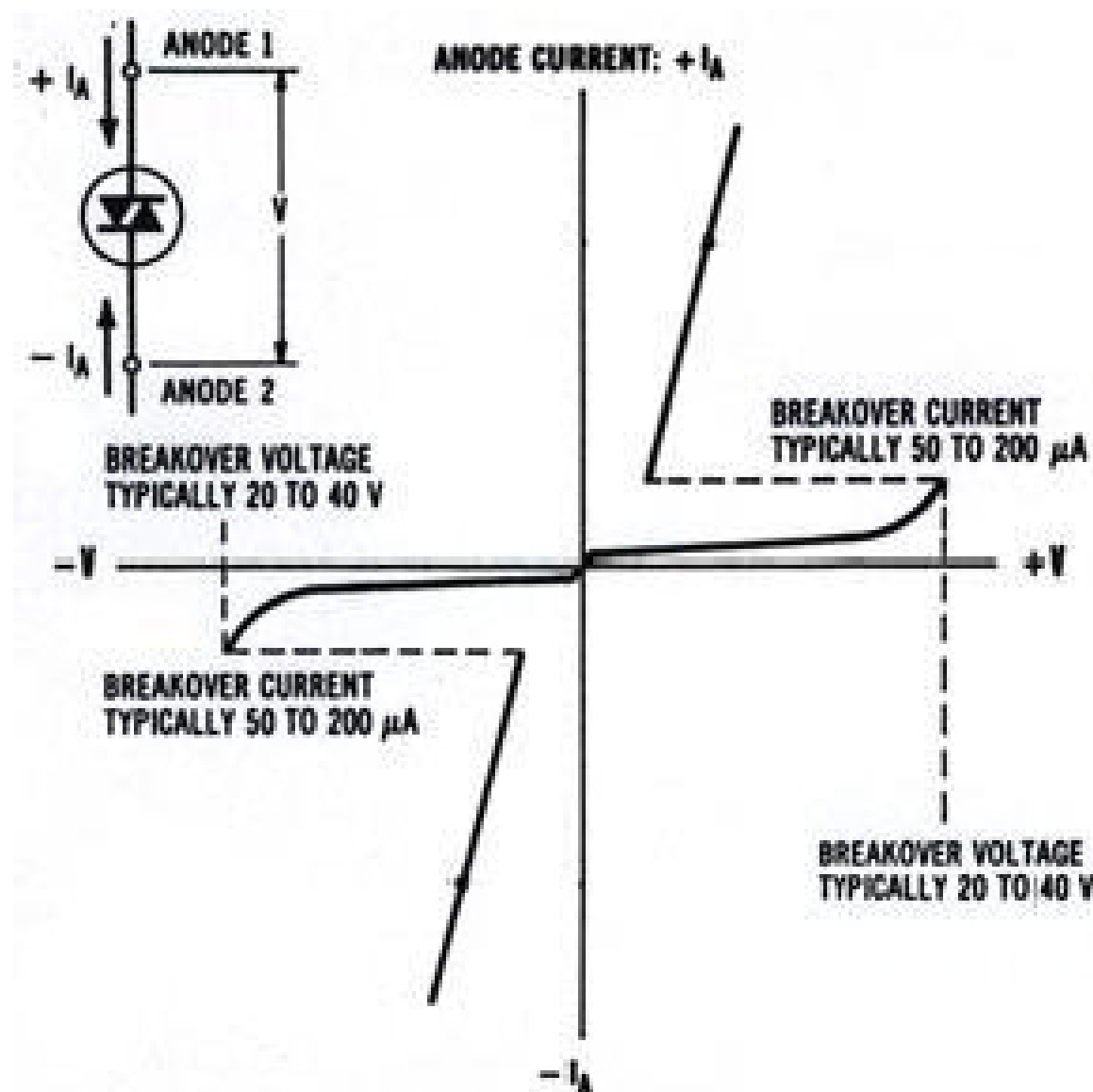
- ▶ **MT2 is negative and G is negative** : The current flows through the switch P2 N1 P1 N4 from MT 1 to MT 2. It is slightly less efficient than mode 1.
- ▶ The mode 1 and mode 4 are efficient mode in triac operations. Therefore these two modes are called normal mode in triac operations. However, the SCR's the triac is triggered with the gate pulse of proper polarity of DC voltages. The triac can be turned OFF only by reducing the device current below the holding value of the current.

5.10.2 VI CHARACTERISTICS OF A TRIAC:

- ▶ It gains the relationship between the triac current and the voltage applied across its two main terminals. We know about all the four modes of operation. Where the triac is operated, with its main terminal 2 and gate both positive with respect to main terminal 1. VI characteristics are shown in figure (The curve OABC). Similarly, where the triac is operated with MT2 and gate are both negative with respect to MT1 the characteristic is shown in figure (ODEF curve).

The VI characteristic has following important points:

- ✓ The curves OABC and ODEF are symmetrical and identical to the forward characteristic of SCR. The triac is off with the applied voltage of either (MT2 is positive with respect to MT1 or MT2 is negative with respect to MT1) exceeds the break over voltage.
- ✓ As the applied voltage of either polarity exceeds the break over voltage the Triac turns ON and the voltage drop across the triac decreases to a low value (V_H). The TRIAC current increases to a value determined by the supply voltage and load resistance.
- ✓ As the value of I_G is increased above zero the break over voltage is lowered. Like SCR the triac is never operated with $I_G = 0$. Where I_G of a suitable value is applied, it turns on at much lower break over voltage.



5.10.3 TRIAC RATINGS:

- ▶ **Latching current (i_l):** It is the amount of current the triac needs to stay in conduction after it has been initially turned on.
- ▶ **Holding current (i_h):** It is the main terminal current required after the triac has been turned ON.
- ▶ **Gate trigger current (i_{gt}):** It is the amount current that is necessary to go into conduction initially. It may be carefully noted that since the gate must be specific polarity for the triac to fire in a specific quadrant, it is possible to apply bias voltage of similar polarity to help the triac trigger with smaller gate current. The bias voltage of the opposite polarity can also be applied but this requires the application of a stronger gate current to get the triac into conduction.
- ▶ **Maximum average gate power (p_{gav}):** It is the average amount of power the gate required to put the gate required to put the triac into conduction.

5.10.4 MAXIMUM PEAK GATE POWER ($P_{GAV(max)}$):

It is the average amount of power the gate requires to put the triac into conduction.

- ▶ **Main terminal rms current ($i_{t rms}$):** It is the maximum amount of current that the triac can control. The value is listed as an RMS value because the triac is primarily used in AC circuits.
- ▶ **Break over voltage (v_{drom}):** It is not listed in forward breakover like the scr because the triac will go into conduction if the breakover voltage is either direction exceeds the

specified level. Like the SCR the breakover voltage rating to the triac should exceed the maximum peak circuit voltage. So it does not go into conduction if a spike of voltage occurs.

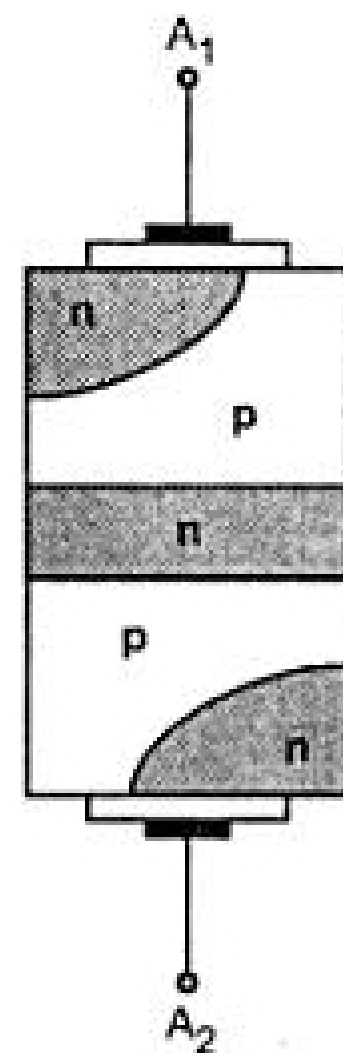
- ▶ Triacs are available commercially with typical breakover voltage rating (V_{DRM}) i.e. (100-1000) V in 100 volts increment. This ON state voltage is the voltage measured across MT1 and MT2 when the triac is in condition.
 - ✓ Reverse current (I_R)
 - ✓ Gate to cathode voltage or gate voltage (V_{GT})
 - ✓ ON state voltage across MT1 and MT 2 (V_{TH}) also called forward voltage (V_F).

5.10.5 APPLICATIONS OF TRIAC:

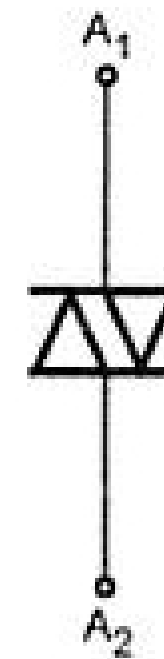
- ▶ The triac has an important property that it can conduct current in either forward or reverse direction, depending upon the polarity of the voltage across its terminal. This property makes the triac very useful in a large number of industrial applications.
 - ✓ Phase control
 - ✓ Motor speed control
 - ✓ Heater control
 - ✓ Light dimming control
 - ✓ Static switch to turn AC power ON and OFF

5.11 DIAC:

- The diac is a two terminal device, which can pass current in either direction where the breakover voltage is reached in either polarity across the two terminals. The basic construction of a diac is similar to a triac, but without a gate terminal as shown in figure.



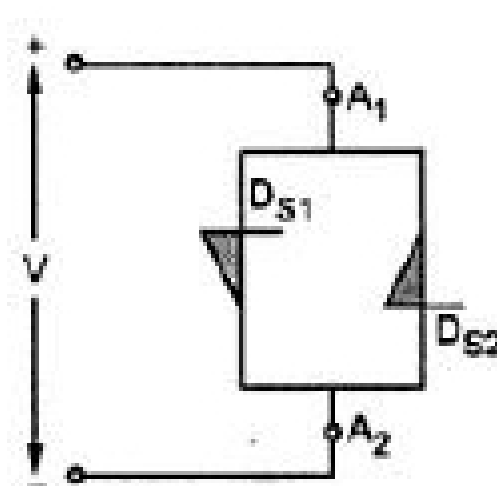
(a) Construction



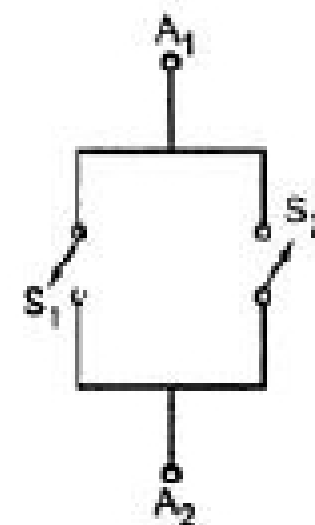
(b) Symbol

- It may be noted from the figure that a diac consist of two 4 layer diodes connected in parallel but is opposite directions. The four layer diodes are $P_1N_1P_2N_2$ AND $P_2N_1P_1N_3$ as shown by the broken lines in the figure since the diac conducts in both directions, therefore the terminal are designated by number instead of anode and cathode. The diac has two main terminals namely MT1 and MT2. The equivalent circuit is shown in figure.
- It consists of two 4 layer diodes connected in parallel but opposite direction.

5.11.1 OPERATION:



(a)

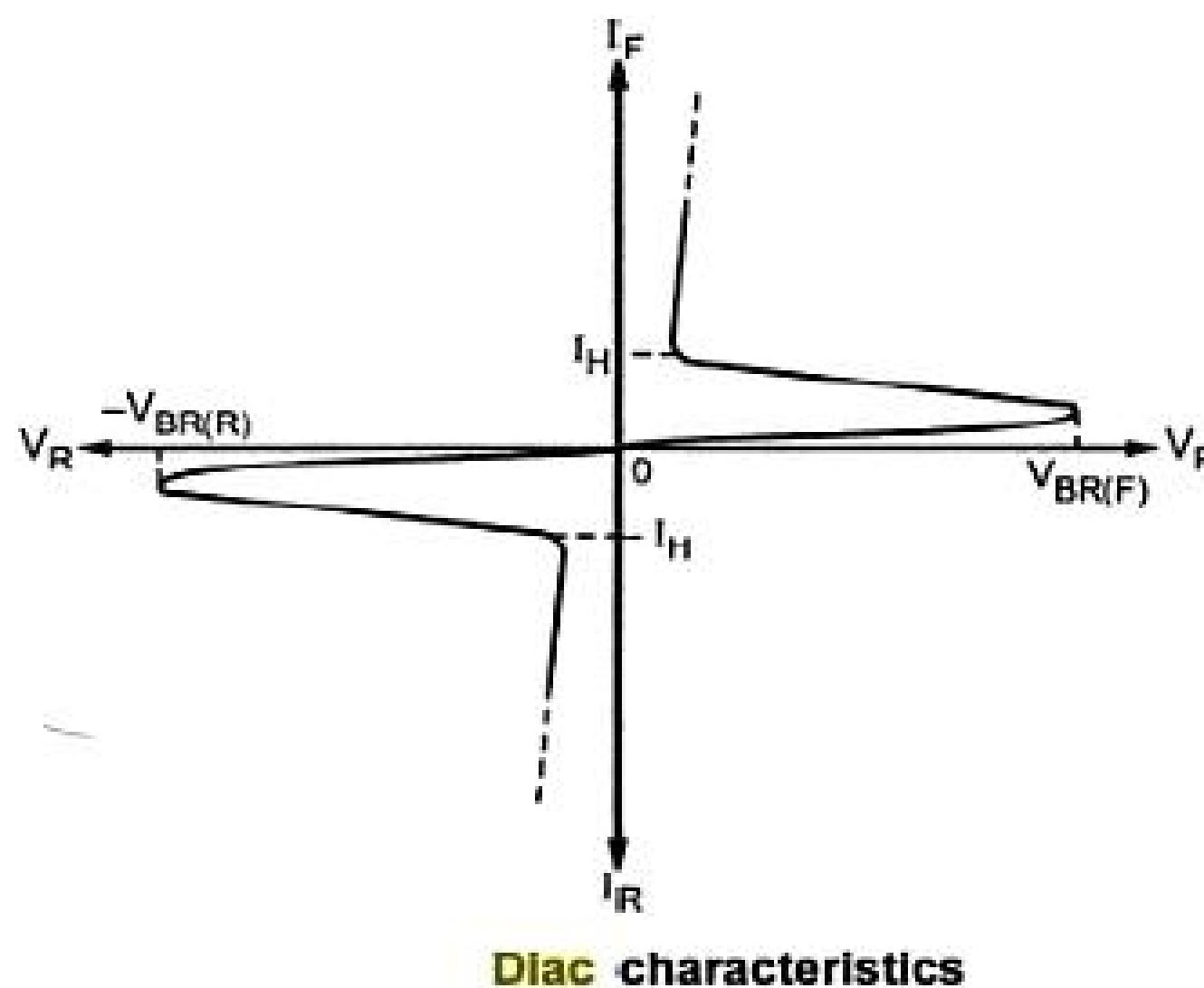


(b)

Diac operation

- ▶ The diac can pass current in either direction depending upon the polarity of voltage across its main terminals. It can be turned ON only when the applied voltage across its main terminals reaches the breakover voltage. When the applied voltage makes the MT2 positive with respect to MT1 the diac passed current through the diode P1N1P2N2 from MT2 to MT1 as shown in figure.
- ▶ However if the applied voltage makes MT2 negative with respect to MT1, the diac passes current through the diode P2N1P1N3 from MT1 to MT2 as shown in figure. The diac turns OFF when the current drops below the holding value. A diac is used as a triggering device for triac in phase control circuits such as light dimming, heat control and motor speed control etc.

5.11.2 VI CHARACTERISTICS OF DIAC:



- ▶ A diac can conduct in other directions, depending upon the polarity of the applied voltage across its main terminals. When a diac is operated with MT2 positive with respect to MT1 the VI characteristics attain as a curve OAB. Similarly when a diac is operated as a curve OCD.

It gives the important information in following points.

- ✓ The curve OAB and OCD are symmetrical and identical. It is true because the operation of a diac is identical in both directions of conduction.
- ✓ The diac does not pass any current until the applied voltage of either polarity reaches the breakover voltage.

- ✓ At the breakover voltage, the diac turns ON and the current through the diac increases rapidly as shown in figure.
- ✓ The operating voltage and current are the same in either direction therefore the diac is a symmetrical bilateral diode.
- ✓ The diac are available with a breakover voltage range from (20-36) v and maximum pulse current of 2A.

5.11.3 APPLICATIONS:

- ✓ Phase control.
- ✓ Motor speed control.
- ✓ Heater control.
- ✓ Lamp dimming control.